Co-ordinate Geometry

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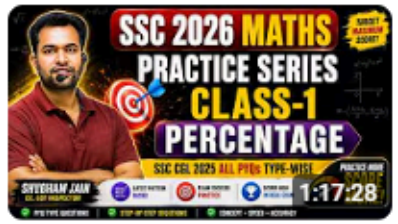

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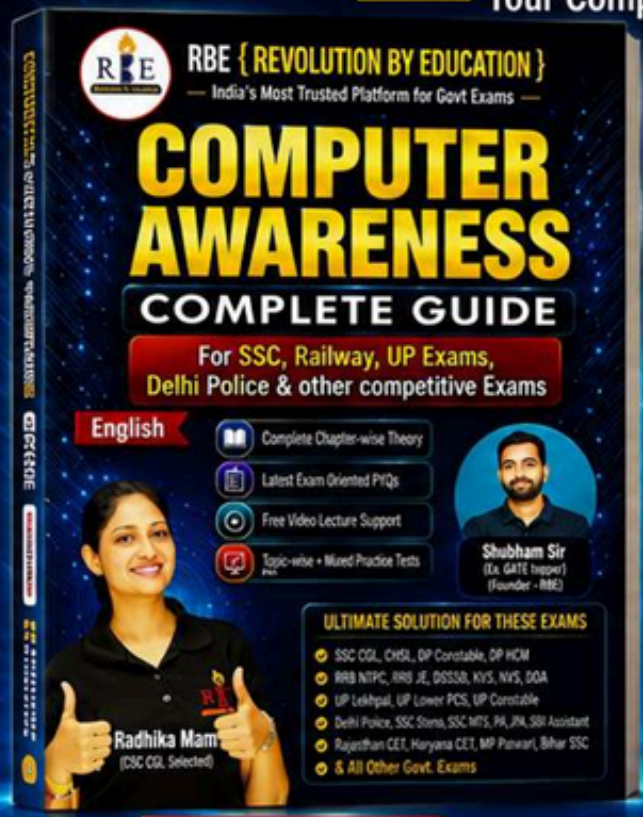

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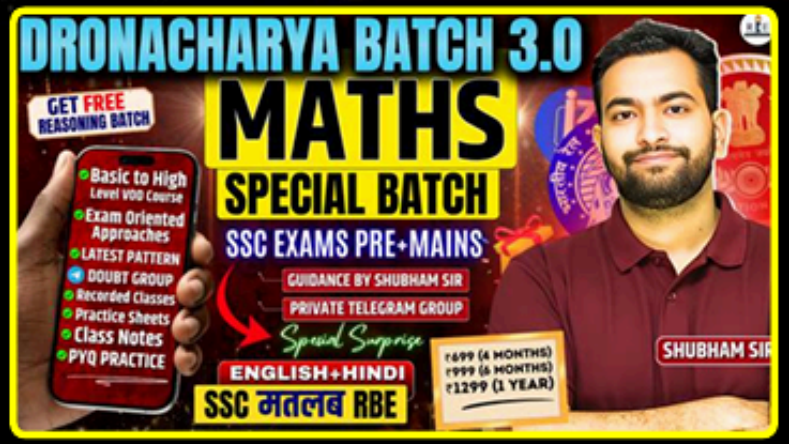

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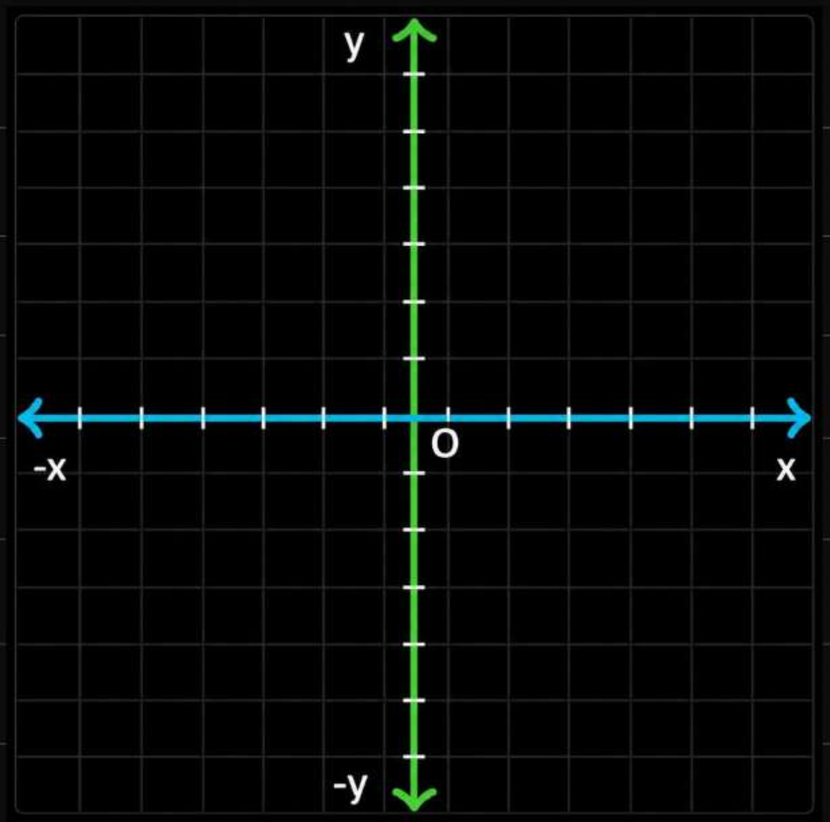
Co-ordinate Geometry

It is one of the branches of geometry where the position of a point is defined using coordinates.

यह ज्यामिति की उन शाखाओं में से एक है, जिसमें किसी बिंदु की स्थिति को निर्देशांकों (coordinates) का उपयोग करके परिभाषित किया जाता है।



Co-ordinate geometry provides the link between geometry & algebra
Where equations are presented in the form of lines and curves.

कोऑर्डिनेट ज्योमेट्री, ज्योमेट्री और अलजेब्रा के बीच संबंध बनाती है, जहाँ समीकरणों को रेखाओं और वक्रों के रूप में दिखाया जाता है।

WORKING OF GPS
GPS कैसे काम करता है

Latitude (अक्षांश)
Distance north or south of the Equator.
भूमध्य रेखा के उत्तर या दक्षिण की दूरी।

Longitude (देशांतर)
Distance east or west of the Prime Meridian.
प्रधान साम्य रेखा के पूर्व या पश्चिम की दूरी।

YOUR PRESENT LOCATION?
आपकी वर्तमान लोकेशन?

The **geographic coordinate system** (GCS) is a **spherical** or **ellipsoidal** coordinate system for measuring and communicating **positions** directly on the **Earth** as **latitude** and **longitude**.

अ जियोग्राफ़िक कोऑर्डिनेट सिस्टम (GCS) एक स्फ़ेरिकल (गोलाकार) या एलिप्सॉइडल (अंडाकार) कोऑर्डिनेट सिस्टम है, सिसका इस्तेमाल पृथ्वी पर सीधे अक्षांश (latitude) और देशांतर (longitude) के रूप में जगहों की स्थिति को मापने और बताने के लिए किया जाता है।

HOW GPS WORKS — GPS कैसे काम करता है

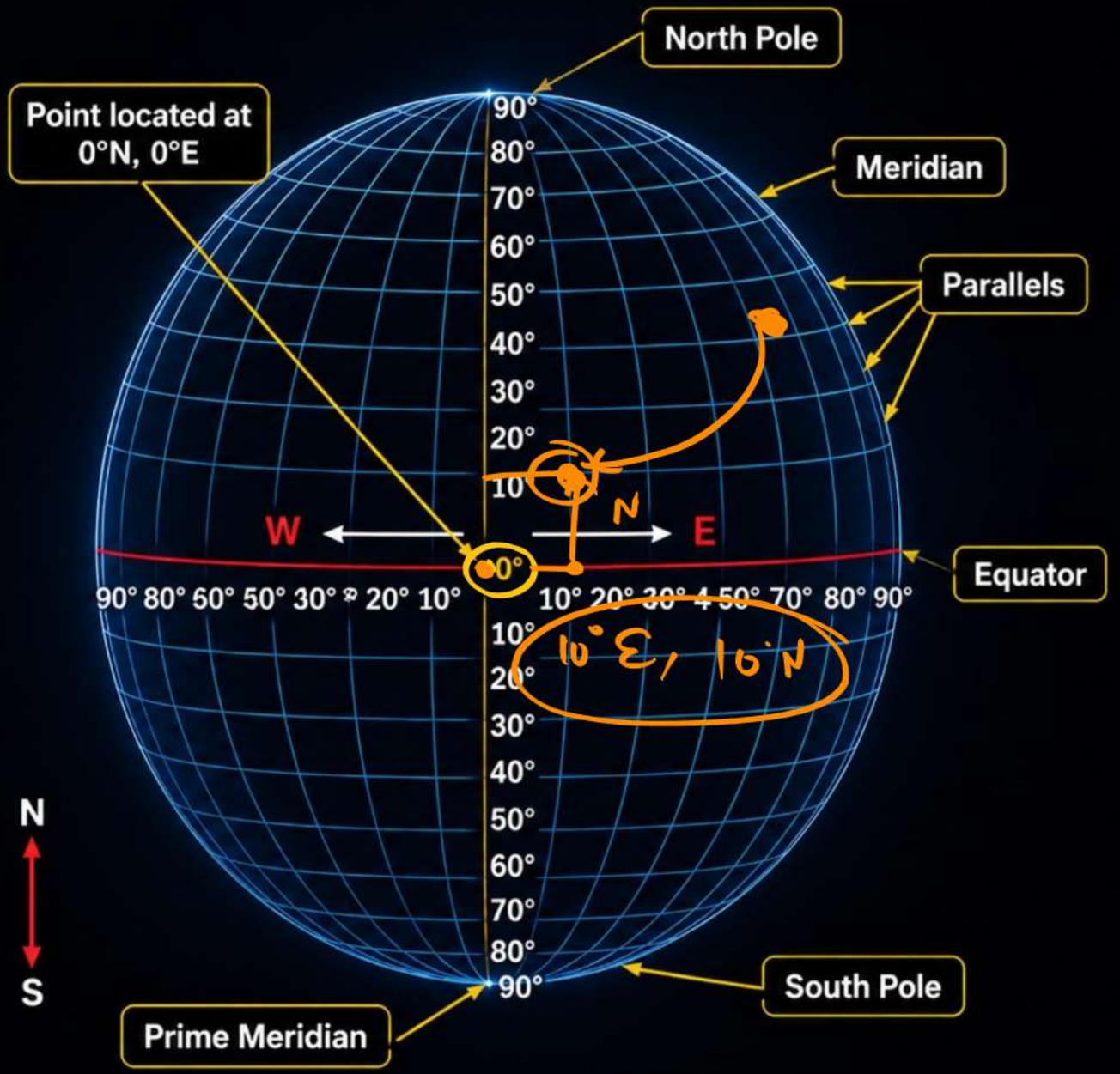
1. **Satellites** send signals continuously.
   सैटेलाइट लगातार संकेत भेजते हैं।
2. Your GPS receiver receives these signals.
   आपका GPS रिसीवर इन संकेतों को प्राप्त करता है।
3. It calculates the distance from multiple satellites.
   यह कई सैटेलाइट से दूरी की गणना करता है।
4. Using this data, your exact location (latitude & longitude) is determined.
   इस डेटा का उपयोग करके आपकी सटीक लोकेशन (अक्षांश और देशांतर) निर्धारित की जाती है।

USES OF GPS — GPS के उपयोग

- Navigation & Route Guidance — नेविगेशन और रूट गाइडेंस
- Agriculture & Precision Farming — कृषि और प्रीसीजन फार्मिंग
- Fleet Management & Tracking — फ्लीट मैनेजमेंट और ट्रैकिंग
- Surveying & Mapping — सर्वेक्षण और मैपिंग
- Disaster Management & Safety — आपदा प्रबंधन और सुरक्षा

Null Island is the point on the Earth's surface at zero degrees latitude and zero degrees longitude **(0°N 0°E),** i.e., where the prime meridian and the equator intersect.

'नल आइलैंड' पृथ्वी की सतह पर वह बिंदु है जो जीरो डिग्री अक्षांश और जीरो डिग्री देशांतर (0°N 0°E) पर स्थित है, यानी जहाँ प्राइम मेरिडियन और भूमध्य रेखा एक-दूसरे को काटती हैं।



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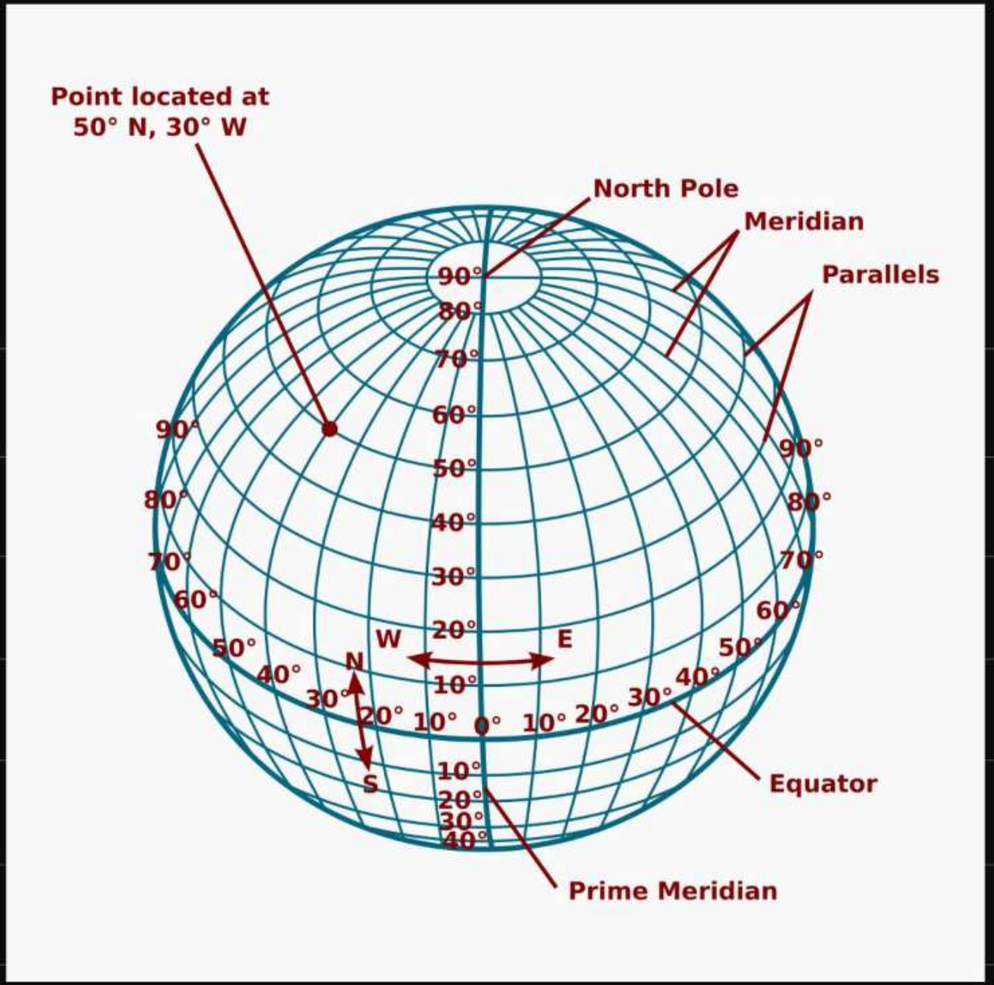


2D

3D

Sphere

Null

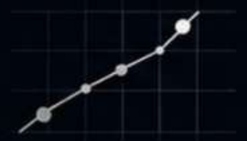
(MSL 9.124

Taj Mahal Co-ordinates

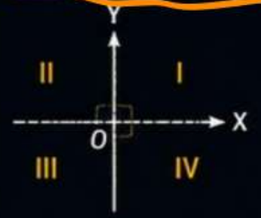
27.1751° N, 78.0421° E

CO-ORDINATE GEOMETRY

COMPLETE TOPIC LIST



1 COORDINATE PLANE & BASICS



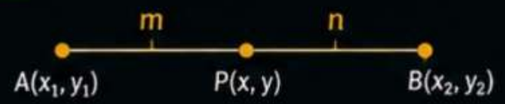
- Cartesian coordinate system
- X-axis, Y-axis, Origin
- Quadrants
- Coordinates of a point
- Signs of coordinates in different quadrants

2 DISTANCE FORMULA

$$d = \sqrt{(x_2-x_1)^2 + (y_2-y_1)^2}$$

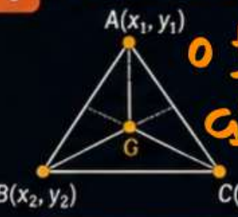
- Distance between two points
- Distance from origin
- Applications of distance formula

3 SECTION FORMULA



- Internal division
- External division
- Midpoint formula
- Finding coordinates in a given ratio

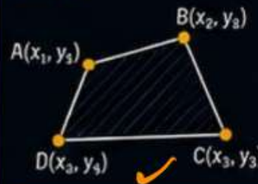
4 CENTROID OF A TRIANGLE



- Coordinates of centroid
- Finding missing vertex using centroid

$$G\left(\frac{x_1+x_2+x_3}{3}, \frac{y_1+y_2+y_3}{3}\right)$$

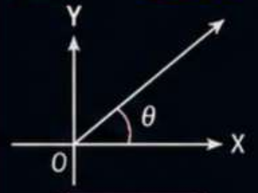
5 AREA USING COORDINATES



- Area of triangle
- Shoelace / determinant method
- Collinearity of three points
- Area of quadrilateral / polygon

$$\text{Area of Triangle} = \frac{1}{2}\left|x_1(y_2-y_3) + x_2(y_3-y_1) + x_3(y_1-y_2)\right|$$

6 SLOPE OF A LINE



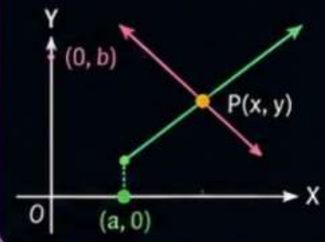
$$m = \tan\theta$$

- Slope formula
- Slope from angle of inclination
- Parallel lines
- Perpendicular lines
- Collinearity using slopes

7 EQUATION OF A STRAIGHT LINE

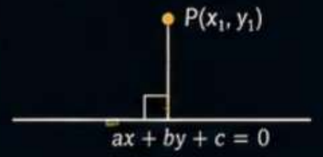
- Slope-intercept form $y = mx + c$
- Point-slope form $y - y_1 = m(x - x_1)$
- Two-point form $\frac{y-y_1}{y_2-y_1} = \frac{x-x_1}{x_2-x_1}$
- Intercept form $\frac{x}{a} + \frac{y}{b} = 1$
- General form $ax + by + c = 0$
- Equation of line parallel / perpendicular to another line

8 INTERCEPTS & INTERSECTION OF LINES



- X-intercept and Y-intercept
- Point of intersection of two lines
- Angle between two lines

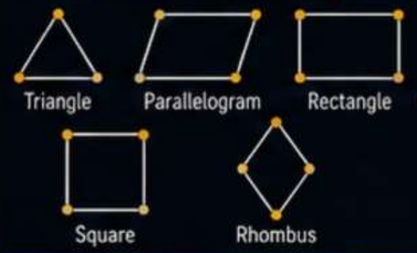
9 DISTANCE RELATED TO LINES



$$\text{Distance} = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

- Distance of a point from a line
- Distance between two parallel lines

10 COORDINATE GEOMETRY OF GEOMETRICAL FIGURES

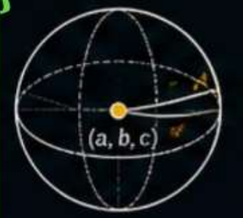


- Identifying figures using coordinates

11 CIRCLE

- Standard equation of circle $(x-h)^2 + (y-k)^2 = r^2$
- Centre and radius from equation
- Circle with given centre and radius
- Circle with diameter endpoints given
- General equation of circle $x^2 + y^2 + 2gx + 2fy + c = 0$

12 SPHERE



- Standard equation of sphere $(x-a)^2 + (y-b)^2 + (z-c)^2 = r^2$
- Centre and radius
- Finding equation from given centre and radius
- General equation of sphere $x^2+y^2+z^2+2gx+2fy+2hz+c=0$

IMPORTANT FOR EXAMS: • DISTANCE → SECTION FORMULA → AREA → SLOPE → STRAIGHT LINE → CIRCLE → SPHERE

CO-ORDINATE GEOMETRY

निर्देशांक ज्यामिति

2D

It is one of the branches of geometry where the position of a point is defined using coordinates.
यह ज्यामिति की एक शाखा है जिसमें बिंदु की स्थिति को निर्देशांकों (Coordinates) की सहायता से निर्धारित किया जाता है।

$2x+3y=4$
$7x+10=9$

x, y

WHAT IS CO-ORDINATE GEOMETRY?

Co-ordinate geometry provides the link between geometry and algebra.
Where equations are presented in the form of lines and curves.

निर्देशांक ज्यामिति ज्यामिति और बीजगणित के बीच एक सेतु का कार्य करती है।
इसमें समीकरणों को रेखाओं और वक्रों के रूप में प्रस्तुत किया जाता है।

WHY IS IT IMPORTANT?

- Helps in locating points precisely.
  बिंदुओं को सटीक रूप से खोजने में मदद करती है।
- Used in real-life applications like GPS, maps, navigation, engineering, etc.
  GPS, मानचित्र, नेविगेशन, इंजीनियरिंग आदि में उपयोगी है।
- Forms the base for higher mathematics and science subjects.
  उच्च गणित और विज्ञान विषयों की नींव है।

CO-ORDINATES (निर्देशांक)

Coordinates are a set of values which helps to show the exact position of a point in the coordinate plane.

P (x, y)

A (1,2)

Here, 'x' is the x-coordinate and 'y' is the y-coordinate.

निर्देशांक मूल्यों का एक समूह है, जो निर्देशांक तल में किसी बिंदु की सटीक स्थिति को दर्शाता है।

P (x, y)

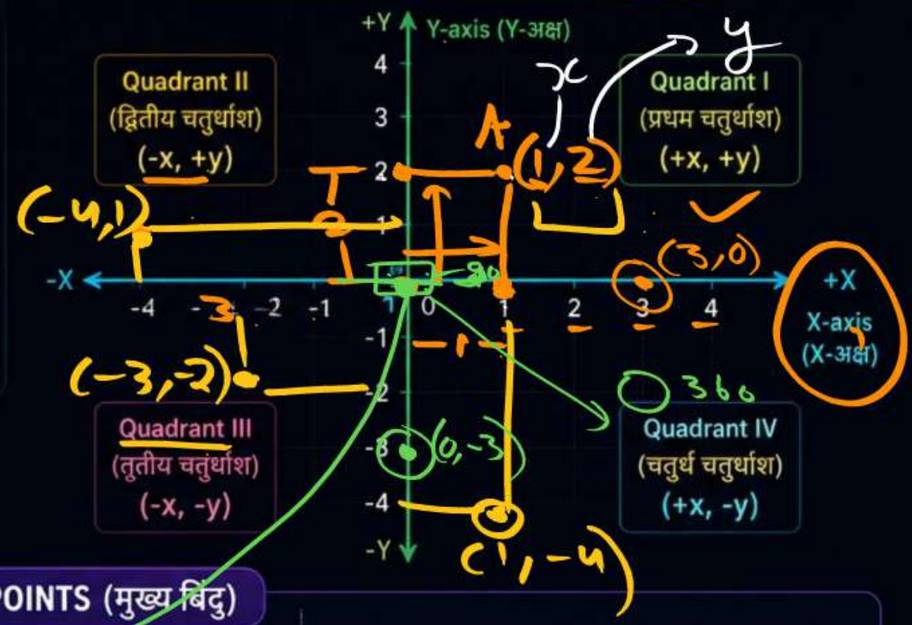
यहाँ, 'x' x-निर्देशांक (x-coordinate) और 'y' y-निर्देशांक (y-coordinate) है।

COORDINATE PLANE (निर्देशांक तल)

A coordinate plane is a 2D plane which is formed by the intersection of two perpendicular lines known as the x-axis and y-axis.

निर्देशांक तल एक 2D तल है, जो दो लम्बवत रेखाओं — x-अक्ष और y-अक्ष के प्रतिच्छेद से बनता है।

THE COORDINATE PLANE (निर्देशांक तल)



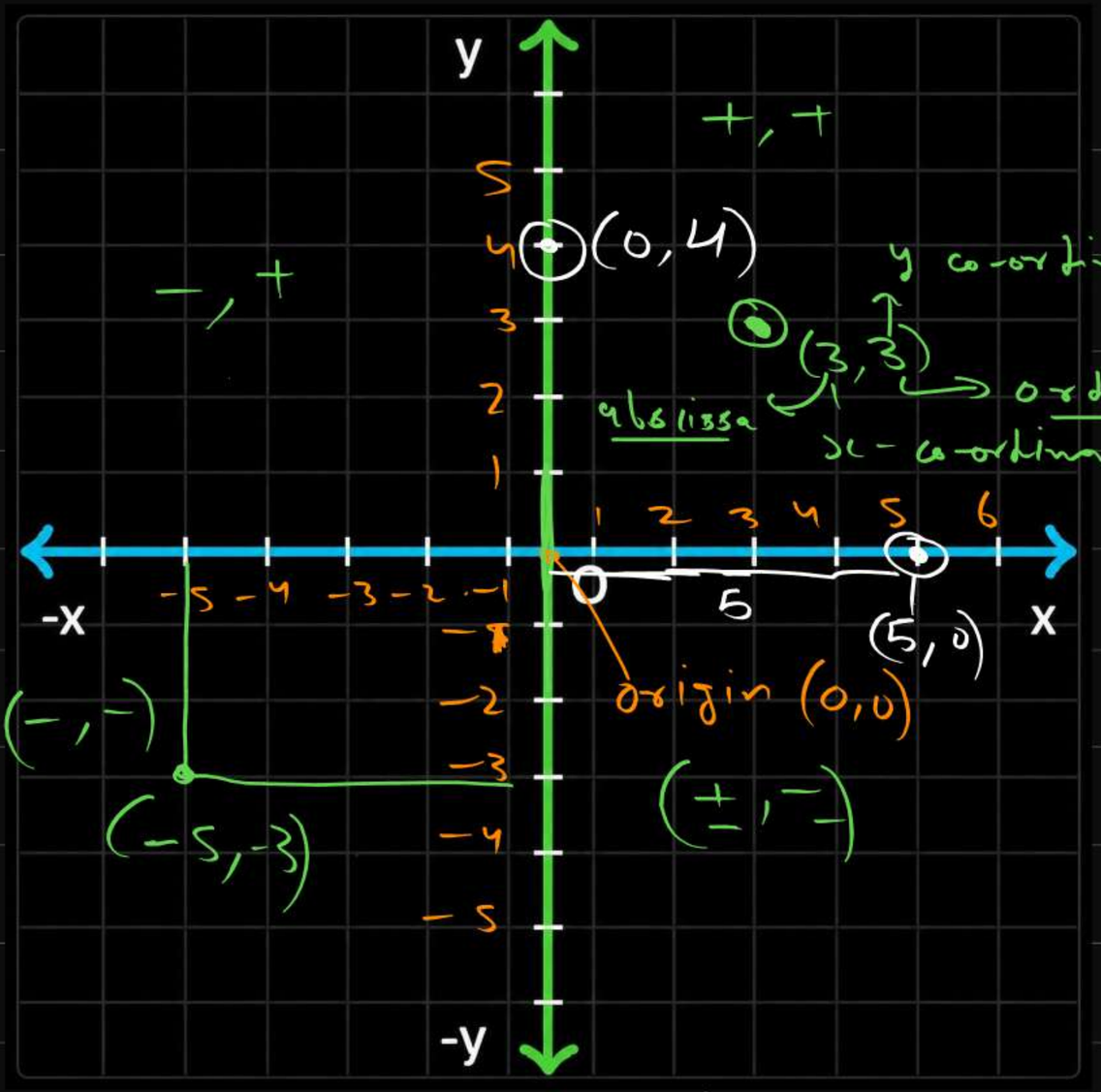
KEY POINTS (मुख्य बिंदु)

- Origin (मूल बिंदु): (0, 0) (0,0)
  जहाँ दोनों अक्ष एक-दूसरे को काटते हैं।
- Points on X-axis (x-अक्ष पर बिंदु): (x, 0)
  y = 0 होता है।
- Points on Y-axis (y-अक्ष पर बिंदु): (0, y)
  x = 0 होता है।
- In Quadrant I (प्रथम चतुर्थांश): (+x, +y)
  दोनों निर्देशांक धनात्मक होते हैं।
- In Quadrant II (द्वितीय चतुर्थांश): (-x, +y)
  x ऋणात्मक, y धनात्मक होता है।
- In Quadrant III (तृतीय चतुर्थांश): (-x, -y)
  दोनों निर्देशांक ऋणात्मक होते हैं।
- In Quadrant IV (चतुर्थ चतुर्थांश): (+x, -y)
  x धनात्मक, y ऋणात्मक होता है।

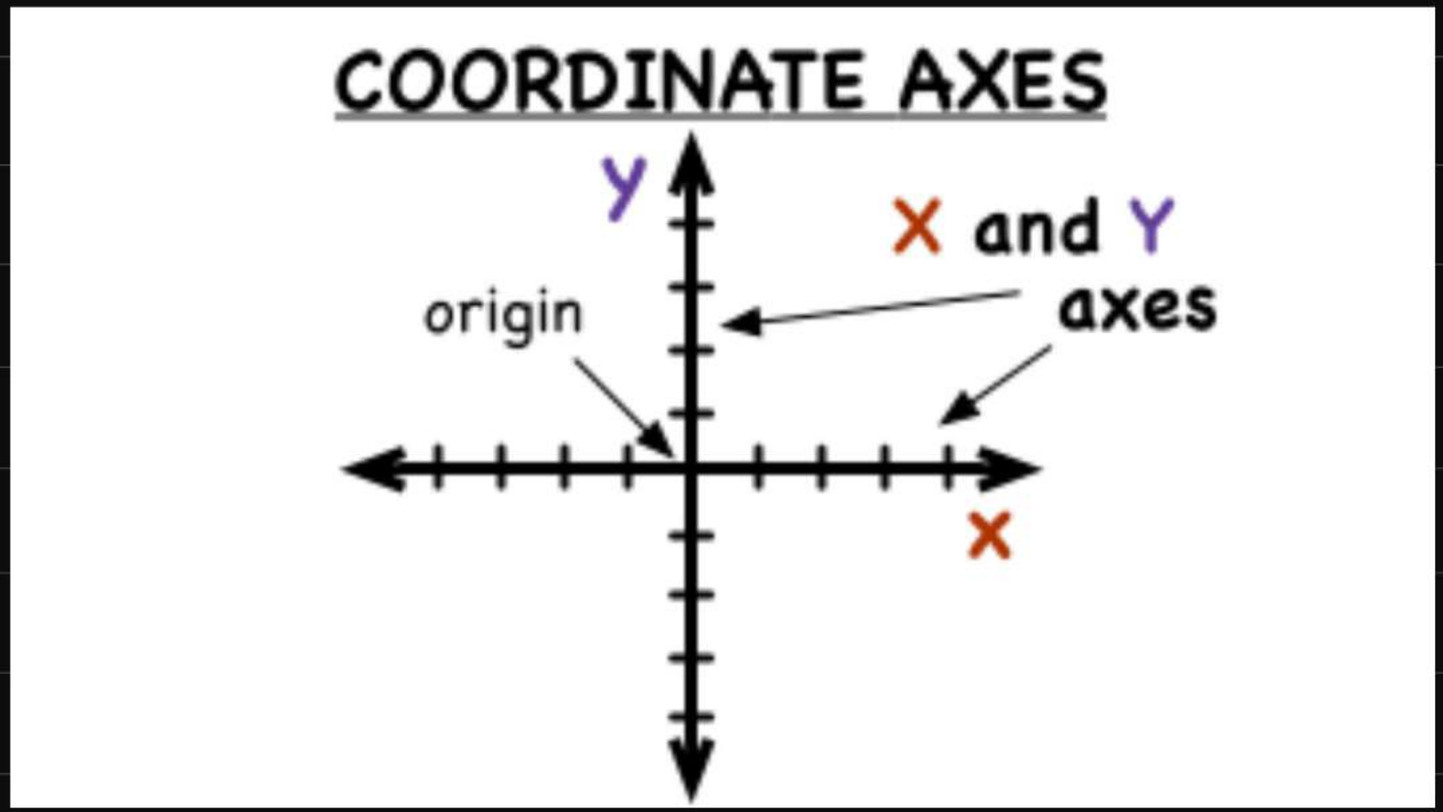
IN SHORT (संक्षेप में): Coordinate Geometry is the art of locating points and representing shapes using numbers (coordinates) on a plane.

निर्देशांक ज्यामिति वह कला है जिसमें संख्याओं (निर्देशांकों) की सहायता से तल पर बिंदु और आकृतियों को दर्शाया जाता हैं।

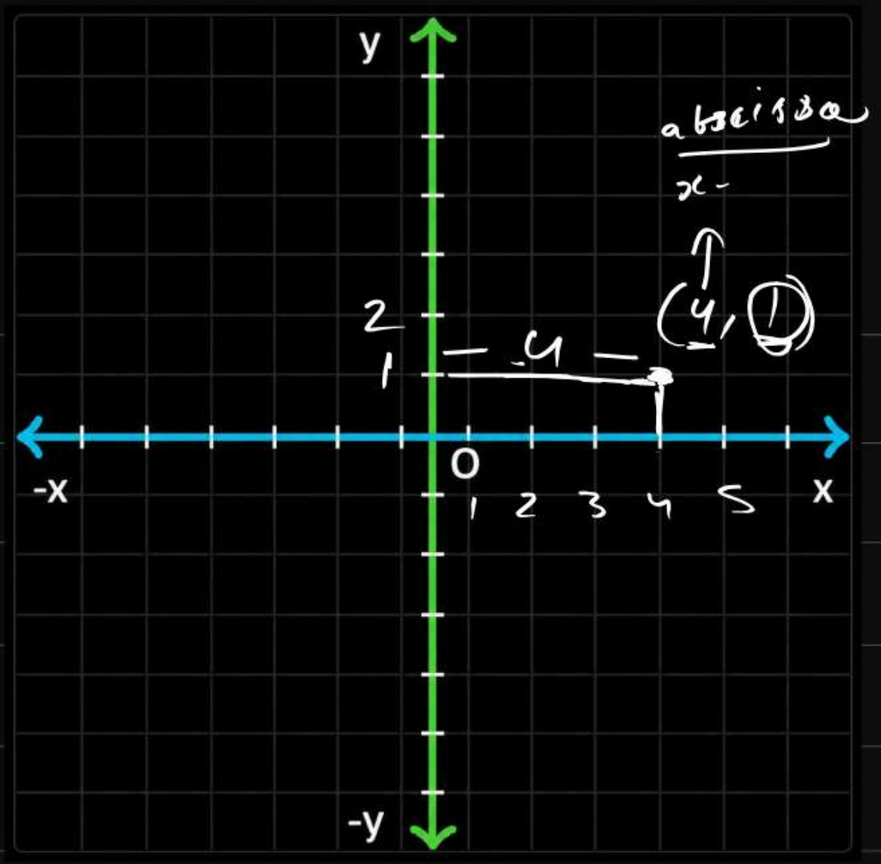
Co-ordinate Geometry



निर्देशांक अक्ष



Mapping of points



The distance of a point from the *y*-axis (y-अक्ष) is called its ***x*-coordinate** **x- निर्देशांक**, or **abscissa** (भुज).

The distance of a point from the *x*-axis (x-अक्ष) is called its ***y*-coordinate** **y- निर्देशांक**, or. **Ordinate** (कोटि)

The coordinates of a point on the *x*-axis are of the form (*x*, 0), and of a point on the *y*-axis are of the form (0, *y*).

$Ax + By = C$

Plotting Line in co-ordinate plane

$y = \frac{24}{8}$

$8x + 3y = 24$

x	0	3	1	−2
y	8	0	16/3 = 5.33	13.33

$8(1) + 3y = 24$

$y = \frac{16}{3}$

$2x + 8 = y$

$8x + 3y = 24$

$(2x + 8 = y) \times 4$

$3y - 32 = 24 - 4y$

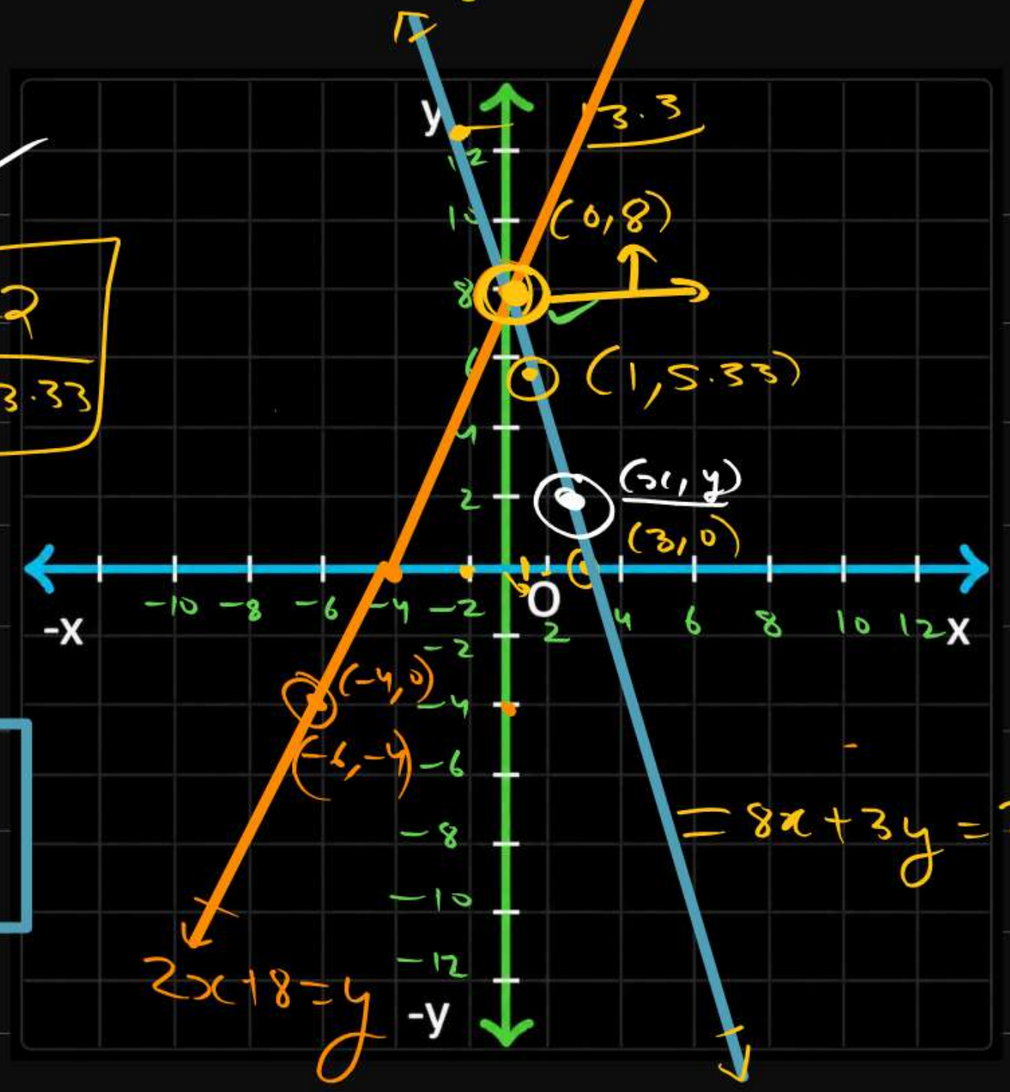
$7y = 56$

$y = 8$

$2x + 8 = 8$

$x = 0$

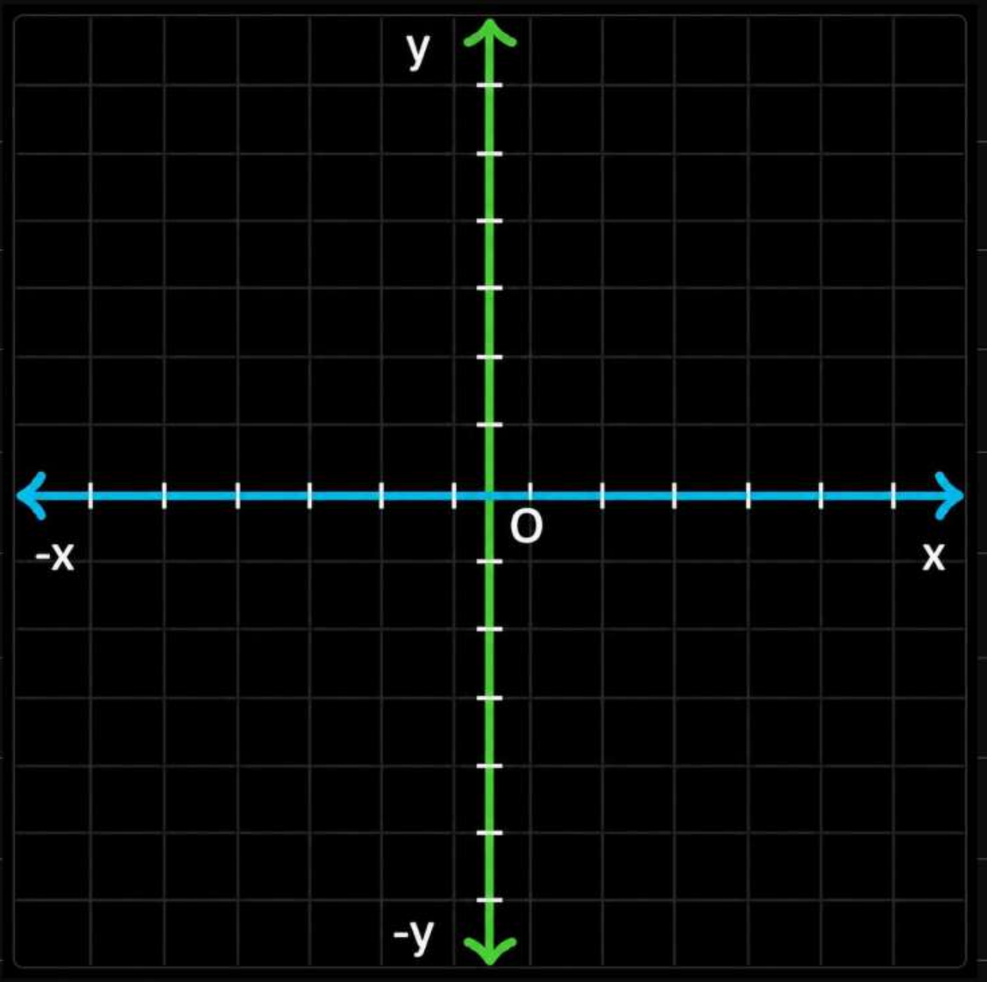
x	0	−4	−6
y	8	0	−4



Plotting Line in co-ordinate plane

$8x+3y=24$

$2x+8=y$



Horizontal and vertical lines
क्षैतिज और ऊर्ध्वाधर लाइनें

(Line segment)

$0x + y = 2$

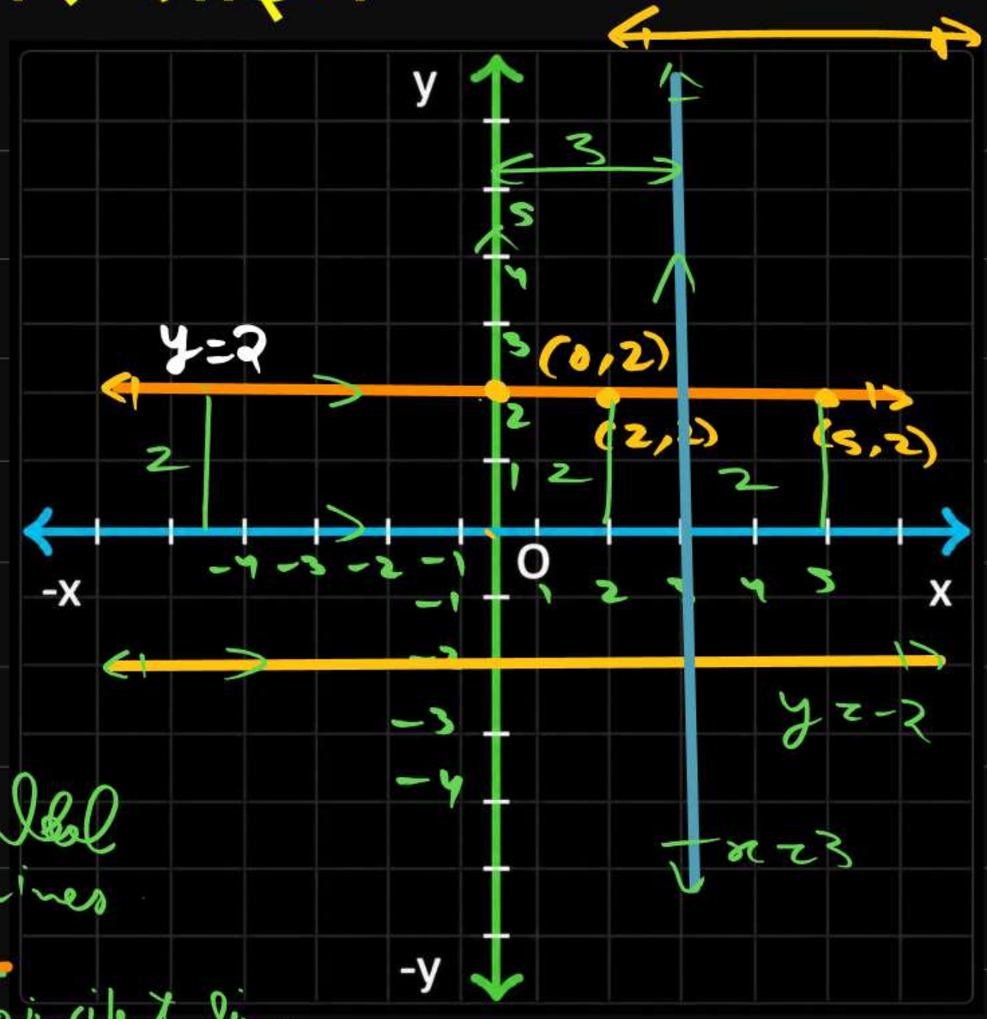
x	0	1	3
y	2	2	2

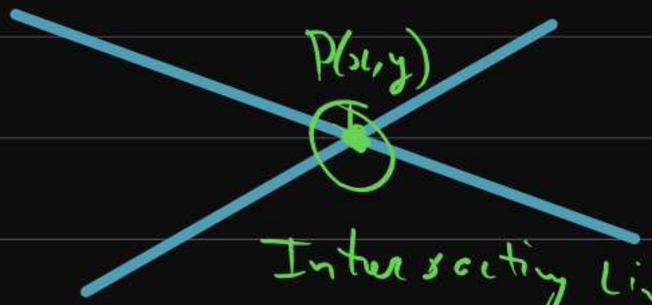
Y= 2 ⇒

Y = -2

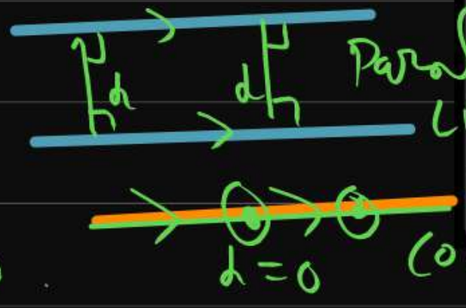
x= 3

Y = -3





Intersecting lines.

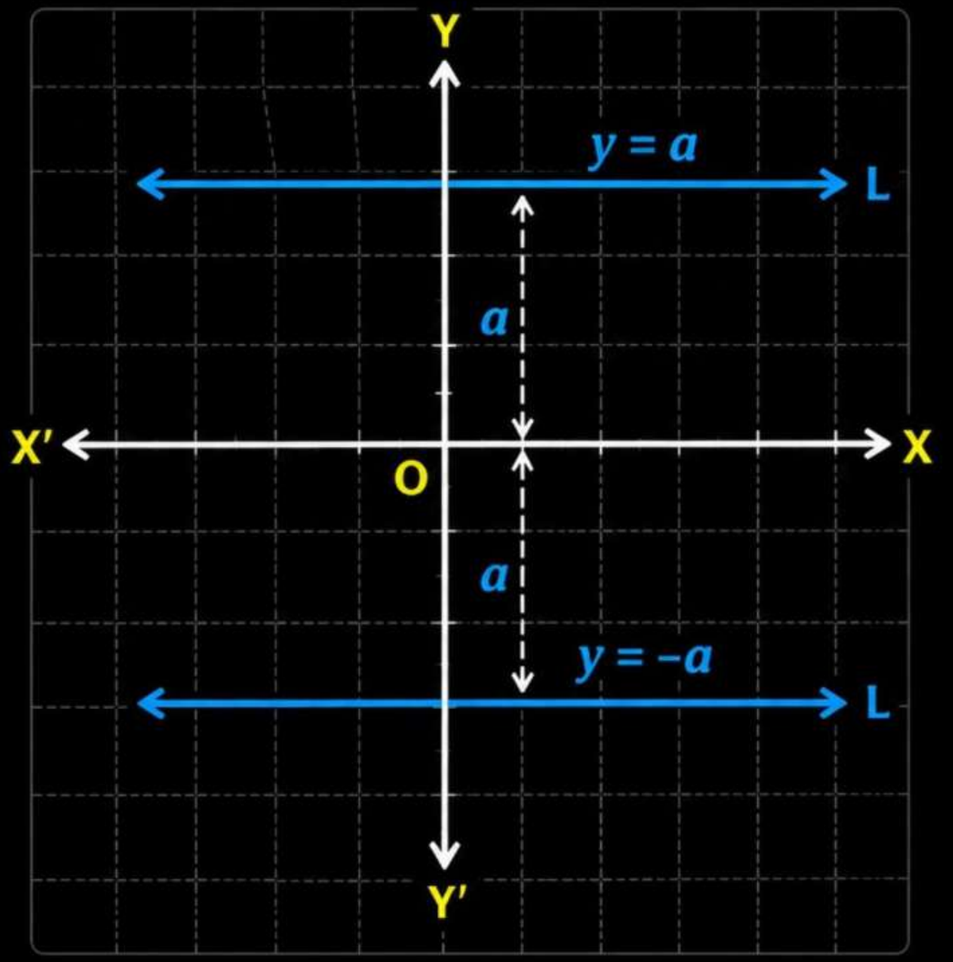


Parallel lines

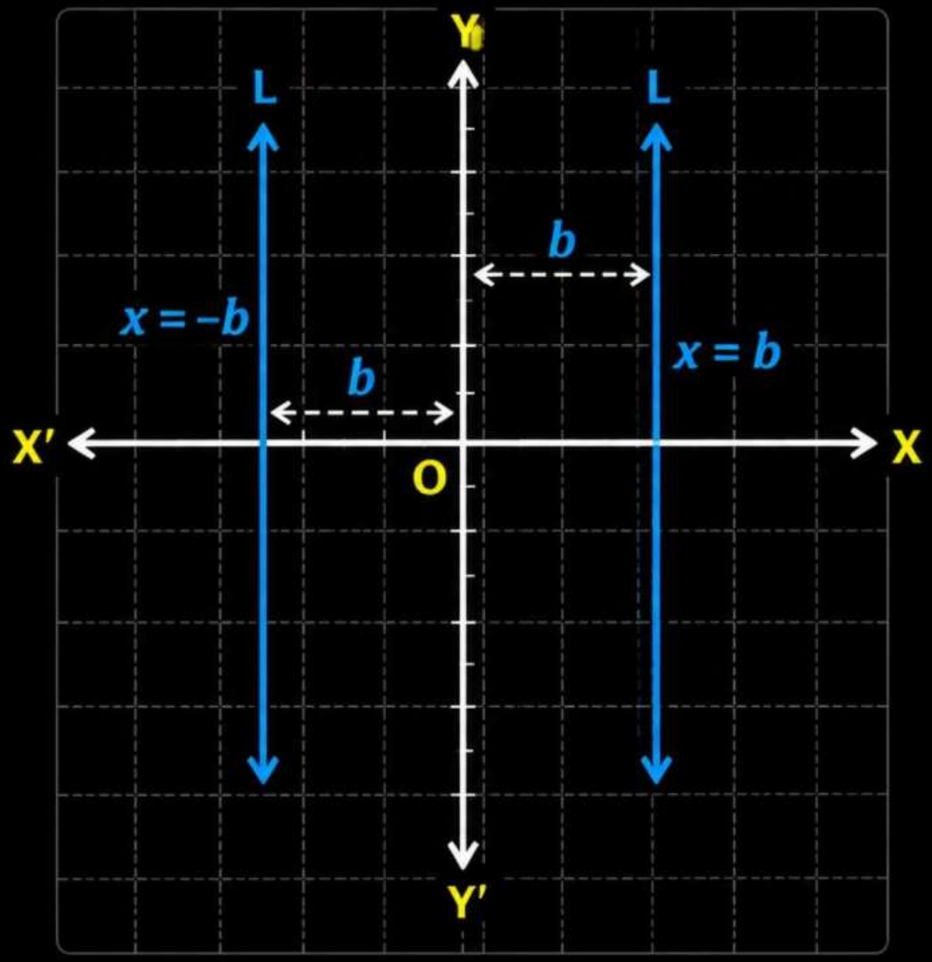
Coincident line.

Horizontal and vertical lines

क्षैतिज और ऊर्ध्वाधर लाइनें



(a)



(b)

Q. Which point lies on $y = -3x + 7$?

कौन सा बिंदु $y = -3x + 7$ पर स्थित है ?

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$y = -3x + 7$

0

$7 = -3(0) + 7 = 7$

$1 = -3(2) + 7 = 1$

LHS = RHS

$-2 = -3(3) + 7$

$-2 = -2$

(a) (0, 7)
(b) (2, 1)
(c) (3, −2)
(d) (0,7), (2,1) and (3,−2) all

Answer: 4

8 3

y = mx + 2

8 = 3x + 2

$x = \frac{6}{3} = 2$

Q. A line y = mx + 2 passes through (3, 8). Find m.

एक रेखा y = mx + 2, (3, 8) से होकर गुजरती है। m ज्ञात करें।

SSC CGL 2025

(a) 5 (b) 4 (c) 3 (d) 2

Answer: 4

Q. The graph of $y = 4x - 5$ passes through which point?

$y = 4x - 5$ का ग्राफ किस बिंदु से होकर गुजरता है?

$y = (4x - 5) \Rightarrow -1$

$= 4(0) - 5 = -5$

SSC CGL 2025

(a) (1, -1) (b) (0, -5) (c) (2, 3) (d) All the above

Answer: 4

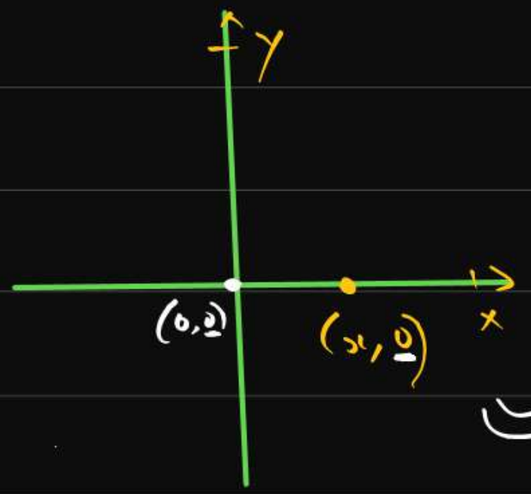
Q. Find the point where y = 5x − 10 crosses the x-axis.

एक त्रिभुज ABC में , भुजा BC को एक बिंदु D तक बढ़ाया जाता है I C पर बाह्य कोण 105° है I A और B पर आंतरिक कोणों का अनुपात 2:5 है I त्रिभुज के सबसे बड़े आंतरिक कोण का माप क्या है ?

$y = 5x - 10$

$0 = 5x - 10$

$x = \frac{10}{5} = 2$



SSC CGL 2025

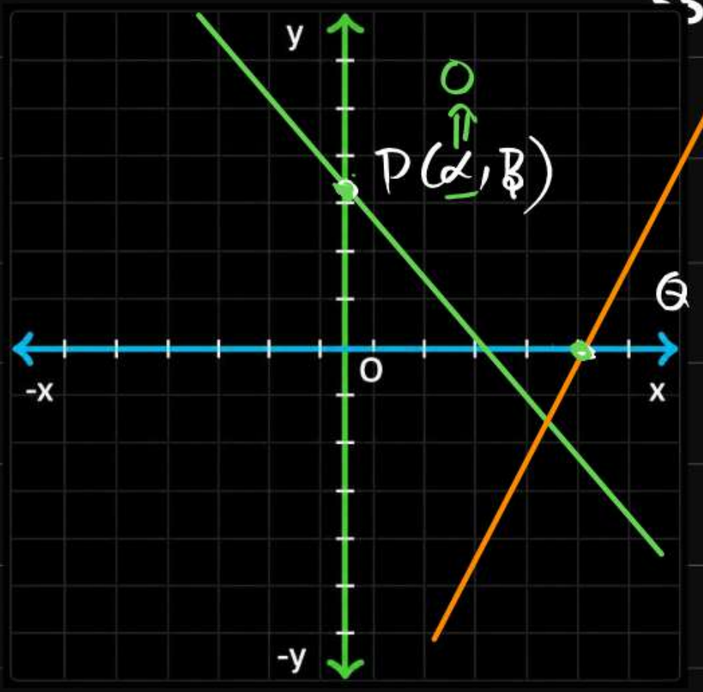
(a) (2, 0) (b) (0, −10) (c) (−2, 0) (d) (2, 10)

Answer: 1

Q. The graph of the equation x - 7y = - 42, intersects the y-axis at P(α, β) and the graph of 6x + y - 15 = 0, intersects the x-axis at Q(γ, δ). What is the value of $\alpha + \beta + \gamma + \delta$?

समीकरण x - 7y = - 42 का आलेख, y-अक्ष को P(α, β) पर प्रतिच्छेद करता है और 6x + y - 15 = 0 का आलेख x-अक्ष को Q(γ, δ). पर प्रतिच्छेद करता है।तो $\alpha + \beta + \gamma + \delta$ का मान क्या है

SSC CGL 2018 tier-2



(a) $\frac{17}{2}$
(b) 6
(c) $\frac{9}{2}$
(d) 5

x - 7y = - 42

$0 - 7\beta = -42$

$\beta = 6$

6x + y - 15 = 0

$6\gamma + 0 = 15$

$\gamma = 2.5$

$\alpha + \beta + \gamma + \delta$

$0 + 6 + 2.5 + 0 = 8.5 = \frac{17}{2}$

Q. Find the solution to the system: y = 3x - 4 and y = -x + 2.

समीकरणों के इस युग्म का हल ज्ञात कीजिए: y = 3x - 4 तथा y = -x + 2.

SSC CGL 2025

(a) (1, -1) (b) (0, 2)
(c) (1.5, 0.5) (d) (1.5, -3.5)

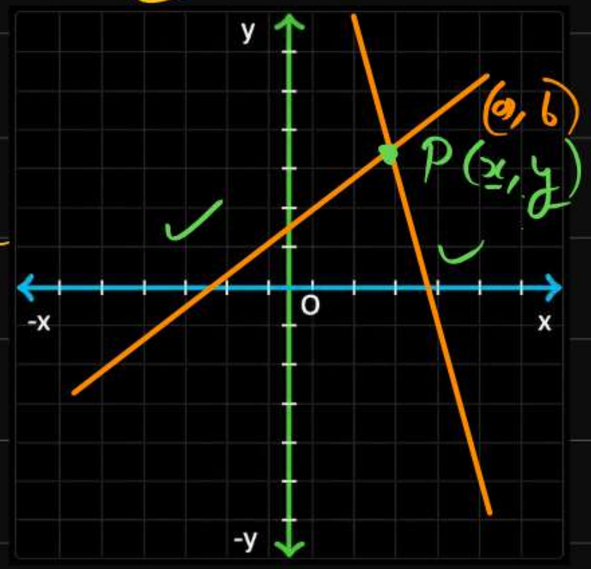
$y = 3x - 4$ $\quad b = 3a - 4$

$y = -x + 2$ $\quad (b = -a + 2)$ (−)

$0 = 4a - 6$

$a = \frac{3}{2} = 1.5$

$b = \frac{1}{2} = 0.5$



Answer: 3

Q. A drone flies along $y = -x + 10$. Another follows $y = x - 4$. At what point do they intersect?

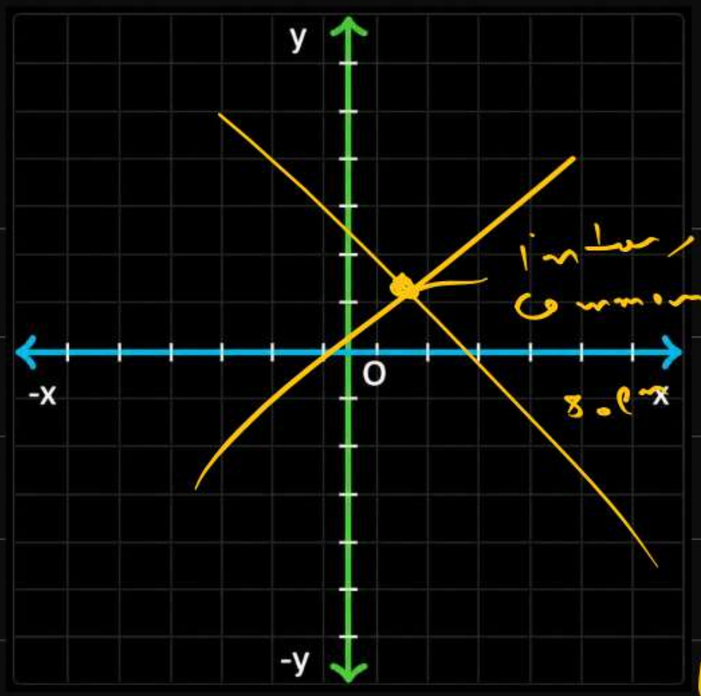
एक ड्रोन $y = -x + 10$ के अनुदिश उड़ता है। दूसरा $y = x - 4$ के अनुदिश उड़ता है। वे किस बिंदु पर प्रतिच्छेद करते हैं?

SSC CHSL 2025

(a) (7,3) (b) (6,4) (c) (5,5) (d) (4,6)

$$y = -x + 10$$
$$y = x - 4$$

$$0 = -2x + 14$$
$$x = 7$$
$$y = 3$$



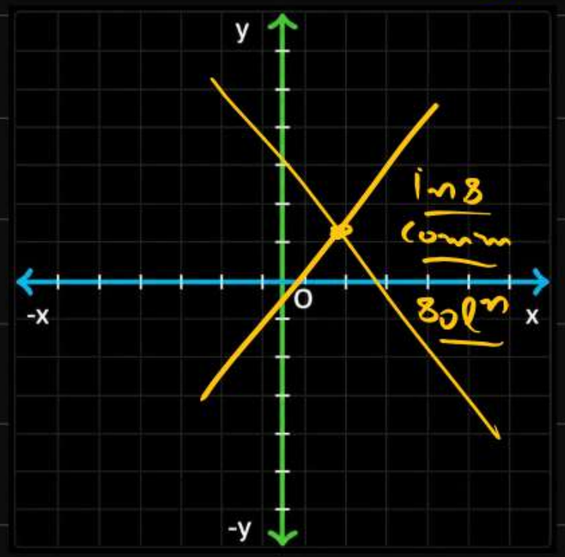
Answer: 1

Q. If equation of line p is x + y = 5 and that of line q is x – y = 3, what are the coordinates of the point common to both the lines?

यदि रेखा p का समीकरण x + y = 5 है और रेखा q का समीकरण x - y = 3 है, तो दोनों रेखाओं में उभयनिष्ठ बिंदु के निर्देशांक क्या हैं?

$$x + y = 5 \quad \text{(4)}$$
$$x - y = 3 \quad \text{(1)} \quad +$$

$$2x = 8$$
$$x = 4$$
$$y = 1$$



SSC CHSL 2024 T-1

(a) (2, 1)
(b) (2, 3)
(c) (1, 4)
(d) (4, 1)
Ans: d

Q. If the intersection point of the equations 3p – 2q + 23 = 0 and –2p + q – 13 = 0 on the graph is (p, q), then the value of (3p + q) is equal to:

यदि ग्राफ पर समीकरण 3p – 2q + 23 = 0 और –2p + q – 13 = 0 का प्रतिच्छेदन बिंदु (p, q) है, तो (3p + q) का मान ___ होगा।

$3p - 2q = -23$

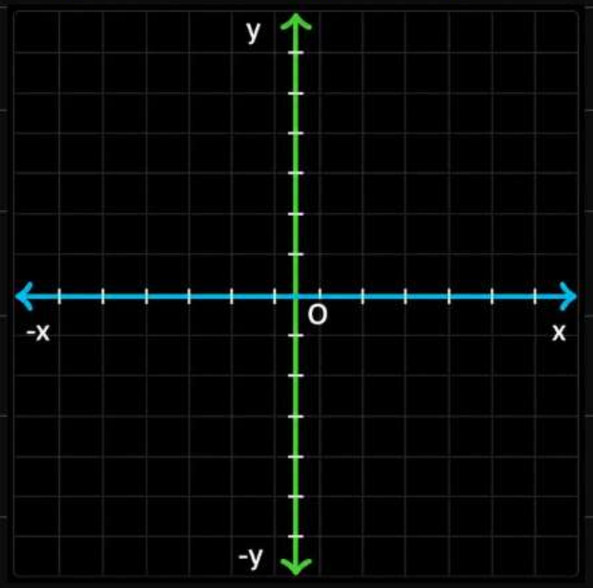
$(-2p + q - 13 = 0) \times 2$

$-p \Rightarrow 3$

$p = -3$

$q = 7$

$3p + q = 3(-3) + 7 = -2$



(a) –2
(b) –1
(c) 2
(d) 1
Ans: a

SSC CHSL 2024 T-1

a, b ⇒ x, y

$$3x - 20y = 2 \quad \times(-6)$$
$$11x - 5y = -61 \quad \times 4$$

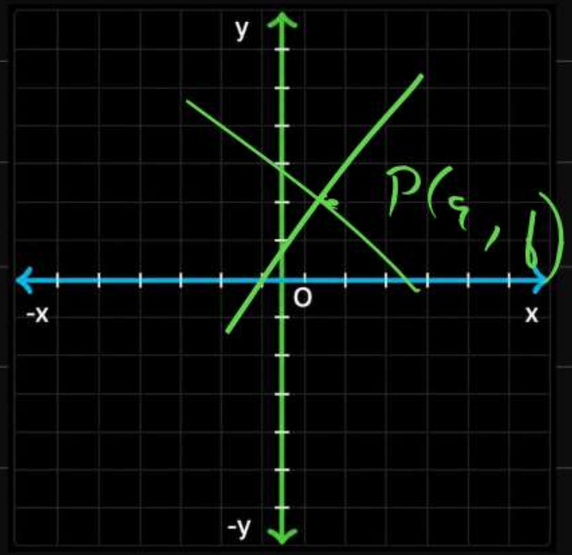
$$-41x = 2 + 244 = 246$$

$$x = -6$$
$$y = -1$$

$$\frac{36 + 1 - 6}{36 - 1 + 6} = \frac{31}{41}$$

Q. The graphs of the equations 3x-20y-2=0 and 11x-5y+61=0 intersect at P(a,b). What is the value of $(a^2+b^2-ab)/(a^2 - b^2 + ab)$?
समीकरण 3x-20y-2=0 और 11x-5y+61 = 0 का ग्राफ, P (a, b) पर प्रतिच्छेद करते हैं। $(a^2+b^2-ab)/(a^2 - b^2 + ab)$का मान क्या है?

SSC CGL 2019 Tier-2



A. $\frac{37}{35}$
B. $\frac{41}{31}$
C. $\frac{5}{7}$
D. $\frac{31}{41}$

Q. If the point of intersection of lines $x + y = 2$ and $2x - y = 1$ lies on $y = Kx + 5$, what is the value of K?

यदि रेखाओं x + y = 2 और 2x - y=1 का प्रतिच्छेद बिंदु y = Kx + 5 पर स्थित है, तो K का मान ज्ञात कीजिए।

$$\begin{matrix} x + y = 2 \\ 2x - y = 1 \end{matrix} \quad +$$

$$3x = 3$$

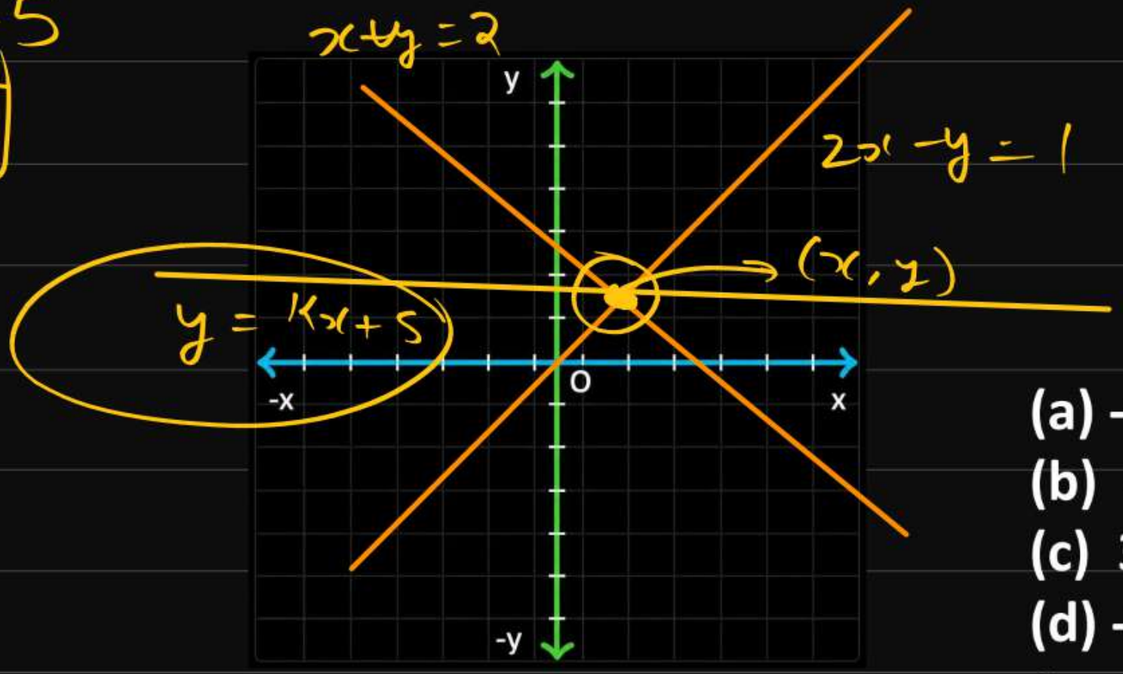
$$x = 1$$

$$y = 1$$

$$y = Kx + 5$$

$$1 = K + 5$$

$$-4 \in K$$



(a) -4
(b) 4
(c) 3
(d) -3
Ans: a

SSC CHSL 2024 T-1

Q. The graphs of the equations 7x+11y=3 and 8x+y=15 intersect at the point P, which also lies on the graphs of the equation:

समीकरणों के ग्राफ़ 7x+11y=3 और 8x+y=15 बिंदु P पर प्रतिच्छेद करते हैं, जो समीकरण के ग्राफ़ पर भी स्थित है:

SSC CGL 2020 Tier-2

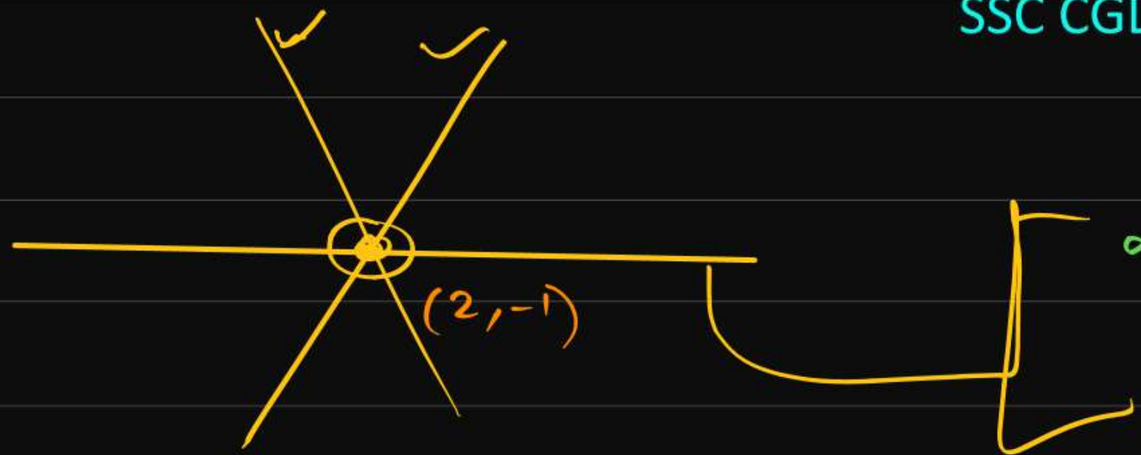
$7x+11y=3$

$8x+y=15$ $\times 11$ $(-)$

$81x = 162$

$x = 2$

$y = -1$



$2 - 1 = 5$

(a) $3x+y=3$

(b) $2x+y=2$

(c) $2x-y=1$

(d) $3x+5y=1$

$3(2) + 5(-1)$

$6 - 5 = 1$

Q. The graphs of the linear equation 4x-2y = 10 and 4x+ky = 2 intersect at a point (a,4). The value of k is equal to:

रैखिक समीकरण 4x-2y = 10 और 4x + ky = 2 के ग्राफ बिंदु (a, 4) पर प्रतिछेदित करते हैं। K का मान बराबर है:

SSC CGL 2019 Tier-2

$4x-2y = 10$

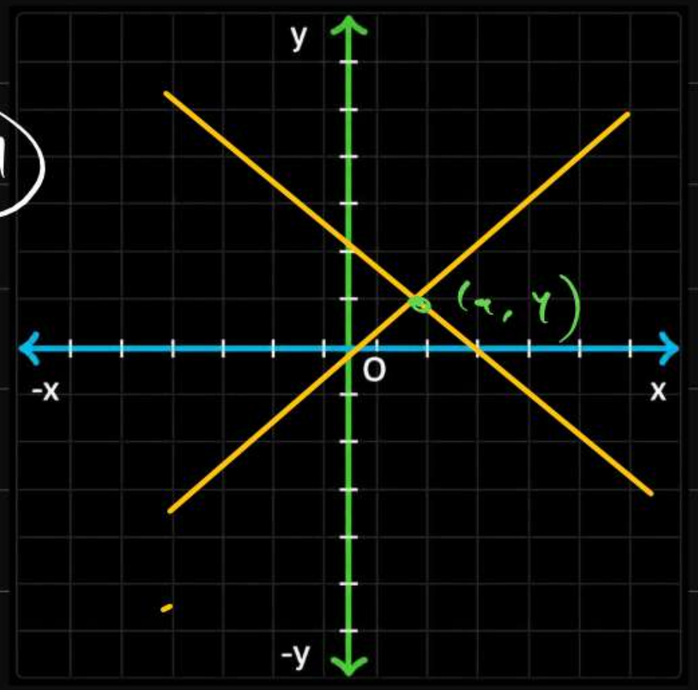
$4x+ky = 2$

$\Rightarrow 4a - 2(4) = 10$

$a = 4.5$

$4(4.5) + K(4) = 2$

$K = \frac{-8}{2} = -4$



(a) 3
(b) -3
(c) -4
(d) 4

Q) . In the x-y coordinate method, two points (m+1, n+3) and (m+4, n+k) lie on a line 4y = 9x - 5, then find the value of k.

xy निदेशांक पद्धति में दो बिंदु (m+1, n +3) और (m+4, n +k) एक रेखा 4y= 9x-5 पर स्थित है तब k का मान ज्ञात करें।

(a) 33/4
(b) 35/4
(c) 39/4
(d) 41/4

$4y = 9x - 5$

$(m+1)$ $(n+3)$

$4n + 12 = 9m + 9 - 5$

$4n - 9m = -8$

$4n - 9m = -4k + 31$

$4(m+k) = 9[m+4) - 5$

$-8 = -4k + 31$

$4k = 39$

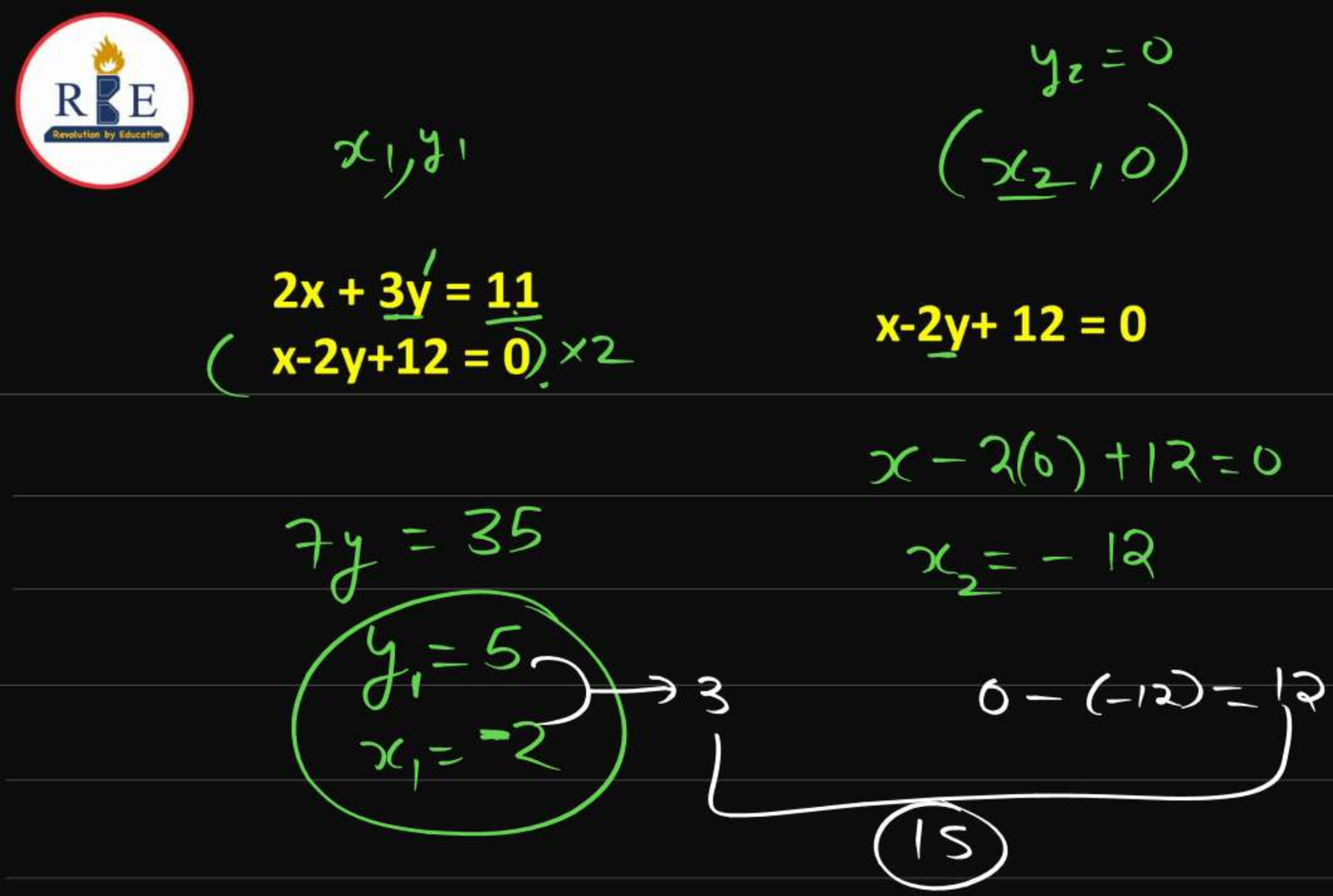
$k = 39/4$

Q-) The graphs of the equations 2x + 3y = 11 and x-2y+12 = 0 intersects at $P(x_1,y_1)$ and the graph of the equation x-2y+ 12 = 0 intersects the x-axis at $Q(x_2,y_2)$. What is the value of $(x_1-x_2 + y_1 + y_2)$

समीकरण 2x + 3y = 11 और x-2y+12 = 0 के रेखांकन P(x1,y1) पर प्रतिच्छेद करते हैं और समीकरण x-2y+ 12 = 0 का रेखांकन x-अक्ष को Q(x2,y2) पर प्रतिच्छेद करता है। . (x1-x2 + y1 + y2) का मान क्या है

SSC CGL 2018 Tier-2

A. 13
B. -11
C. 15
D. -9



Calculation of Area

Q. What is the area of the region created by the two linear equations $2x + y = 6$ and $2x - y = -2$ bounded by these lines and the x-axis?

दो रैखिक समीकरणों $2x + y = 6$ और $2x - y = -2$ द्वारा निर्मित उस क्षेत्र का क्षेत्रफल ज्ञात कीजिए जो इन रेखाओं और x-अक्ष से परिबद्ध (bounded) है?

(a) 3 वर्ग इकाई
(b) 5 वर्ग इकाई
(c) 8 वर्ग इकाई
(d) 6 वर्ग इकाई

Ans: c

SSC CHSL 2024 T-1

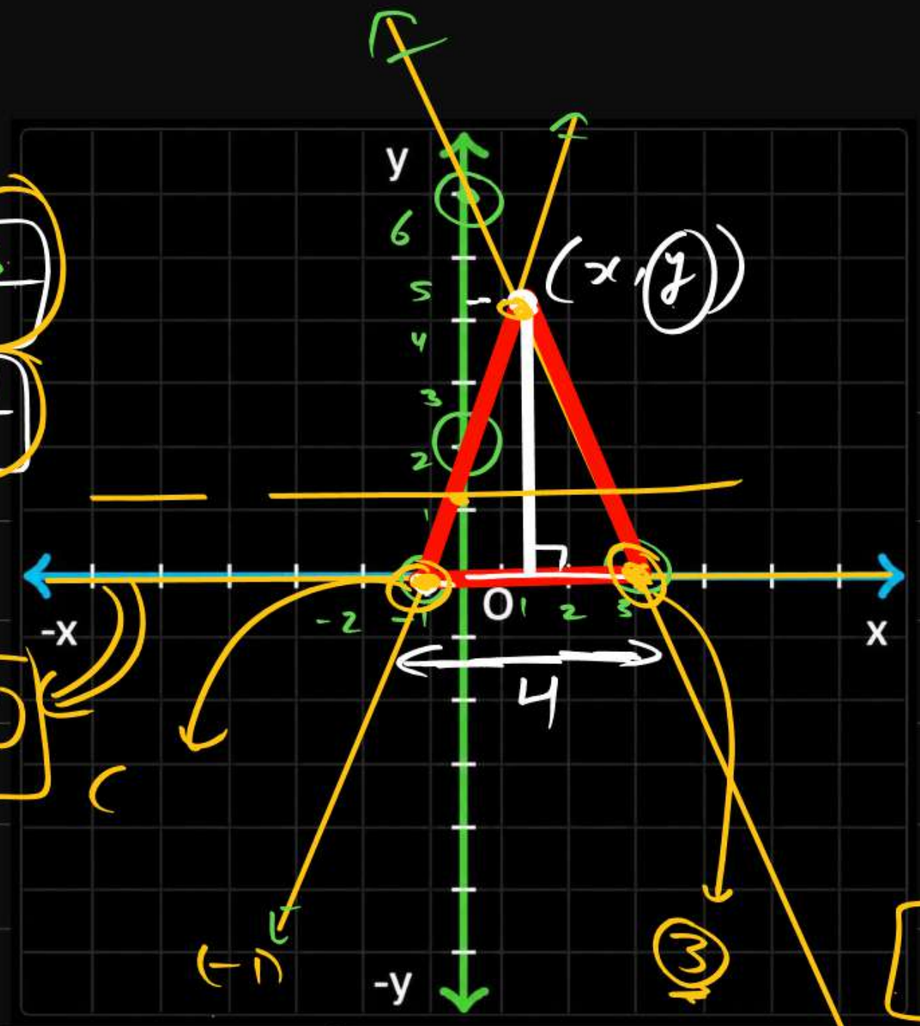
$2x + y = 6$
$2x - y = -2$

x	0	3
y	6	0

x	0	-1
y	2	0

$2y = 8$
$y = 4$

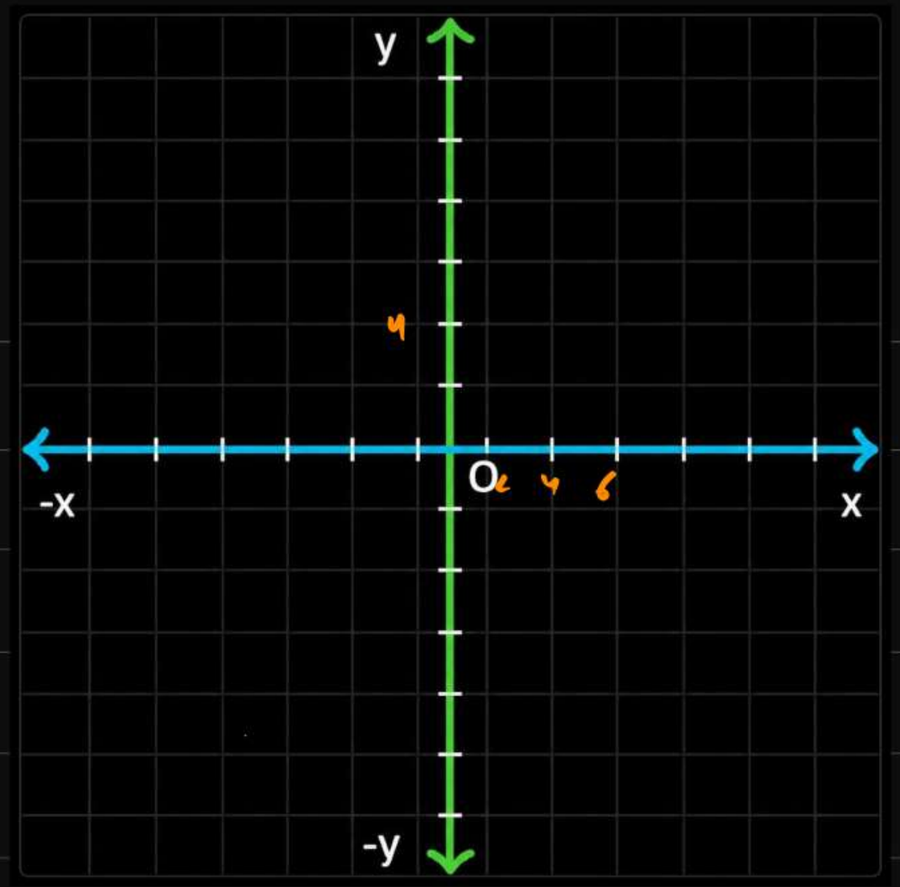
$y = 0$



$A = \frac{1}{2} \times B \times H$

$= \frac{1}{2} \times |3 - (-1)| \times y$

$= \frac{1}{2} \times 4 \times 4 = 8$



Q. What is the area (in unit square) of the triangle enclosed by the graphs of 2x+5y=12, x+y=3 and x-axis?

2x+5y=12, x+y=3 और x-अक्ष के आलेखों से घिरे त्रिभुज का क्षेत्रफल (इकाई वर्ग में) क्या है?

(a) 4
(b) 2.5
(c) 3.5
(d) 3

SSC CGL 2020 Tier-2

$(2x+5y=12,\ x+y=3)\times 2$

$x_1 = 6, \quad y = 0$

$x_2 = 3$

$3y = 6$

$y = 2$

$= \frac{1}{2}|x_1 - x_2| \times y$

$\Rightarrow \frac{1}{2}|6-3| \times 2 \Rightarrow 3$

Q. What is the area (in sq. units) of the triangle formed by the graphs of the equations 2x+5y-12=0, x+y=3 and y=0?

2x+5y-12 = 0, x+y = 3 and y = 0 के समीकरणों के द्वारा गठित त्रिभुज का क्षेत्रफल (वर्ग इकाइयों में) क्या है?

SSC CGL 2019 Tier-2

(a) 6
(b) 5
(c) 2
(d) 3

$2x+5y=12$

$(x+y=3) \times 2$

$x_1 = 6$

$y = 0$

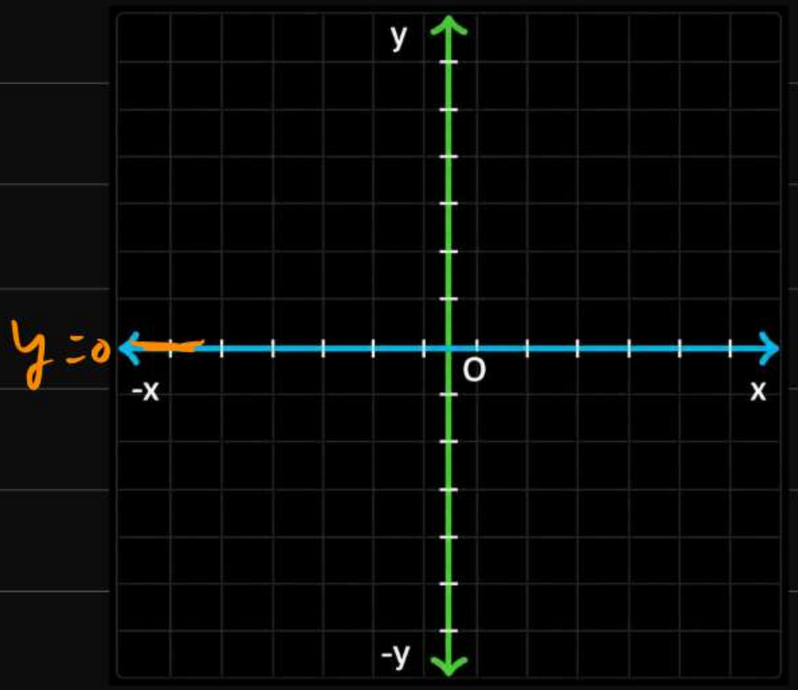
$x_2 = 3$

$3y = 6$

$y = 2$

$\Rightarrow \frac{1}{2} | 6-3 | \times 2$

$\Rightarrow 3$



Q) Find the area of a triangle made up of simple lines x+2y-6=0, x+4y-20=0 and x=0.
सरल रेखाओं x+2y-6=0, x+4y-20=0 तथा y=0 से बने त्रिभुज का क्षेत्रफल ज्ञात करें।

(a) 8
(b) 10
(c) 16
(d) 12

$x+2y=6$ ×2
$x+4y=20$ ×1

$x=0$

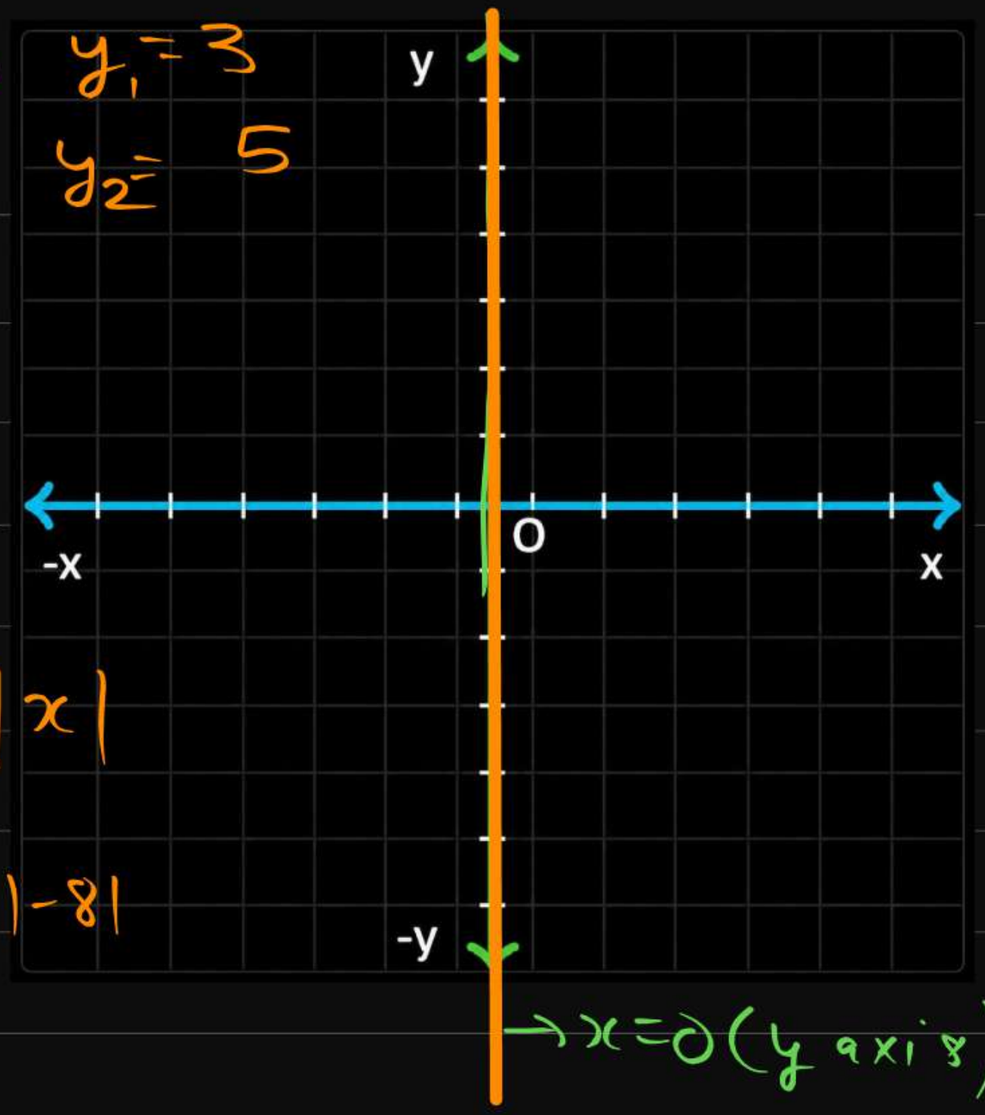
$y_1=3$
$y_2=5$

$-x=8$
$x=-8$

$A=\frac{1}{2}|y_1-y_2|\times|x|$

$\Rightarrow \frac{1}{2}\times|3-5|\times|-8|$

$\Rightarrow 8$



$\rightarrow x=0$ (y axis)

Q) Find the area of a triangle made up of simple lines x+2y-6=0, x+4y-20=0 and x=0.

सरल रेखाओं x+2y-6=0, x+4y-20=0 तथा y=0 से बने त्रिभुज का क्षेत्रफल ज्ञात करें।

(a) 8
(b) 10
(c) 16
(d) 12

$x+2y=6$

$x+4y=20$

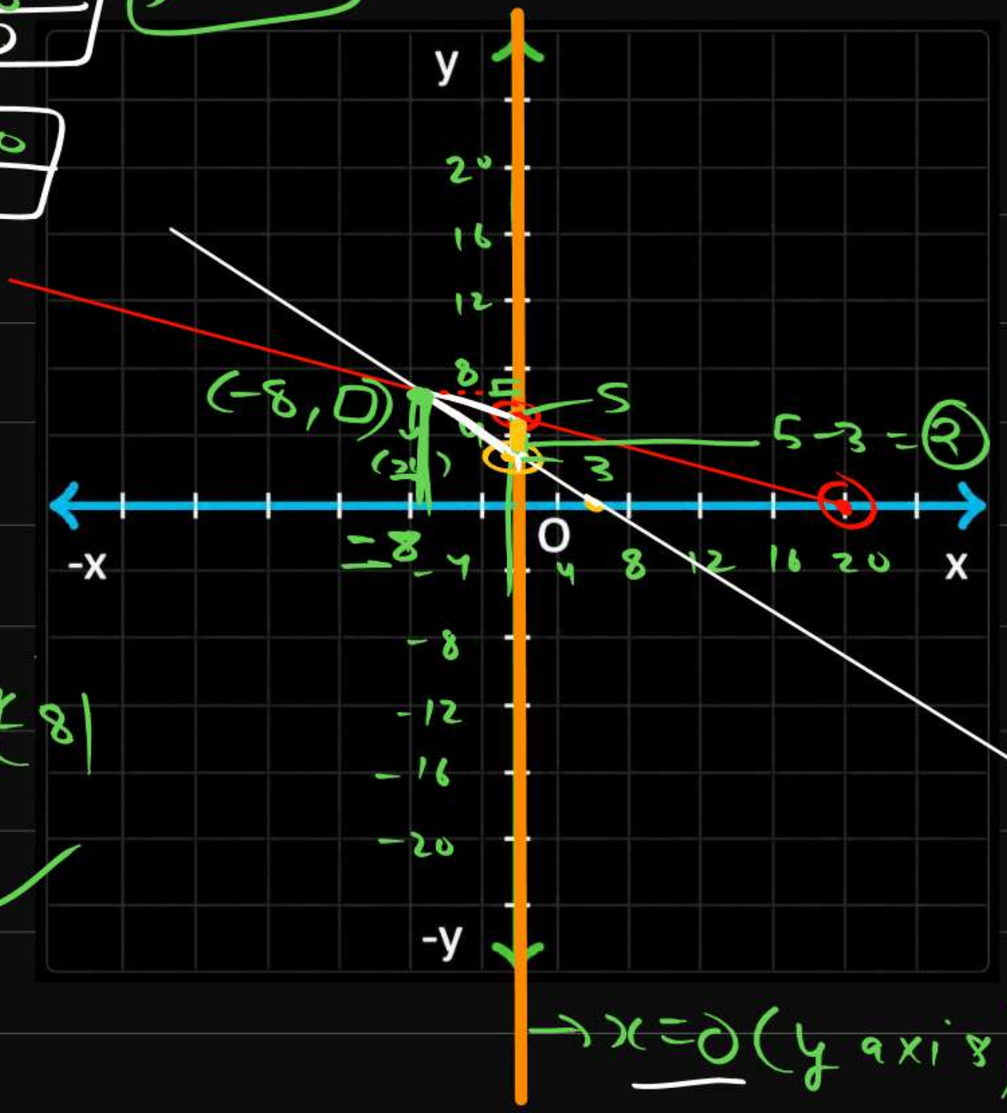
x	0	6
y	3	0

x	0	20
y	5	0

$y=0$

Solve

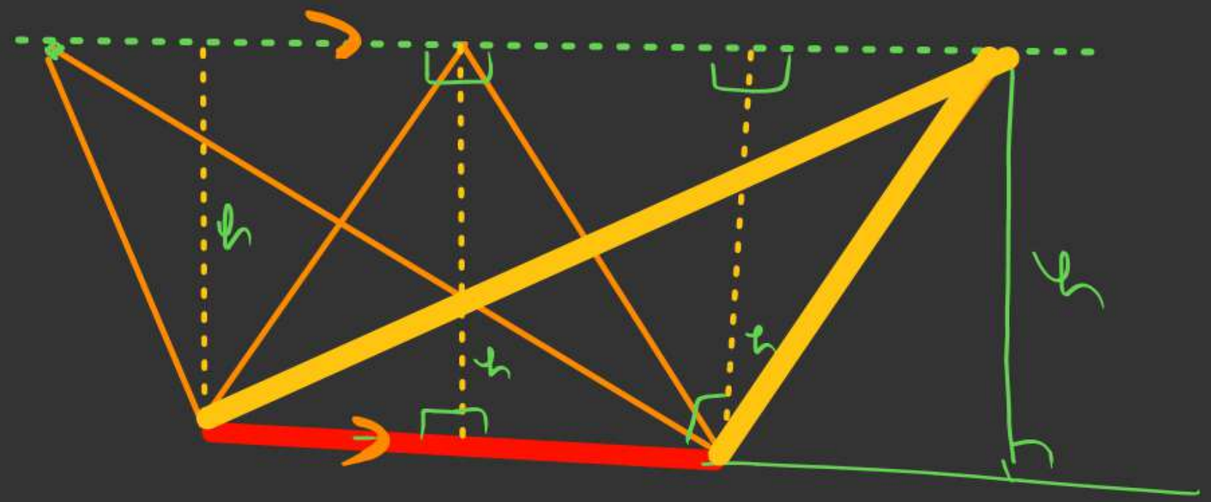
$x=-8$



$A=\frac{1}{2}\times(5-3)\times|8|$

$\Rightarrow 8$ ✓

$\rightarrow x=0$ (y axis)



$$A = \frac{1}{2} \times b \times h$$

$x = 0$

$y = 0$

$5x+3y=15$ ⇒

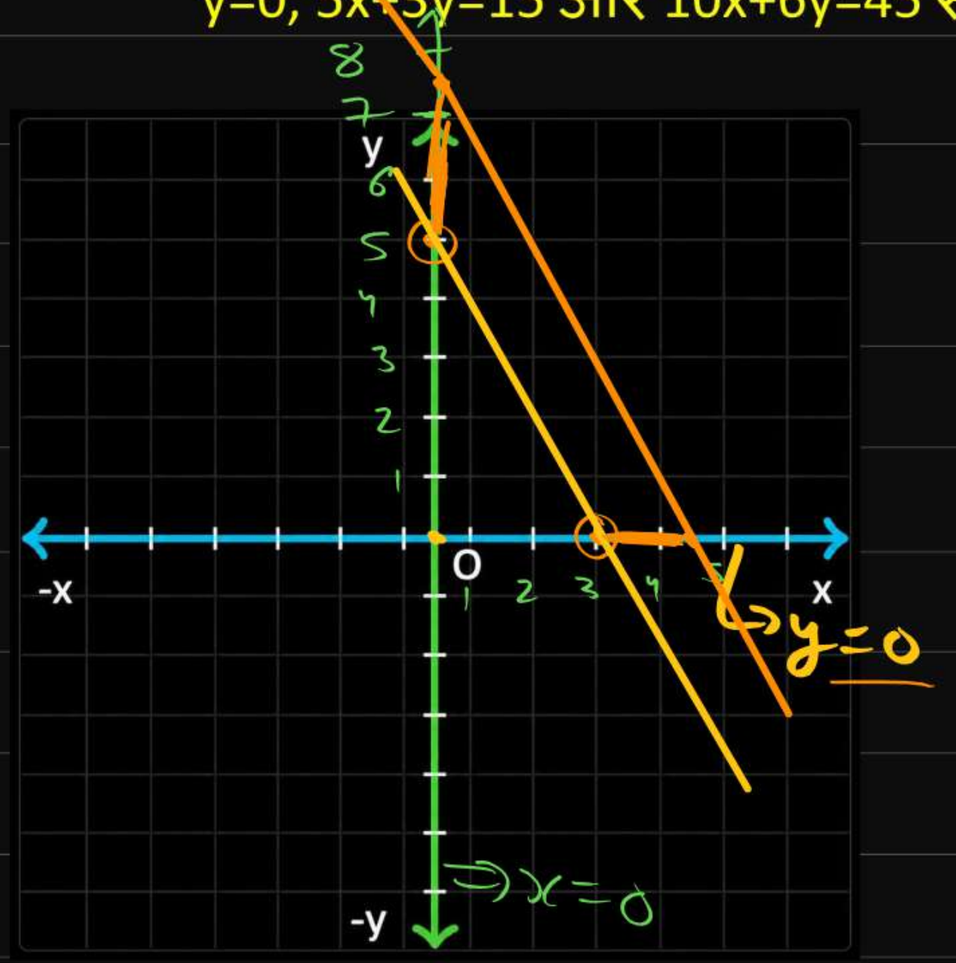
x	0	3
y	5	0

$10x+6y=45$ ⇒

x	0	4.5
y	7.5	0

Q) Find the area of trapezium which is formed by lines x=0, y=0, 5x+3y=15 and 10x+6y=45.

समलम्ब का क्षेत्रफल ज्ञात करें जो कि x =0, y=0, 5x+3y=15 और 10x+6y=45 से बनता है।



(a) 71/8
(b) 67/8
(c) 75/8
(d) 85/8

Q) Find the area of trapezium which is formed by lines x=0, y=0, 5x+3y=15 and 10x+6y=45.

समलम्ब का क्षेत्रफल ज्ञात करे जो कि x =0, y=0, 5x+3y=15 और 10x+6y=45 से बनता है।

(a) 71/8
(b) 67/8
(c) 75/8
(d) 85/8

$x = 0$

$y = 0$

$5x+3y=15$ ⇒

x	0	3
y	5	0

$10x+6y=45$ ⇒

x	0	4.5
y	7.5	0

$\frac{5}{10} = \frac{3}{6} \neq \frac{-15}{-45}$

$A_1x + B_1y + C_1 = 0$

$A_2x + B_2y + C_2 = 0$

II line

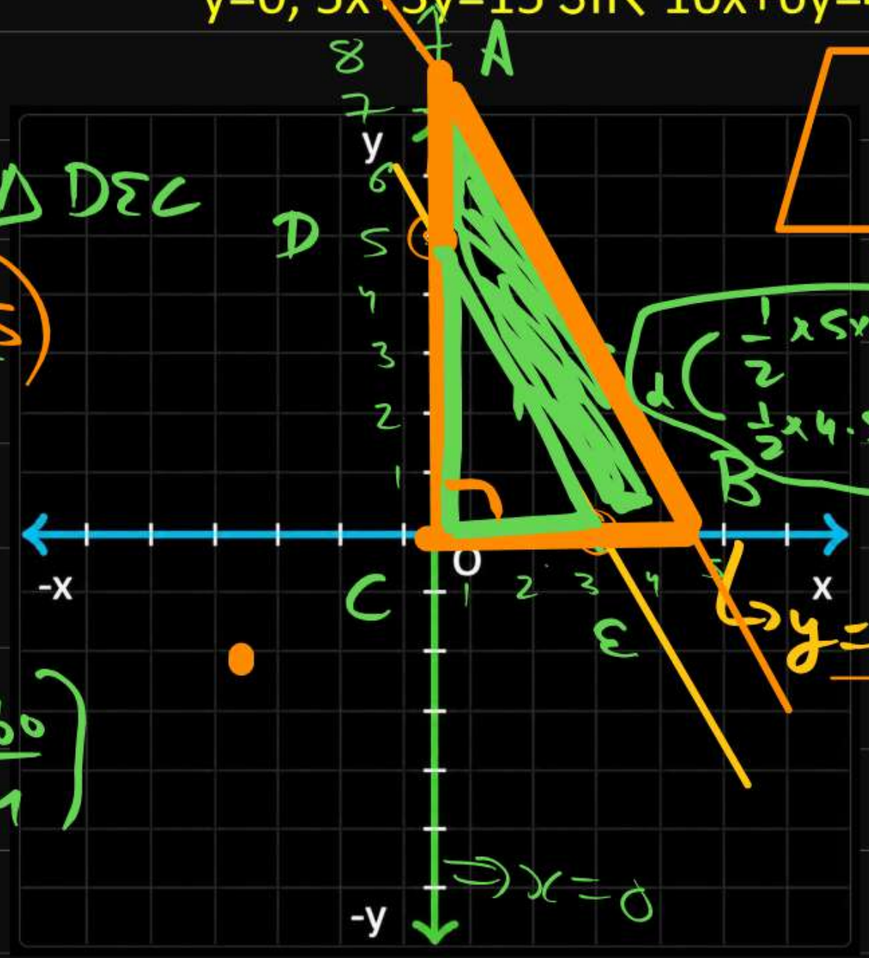
$\frac{A_1}{A_2} = \frac{B_1}{B_2} \neq \frac{C_1}{C_2}$.

Area = $\Delta ABC - \Delta DEC$

$= \left(\frac{1}{2} \times 4.5 \times 7.5\right) - \frac{1}{2}(3 \times 5)$

$\Rightarrow \frac{1}{2}\left[\frac{135}{4} - \frac{60}{4}\right]$

$\Rightarrow 75/8$



$\frac{1}{2} \times S_1 \times S_2$

$\frac{1}{2} \times 5 \times 3$

$\frac{1}{2} \times 4.5 \times 7.5$

Q) Find the area of trapezium which is formed by lines x=0, y=0, 5x+3y=15 and 10x+6y=45.

समलम्ब का क्षेत्रफल ज्ञात करे जो कि x =0, y=0, 5x+3y=15 और 10x+6y=45 से बनता है।

(a) 71/8
(b) 67/8
(c) 75/8
(d) 85/8

$5x+3y=15$

$10x+6y=45$

x	0	3
y	5	0

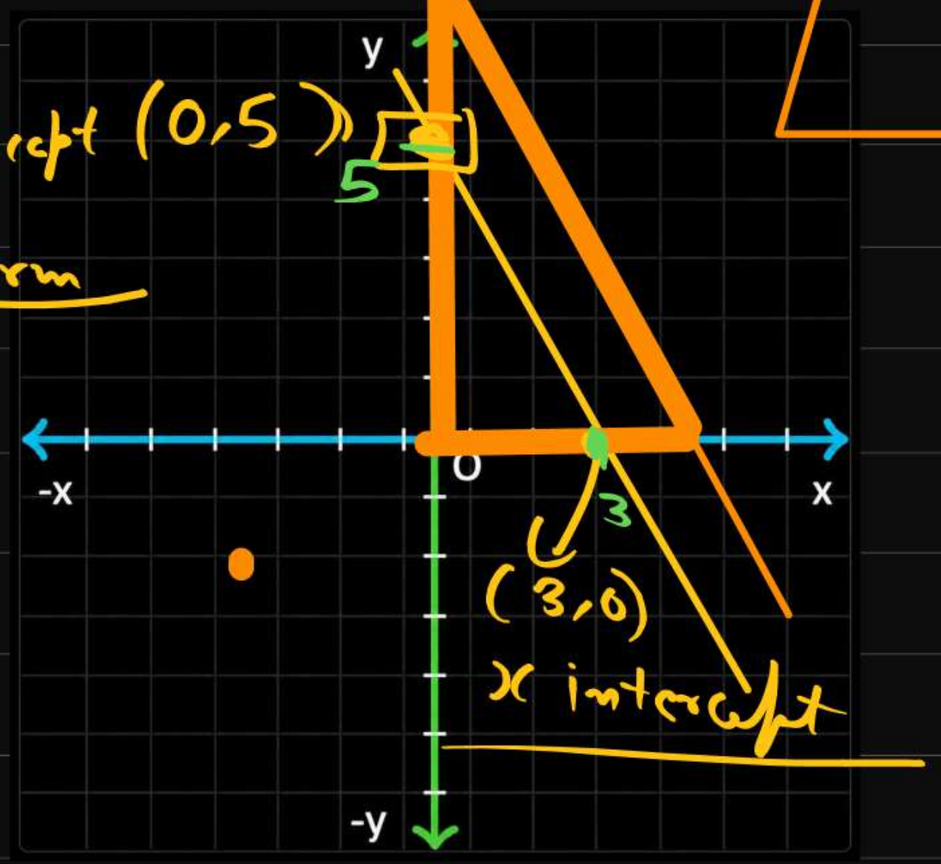
$\frac{x}{4.5}+\frac{y}{7.5}=1$

$\frac{x}{x_i}+\frac{y}{y_i}=1 \Rightarrow$ intercept form

$5x+3y=15$

$\frac{5x}{15}+\frac{3y}{15}=1$

$\frac{x}{3}+\frac{y}{5}=1$



Q). Find the area of the quadrilateral PQRS whose diagonal PR and QS are the x-axis and y-axis respectively. The equation of the line PQ is 9x+12y = -108 and the equation of RS is 6x+8y = 48.

चतुर्भुज PQRS का क्षेत्रफल ज्ञात करें जिसके विकर्ण PR और QS क्रमशः X अक्ष तथा y अक्ष है। रेखा PQ का समीकरण 9x+12y=-108 और RS का समीकरण 6x+8y=48 है।

(a) 144 (b) 136
(c) 150. (d) 125

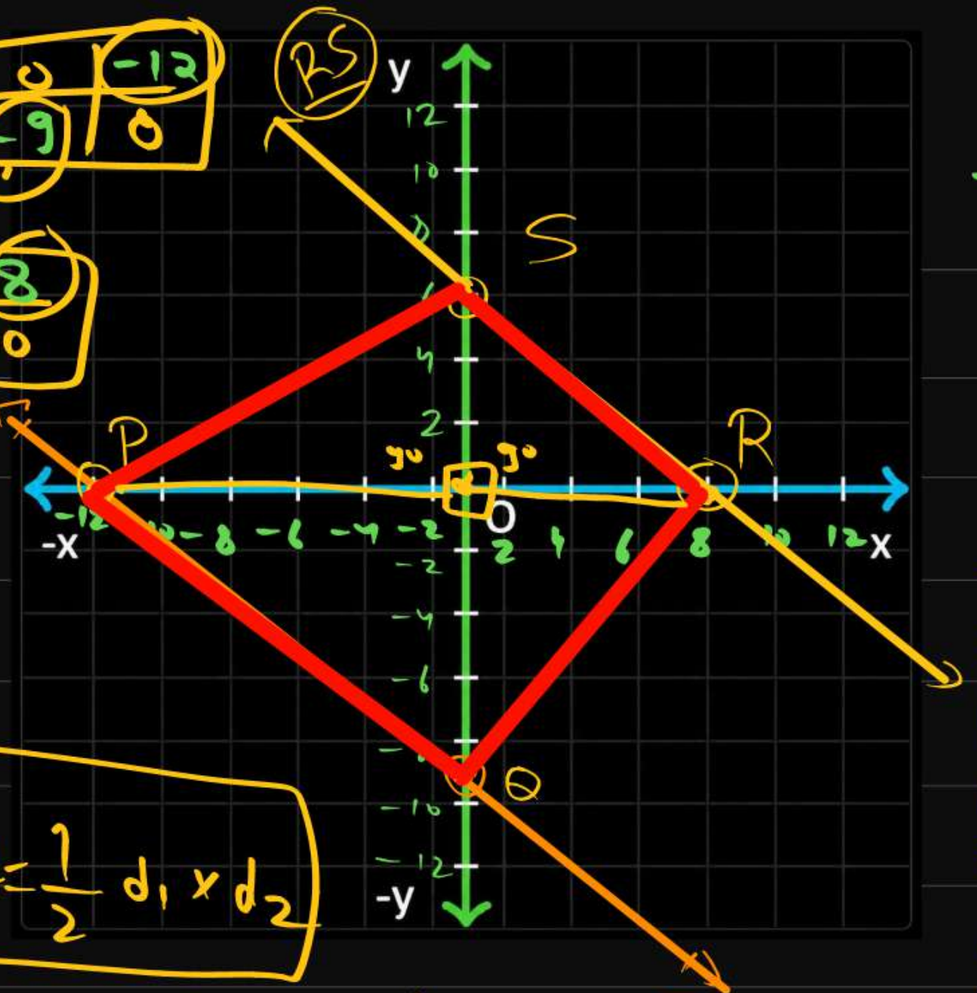
PQ → 9x+12y = -108 ⇒

x	0	-12
y	-9	0

RS — 6x+8y = 48

x	0	8
y	6	0

$\frac{x}{8} + \frac{y}{6} = 1$



Area of quad $= \frac{1}{2} d_1 \times d_2$

$\Rightarrow \frac{1}{2} \times |8-(-12)| \, |6-(-9)|$

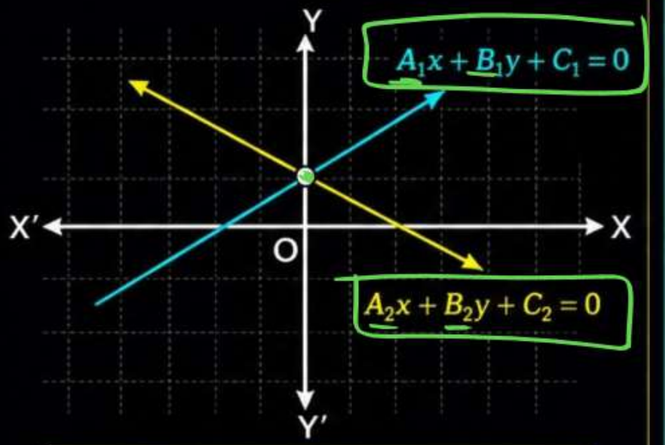
$\Rightarrow \frac{1}{2} \times 20 \times 15 = 150$

Intersecting, Parallel and Co-incident Lines

$A_1x + B_1y + C_1 = 0$ $A_2x + B_2y + C_2 = 0$

1. INTERSECTING LINES

Lines intersect at one point.



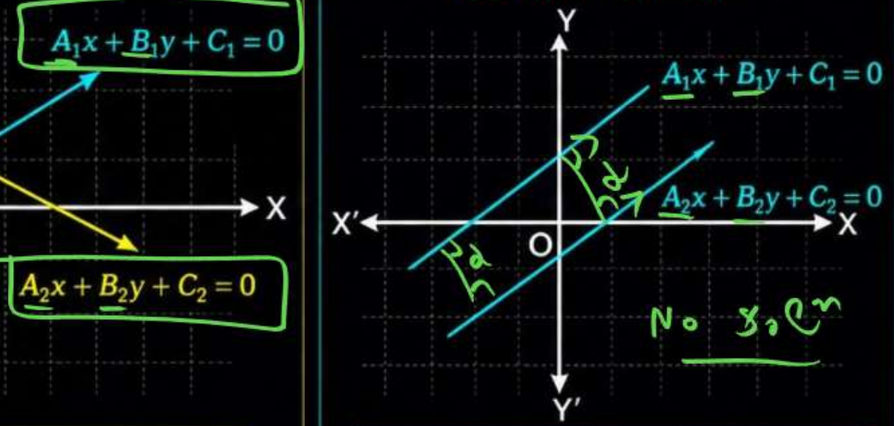
CONDITION

$$\frac{A_1}{A_2} \neq \frac{B_1}{B_2}$$

The ratios of coefficients of x and y are not equal.

2. PARALLEL LINES

Lines are parallel and do not intersect.



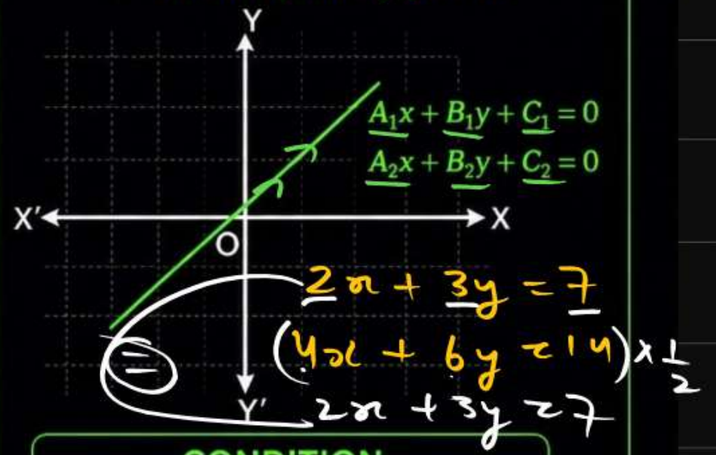
CONDITION

$$\frac{A_1}{A_2} = \frac{B_1}{B_2} \neq \frac{C_1}{C_2}$$

The ratios of coefficients of x and y are equal but the ratio of constants is not equal.

3. CO-INCIDENT LINES

Lines coincide and have infinitely many points.



CONDITION

$$\frac{A_1}{A_2} = \frac{B_1}{B_2} = \frac{C_1}{C_2}$$

The ratios of coefficients of x, y and constants are all equal.

NOTE: If $A_1 = A_2$, $B_1 = B_2$ and $C_1 = C_2$, then both the equations represent the same line.

Co-ordinate Geometry

निर्देशांक ज्यामिति

(From Basics)

Distance Formula- दूरी सूत्र

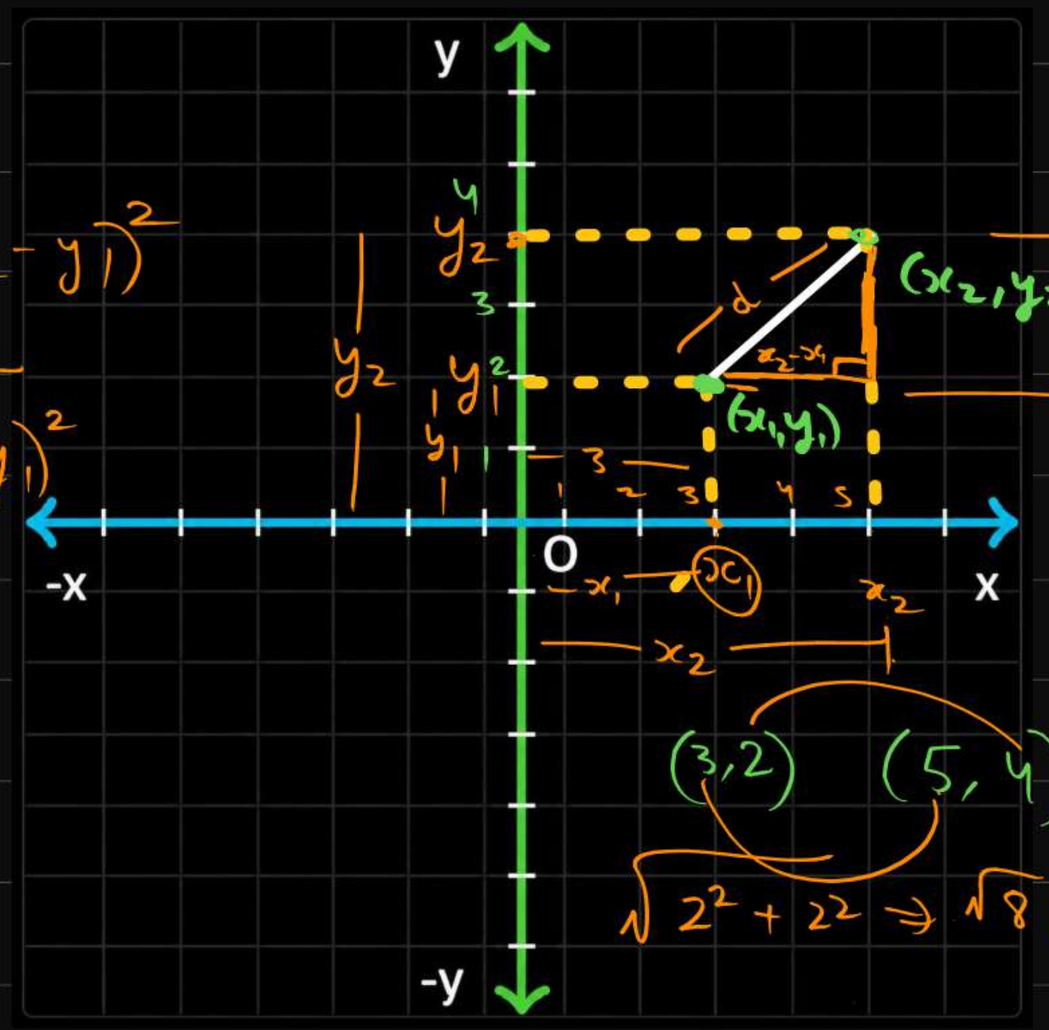
Distance between 2 points

$H^2 = B^2 + P^2$

$d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$

$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

+ve



(3, 2) (5, 4)

$\sqrt{2^2 + 2^2} \Rightarrow \sqrt{8} = 2\sqrt{2}$

Distance Formula- दूरी सूत्र

Distance between 2 points

$H^2 = B^2 + P^2$

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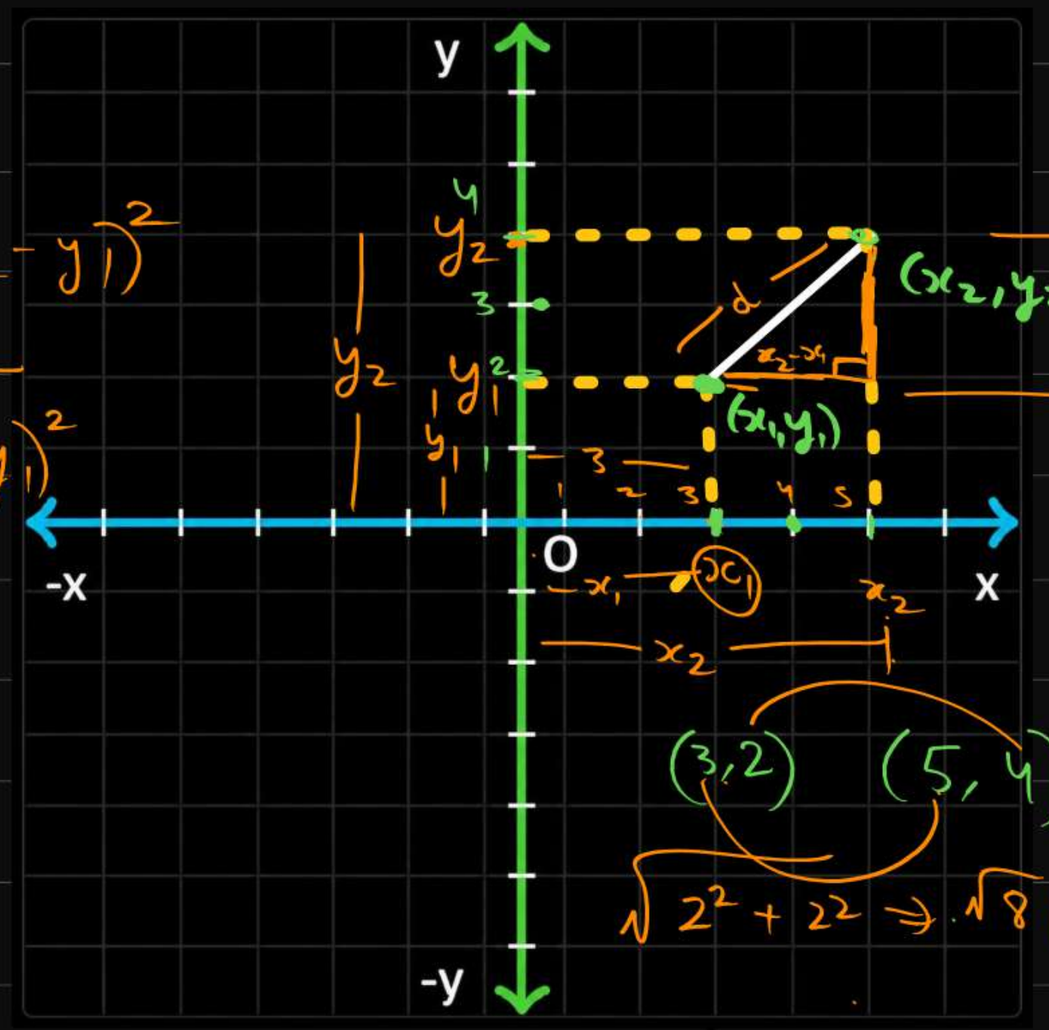
+ve

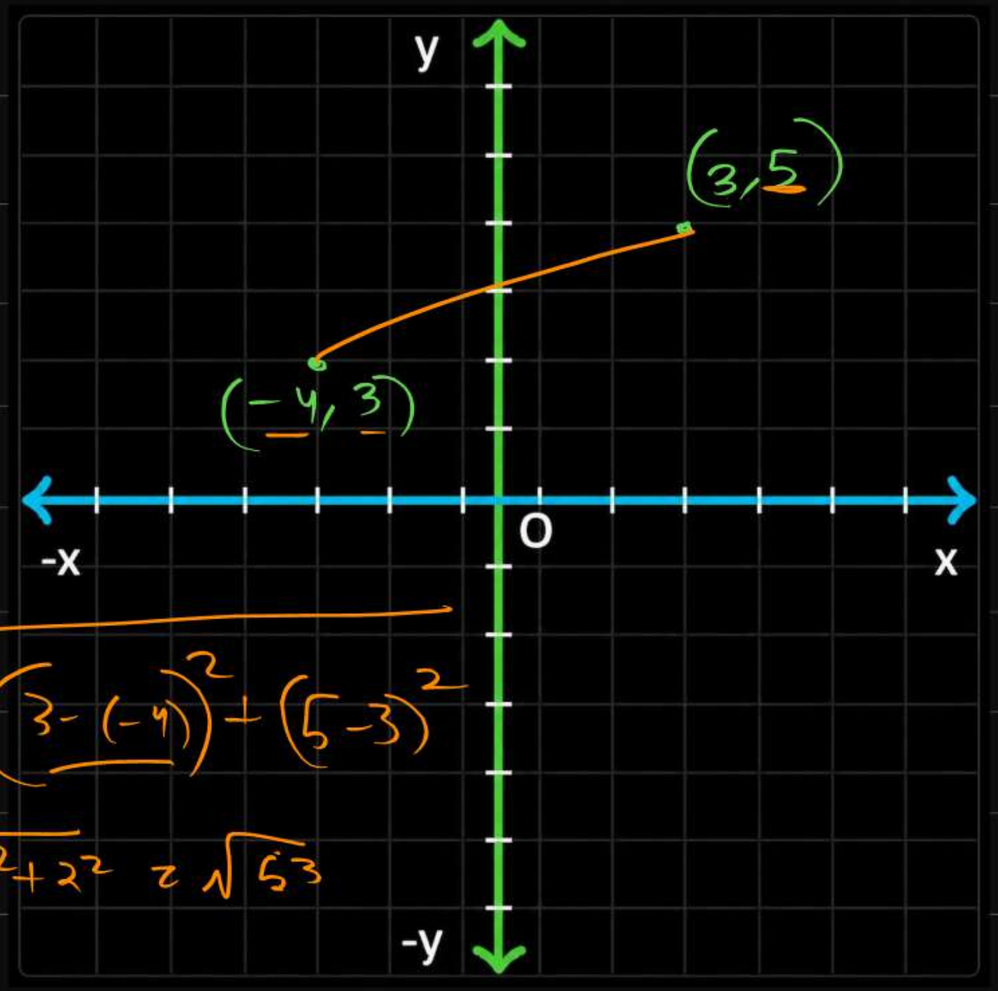


(3, 2) (5, 4)

$\sqrt{2^2 + 2^2} \Rightarrow \sqrt{8} = 2\sqrt{2}$

Distance Formula- दूरी सूत्र

Distance of a point from origin



$$d = \sqrt{(3-(-4))^2+(5-3)^2}$$

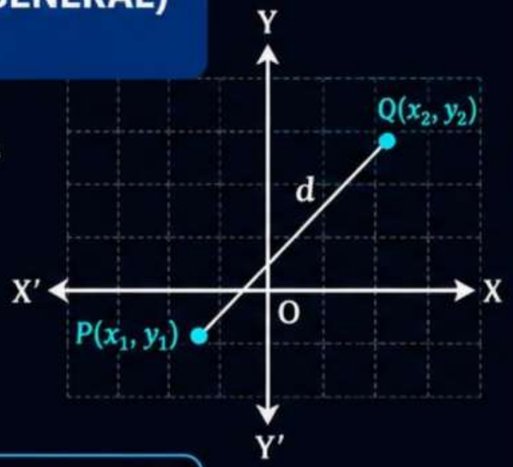
$$\Rightarrow \sqrt{7^2+2^2} = \sqrt{53}$$

DISTANCE FORMULA दूरी सूत्र

1. DISTANCE FORMULA (GENERAL)
दूरी सूत्र (सामान्य)

The distance between two points $P(x_1, y_1)$ and $Q(x_2, y_2)$ is

दो बिंदुओं $P(x_1, y_1)$ और $Q(x_2, y_2)$ के बीच की दूरी



$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Where | जहाँ

x_1, y_1	= Coordinates of point P	x_1, y_1	= बिंदु P के निर्देशांक
x_2, y_2	= Coordinates of point Q	x_2, y_2	= बिंदु Q के निर्देशांक
d	= Distance between P and Q	d	= P और Q के बीच की दूरी

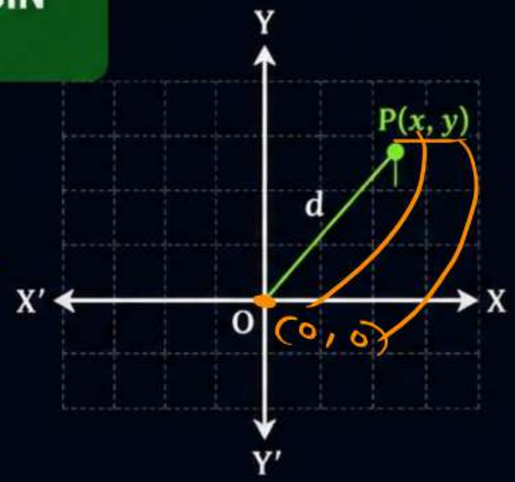
Key Points | मुख्य बातें

- Always take the difference of corresponding coordinates.
- Square the differences, add them and then take square root.
- सदैव संबंधित निर्देशांकों का अंतर लें।
- अंतरों का वर्ग करें, उन्हें जोड़ें और फिर वर्गमूल निकालें।

2. DISTANCE FROM ORIGIN
मूल बिंदु से दूरी

The distance of a point $P(x, y)$ from the origin $O(0, 0)$ is

बिंदु $P(x, y)$ की मूल बिंदु $O(0, 0)$ से दूरी



$$d = \sqrt{x^2 + y^2}$$

Where | जहाँ

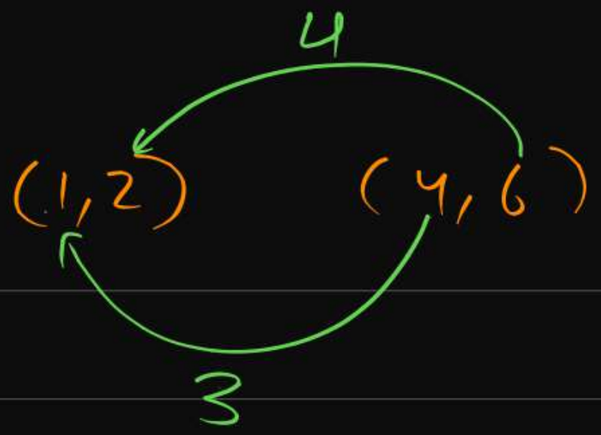
x, y	= Coordinates of point P	x, y	= बिंदु P के निर्देशांक
d	= Distance from origin	d	= मूल बिंदु से दूरी

- It is a special case of the distance formula with $P(x, y)$ and $Q(0, 0)$.
- Always positive or zero.
- यह दूरी सूत्र का विशेष रूप है, जहाँ $P(x, y)$ और $Q(0, 0)$ हैं।
- यह सदैव धनात्मक या शून्य होती है।

Q. What is the length of the segment joining (1,2) and (4,6)?
(1,2) और (4,6) को जोड़ने वाले रेखाखंड की लंबाई क्या है?

SSC CGL 2025

(a) 5 (b) 2 (c) 4 (d) 3



$$d = \sqrt{3^2 + 4^2}$$

$$\Rightarrow \sqrt{25} = 5$$

Q. The lines x+y = 10 and -3x+y = 2 have a unique solution. What is the distance (in units) between the unique solution and the point of intersection of the line x+y = 10 and the x-axis?

रेखाओं x+y = 10 और -3x+y = 2 का एक अद्वितीय हल है। रेखा x+y = 10 के अद्वितीय हल और प्रतिच्छेदन बिंदु और x-अक्ष के बीच की दूरी (इकाई में) कितनी है?

SSC CHSL 2024 T-1

$x+y=10$ ... (i)

$-3x+y=2$ ... (ii)

(i) − (ii):

$4x = 8$

$x = 2$

Put $x = 2$ in (i): $2+y=10 \Rightarrow y = 8$

$(2, 8)$

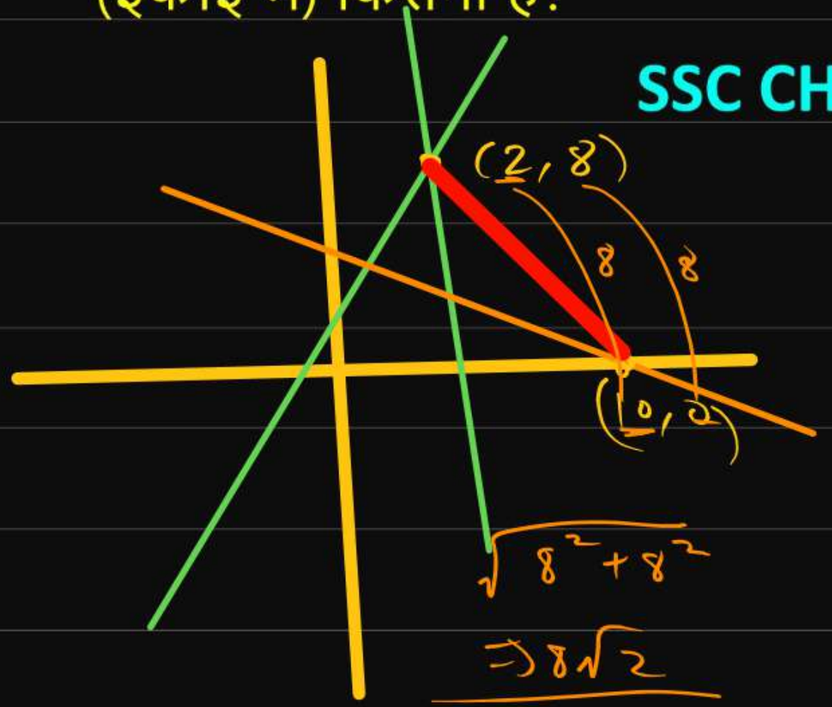
$0 = x+y-10$

$x+y=10$

$x = 10$

$y = 0$

$(10, 0)$



$d = \sqrt{8^2+8^2}$

$\Rightarrow d = 8\sqrt{2}$

(a) $10\sqrt{2}$

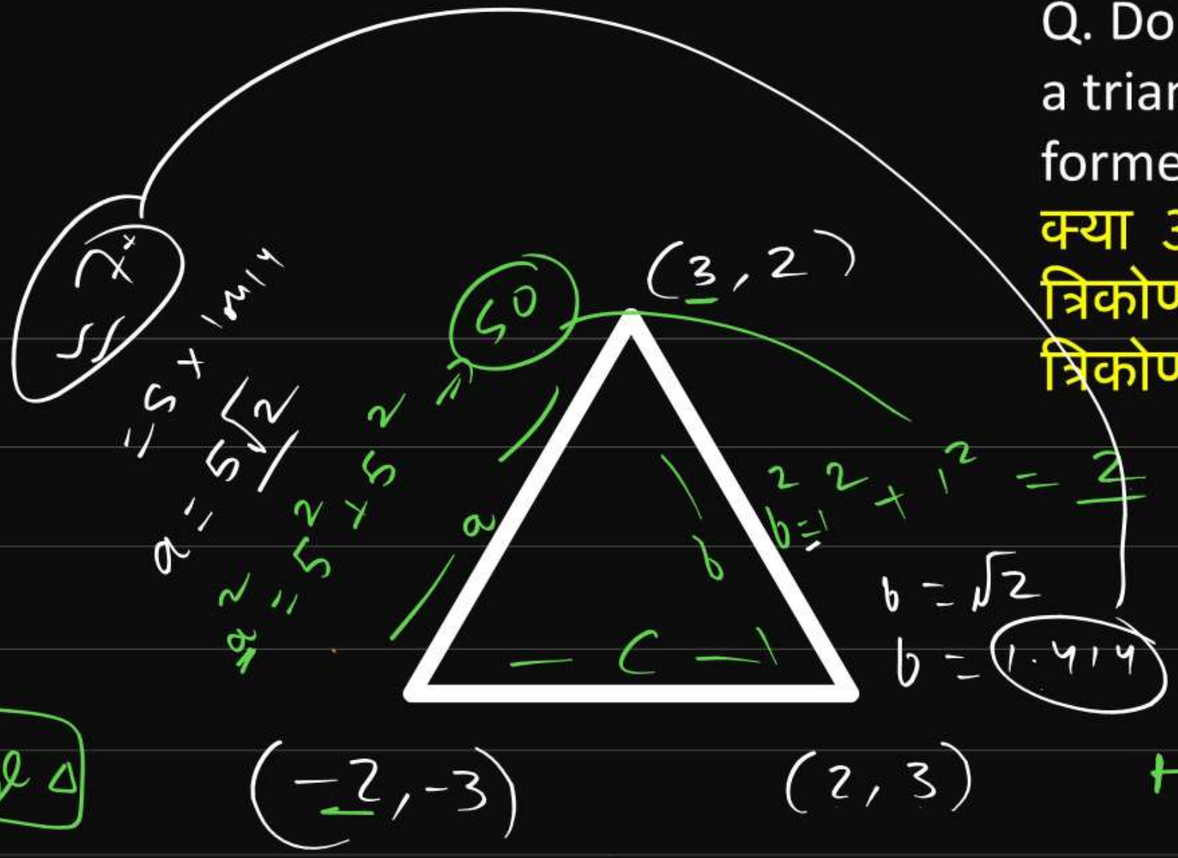
(b) $8\sqrt{2}$

(c) 8

(d) 10

Q. Do the points (3, 2), (–2, –3) and (2, 3) form a triangle? If so, name the type of triangle formed

क्या अंक (3, 2), (-2, –3) और (2, 3) एक त्रिकोण बनाते हैं? यदि ऐसा है, तो गठित त्रिकोण के प्रकार का नाम दें



Rt. angle Δ

$c^2 = a^2 + b^2$

$52 = 50 + 2 = 52$

$c^2 = 4^2 + 6^2$

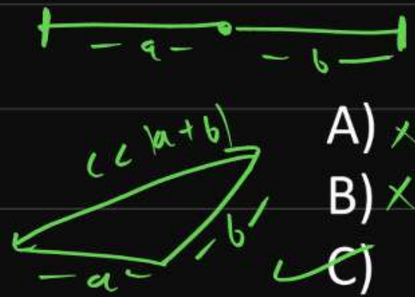
$c^2 = \Rightarrow 52$

$c = \sqrt{52} \Rightarrow 7 - 8$

49 64

$a + b > c$ ✓

Δ will be formed



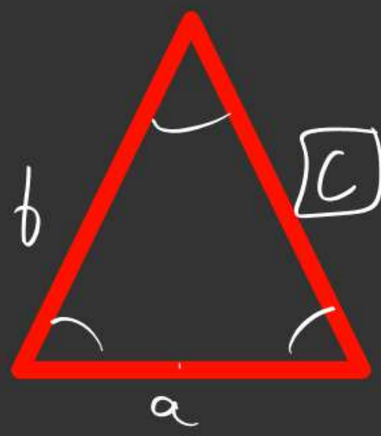
A) ✗ Acute angled triangle

B) ✗ Obtuse angled triangle

C) Right angled triangle

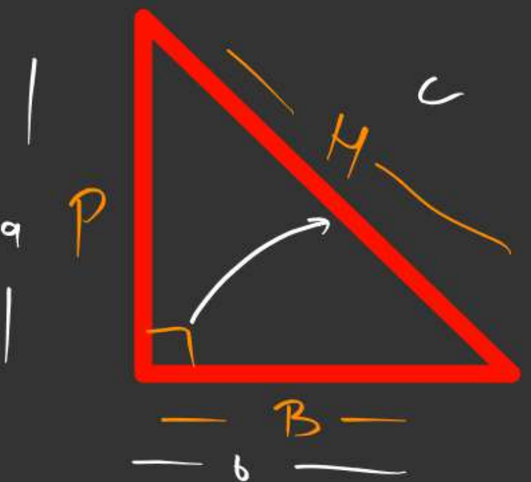
D) Triangle won't be formed

All angles less than 90°



C ⇒ Biggest side

$C^2 < a^2 + b^2$

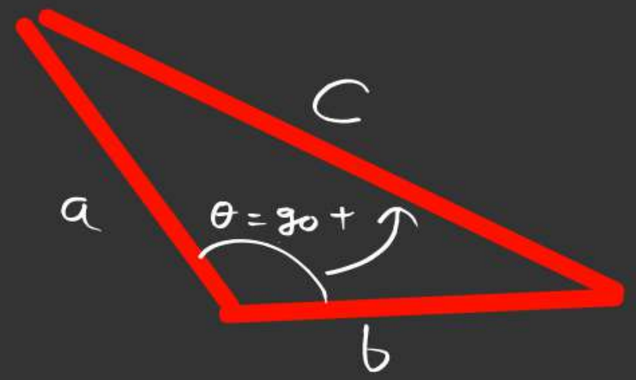


Rt. Δ

$M^2 = P^2 + B^2$

$c^2 = a^2 + b^2$

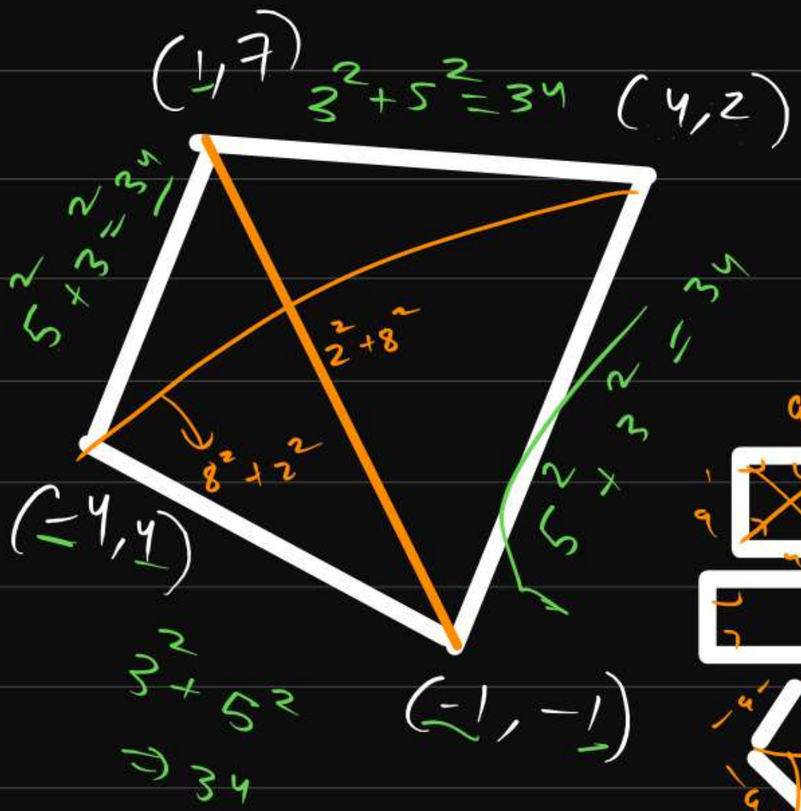
1 angle > 90°



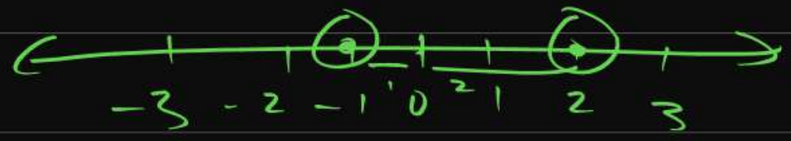
$C^2 > a^2 + b^2$

Q. The points (1, 7), (4, 2), (–1, –1) and (– 4, 4) are the vertices of.

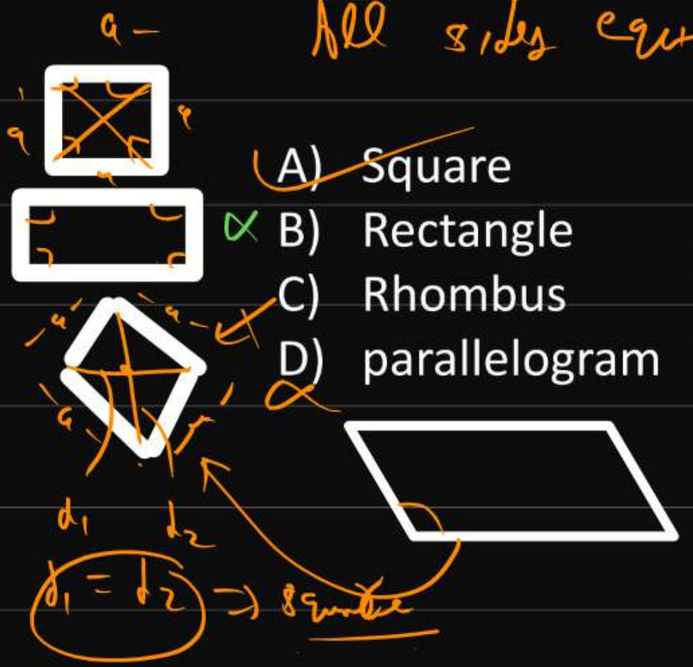
अंक (1, 7), (4, 2), (–1, –1) और (– 4, 4) कोने हैं



$d_1 = d_2$



All sides equal; Both diagonals equal



A) Square वर्ग

B) Rectangle आयत

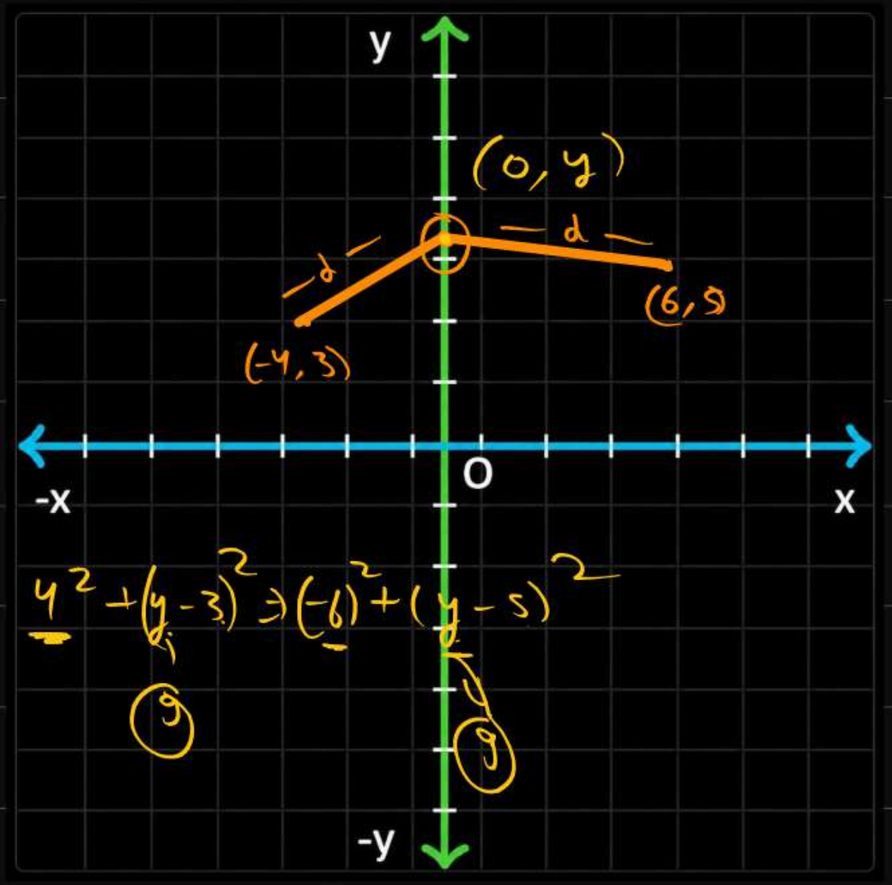
C) Rhombus समचतुर्भुज

D) parallelogram समानांतर चतुर्भुज

$d_1 = d_2$ ⇒ Square

Q. Find a point on the *y*-axis which is equidistant from the points A(6, 5) and B(– 4, 3).

Y- अक्ष पर एक बिंदु खोजें जो कि बिंदु A (6, 5) और B (- 4, 3) से समान दूरी पर है।



A) (0,9)
B) (0,4)
C) (1,9)
D) (1,4)

Q. Find a relation between x and y such that the point (x, y) is equidistant from the points $(7, 1)$ and $(3, 5)$.

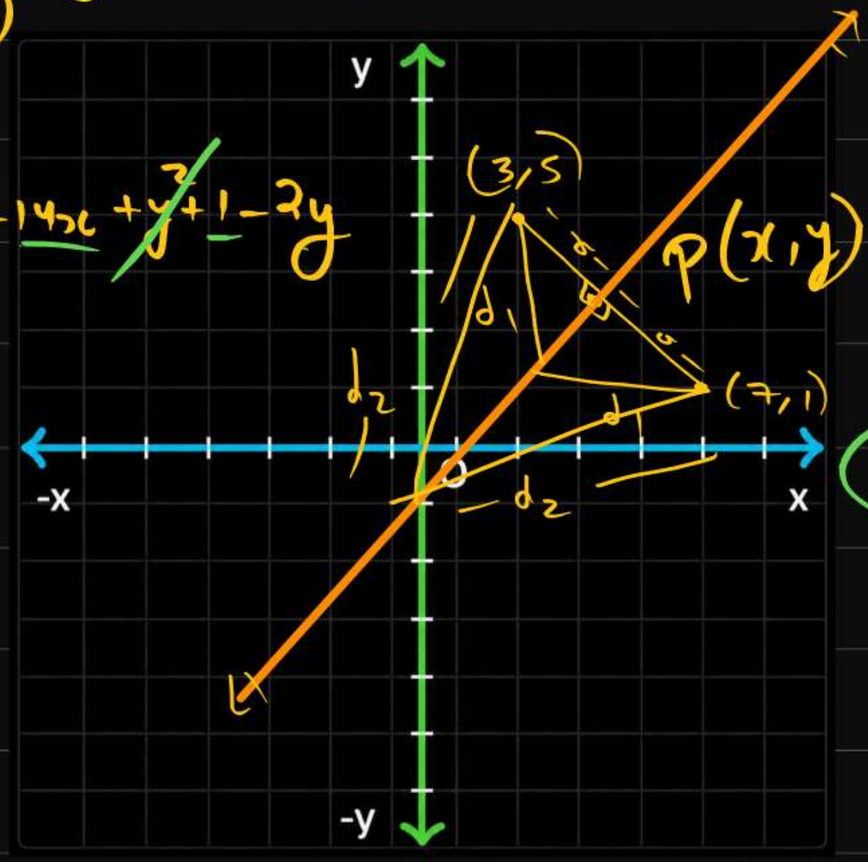
X और y के बीच एक संबंध ज्ञात करें जैसे कि बिंदु (x, y) बिंदुओं (7, 1) और (3, 5) से समान दुरी पर हो है।

$$D^2 = (x-3)^2 + (y-5)^2 = (x-7)^2 + (y-1)^2$$

$$x^2 + 9 - 6x + y^2 + 25 - 10y = x^2 + 49 - 14x + y^2 + 1 - 2y$$

$$8x - 8y = 16$$

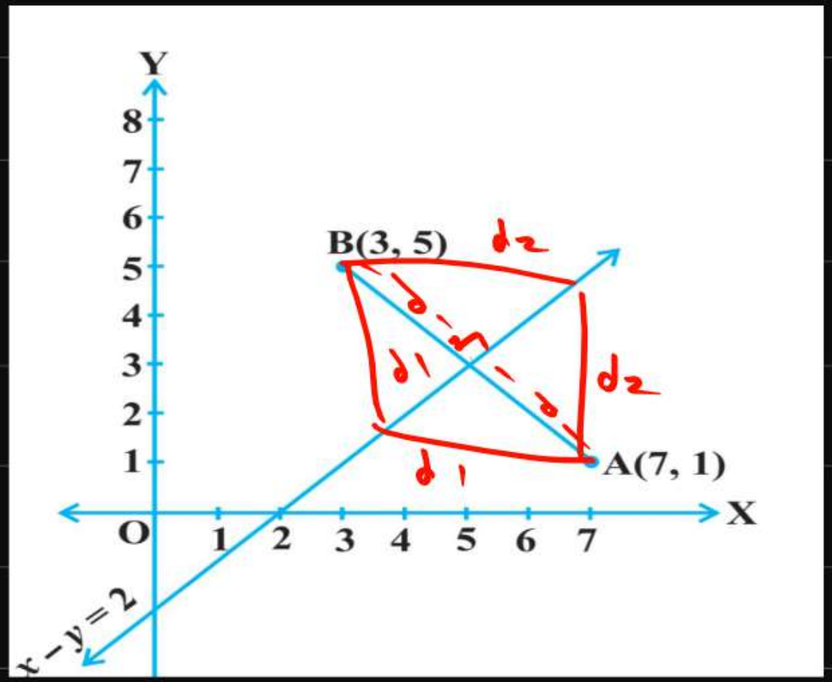
$$\boxed{x - y = 2}$$



A) 2x-y =4
B) 3x-4y= 7
C) 5x+3y =5
D) x-y = 2

Q-4) Find a relation between *x* and *y* such that the point (*x* , *y*) is equidistant from the points (7, 1) and (3, 5).

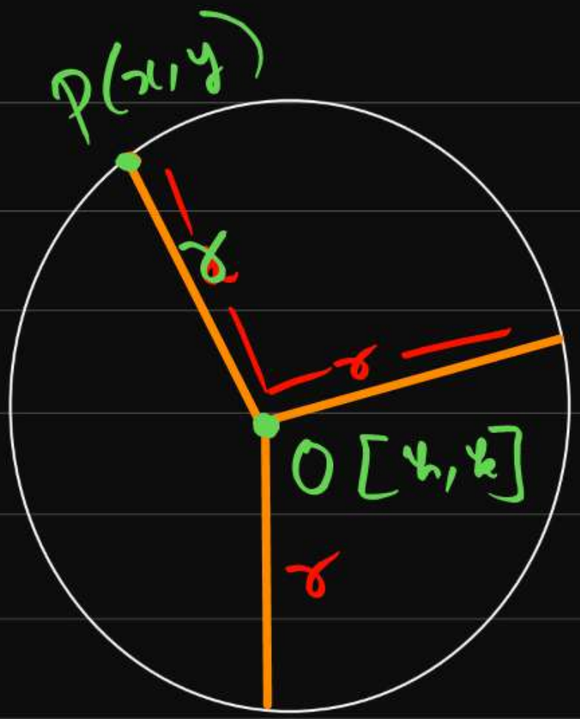
X और y के बीच एक संबंध ज्ञात करें जैसे कि बिंदु (x, y) बिंदुओं (7, 1) और (3, 5) से समान दुरी पर हो है।



A) 2x-y =4
B) 3x-4y= 7
C) 5x+3y =5
D) x-y = 2

Circle- वृत्त

$$r^2 = (x-h)^2 + (y-k)^2$$



If point O is origin

O [0, 0]

h k

$$r^2 = x^2 + y^2$$

Circle- वृत्त

A circle is the set of all points in a plane that are equidistant from a fixed point in the plane.

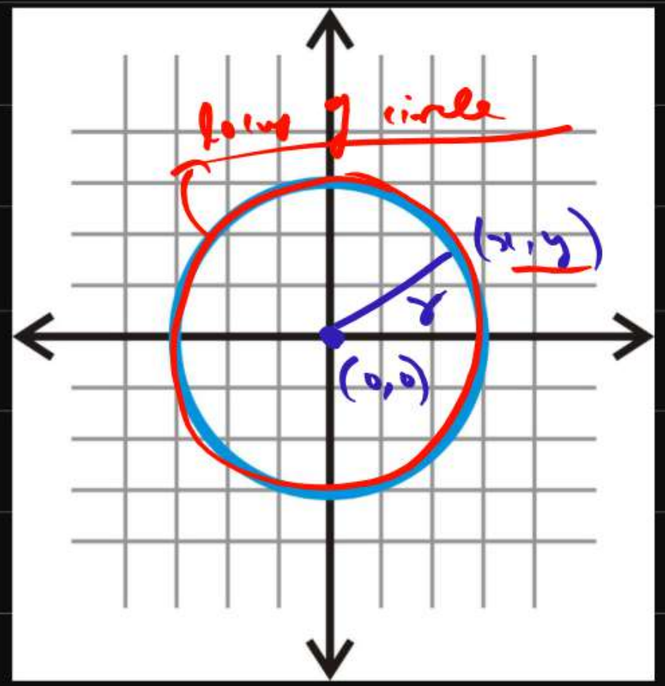
एक वृत्त एक सतह में सभी बिंदुओं का सेट होता है जो विमान में एक निश्चित बिंदु से समान होते हैं।

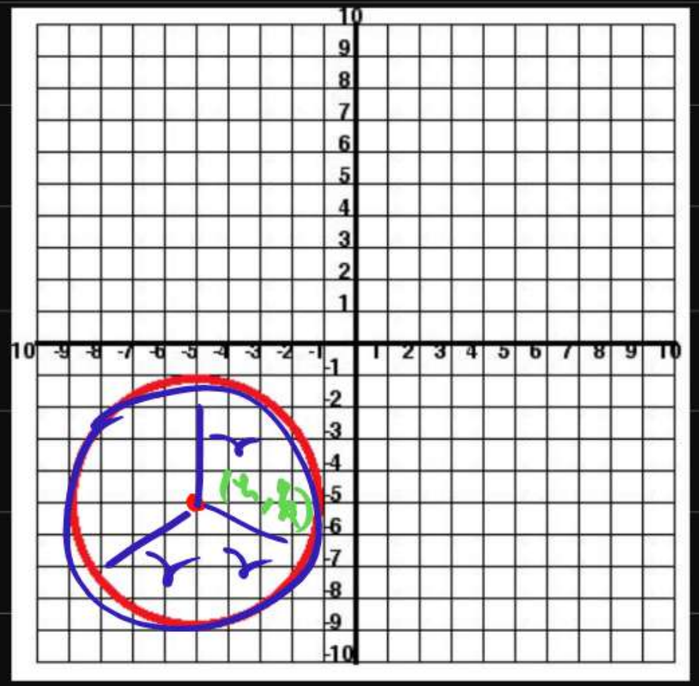
The locus(बिन्दुपथ) of a circle is defined as a set of points on a plane at the same distance from the centre point.

In Mathematics, a locus is a curve or other shape made by all the points satisfying a particular equation of the relation between the coordinates, or by a point, line, or moving surface.

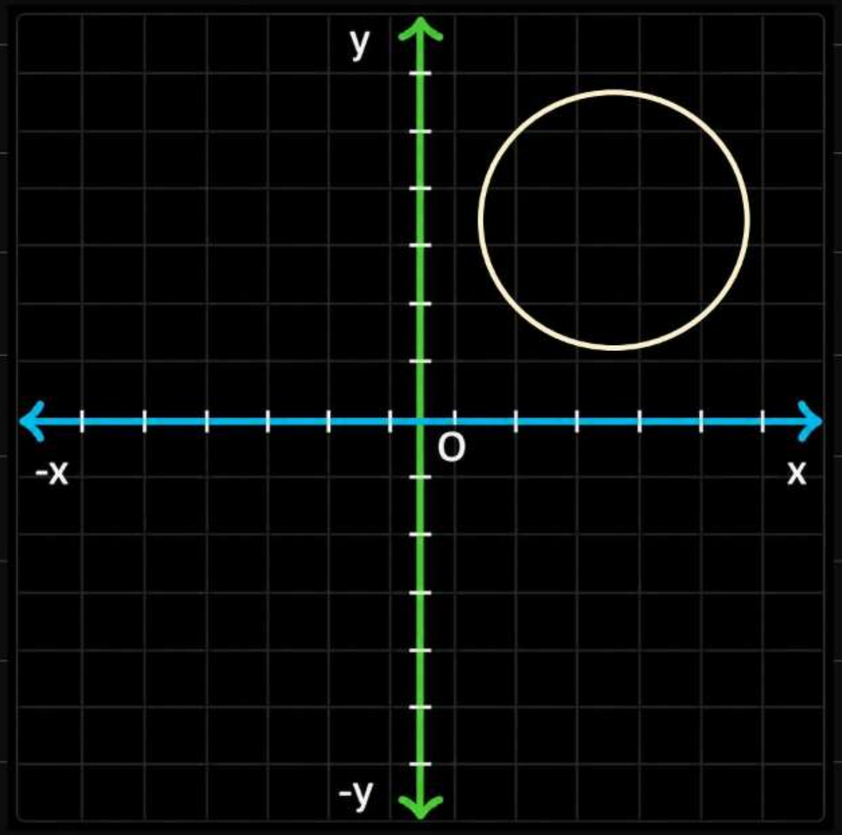
$$(x-h)^2+(y-k)^2=r^2$$

$$r^2=x^2+y^2$$

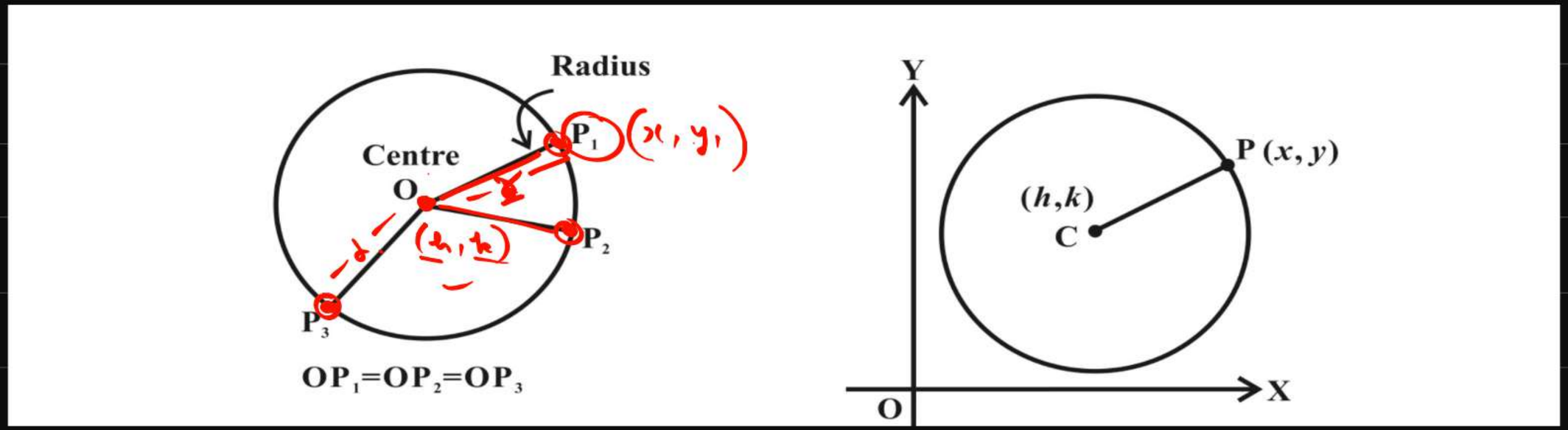




Equation of Circle-वृत्त का समीकरण



Equation of Circle-वृत्त का समीकरण

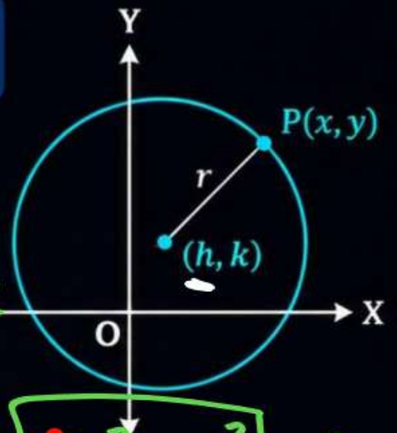


CIRCLE (वृत्त)

1. STANDARD EQUATION OF A CIRCLE
वृत्त का मानक समीकरण

The equation of a circle with centre (h, k) and radius r is

जिस वृत्त का केंद्र (h, k) तथा त्रिज्या r हो, उसका समीकरण है



$$(x-h)^2+(y-k)^2=r^2$$

Where | जहाँ

(h, k)	= Coordinates of centre	(h, k)	= केंद्र के निर्देशांक
r	= Radius of the circle	r	= वृत्त की त्रिज्या

KEY POINTS | मुख्य बातें

- If centre is $(0, 0)$, equation becomes $x^2+y^2=r^2$
- All points on the circle are at a distance r from the centre.

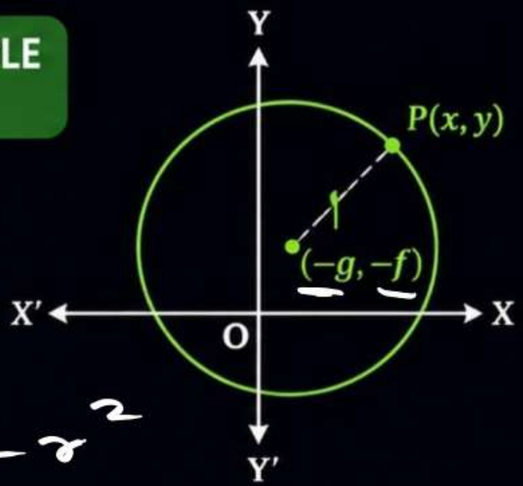
- यदि केंद्र $(0, 0)$ हो, तो समीकरण होगा $x^2+y^2=r^2$
- वृत्त पर स्थित सभी बिंदु केंद्र से r दूरी पर होते हैं।

2. GENERAL EQUATION OF A CIRCLE
वृत्त का सामान्य समीकरण

The general equation of a circle is

वृत्त का सामान्य समीकरण है

$$x^2+y^2+2gx+2fy+c=0$$



Where | जहाँ

Centre $=(-g, -f)$

Radius $=\sqrt{g^2+f^2-c}$

केंद्र $=(-g, -f)$

त्रिज्या $=\sqrt{g^2+f^2-c}$

3. IMPORTANT CONDITIONS | महत्वपूर्ण स्थितियाँ

(i) REAL CIRCLE
(i) वास्तविक वृत्त

$g^2+f^2-c>0$

Radius is real

त्रिज्या वास्तविक है

(ii) POINT CIRCLE
(ii) बिंद वृत्त

$g^2+f^2-c=0$

Radius is zero

त्रिज्या शून्य है

(iii) IMAGINARY CIRCLE
(iii) काल्पनिक वृत्त

$g^2+f^2-c<0$

Radius is imaginary

त्रिज्या काल्पनिक है

Remember | याद रखें: Centre (h, k) और Radius r जानने के लिए मानक समीकरण का उपयोग करें। | यदि समीकरण general form में हो, तो g, f, c से केंद्र और त्रिज्या ज्ञात करें।

Q. The equation of circle with centre (1, -2) and radius 4cm is:

केंद्र (1, -2) और त्रिज्या 4 सेमी के साथ वृत्त का समीकरण है:

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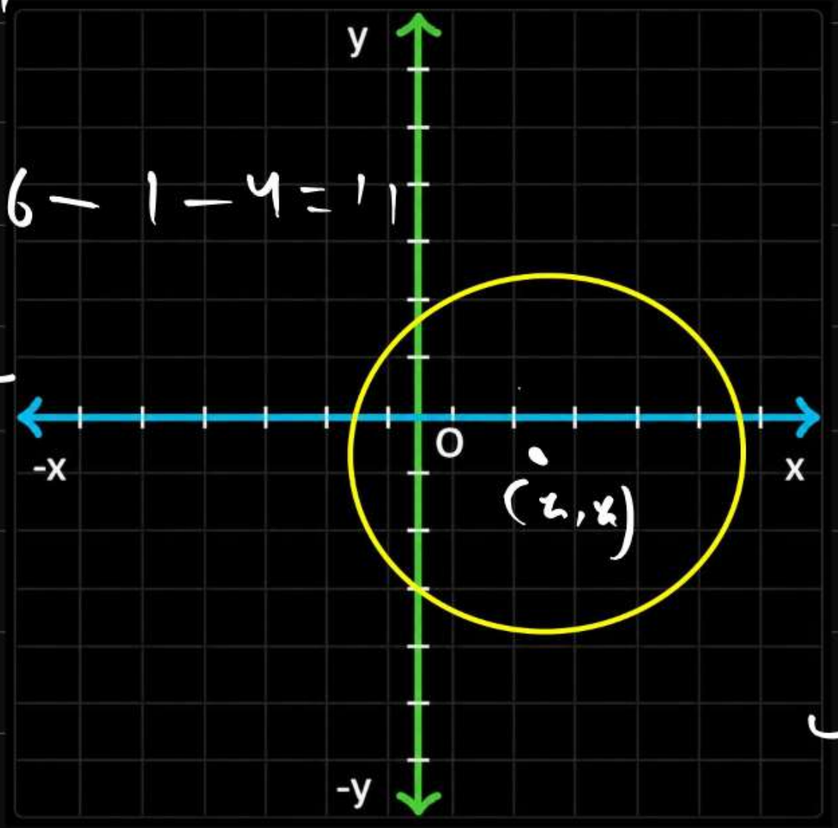
$$(x-h)^2+(y-k)^2=r^2$$

$$(x-1)^2+(y-(-2))^2=4^2$$

$$x^2+y^2-2x+4y=16-1-4=11$$

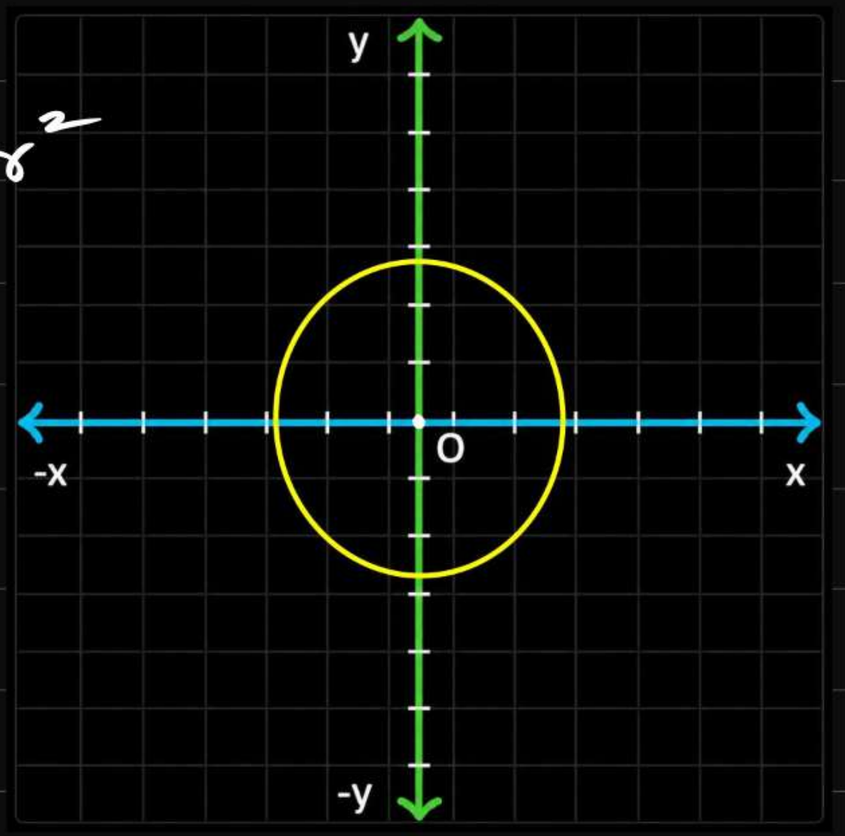


(a) $x^2+y^2+2x-4y=16$
(b) $x^2+y^2-2x+4y=16$
(c) $x^2+y^2+2x-4y=11$
(d) $x^2+y^2-2x+4y=11$

Q. Find an equation of the circle with centre at (0,0) and radius *r*.

केंद्र में (0,0) और त्रिज्या r के साथ वृत्त का एक समीकरण ज्ञात कीजिए।

$x^2+y^2=r^2$



Q. What is the equation of a circle with centre at origin and radius 6 cm?

उस वृत्त का समीकरण क्या है जिसका केंद्र मूल बिंदु पर है और त्रिज्या 6 cm है?

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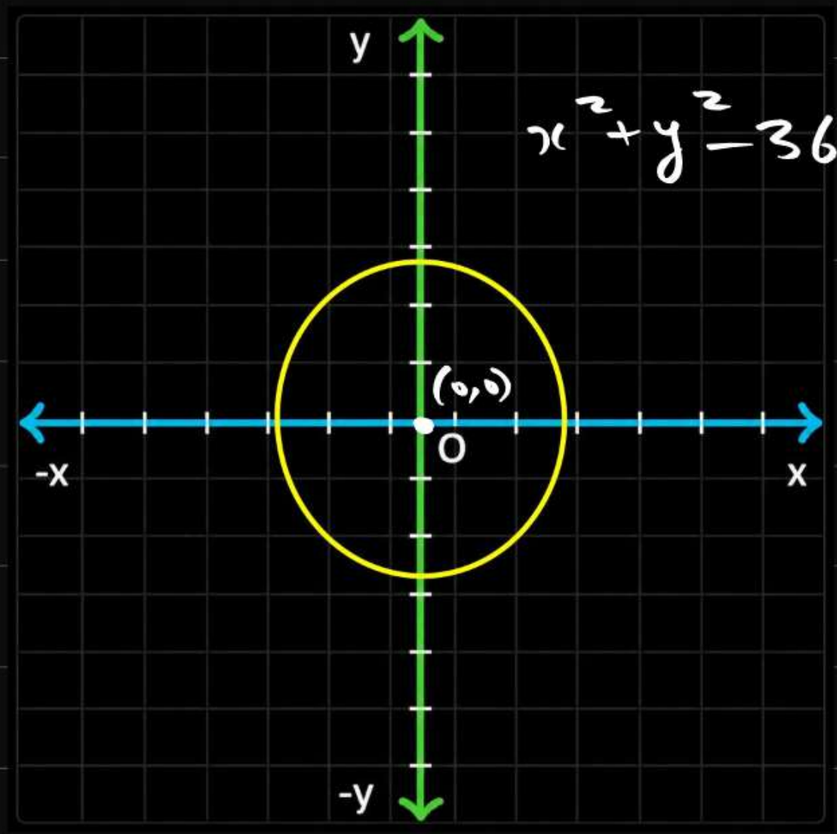

(a) $x^2 + y^2 - x - y = 36$

(b) $x^2 + y^2 - x = 36$

(c) $x^2 + y^2 - y = 36$

(d) $x^2 + y^2 - 36 = 0$

Q. Find the equation of a circle whose diameter has endpoints (4, 3) and (-2, 1).
एक वृत्त के समीकरण का पता लगाएं जिसके व्यास के अंत बिंदु (4, 3) और (-2, 1) है।

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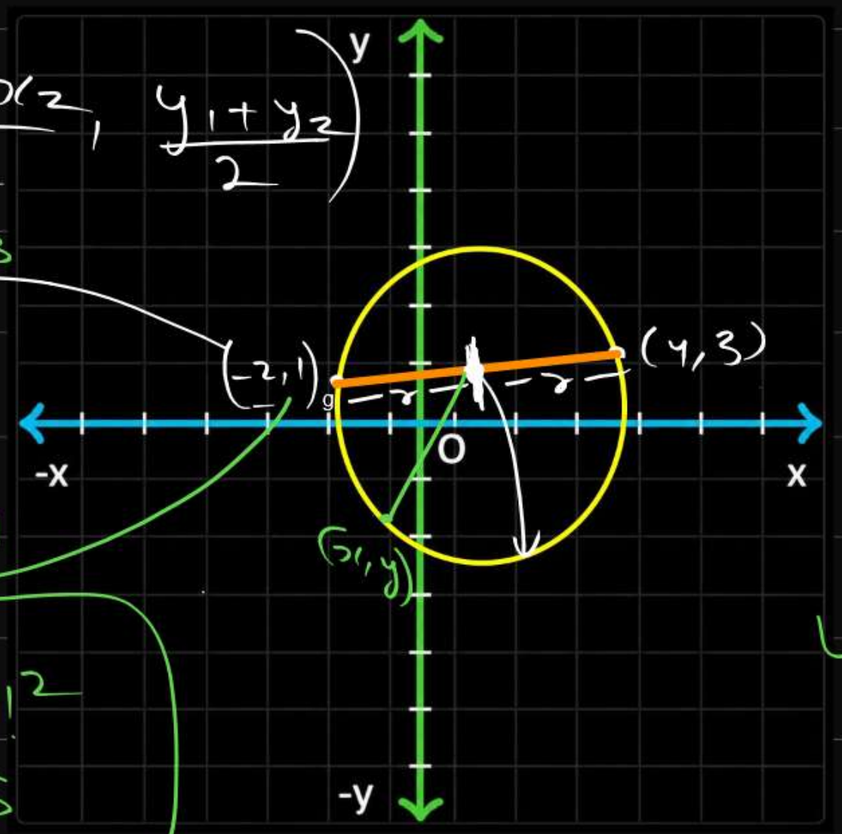
$$\text{Midpoint} = \left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$$

$$h = \frac{-2+4}{2} \Rightarrow 1$$

$$k = \frac{1+3}{2} = 2$$

$$(x-1)^2+(y-2)^2 = 3^2+1^2$$

$$x^2+y^2-2x-4y \Rightarrow 5$$



(a) $x^2 + y^2 - 2x - 4y = 3$
(b) $x^2 + y^2 - 2x - 4y = 5$
(c) $x^2 + y^2 - 6x + 2y = 3$
(d) $x^2 + y^2 - 2x + 4y = 5$

Q. Find the co-ordinates of the centre of the circle which passes through the points (2, – 2), and (3,4) and whose centre lies on the line $x + y = 1$.

वृत्त के केंद्र के निर्देशांक ज्ञात कीजिए जो बिंदु (2, - 2), और (3,4) से होकर गुजरता है और जिसका केंद्र लाइन $x + y = 1$ पर है।

$2h + 2k = 1 \times 2$

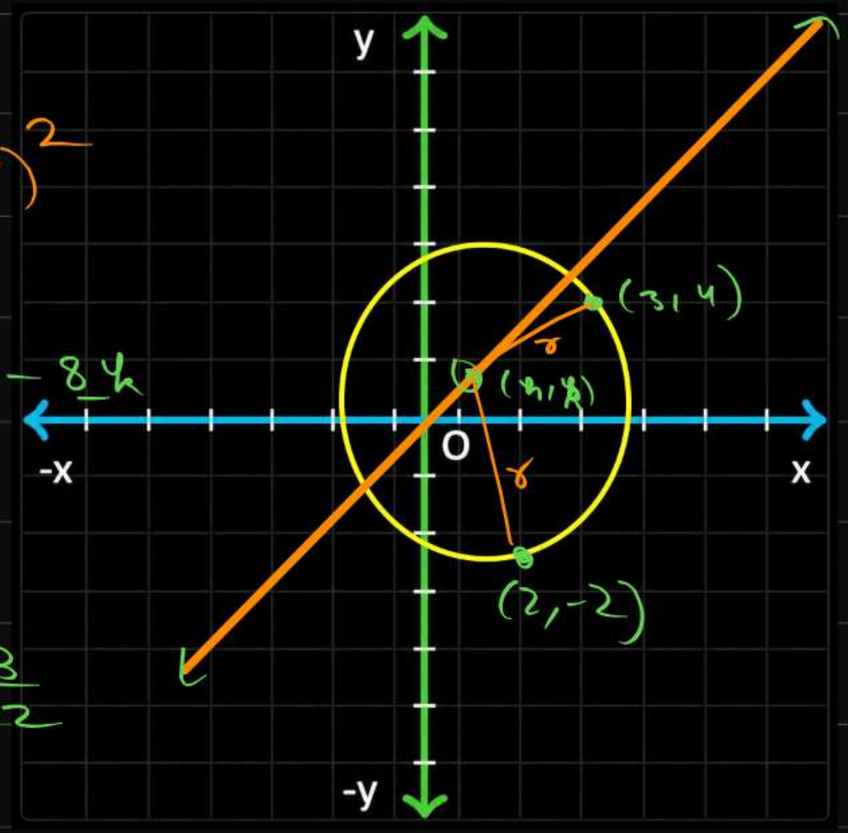
$(h-2)^2 + (k+2)^2 = (h-3)^2 + (k-4)^2$

$(4) - 4h + (4) + 4k = 9 - 6h + 16 - 8k$

$2h + 12k = 17$

$10k = 15 \; ; \; k = \frac{3}{2}$

$h = \frac{-1}{2}$



Expected Question

$\left(\frac{-1}{2}, \frac{3}{2}\right)$

Q. A circle passing through the points (0,0), (-4,2) and (-6,4). Find the radius of the circle.

बिंदु (0,0), (-4,2) और (-6,4) से होकर गुजरने वाला एक वृत है। वृत की त्रिज्या ज्ञात कीजिए।

Expected Question

$r^2 = a^2 + b^2 = (a+4)^2 + (b-2)^2 = (a+6)^2 + (b-4)^2$

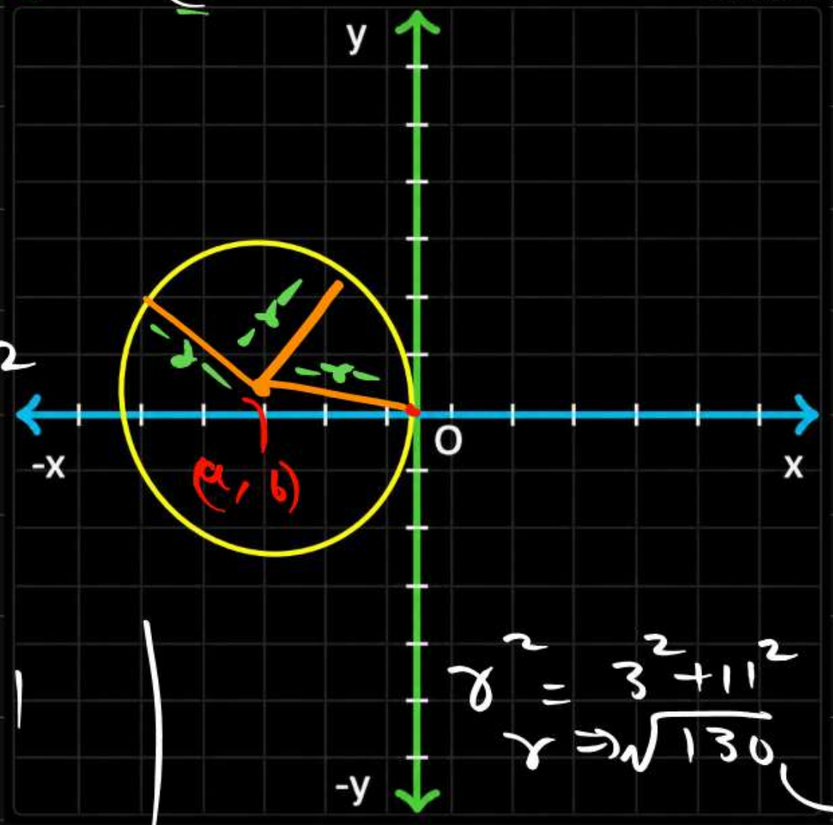
$-16 - 4 = 8a - 4b = -20$

$(2a - b = -5) \times 2$

$12a - 8b = -52$

$3a - 2b = -13$

$a = 3 \; ; \; b = 11$



$r^2 = 3^2 + 11^2$

$r = \sqrt{130}$

(a) √145
(b) √139
(c) √122
(d) √130

$$x^2 + y^2 + 24x + 36y - 13 = 0$$

$2 \times 12 \times x$ $\quad$ $2 \times 18 \times y$

g $\quad$ f $\quad$ c

$$r = \sqrt{(12)^2 + (18)^2 - (-13)} = \sqrt{f^2 + g^2 - c}$$

$$\Rightarrow \sqrt{144 + 324 + 13}$$

$$\Rightarrow \sqrt{481} \leftrightharpoons 22^{4}$$

$$(x-h)^2+(y-k)^2=r^2$$

$2\times12\times x$ $\qquad$ $2\times18\times y$

$$x^2+y^2+24x+36y-13=0$$

$$x^2+24x+12^2-12^2$$

$$y^2+36y+18^2-18^2$$

$$-13=0$$

$$\left[(x+12)^2+(y+18)^2-12^2-18^2\right]-13=0$$

$$(x+12)^2+(y+18)^2=12^2+18^2+13$$

$$\|$$

$$r^2$$

$(x_1, y_1) \leftrightarrow (x_2, y_2)$

$d = \sqrt{(x_2-x_1)^2 + (y_2-y_1)^2}$

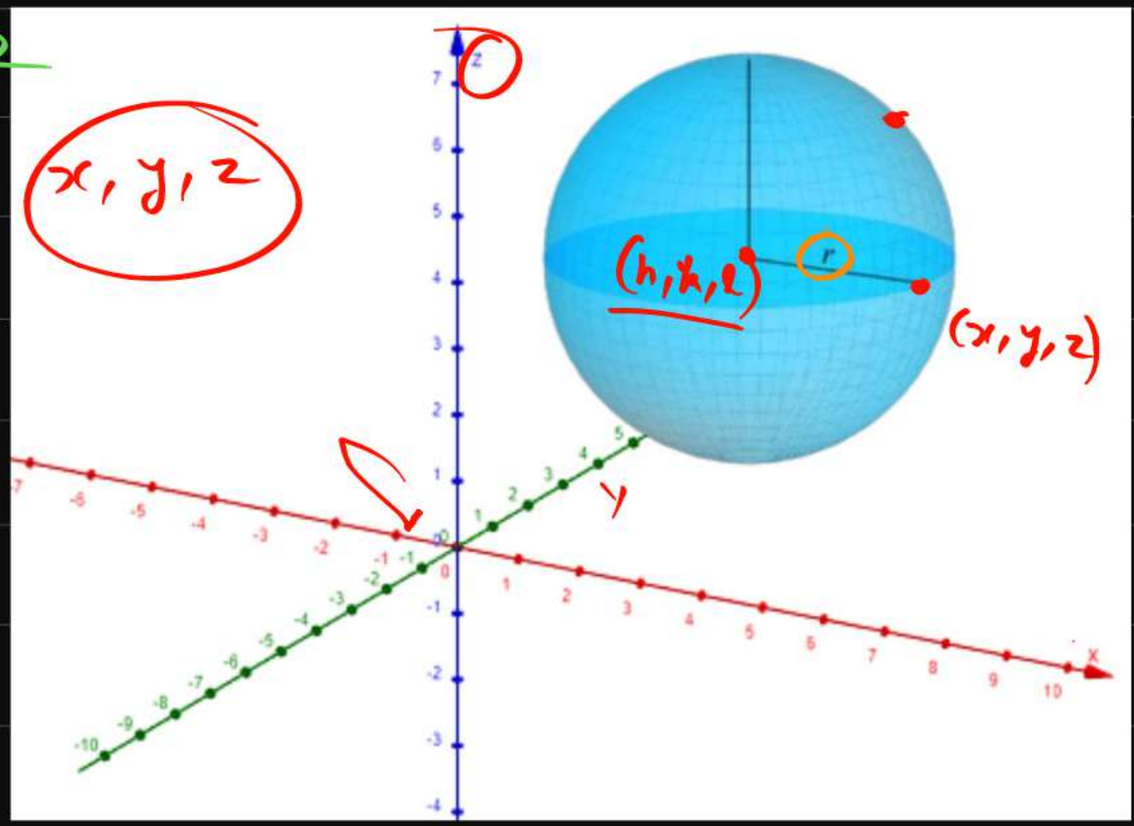
Equation of sphere

(x_1, y_1, z_1) (x_2, y_2, z_2)

$$D^2 = (x_2-x_1)^2 + (y_2-y_1)^2 + (z_2-z_1)^2$$

x, y, z

$$r^2 = (x-h)^2 + (y-k)^2 + (z-l)^2$$



3 D cartesian plane

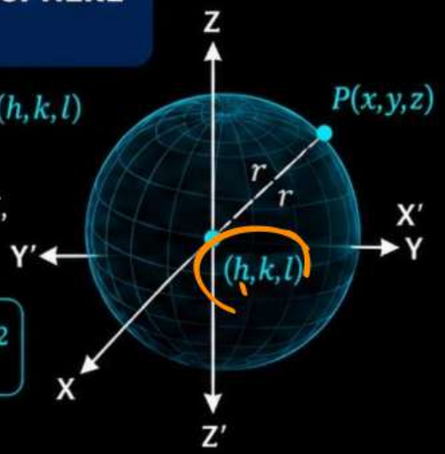
SPHERE (गोला)

1. STANDARD EQUATION OF SPHERE
गोले का मानक समीकरण

The equation of a sphere with centre (h,k,l) and radius r is

जिस गोले का केंद्र (h, k, l) और त्रिज्या r हो, उसका समीकरण है

$$(x-h)^2+(y-k)^2+(z-l)^2=r^2$$



Where | जहाँ

(h, k, l)	= Coordinates of centre	(h, k, l)	= केंद्र के निर्देशांक
r	= Radius of the sphere	r	= गोले की त्रिज्या

KEY POINTS | मुख्य बातें

- If centre is $(0, 0, 0)$, equation becomes $x^2+y^2+z^2=r^2$
- All points on the sphere are at a distance r from the centre.

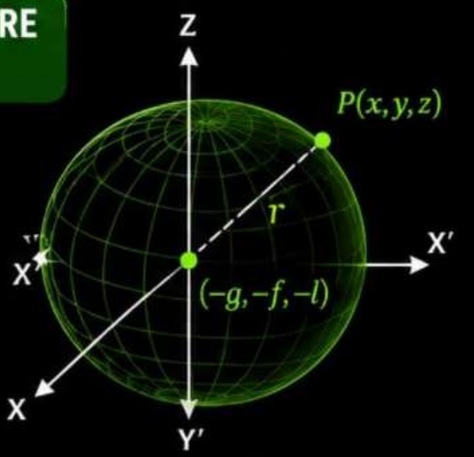
- यदि केंद्र $(0, 0, 0)$ हो, तो समीकरण होगा $x^2+y^2+z^2=r^2$
- गोले पर स्थित सभी बिंदु केंद्र से r दूरी पर होते हैं।

2. GENERAL EQUATION OF SPHERE
गोले का सामान्य समीकरण

The general equation of a sphere is

गोले का सामान्य समीकरण है

$$x^2+y^2+z^2+2gx+2fy+2lz+c=0$$



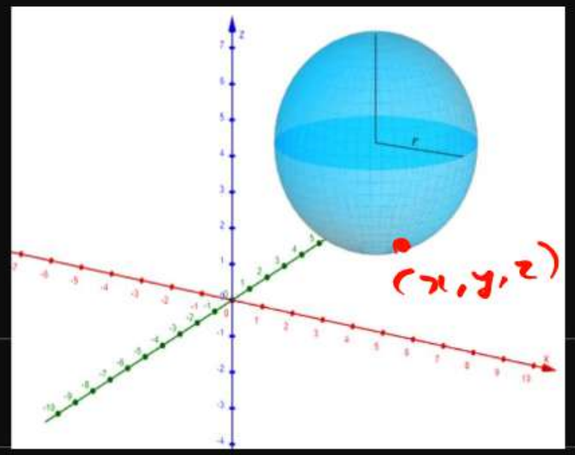
Where | जहाँ

Centre	$= (-g, -f, -l)$	केंद्र	$= (-g, -f, -l)$
Radius	$= \sqrt{g^2+f^2+l^2-c}$	त्रिज्या	$= \sqrt{g^2+f^2+l^2-c}$

3. IMPORTANT CONDITIONS | महत्वपूर्ण स्थितियाँ

(i) REAL CPHERE (i) वास्तविक गोला	(ii) POINT SPHERE (ii) बिंदु गोला	(iii) IMAGINARY SPHERE (iii) काल्पनिक गोला
$g^2+f^2+l^2-c>0$	$g^2+f^2+l^2-c=0$	$g^2+f^2+l^2-c<0$
Radius is real त्रिज्या वास्तविक है	Radius is zero त्रिज्या शून्य है	Radius is imaginary त्रिज्या काल्पनिक है

Remember | याद रखें: Centre (h, k, l) और Radius r जानने के लिए मानक समीकरण का उपयोग करें।
केंद्र और त्रिज्या ज्ञात करने के लिए सामान्य समीकरण में g, f, l, c का उपयोग करें।



Q. Find the equation of the sphere having centre (-1,2,-3) and the radius of 3 units.

केंद्र (-1,2,-3) और 3 इकाइयों की त्रिज्या वाले गोले का समीकरण ज्ञात करें।

$(x+1)^2 + (y-2)^2 + (z+3)^2 = 3^2$

$1 \qquad 4 \qquad 9$

$x^2 + y^2 + z^2 + 2x - 4y + 6z + 5 = 0$

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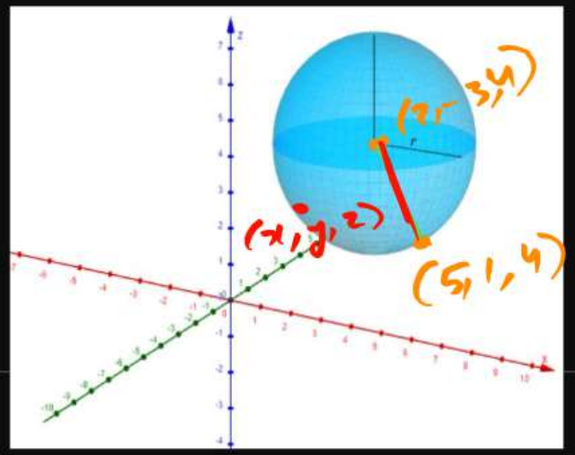
(a) $x^2 + y^2 + z^2 + x - 2y + 3z + 5 = 0$

(b) $x^2 + y^2 + z^2 - x + 2y - 3z + 5 = 0$

(c) $x^2 + y^2 + z^2 + 2x - 4y + 6z + 5 = 0$

(d) $x^2 + y^2 + z^2 - 2x + 4y - 6z + 5 = 0$



Q. Find the equation of the sphere having centre (2, –3, 4) and passing through the point (5, 1, 4).

केंद्र (2, –3, 4) वाले तथा बिंदु (5, 1, 4) से होकर गुजरने वाले गोले का समीकरण ज्ञात करें।

$(x-2)^2+(y+3)^2+(z-4)^2 = 3^2+4^2+0^2 = 25$

2^2 3^2 4^2

$x^2+y^2+z^2-4x+6y-8z+4=0$

(a) $(x-2)^2+(y+3)^2+(z-4)^2=25$ ✓

(b) $(x+2)^2+(y-3)^2+(z+4)^2=25$

(c) $(x-2)^2+(y+3)^2+(z-4)^2=16$

(d) $(x-2)^2+(y-3)^2+(z-4)^2=25$

Co-ordinate Geometry

निर्देशांक ज्यामिति

(Basic to high Level)

Latest Pattern

Co-ordinate Geometry

Part-3

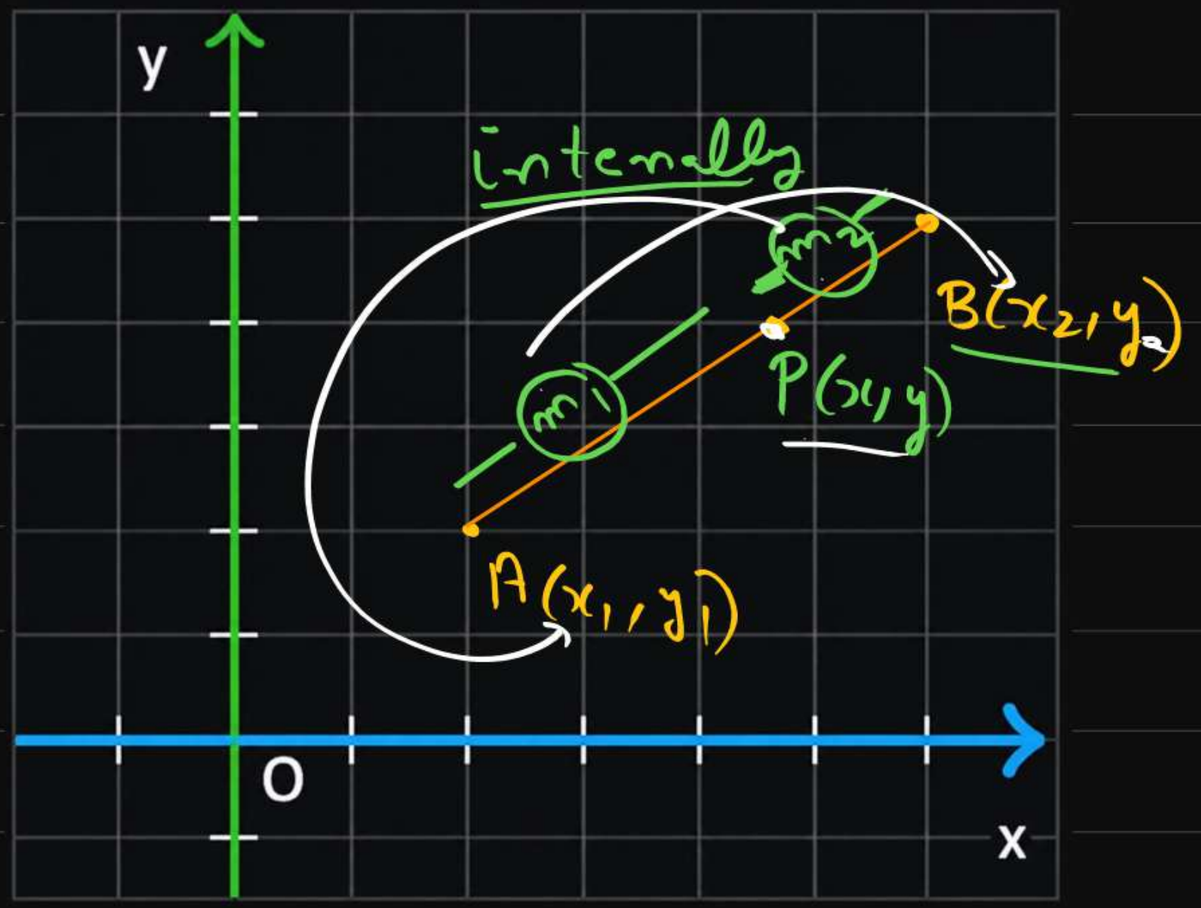
Section formula and Area of a triangle | Quad

Section Formula विभाजन सूत्र

(internally)

$$x = \frac{m_1 x_2 + m_2 x_1}{m_1 + m_2}$$

$$y = \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2}$$

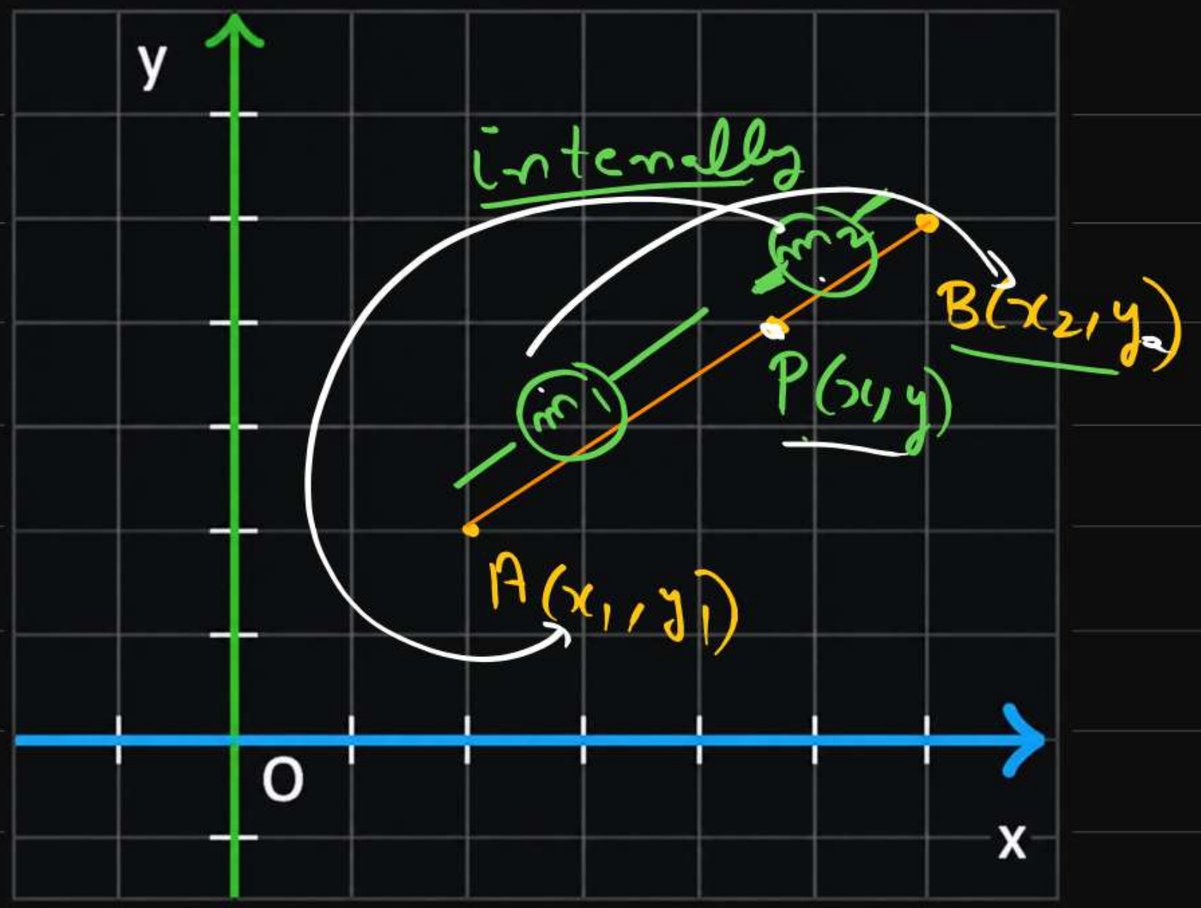


Section Formula विभाजन सूत्र

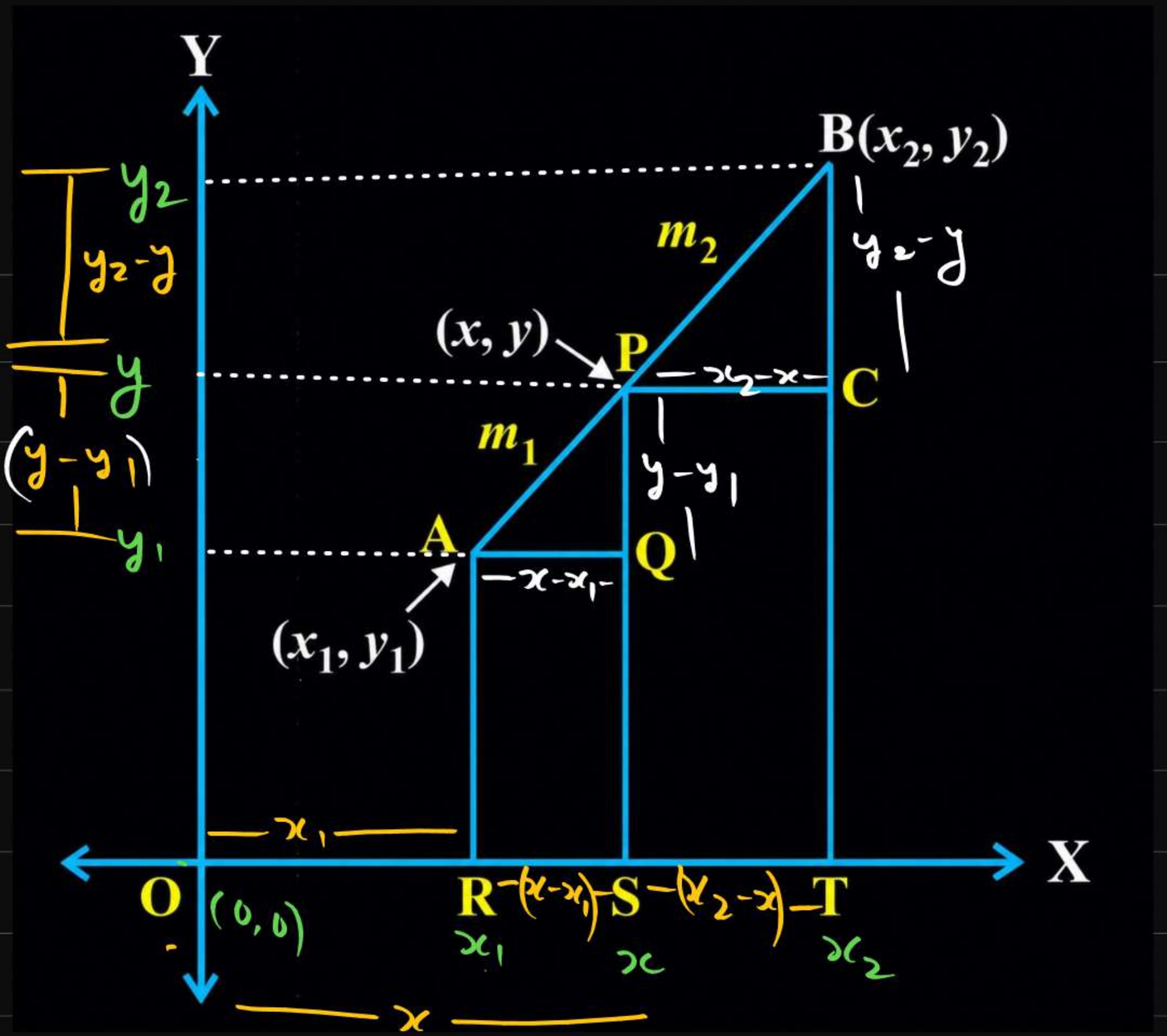
(internally)

$$x = \frac{m_1 x_2 + m_2 x_1}{m_1 + m_2}$$

$$y = \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2}$$



Section Formula विभाजन सूत्र
(Internally)



Section Formula विभाजन सूत्र
(Internally)

$$\cos\theta = \frac{B}{H} = \frac{x - x_1}{m_1 k} = \frac{x_2 - x}{m_2 k}$$

$$x(m_2 + m_1) = m_1 x_2 + m_2 x_1$$

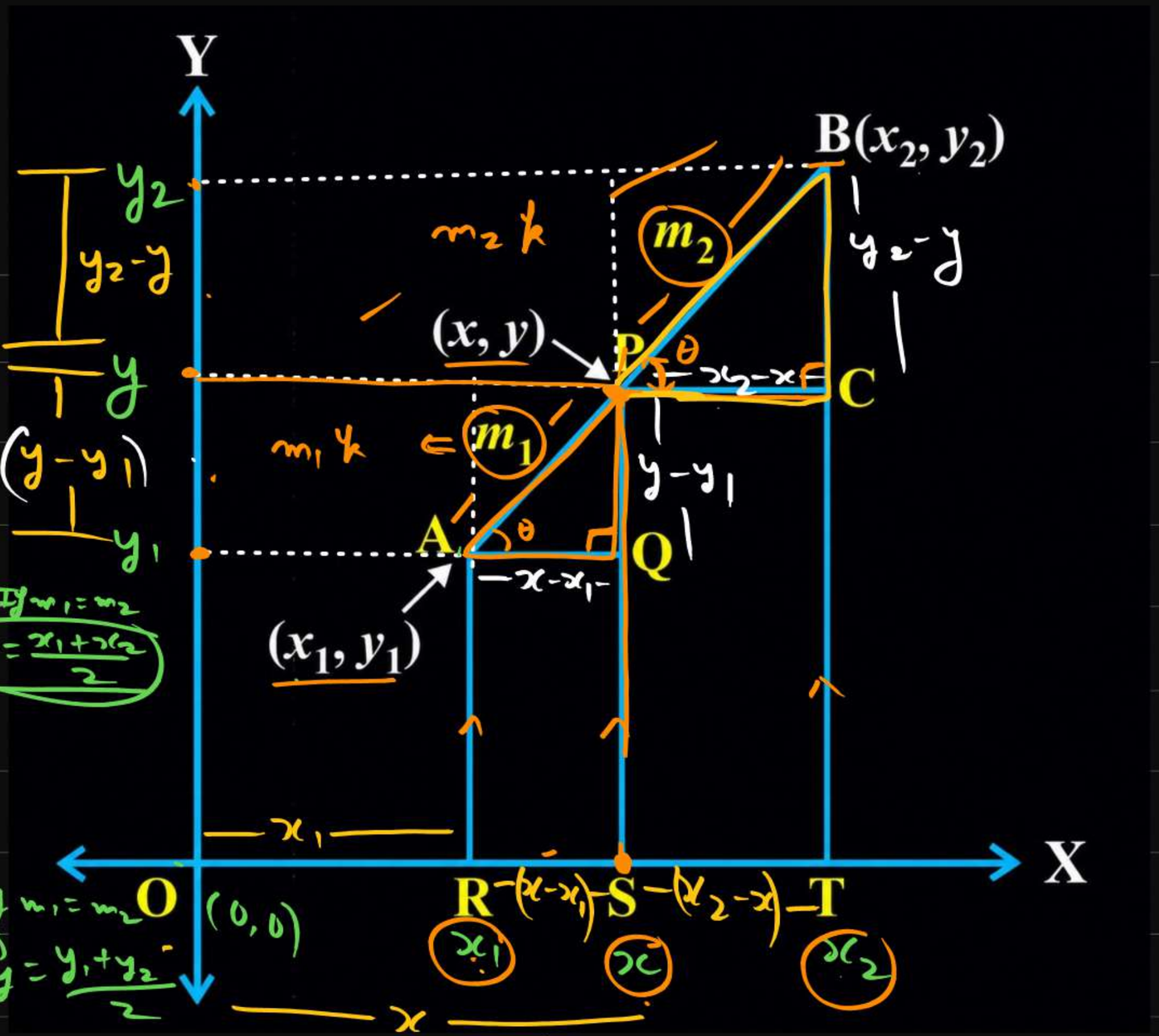
$$x = \frac{m_1 x_2 + m_2 x_1}{m_1 + m_2}$$

If $m_1 = m_2$, $x = \frac{x_1 + x_2}{2}$

$$\sin\theta = \frac{P}{H} = \frac{y - y_1}{m_1 k} = \frac{y_2 - y}{m_2 k}$$

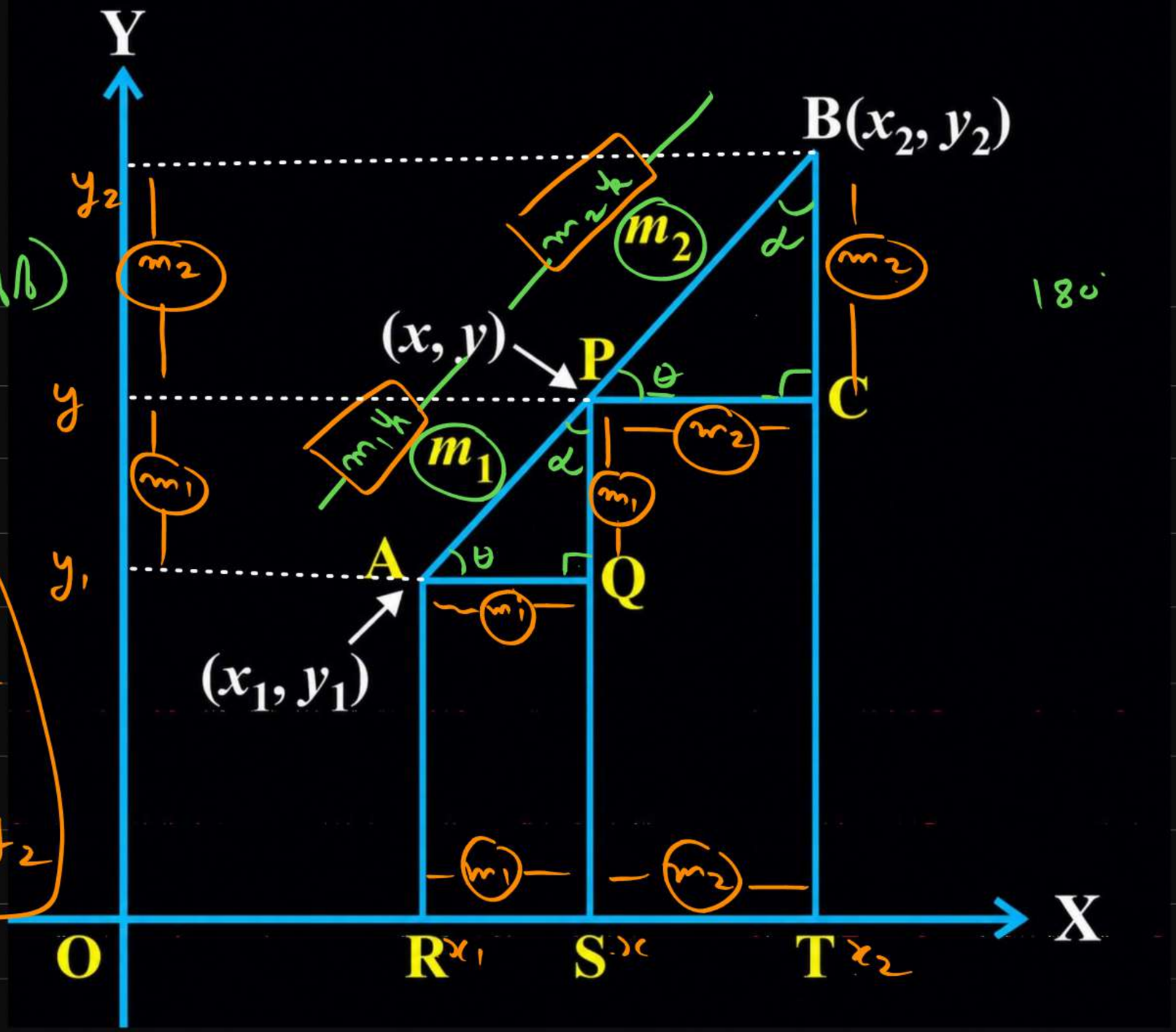
$$y = \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2}$$

If $m_1 = m_2$, $y = \frac{y_1 + y_2}{2}$



$\triangle AQP \sim \triangle PCB$ (AAA)

$$\frac{AQ}{PC} = \frac{QP}{CB} = \frac{AP}{PB} = \frac{m_1}{m_2}$$

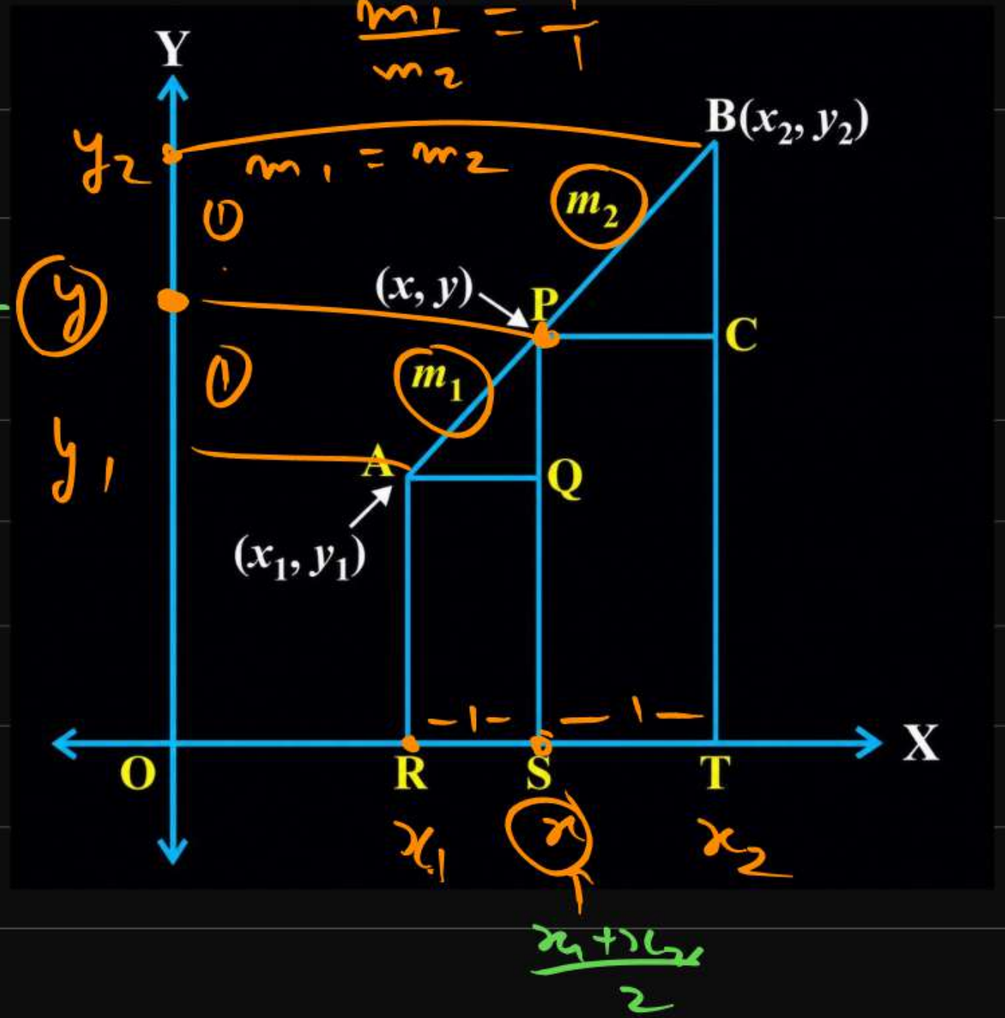


If m:n = 1:1. Put these values in section formula, you will get Mid point formula

Mid Point formulae

The mid-point of the line segment joining the points $P(x_1,y_1)$ and $Q(x_2,y_2)$ is

$\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$



Section Formula विभाजन सूत्र (Externally)

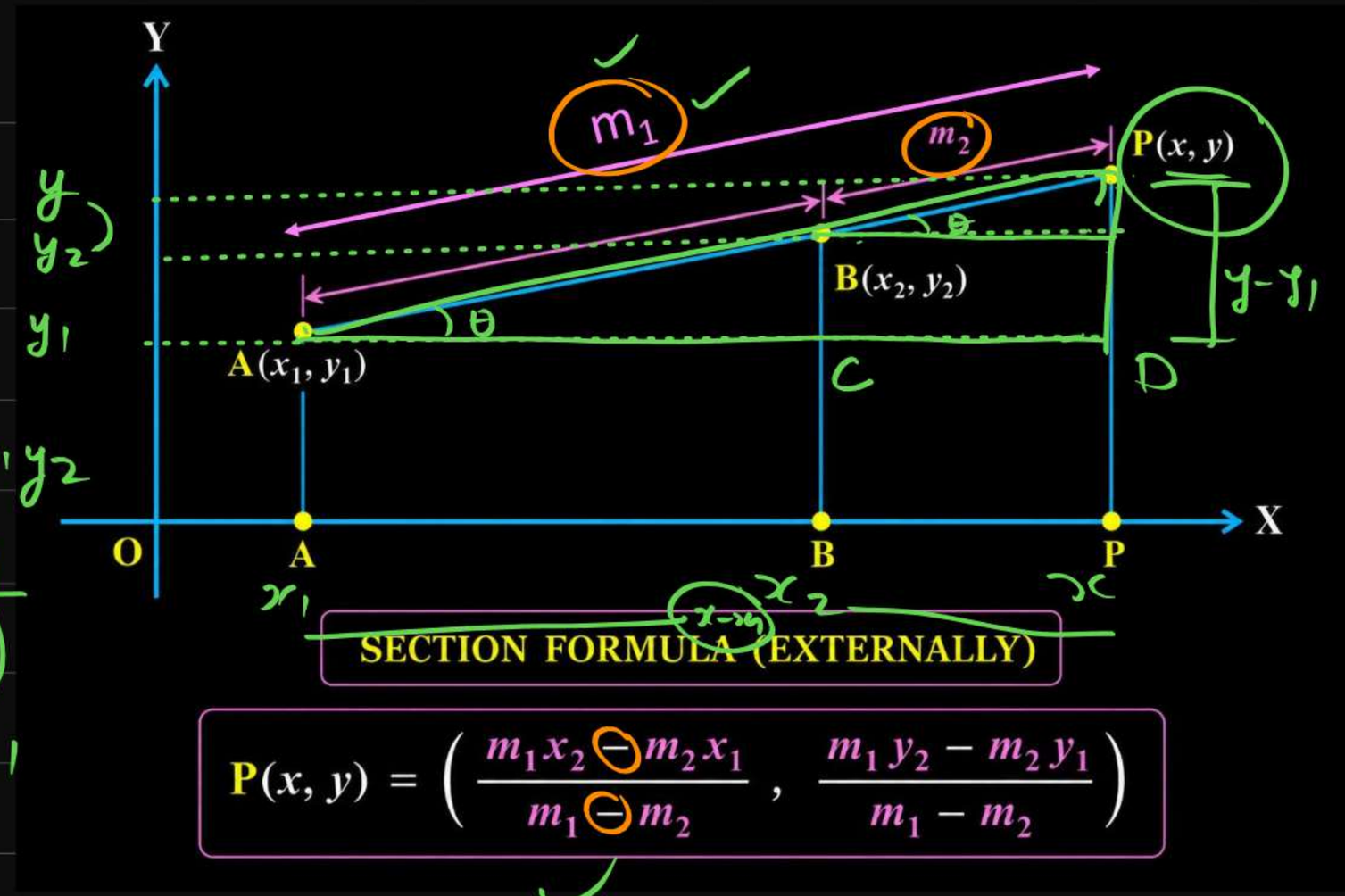
$m_1 : m_2$

$\triangle APD = \frac{y - y_1}{m_1} = \frac{y - y_2}{m_2}$

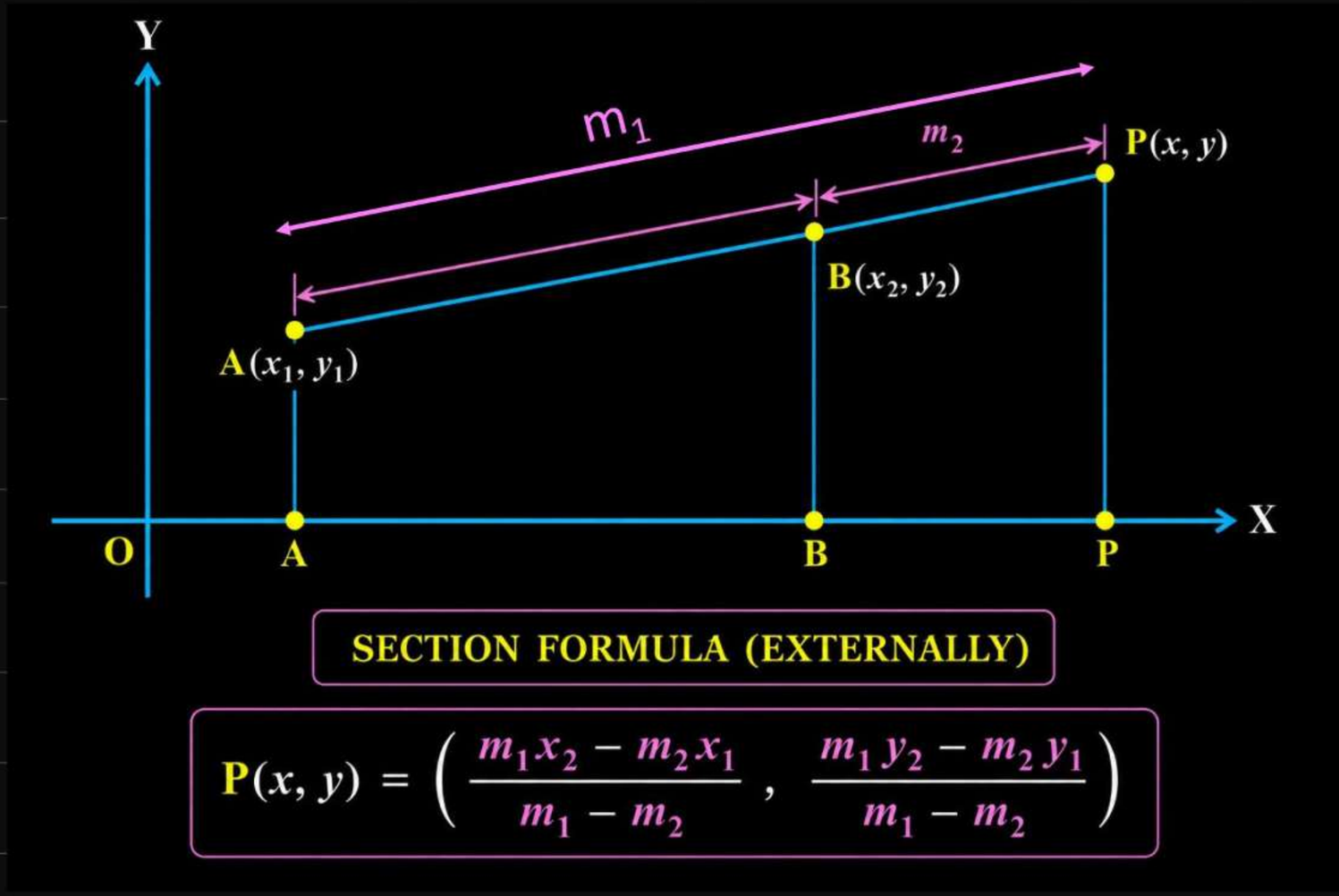
$y[m_2 - m_1] = m_2 y_1 - m_1 y_2$

$y = \frac{(m_2 y_1 - m_1 y_2)}{-(m_2 - m_1)}$

$y \Rightarrow \frac{m_1 y_2 - m_2 y_1}{m_1 - m_2}$



Section Formula विभाजन सूत्र
(Externally)



SECTION FORMULA (EXTERNALLY)

$$\mathbf{P}(x, y) = \left(\frac{m_1 x_2 - m_2 x_1}{m_1 - m_2} , \frac{m_1 y_2 - m_2 y_1}{m_1 - m_2} \right)$$

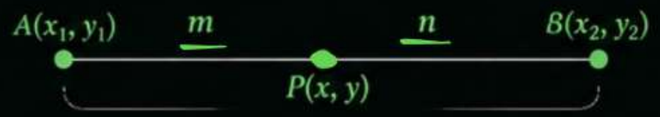
SECTION FORMULA & MID POINT FORMULA

निर्देशांक ज्यामिति का महत्वपूर्ण सूत्र (Important Formulas of Coordinate Geometry)

1 SECTION FORMULA – INTERNAL DIVISION
(आंतरिक विभाजन सूत्र)

यदि बिंदु $P(x, y)$ रेखा खंड AB को आंतरिक रूप से $m:n$ के अनुपात में विभाजित करता है, जहां $A(x_1, y_1)$ और $B(x_2, y_2)$ हैं।

If point $P(x, y)$ divides the line segment AB internally in the ratio $m:n$, where $A(x_1, y_1)$ and $B(x_2, y_2)$ are the end points.



$$P(x, y) = \left(\frac{mx_2 + nx_1}{m + n}, \frac{my_2 + ny_1}{m + n} \right)$$

जहाँ / Where,

- $A(x_1, y_1)$ = पहला बिंदु / First point
- $B(x_2, y_2)$ = दूसरा बिंदु / Second point
- $m:n$ = विभाजन का अनुपात / Ratio of division
- $P(x, y)$ = विभाजित बिंदु / Point of division

★ नोट / Note:
यह सूत्र केवल तब उपयोग करते हैं जब बिंदु रेखा खंड के बीच (बीच के अंदर) हो।
Use this formula only when the point lies between the two end points.

2 SECTION FORMULA – EXTERNAL DIVISION
(बाह्य विभाजन सूत्र)

यदि बिंदु $P(x, y)$ रेखा खंड AB को बाह्य रूप से $m:n$ के अनुपात में विभाजित करता है, जहां $A(x_1, y_1)$ और $B(x_2, y_2)$ हैं।

If point $P(x, y)$ divides the line segment AB externally in the ratio $m:n$, where $A(x_1, y_1)$ and $B(x_2, y_2)$ are the end points.

$$P(x, y) = \left(\frac{mx_2 - nx_1}{m - n}, \frac{my_2 - ny_1}{m - n} \right)$$

जहाँ / Where,

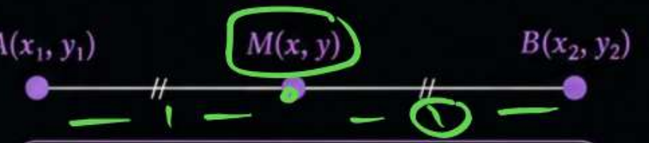
- $A(x_1, y_1)$ = पहला बिंदु / First point
- $B(x_2, y_2)$ = दूसरा बिंदु / Second point
- $m:n$ = बाह्य विभाजन का अनुपात / Ratio of external division
- $P(x, y)$ = विभाजित बिंदु / Point of division

★ नोट / Note:
यह सूत्र तब उपयोग करते हैं जब बिंदु रेखा खंड के बाहर किसी ओर स्थित हो।
Use this formula when the point lies outside the line segment.

3 MID POINT FORMULA
(मध्य बिंदु सूत्र)

यदि $A(x_1, y_1)$ और $B(x_2, y_2)$ रेखा खंड के दो सिर बिंदु हैं, तो रेखा खंड AB का मध्य बिंदु $M(x, y)$ होगा।

If $A(x_1, y_1)$ and $B(x_2, y_2)$ are the end points of a line segment, then the mid point $M(x, y)$ of AB is:



$$M(x, y) = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

जहाँ / Where,

- $A(x_1, y_1)$ = पहला सिर बिंदु / First end point
- $B(x_2, y_2)$ = दूसरा सिर बिंदु / Second end point
- $M(x, y)$ = मध्य बिंदु / Mid point

★ विशेष बात / Special Point:
मध्य बिंदु वह बिंदु होता है जो रेखा खंड को दो बराबर भागों में बाँटता है।
The mid point divides the line segment into two equal parts.

EXAMPLE / उदाहरण

Example: A(2, 3), B(8, 7) and P divides AB internally in the ratio 2:3

$$P(x, y) = \left(\frac{2(8) + 3(2)}{2 + 3}, \frac{2(7) + 3(3)}{2 + 3} \right) = \left(\frac{22}{5}, \frac{23}{5} \right)$$

$$P(x, y) = (4.4, 4.6)$$

Example: A(2, 3), B(8, 7) and P divides AB externally in the ratio 2:1

$$P(x, y) = \left(\frac{2(8) - 1(2)}{2 - 1}, \frac{2(7) - 1(3)}{2 - 1} \right) = (14, 11)$$

$$P(x, y) = (14, 11)$$

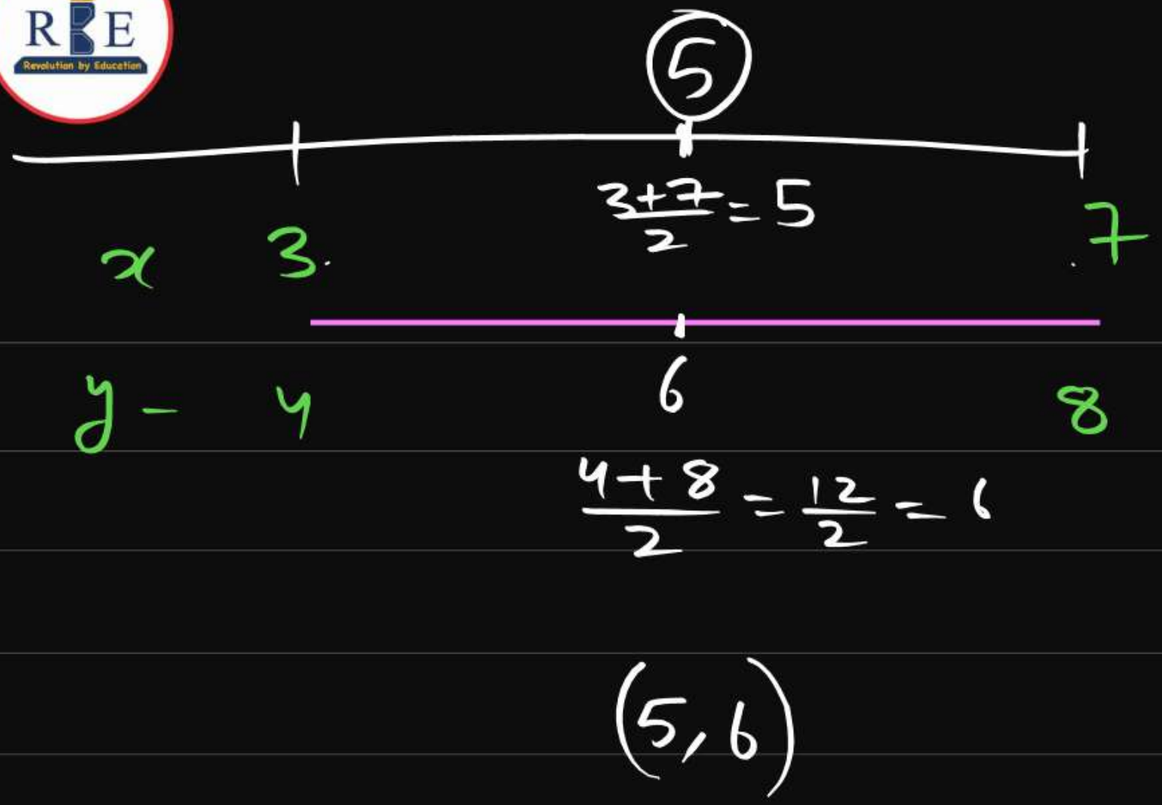
Example: A(2, 3), B(8, 7)

$$M(x, y) = \left(\frac{2 + 8}{2}, \frac{3 + 7}{2} \right) = (5, 5)$$

$$M(x, y) = (5, 5)$$

QUICK TIP / त्वरित टिप
इन सूत्रों का उपयोग ज्यामिति, दूरी, क्षेत्रफल, और कई अन्य प्रश्नों को हल करने में होता है।
These formulas are widely used in geometry, distance, area and many other problems.

☆ याद रखें / Remember: Practice Makes Perfect! ☆



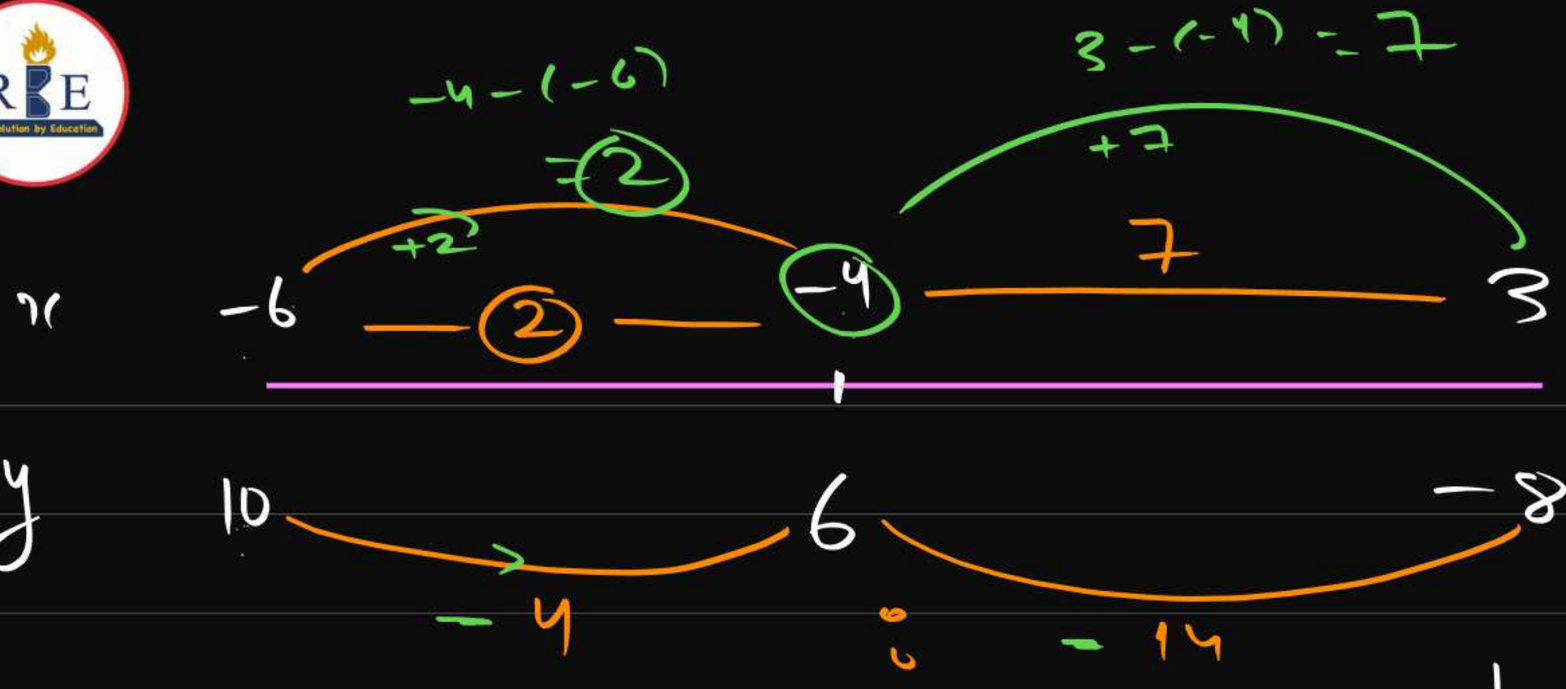
Q. Find the midpoint of segment joining (3,4) and (7,8).

(3,4) और (7,8) को जोड़ने वाला रेखाखंड का मध्यबिंदु ज्ञात कीजिए।

SSC CGL 2025

(a) (3, 4) (b) (5, 6)
(c) (4, 5) (d) (6, 7)

Answer: 2



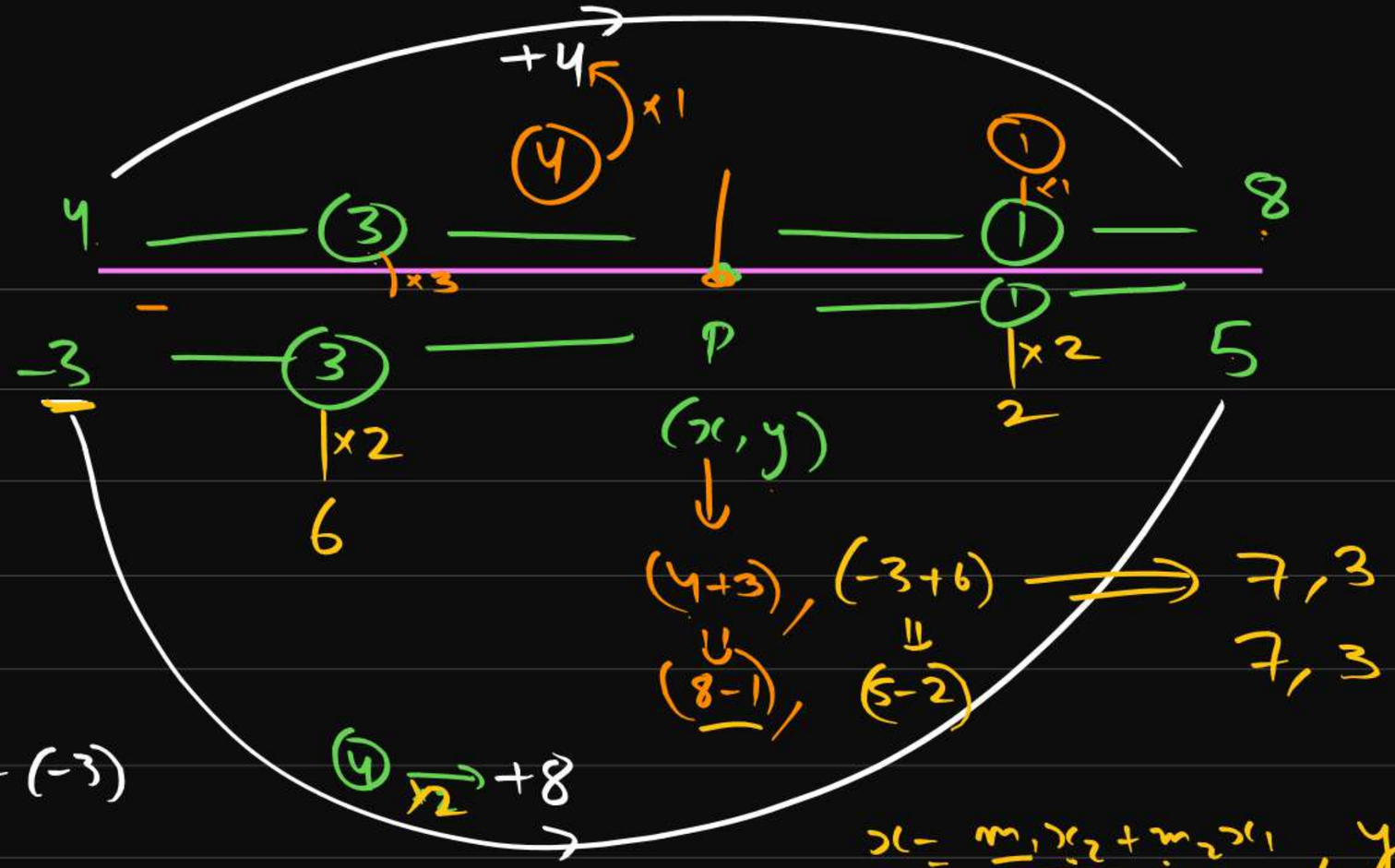
Q-) In what ratio does the point (– 4, 6) divide the line segment joining the points A(– 6, 10) and B(3, – 8)?

किस अनुपात में बिंदु (- 4, 6); बिंदु A (- 6, 10) और B (3, - 8) से जुड़ने वाले रेखा खंड को विभाजित करता है?

A) 2:7
B) 7:5
C) 7:2
D) 5:7

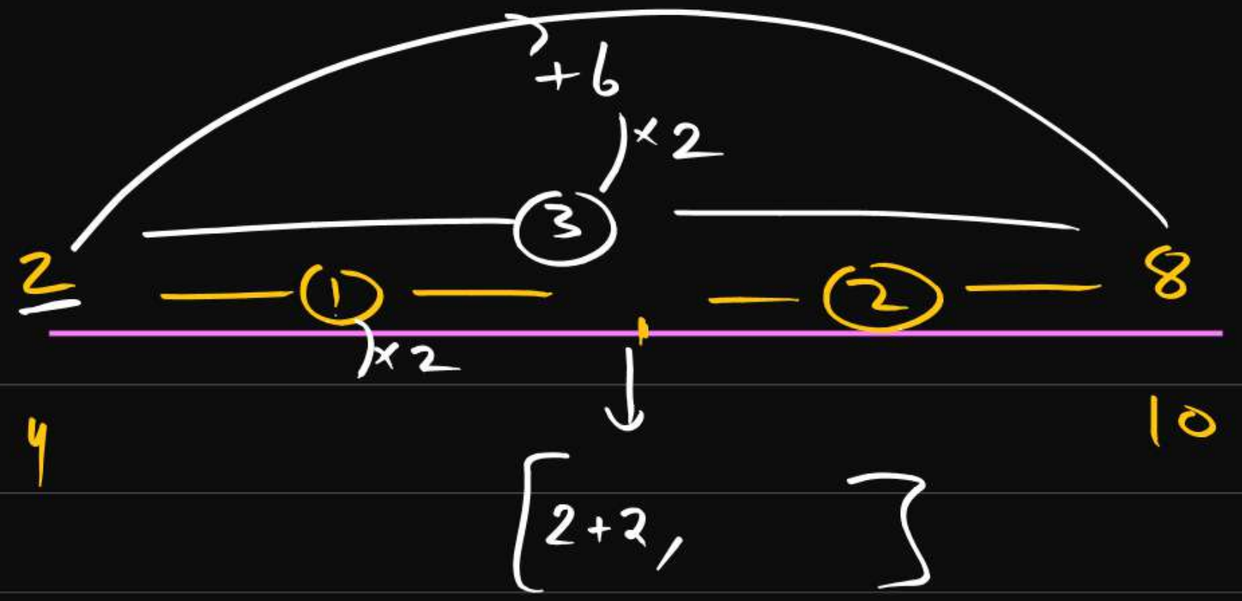
Q-) Find the coordinates of the point which divides the line segment joining the points. (4, – 3) and (8, 5) in the ratio 3 : 1 internally.

उस बिंदु के निर्देशांक का पता लगाएं जो अंकों को जोड़ने वाले रेखा खंड (4, -3) और (8, 5) को 3: 1 के अनुपात में आंतरिक रूप से विभाजित करता है।



$$x = \frac{m_1x_2 + m_2x_1}{m_1 + m_2}, \quad y = \frac{m_1y_2 + m_2y_1}{m_1 + m_2}$$

A) (3,7)
B) (7,3)
C) (4,-3)
D) (-7,3)



$② = \frac{6}{3}$

Q. Find the coordinates of the point which divides the line joining the points **(2, 4)** and **(8, 10)** internally in the ratio **1 : 2.**

उस बिंदु के निर्देशांक ज्ञात कीजिए जो बिंदुओं (2, 4) तथा (8, 10) को मिलाने वाले रेखाखंड को 1 : 2 के अनुपात में अन्तः विभाजित करता है।

$$x = \frac{1\times 8 + 2\times 2}{1+2} = 4$$

$$y = \frac{1\times 10 + 2\times 4}{1+2} = 6$$

(a) (3, 5)
(b) (4, 6)
(c) (5, 7)
(d) (6, 8)

$① \rightarrow \frac{6}{3} = 2$

$4 + 4 = 8$

Internally

(9) ⇒ −5

(4) ⇒ $\frac{-5}{9} \times 4 \Rightarrow \frac{-20}{9}$

(4) ⇒ $\frac{8}{9} \times 4 \Rightarrow \frac{32}{9}$

$$\frac{-5}{y = \frac{-13}{9}}$$

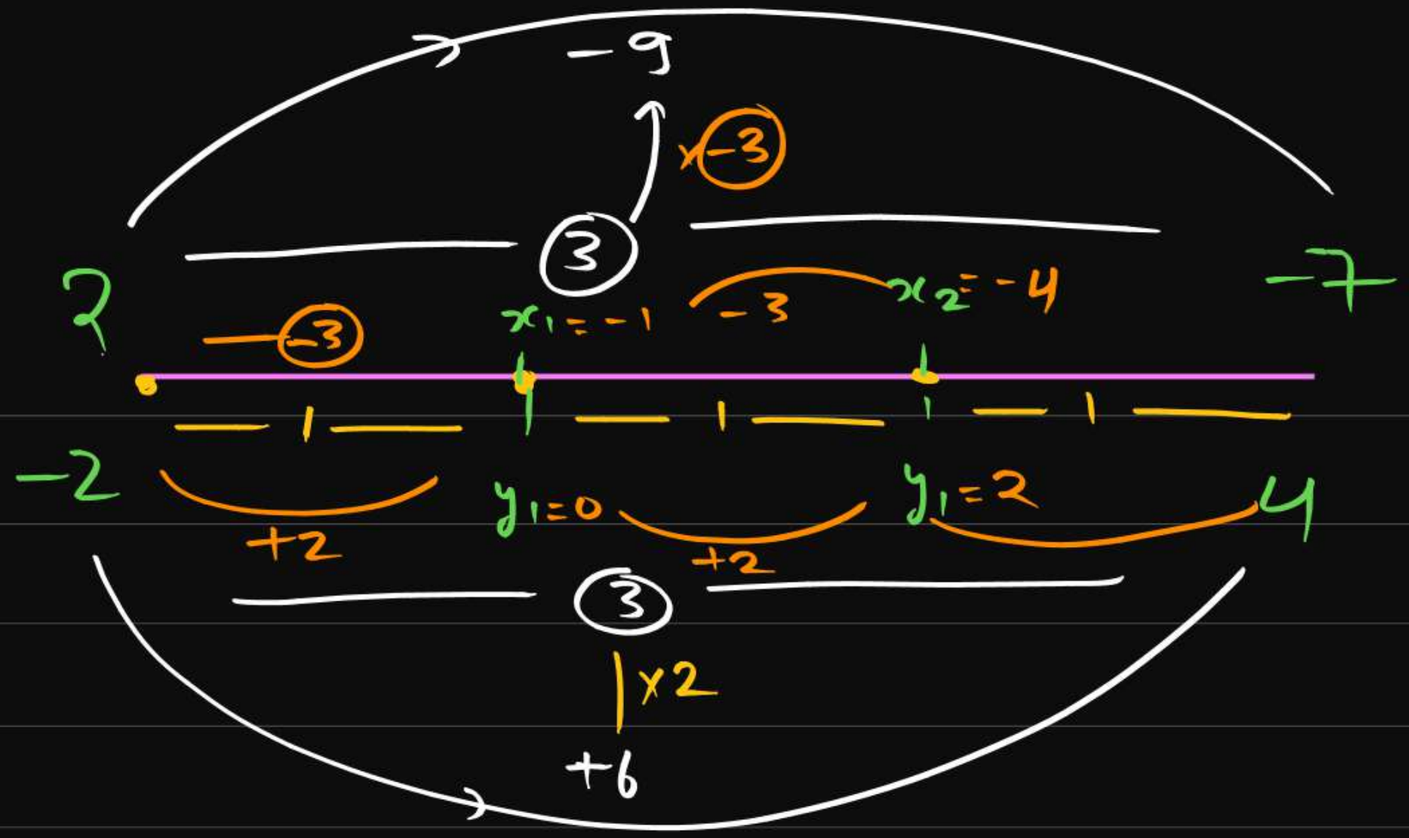
$$+3$$

$$x = \frac{7}{9}$$

Q) Find the coordinates of the point which divides the line joining points (3,-5) and (-2,3) into 4: 5.

उस बिंदु के निदेशांक ज्ञात करो जो बिंदु (3,-5) तथा (-2,3) को मिलाने वाली रेखा को 4:5 में अन्तः विभाजित करता है।

(a) (-5/3, -11/3)
(b) (7/9, -13/9)
(c) (-5/9, -11/3)
(d) (-5/3, -11/9)

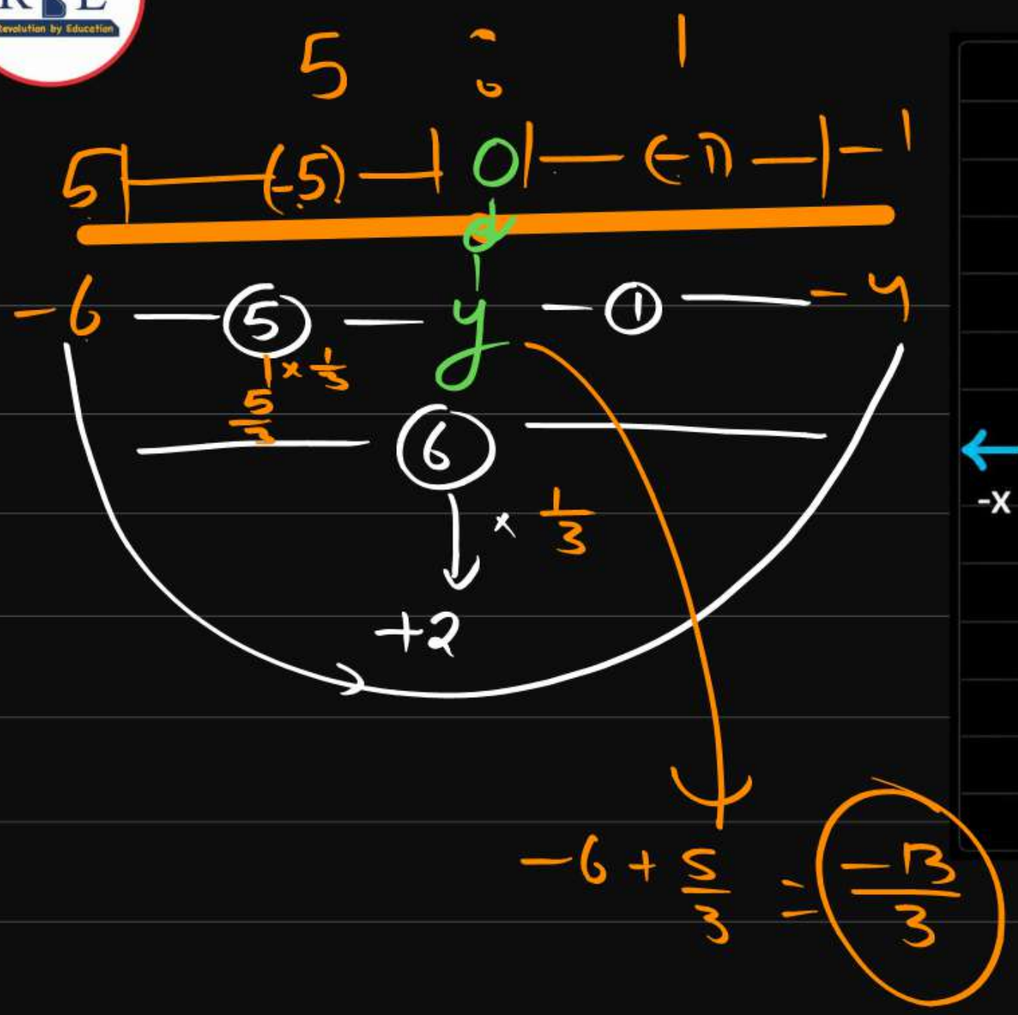


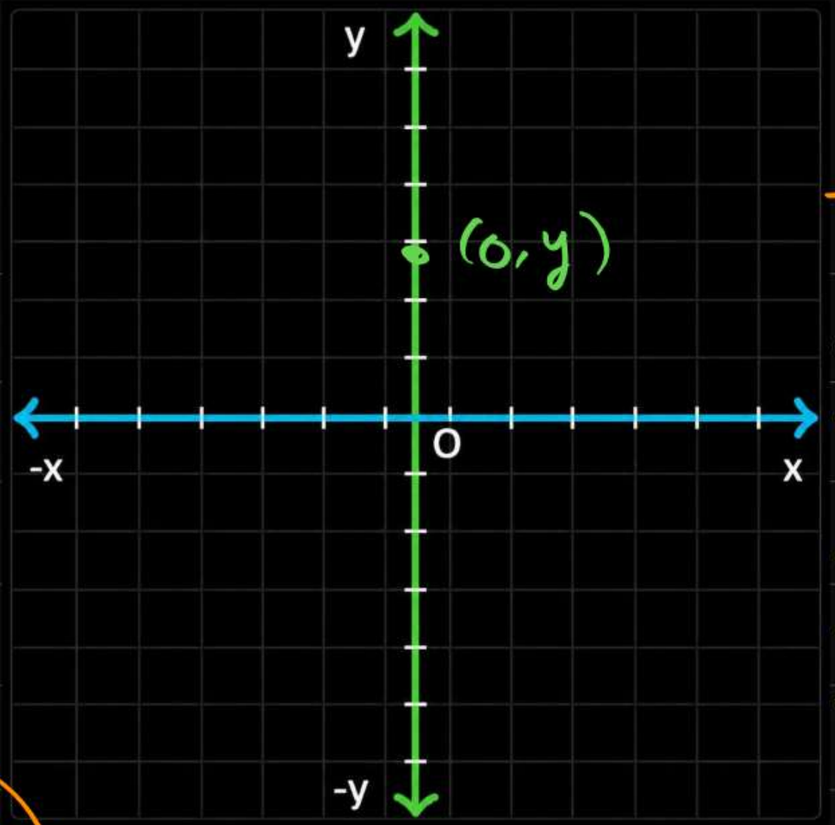
Q-) Find the coordinates of the points of trisection (i.e., points dividing in three equal parts) of the line segment joining the points A(2, – 2) and B(– 7, 4).

बिंदु A (2, - 2) और B (- 7, 4) को मिलाने वाले रेखाखंड को तीन समान भागों में विभाजित करने वाले बिंदुओं के निर्देशांक ज्ञात करें।

$(-1, 0)$ & $(-4, 2)$

(a) **(-1, 0) and (-4, 2)**
(b) **(1, -1) and (-5, 3)**
(c) **(-2, 0) and (-5, 2)**
(d) **(-1, 1) and (-4, 3)**

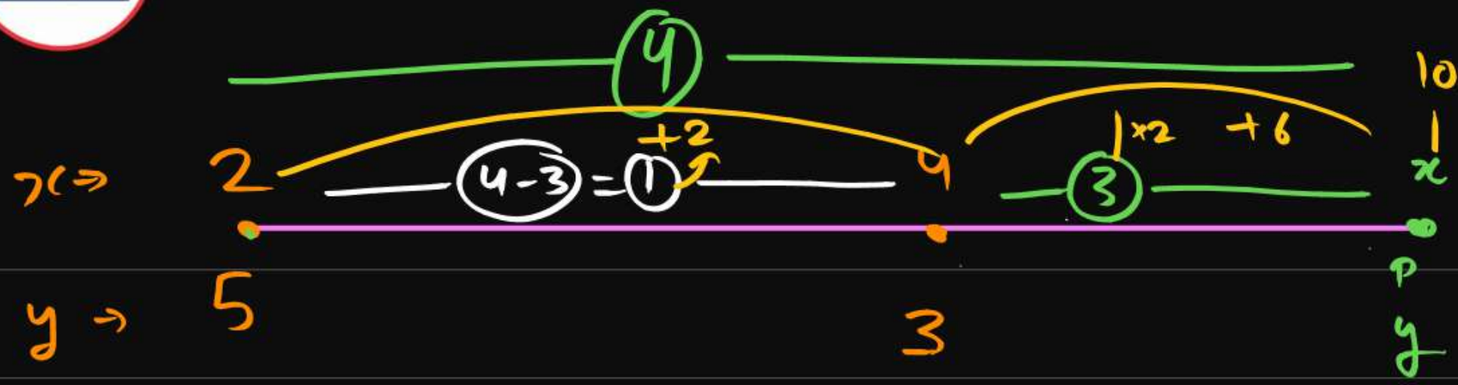




Q-) Find the ratio in which the *y*-axis divides the line segment joining the points (5, – 6) and (–1, – 4). Also find the point of intersection.

वह अनुपात ज्ञात करें जिसमें y- अक्ष बिंदुओं (5, - 6) और (-1, - 4) से जुड़ने वाले रेखाखंड को विभाजित करता है। प्रतिच्छेदन के बिंदु को भी खोजें।

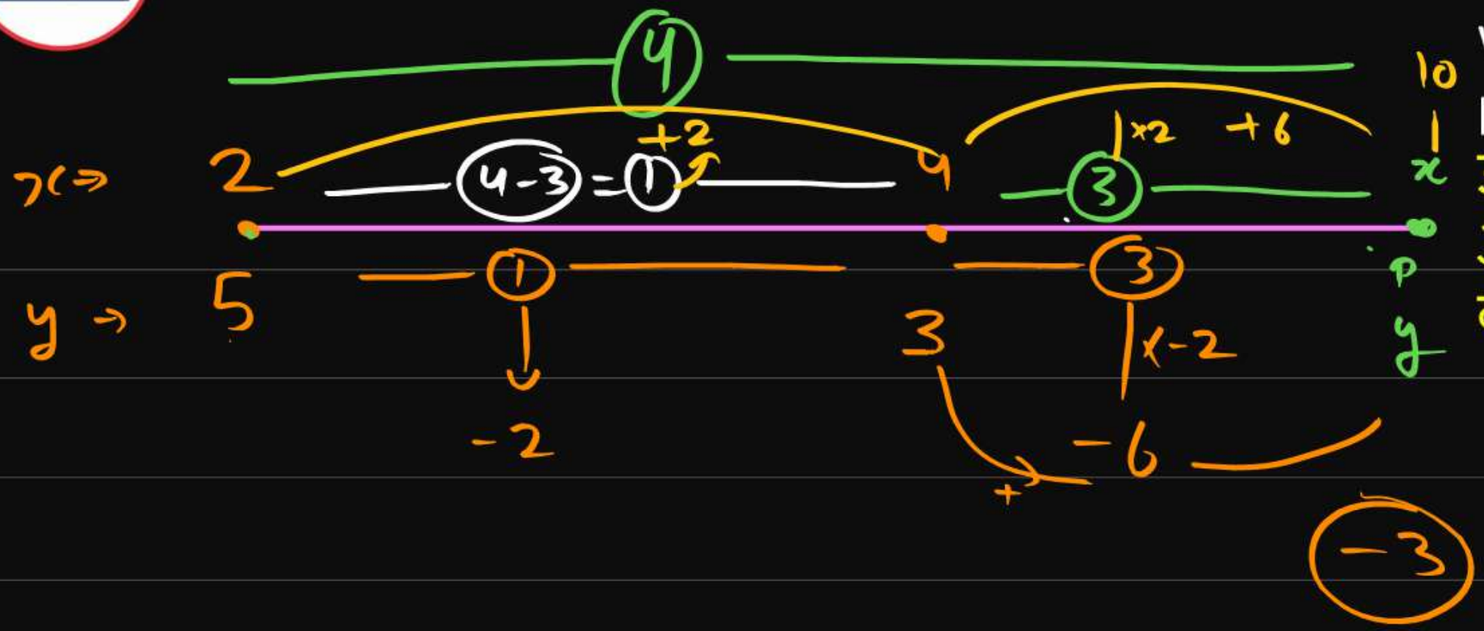
A) 5:1; (0,-13/3)
B) 2:5; (0, -17/3)
C) 5:2; (0, -5)
D) 1:5; (0, -4.5)



Q) . Find the Coordinates of the point which externally divides the line joining points (2,5) and (4,3) in 4:3.

उस बिंदु के निदेशक ज्ञात करें जो बिंदु (2,5) और (4,3) को मिलाने वाली रेखा को 4:3 में बाह्य विभाजित करता है।

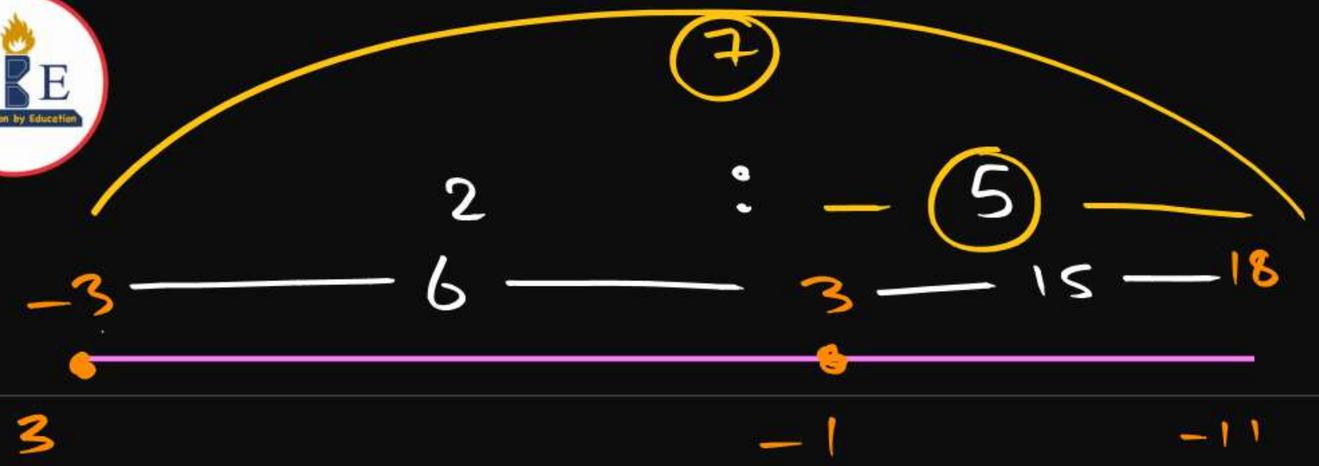
(a) (0,-3)
(b) (5,-1)
(c) (10,-3)
(d) (5,-3)



Q) . Find the Coordinates of the point which externally divides the line joining points (2,5) and (4,3) in 4:3.

उस बिंदु के निदेशक ज्ञात करें जो बिंदु (2,5) और (4,3) को मिलाने वाली रेखा को 4:3 में बाह्य विभाजित करता है।

(a) (0,-3)
(b) (5,-1)
(c) (10,-3)
(d) (5,-3)

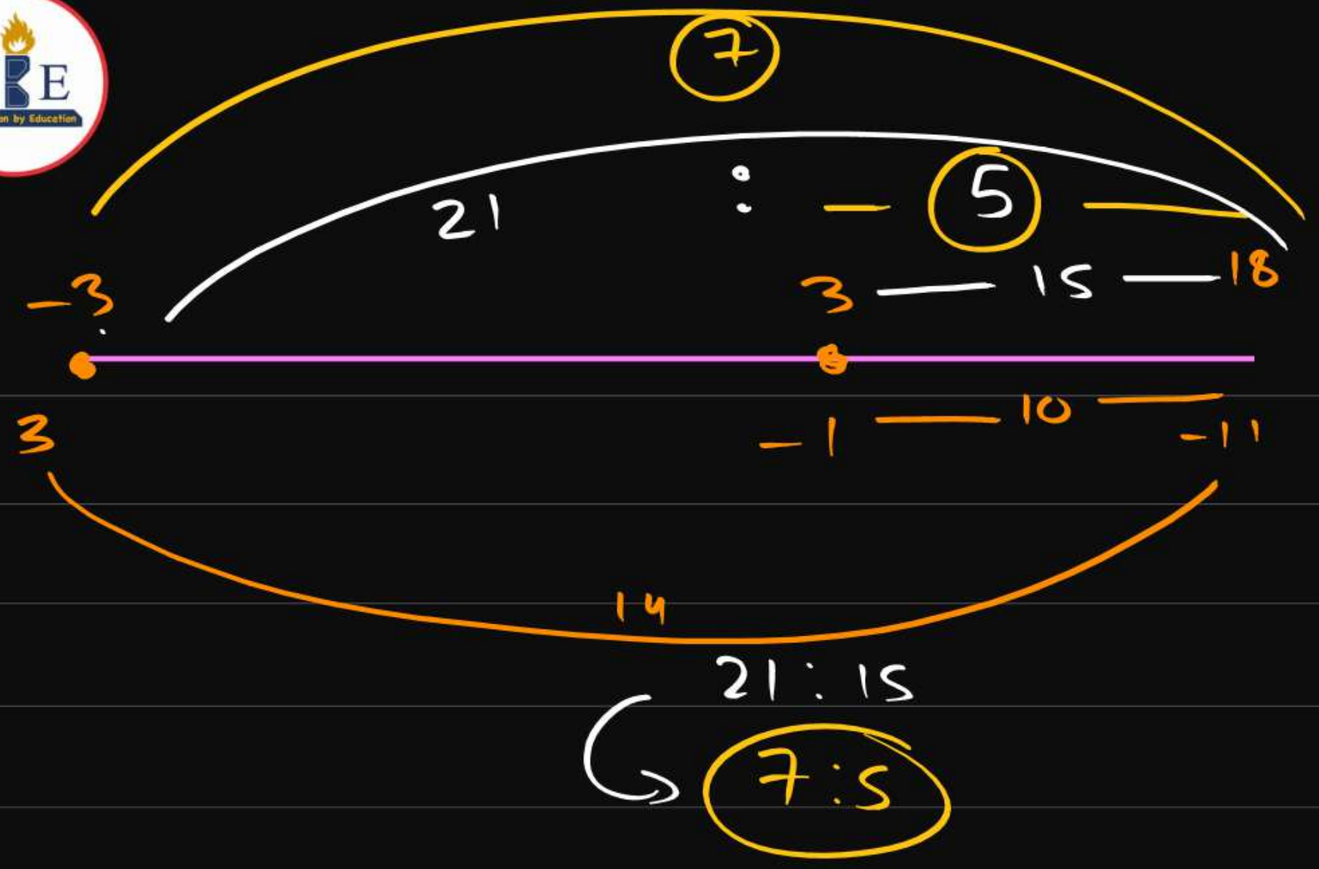


Q-) In what ratio does the point (18, -11) externally divides the line segment joining the points A(– 3, 3) and B(3, – 1)?

बिंदु (18, -11) बिंदुओं A(-3, 3) और B(3, - 1) को मिलाने वाले रेखाखंड को बाह्य रूप से किस अनुपात में विभाजित करता है?

7 : 5

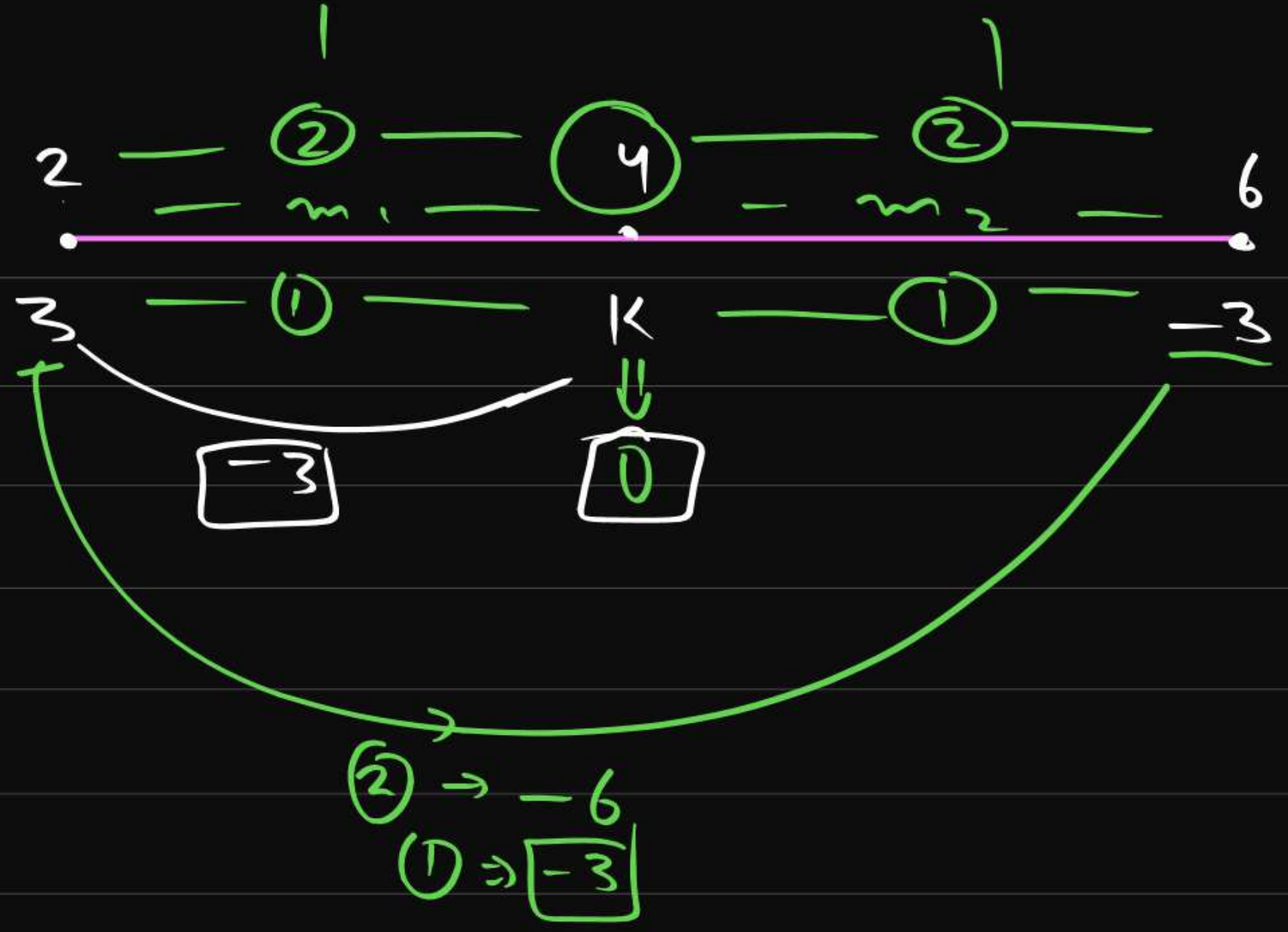
A) 7:5
B) 7:2
C) 5:2
D) 2:5



Q-) In what ratio does the point (18, -11) externally divides the line segment joining the points A(– 3, 3) and B(3, – 1)?

बिंदु (18, -11) बिंदुओं A(-3, 3) और B(3, - 1) को मिलाने वाले रेखाखंड को बाह्य रूप से किस अनुपात में विभाजित करता है?

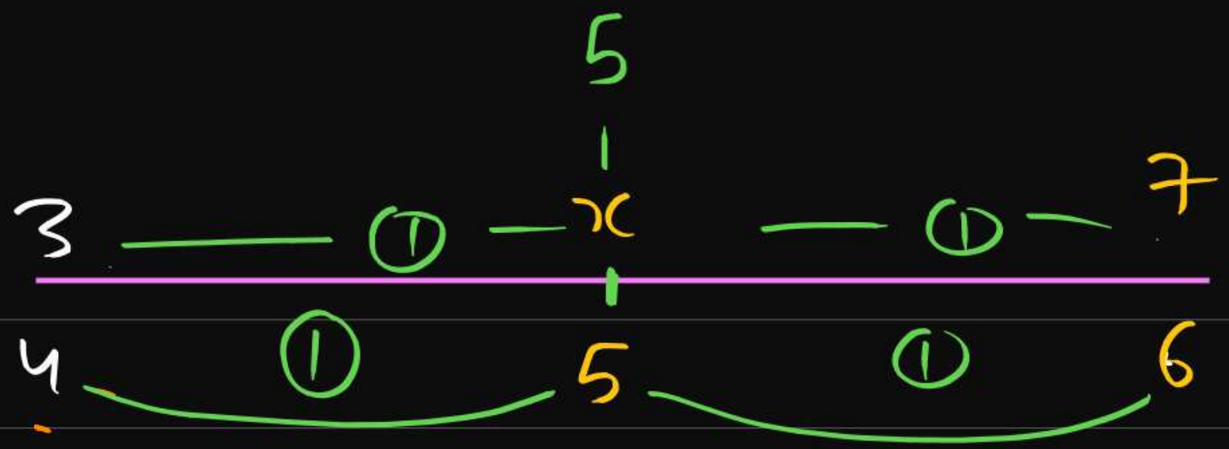
A) 7:5
B) 7:2
C) 5:2
D) 2:5



Q-) Find the value of k if the points A(2, 3), B(4, k) and C(6, –3) are collinear.

K का मान ज्ञात करें यदि अंक A (2, 3), B (4, k) और C (6, –3) आपस में संरेख (एक ही रेखा पर) हैं

A) 0
B) 2
C) -1
D) 3



Q) Find the value of x if (3,4), (7,6) and (x,5) are co-linear.

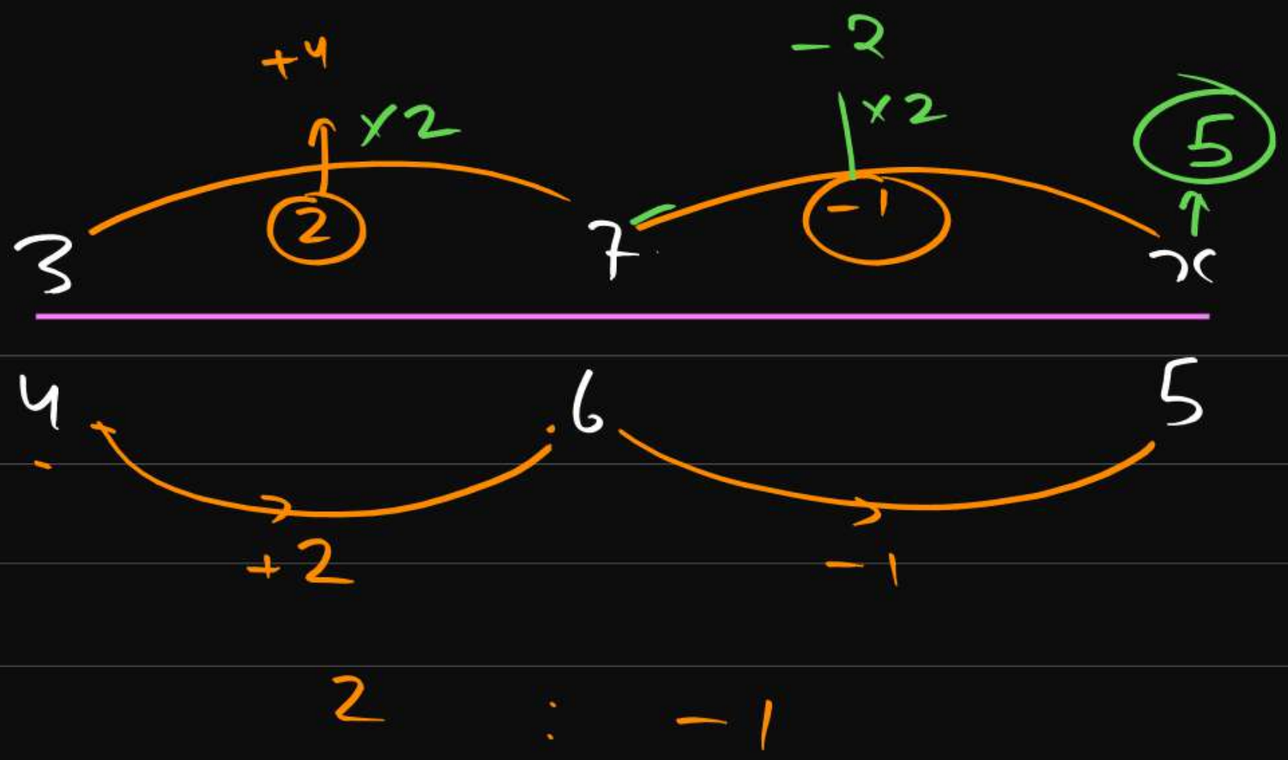
यदि (3,4), (7,6) और (x, 5) सह-रैखिक हैं तो X का मान ज्ञात करें।

(a) 3
(b) 6
(c) 5
(d) 4

(2) → 7 - 3 = 4

(1) = 2

x = 3 + 2 = 5



Q) Find the value of x if (3,4), (7,6) and (x,5) are co-linear.

यदि (3,4), (7,6) और (x, 5) सह-रैखिक हैं तो X का मान ज्ञात करें।

(a) 3
(b) 6
(c) 5
(d) 4

$$P = \frac{(x+y)+(x-y)}{2} \Rightarrow (x)$$

$$Q = \frac{(y-x)+(x+y)}{2} \Rightarrow (y)$$

Q) If the point (P, Q) is equidistant from the points (x+y,y-x) and (x-y,x+y) then which of the following is true?
यदि बिंदु (P, Q) बिंदुओं (x+y, y-x) और (x-y, x+y) से समान है तो निम्नलिखित में से कौन सा सत्य है?

$xy = yx$

$\propto xy = 2yx$

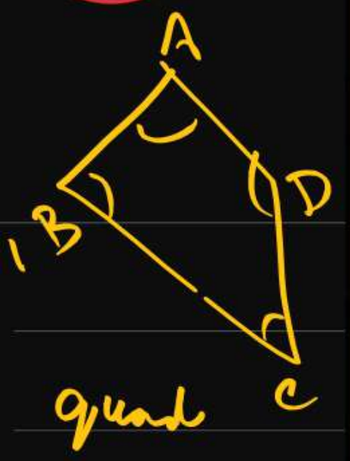
$\propto xy + yx = 0$

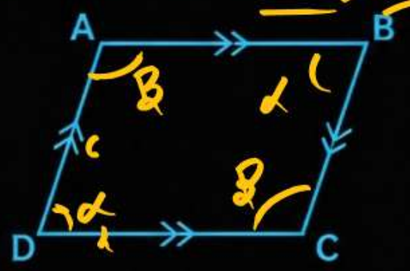
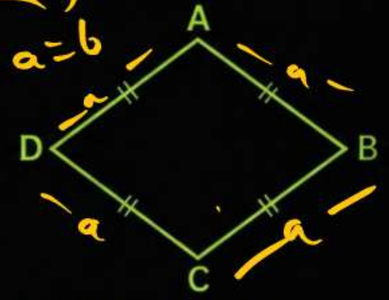
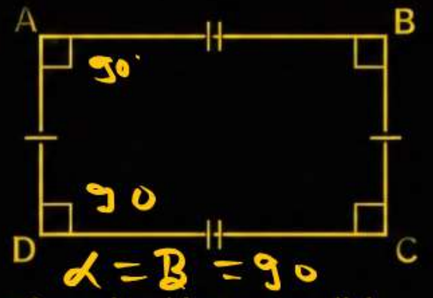
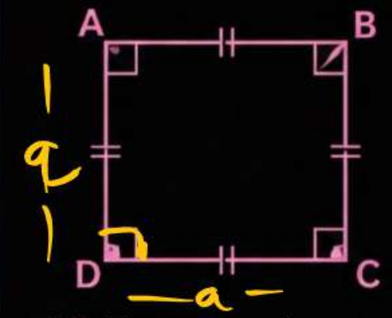
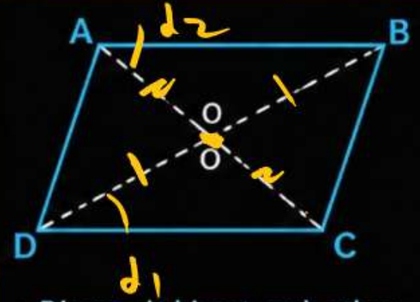
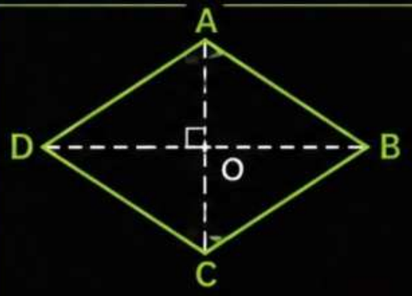
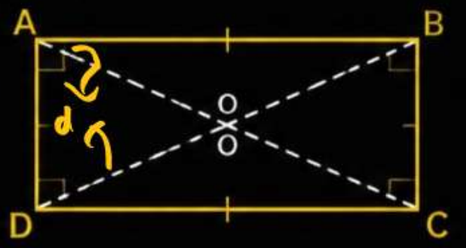
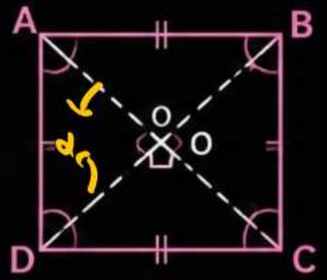
$\propto 2xy = yx$

(a) Py=Qx
(b) Py=2Qx
(c) Py+Qx=0
(d) 2Py=Qx

PROPERTIES OF QUADRILATERALS

BASED ON SIDES & DIAGONALS ONLY

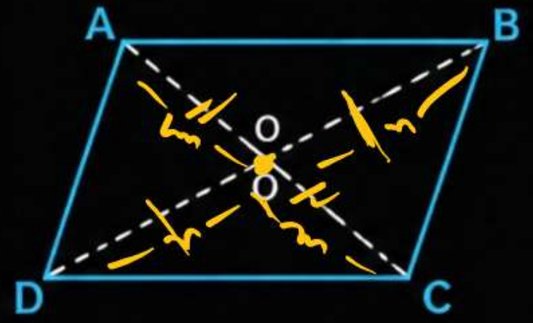


	PARALLELOGRAM	RHOMBUS	RECTANGLE	SQUARE
SIDES	 • Opposite sides are parallel. • Opposite sides are equal. • Adjacent sides are not equal (in general).	 • All sides are equal. • Opposite sides are parallel.	 • Opposite sides are parallel. • Opposite sides are equal. • Adjacent sides are not equal (in general).	 • All sides are equal. • Opposite sides are parallel. • All angles are right angles.
DIAGONALS	 • Diagonals bisect each other. • Diagonals are not equal (in general). • Diagonals do not bisect each other at right angles (in general).	 • Diagonals bisect each other. • Diagonals are perpendicular to each other. • Diagonals bisect the angles. • Diagonals are not equal (in general).	 • Diagonals bisect each other. • Diagonals are equal. • Diagonals are not perpendicular (in general). • Diagonals do not bisect the angles (in general).	 • Diagonals bisect each other. • Diagonals are equal. • Diagonals are perpendicular to each other. • Diagonals bisect the angles.

PROPERTIES OF QUADRILATERALS

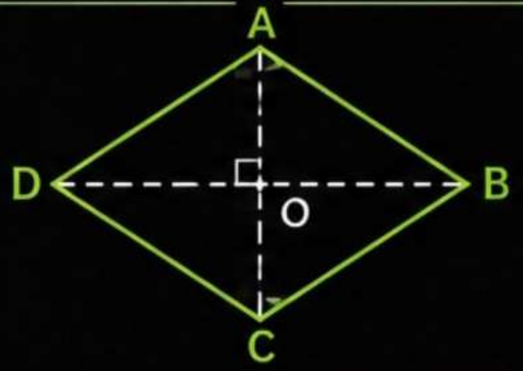
BASED ON SIDES & DIAGONALS ONLY

PARALLELOGRAM



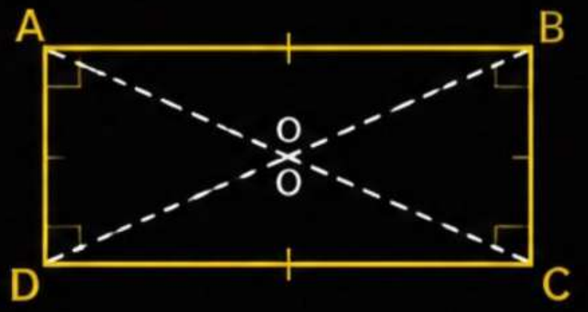
- Diagonals bisect each other.

RHOMBUS



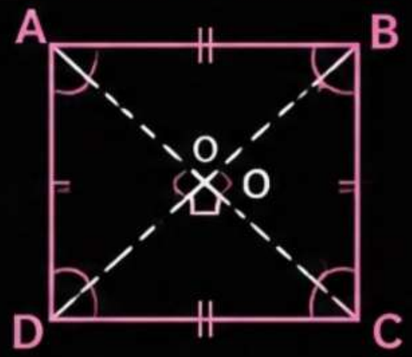
- Diagonals bisect each other.

RECTANGLE

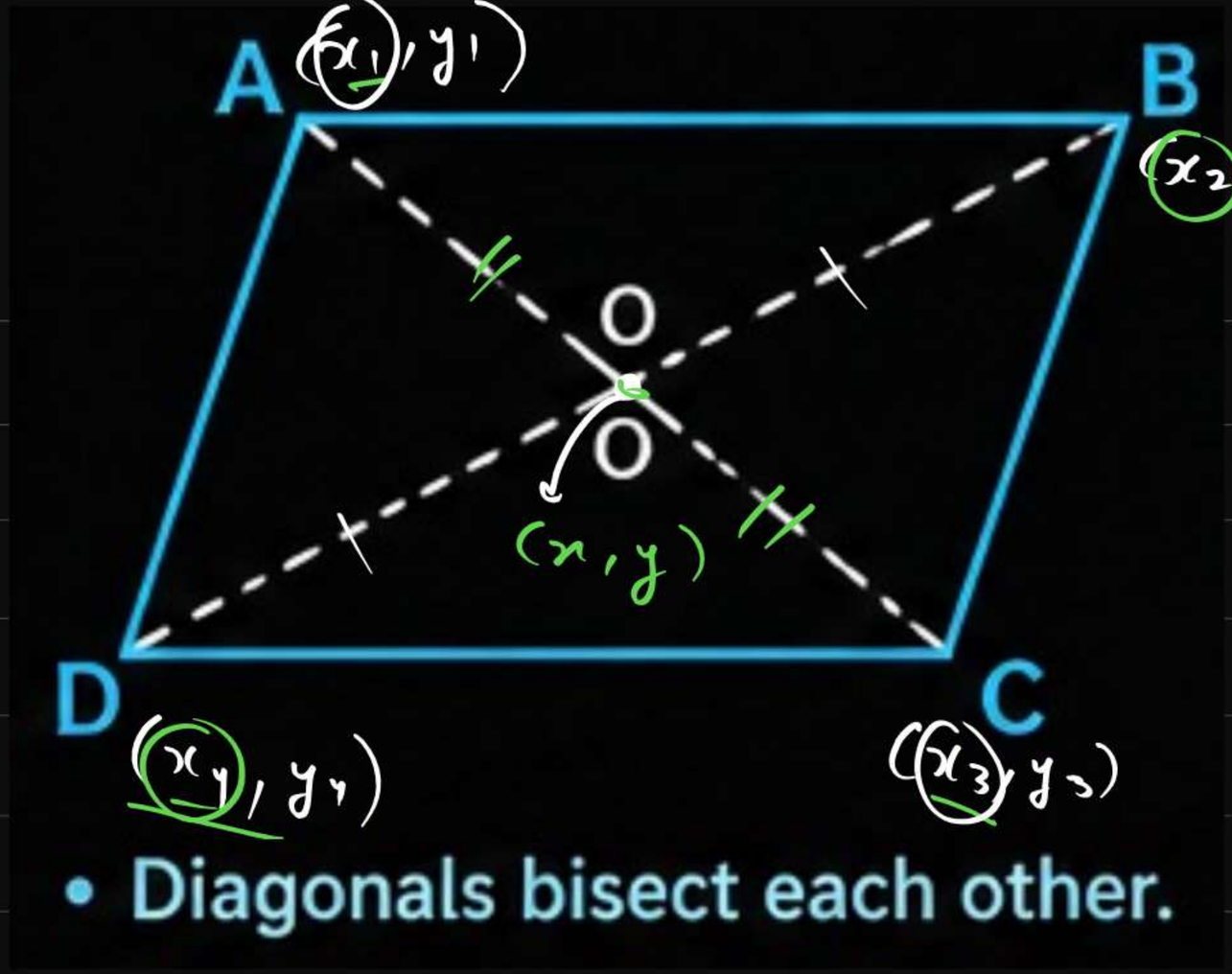


- Diagonals bisect each other.

SQUARE



- Diagonals bisect each other.



$$x = \frac{x_2 + x_4}{2} = \frac{x_1 + x_3}{2}$$

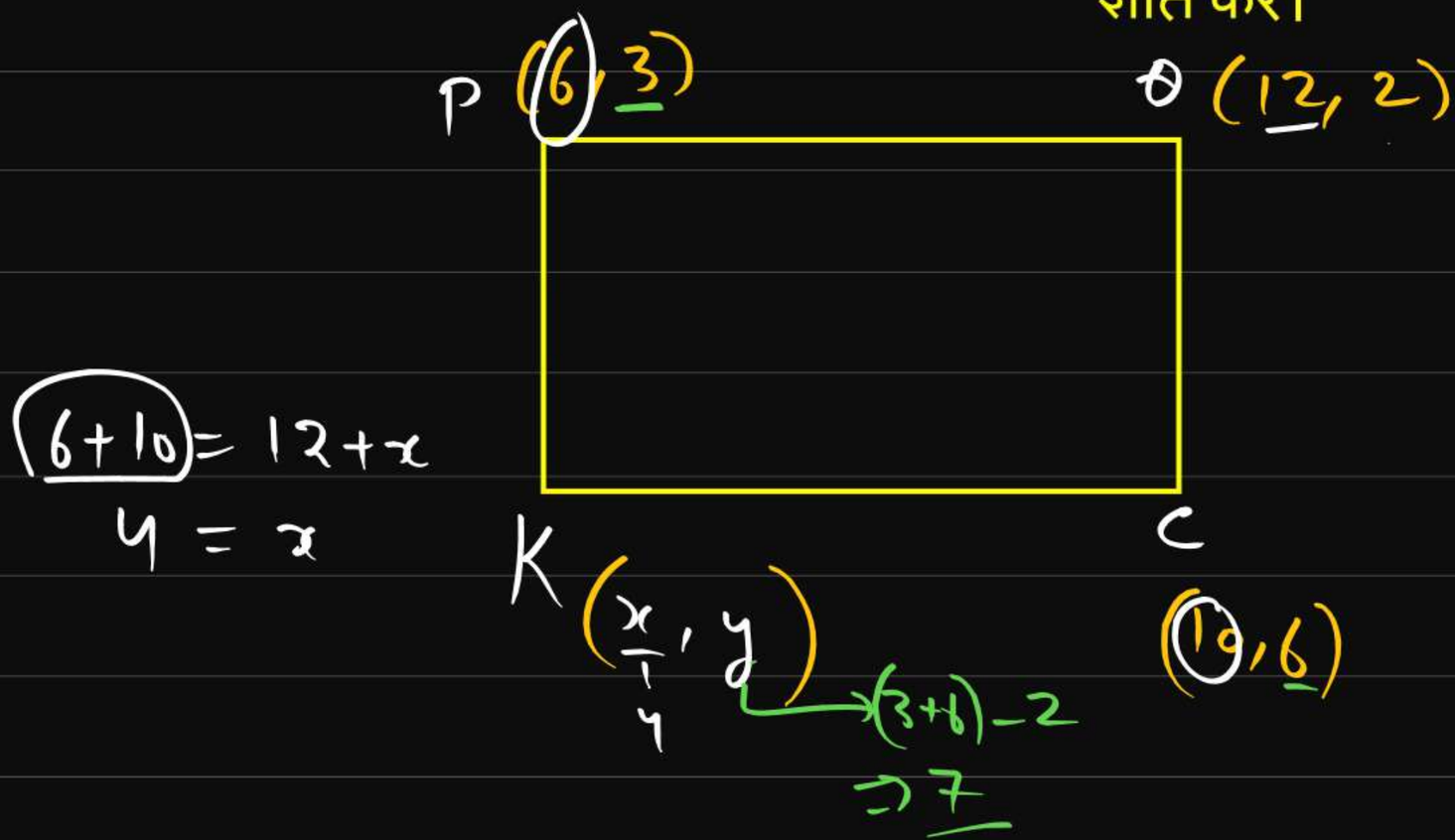
$$x_2 + x_4 = x_1 + x_3$$

$$y = \frac{y_2 + y_4}{2} = \frac{y_1 + y_3}{2}$$

$$y_2 + y_4 = y_1 + y_3$$

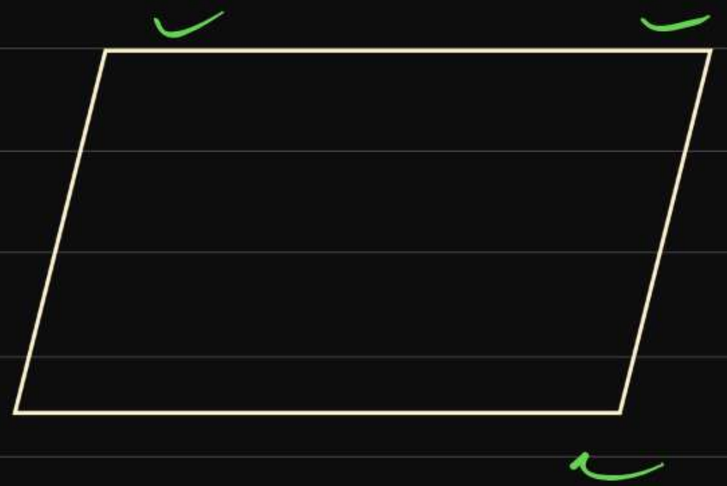
Q) A rectangle whose coordinates are P(6,3), Q(12,2) and C(10,6) then find the fourth coordinates of rectangle.

एक आयत जिसका निर्देशांक P(6,3), Q(12,2) और C(10,6) है, तब आयत के चौथे निर्देशांक ज्ञात करें।



(a) (4,7)
(b) (8,7)
(c) (4,6)
(d) (-4,7)

Q) (-2,0), (6,2) and (4,4) are three vertices of a parallelogram taken in order then what will be the coordinates of the fourth vertex.
(-2,0), (6,2) और (4,4) क्रम में लिए गए एक समांतर चतुर्भुज के तीन शीर्ष हैं तो चौथे शीर्ष के निर्देशांक क्या होंगे।



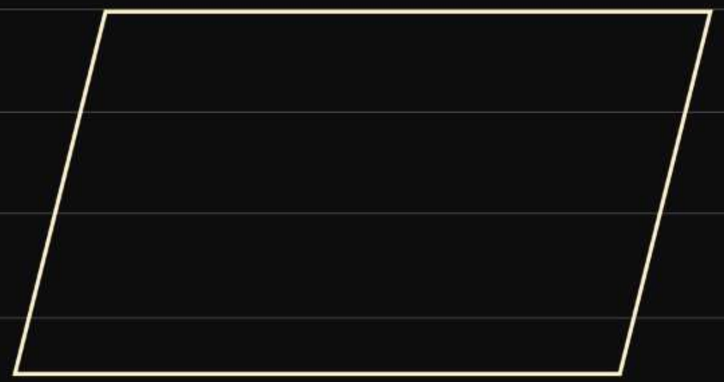
$(4+2)-6 = x = -4$

$(0+4)-2 = y = 2$

(a) (-4,2)
(b) (-3,2)
(c) (-2,1)
(d) (-3,1)

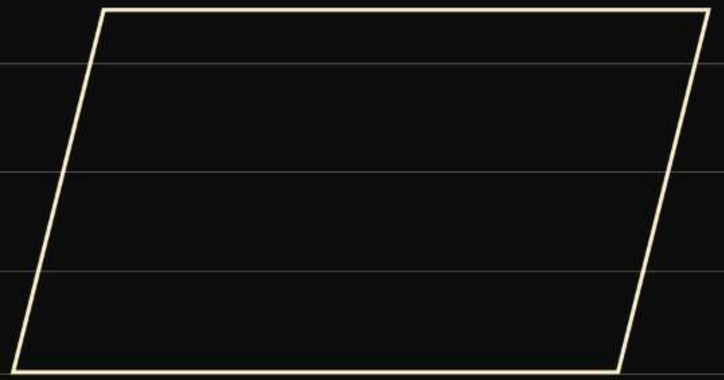
Q. If (1, 2), (4, y), (x, 6) and (3, 5) are the vertices of a parallelogram taken in order. Find x and y.

यदि (1, 2), (4, y), (x, 6) और (3, 5) क्रम में लिए गए एक समांतर चतुर्भुज के शीर्षबिंदु हैं। तो x और y ज्ञात कीजिए।



$x = 6$

$y = 1$



Q. If the points A(1, 2), B(4, 3), C(1, 0) and D(k, -1) are the vertices of a parallelogram then, the value of k is :

यदि बिंदु A(1, 2), B(4, 3), C(1, 0) और D(k, -1) एक समांतर चतुर्भुज के शीर्ष हैं , तो k का मान है :

[RRB Technician 2024 Grade-I]

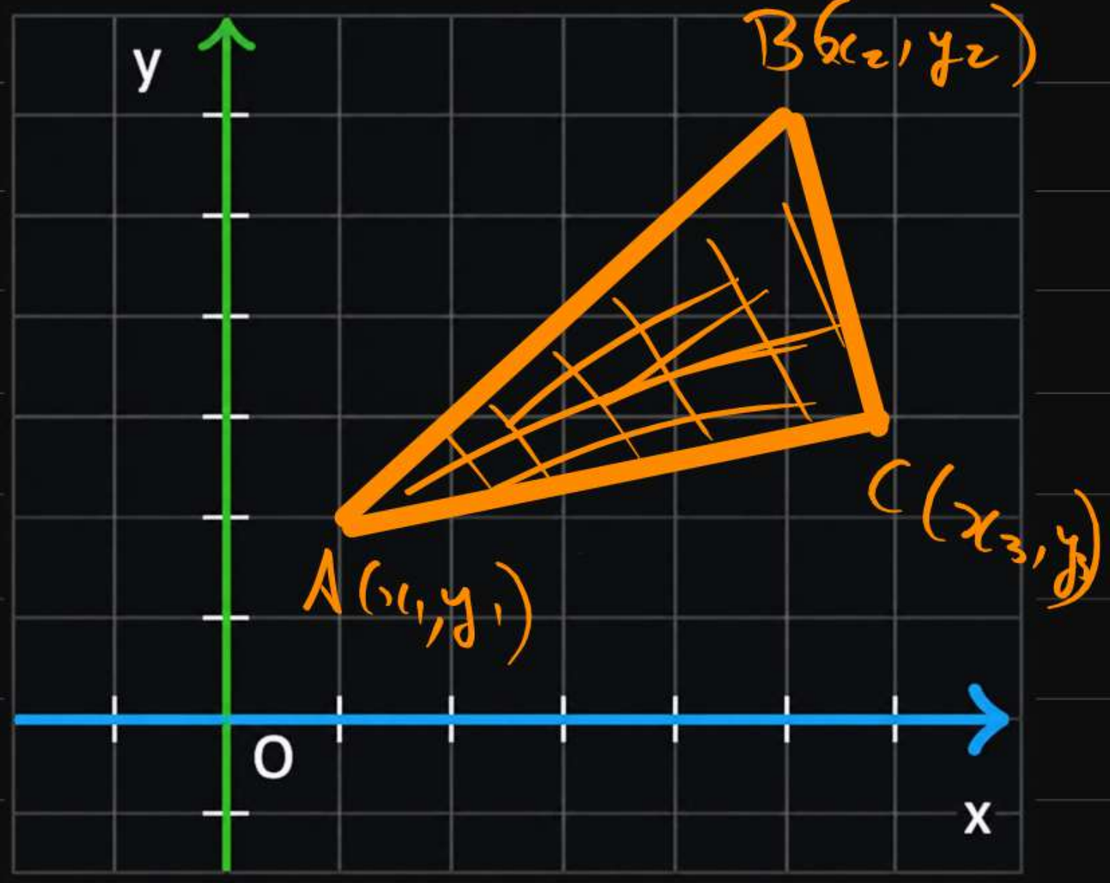
(a) -2 (b) 2 (c) 1 (d) 1

$4+k=(1+1)$

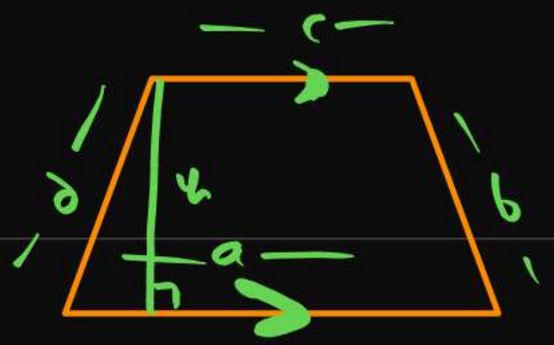
$k=-2$

Answer: 1

Area of a Triangle
त्रिभुज का क्षेत्रफल



Area of a Triangle
त्रिभुज का क्षेत्रफल



$$Ar = \frac{1}{2} \times \text{Sum of 11 sides} \times h$$

$$\Rightarrow \frac{1}{2}(a+c) \times h$$

$$\square QPAB = \frac{1}{2}(y_2 + y_1)(x_1 - x_2)$$

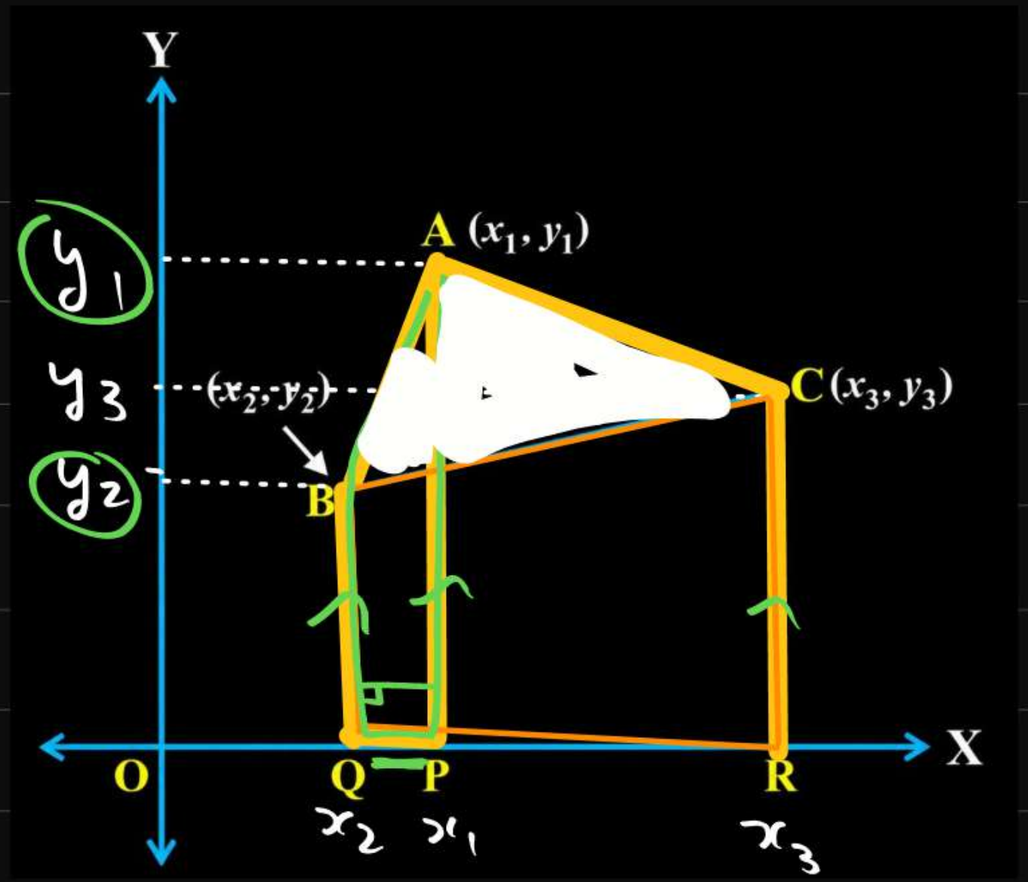
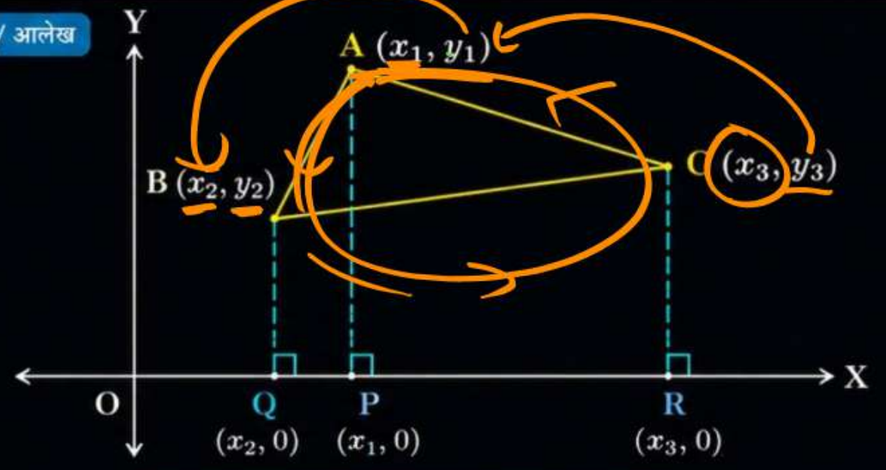


DIAGRAM / आलेख



A (x_1, y_1),	B (x_2, y_2),	C (x_3, y_3)
Q $(x_2, 0)$,	P $(x_1, 0)$,	R $(x_3, 0)$

1 GENERAL FORMULA (सामान्य सूत्र)

area of Δ ABC = area of trapezium ABQP + area of trapezium APRC
− area of trapezium BQRC.

Δ ABC का क्षेत्रफल = समलंब चतुर्भुज ABQP का क्षेत्रफल + समलंब चतुर्भुज APRC का क्षेत्रफल
− समलंब चतुर्भुज BQRC का क्षेत्रफल

We know that / हम जानते हैं कि

area of a trapezium $= \frac{1}{2}$ (sum of parallel sides)(distance between them)
समलंब चतुर्भुज का क्षेत्रफल $= \frac{1}{2}$ (समांन्तर भुजाओं का योग)(उनके बीच की दूरी)

Therefore / अत:

$$\text{Area of } \Delta ABC = \frac{1}{2}(BQ + AP)\,QP + \frac{1}{2}(AP + CR)\,PR - \frac{1}{2}(BQ + CR)\,QR$$

$$= \frac{1}{2}(y_2 + y_1)(x_1 - x_2) + \frac{1}{2}(y_1 + y_3)(x_3 - x_1) - \frac{1}{2}(y_2 + y_3)(x_3 - x_2)$$

$$= \frac{1}{2}\left[x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)\right]$$

Thus, the area of Δ ABC is the numerical value of the expression

$$\frac{1}{2}\left[x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)\right]$$

अत: Δ ABC का क्षेत्रफल इस व्यंजक के संख्यान्मक मान के बराबर है

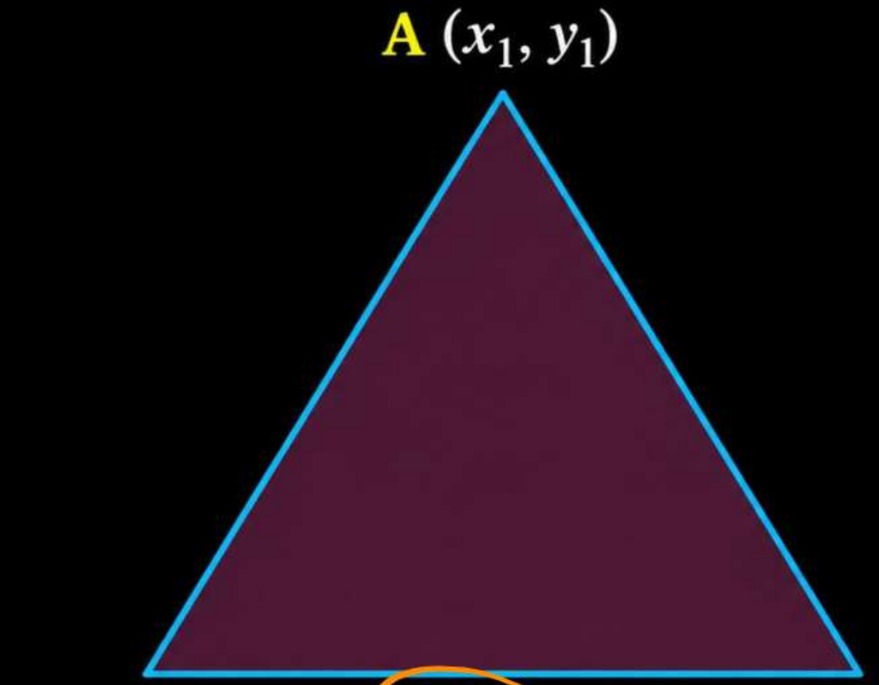
$$\frac{1}{2}\left[x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)\right]$$

★ Note / नोट : यदि बिंदु घड़ी की दिशा (clockwise) में लिए जाएं तो मान ऋणान्मक (negative) होगा ।
We take absolute value. / हम निरपेक्ष मान (absolute value) लेते हैं ।

$$A = \frac{1}{2}\left|x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)\right|$$

If $(x_3, y_3) = (0, 0)$

$$A = \frac{1}{2}\left|x_1 y_2 - x_2 y_1\right|$$

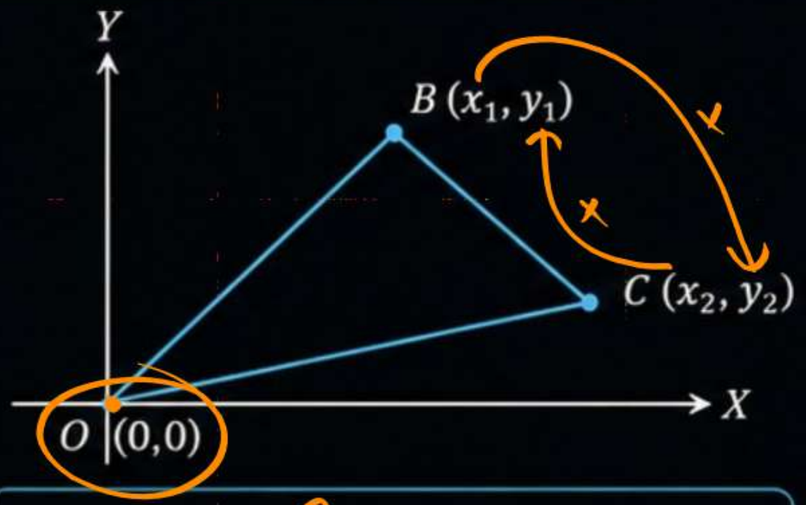


$$\text{Area of } \triangle = \frac{1}{2}\{x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)\}$$

2 AREA OF TRIANGLE WHEN ONE VERTEX IS AT ORIGIN
जब एक शीर्ष मूल बिंदु (0,0) पर हो

Let the vertices of Δ ABC be
मान लीजिए Δ ABC के शीर्ष हैं

$$A\,(0,0),\quad B\,(x_1, y_1),\quad C\,(x_2, y_2)$$

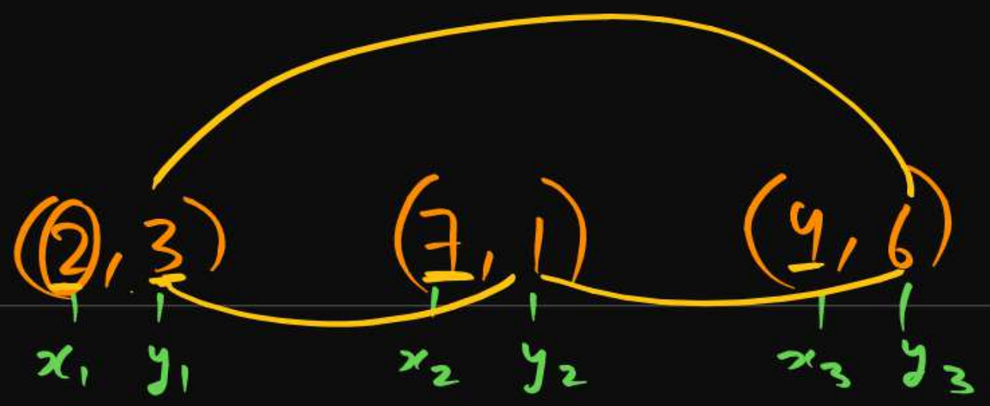


$$\text{Area} = \frac{1}{2}\left| x_1\, y_2 - x_2\, y_1 \right|$$

★ Here, | | denotes absolute value.
यहाँ | | निरपेक्ष मान (absolute value) को दर्शाता हैं।

Q. A triangle has vertices at A(2,3), B(7,1), and C(4,6). What is its area?

किसी संख्या के 20% और उसी संख्या के 10% का अंतर 120 है। वह संख्या क्या है ?



SSC CGL 2025

(a) 11 sq units (b) 9.5 sq units
(c) 15.5 sq units (d) 13 sq units

$$A = \frac{1}{2}\left| x_1(y_2-y_3) + x_2(y_3-y_1) + x_3(y_1-y_2) \right|$$

$$= \frac{1}{2}\left| 2[1-6] + 7(6-3) + 4(3-1) \right|$$

$$\Rightarrow \frac{1}{2}\left| -10 + 21 + 8 \right| \Rightarrow \frac{1}{2}\left| -19 \right| = \frac{19}{2} = 9.5$$

Answer: 2

Solving using Making Origin as Zero

x	y
-2	-3

Q. A triangle has vertices at A(2,3), B(7,1), and C(4,6). What is its area?

किसी संख्या के 20% और उसी संख्या के 10% का अंतर 120 है। वह संख्या क्या है ?

SSC CGL 2025

(a) 11 sq units (b) 9.5 sq units
(c) 15.5 sq units (d) 13 sq units

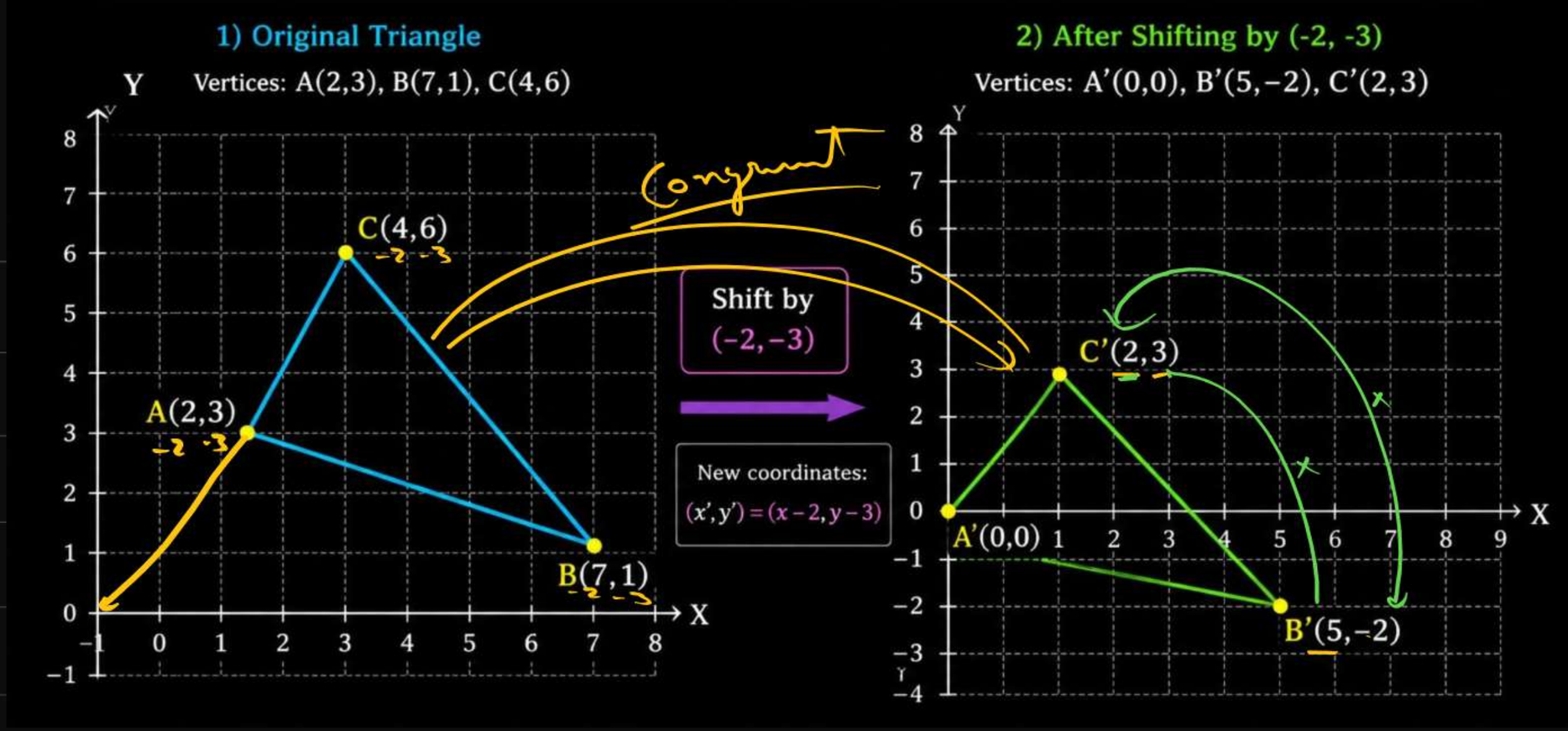
A (0,0) B (5,-2) C (2,3)

$$\frac{1}{2}\left|15 - 2(-2)\right|$$

$$\Rightarrow \frac{1}{2}|19| = 9.5$$

Answer: 2

Q. A triangle has vertices at A(2,3), B(7,1), and C(4,6). What is its area?



2, 3
−2 −3
0, 0

Shifting 1 point Origin

$= \frac{1}{2}|x_1y_2 - y_2x_1|$

$\frac{1}{2}|(-4) - 15| = 9.5$

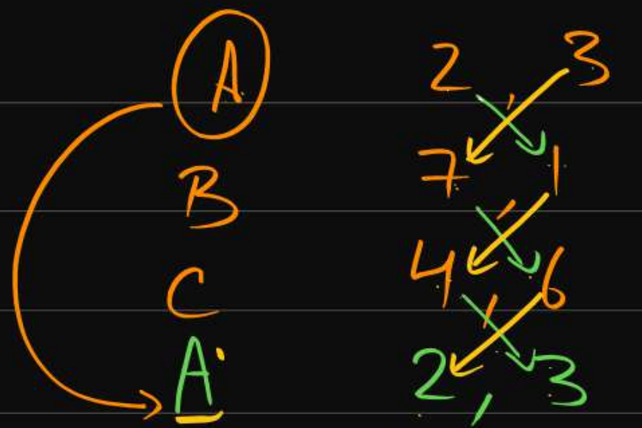
Solving using Shoelace formula

Q. A triangle has vertices at A(2,3), B(7,1), and C(4,6). What is its area?

किसी संख्या के 20% और उसी संख्या के 10% का अंतर 120 है। वह संख्या क्या है?

SSC CGL 2025

(a) 11 sq units (b) 9.5 sq units
(c) 15.5 sq units (d) 13 sq units



$(2\times1)-(3\times7)=-19$

$(7\times6)-(1\times4)=38$

$(4\times3)-(2\times6)=0$

$\frac{1}{2}|S_1-S_2|=\text{Area}=\frac{1}{2}\left|\underbrace{[(2\times1)+(7\times6)+(4\times3)]}_{S_1}-\underbrace{(3\times7+1\times4+6\times2)}_{S_2}\right|$

$=\frac{1}{2}\left|[(2\times1)-(3\times7)]+[(7\times6)-(1\times4)]+[(4\times3)-(6\times2)]\right|$

$=\frac{1}{2}|19|=9.5$

Answer: 2

SHOELACE FORMULA

(Gauss's Area Formula)

Used to find the area of a simple polygon when the coordinates of its vertices are known.

VERTICES IN ORDER

If the vertices are listed in order (clockwise or counterclockwise):

$$(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$$

FORMULA

$$A = \frac{1}{2} \left| \sum_{i=1}^{n} (x_i y_{i+1} - y_i x_{i+1}) \right|$$

where $(x_{n+1}, y_{n+1}) = (x_1, y_1)$

(i.e., the first point is repeated at the end).

SHOELACE (TABULAR) METHOD

1. Arrange the coordinates in a table.
2. Multiply diagonally downward: $x_i \times y_{i+1}$, and add them.
3. Multiply diagonally upward: $y_i \times x_{i+1}$, and add them.
4. Take the absolute difference of the two sums and divide by 2.

Point	x_i	y_i	$x_i y_{i+1}$ (↘ Downward)	$y_i x_{i+1}$ (↗ Upward)
1	x_1	y_1	$x_1 \times y_2$	$y_1 \times x_2$
2	x_2	y_2	$x_2 \times y_3$	$y_2 \times x_3$
⋮	⋮	⋮	⋮	⋮
n	x_n	y_n	$x_n \times y_{n+1}$	$y_n \times x_{n+1}$
n+1	x_1	y_1	(repeat first point)	(repeat first point)
			Sum = S_1	Sum = S_2

Area of Polygon, $A = \frac{1}{2} |S_1 - S_2|$

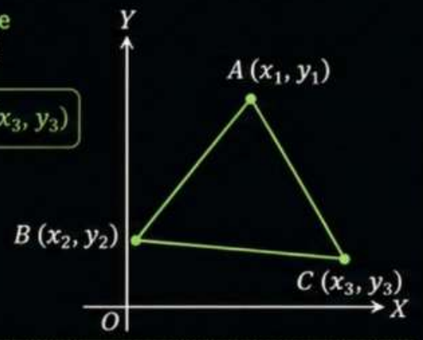
AREA OF A TRIANGLE IN CO-ORDINATE GEOMETRY

निर्देशांक ज्यामिति में त्रिभुज का क्षेत्रफल

1 AREA OF TRIANGLE (GENERAL FORMULA)
त्रिभुज का क्षेत्रफल (सामान्य सूत्र)

Let the vertices of Δ ABC be
मान लीजिए Δ ABC के शीर्ष हैं

$A(x_1, y_1),\ B(x_2, y_2),\ C(x_3, y_3)$



The area of Δ ABC is
Δ ABC का क्षेत्रफल है

$$\text{Area} = \frac{1}{2}\left[x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)\right]$$

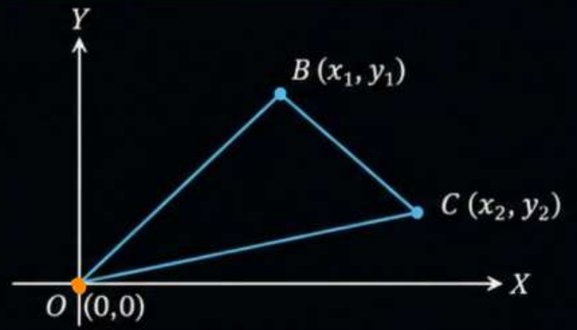
★ Note / नोट

- The area is the numerical value of the above expression.
  क्षेत्रफल उपरोक्त व्यंजक का संख्यात्मक मान होता है।
- If the points are taken in clockwise order, the value will be negative.
  यदि बिंदु घड़ी की दिशा में लिए जाते हैं, तो मान ऋणात्मक (negative) होगा।

2 AREA OF TRIANGLE WHEN ONE VERTEX IS AT ORIGIN
जब एक शीर्ष मूल बिंदु (0,0) पर हो

Let the vertices of Δ ABC be
मान लीजिए Δ ABC के शीर्ष हैं

$A(0, 0),\quad B(x_1, y_1),\quad C(x_2, y_2)$



$$\text{Area} = \frac{1}{2}\left|x_1 y_2 - x_2 y_1\right|$$

★ Here, | | denotes absolute value.
यहाँ | | निरपेक्ष मान (absolute value) को दर्शाता है।

3 SHOELACE METHOD
शूलेस विधि

For Δ ABC with vertices A (x_1, y_1), B (x_2, y_2), C (x_3, y_3)
Δ ABC जिसके शीर्ष A (x_1, y_1), B (x_2, y_2), C (x_3, y_3) हैं

Steps / चरण :

1. Write the coordinates in order and repeat the first point.
   निर्देशांकों को क्रम में लिखें और पहले बिंदु को दोहराएँ।

Point / बिंदु	A	B	C	A (Again)
x	x_1	x_2	x_3	x_1
y	y_1	y_2	y_3	y_1
x × Next y	x_1y_2	x_2y_3	x_3y_1	x_1y_1
y × Next x	y_1x_2	y_2x_3	y_3x_1	y_1x_1

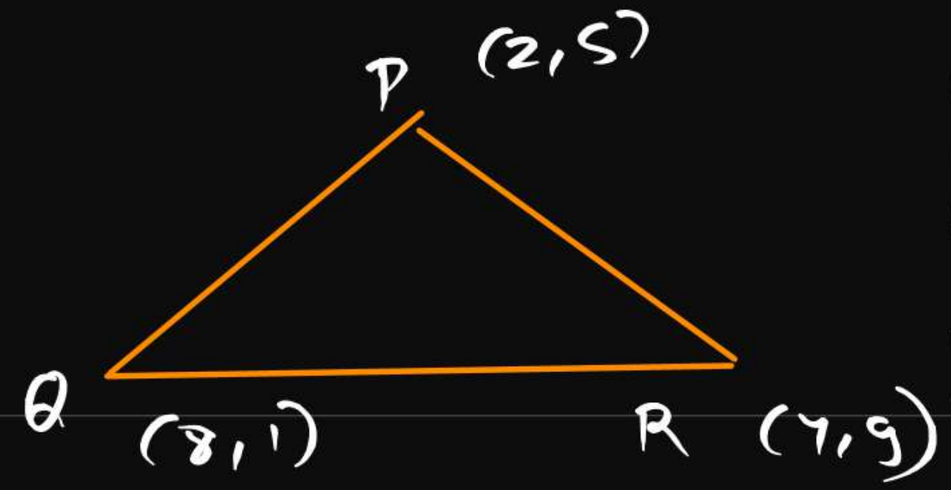
2. Find the sum of (x × next y) and (y × next x).
   (x × next y) और (y × next x) का योग ज्ञात करें।

$S_1 = x_1y_2 + x_2y_3 + x_3y_1$ $\qquad$ $S_2 = y_1x_2 + y_2x_3 + y_3x_1$

3. Area of Δ ABC $= \frac{1}{2}\left|S_1 - S_2\right|$
   Δ ABC का क्षेत्रफल $= \frac{1}{2}\left|S_1 - S_2\right|$

REMEMBER / याद रखें

- Order of points matters.
  बिंदुओं का क्रम महत्वपूर्ण होता है।
- Always take absolute value.
  हमेशा निरपेक्ष मान (absolute value) लें।
- All three methods give the same area.
  तीनों विधियाँ समान क्षेत्रफल देती हैं।



(2,5) (8,1) (4,9)

$$\Rightarrow \frac{1}{2}\left|2\underline{x-8} + \underline{8 \times 4} + \underline{4 \times 4}\right|$$

$$\Rightarrow \frac{1}{2}|32| \Rightarrow \underline{16}$$

Q. A geological survey team has established three research stations forming a triangular monitoring zone. The stations are located at coordinates P(2,5), Q(8,1), and R(4,9) on a coordinate grid where each unit represents 1 kilometer. What is the area of the triangular monitoring zone?

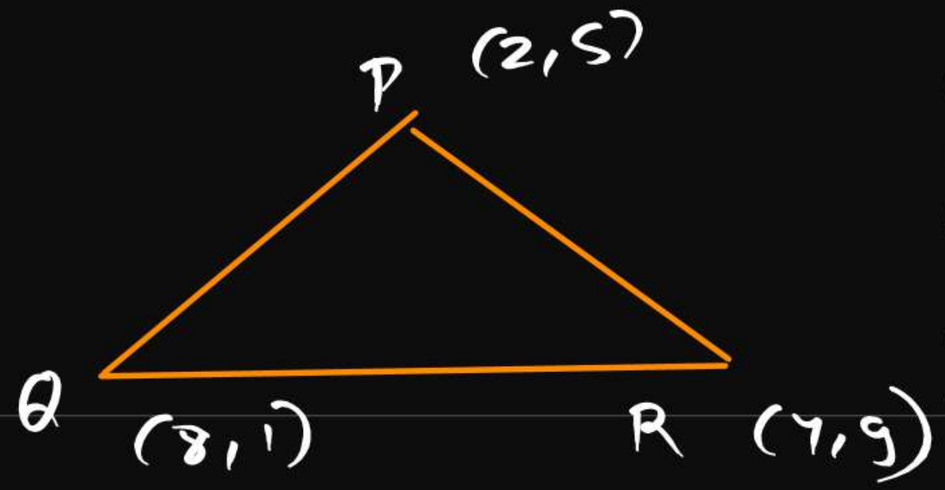
एक भूवैज्ञानिक सर्वेक्षण दल ने त्रिकोणीय मॉनिटरिंग ज़ोन बनाने वाले तीन अनुसंधान केंद्र स्थापित किए हैं। ये केंद्र एक निर्देशांक ग्रिड पर P(2,5), Q(8,1) और R(4,9) निर्देशांकों पर अवस्थित हैं, जहां प्रत्येक इकाई 1 किलोमीटर को दर्शाती है। त्रिकोणीय मॉनिटरिंग ज़ोन का क्षेत्रफल कितना है?

1 unit = 1 Km

RRB NTPC 2025-26 (Mar 2026) GL CBT-1

(a) 14 square kilometers
(b) 10 square kilometers
(c) 12 square kilometers
(d) 16 square kilometers

Answer: 4



$$\begin{array}{c|cc} (2,5) & (8,1) & (4,9) \\ -2\ -5 & -2\ -5 & -2\ ,-5 \\ (0,0) & 6,-4 & 2,4 \end{array}$$

$$\frac{1}{2}\left|24-(-8)\right|$$

$$\frac{1}{2}\times 32 = 16$$

Q. A geological survey team has established three research stations forming a triangular monitoring zone. The stations are located at coordinates P(2,5), Q(8,1), and R(4,9) on a coordinate grid where each unit represents 1 kilometer. What is the area of the triangular monitoring zone?

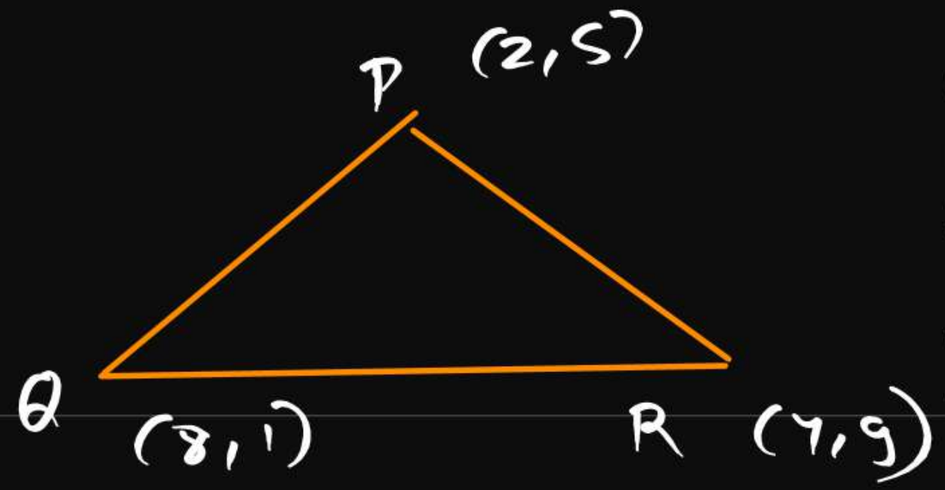
एक भूवैज्ञानिक सर्वेक्षण दल ने त्रिकोणीय मॉनिटरिंग ज़ोन बनाने वाले तीन अनुसंधान केंद्र स्थापित किए हैं। ये केंद्र एक निर्देशांक ग्रिड पर P(2,5), Q(8,1) और R(4,9) निर्देशांकों पर अवस्थित हैं, जहां प्रत्येक इकाई 1 किलोमीटर को दर्शाती है। त्रिकोणीय मॉनिटरिंग ज़ोन का क्षेत्रफल कितना है?

1 unit = 1 Km

RRB NTPC 2025-26 (Mar 2026) GL CBT-1

(a) 14 square kilometers
(b) 10 square kilometers
(c) 12 square kilometers
(d) 16 square kilometers

Answer: 4



(2,5) (8,1) (4,9)

2 5
8 1
4 9
2 5

2 − 40 = −38
72 − 4 = 68
20 − 18 = 2
+32

$\frac{1}{2} \times 32 = 16$

Q. A geological survey team has established three research stations forming a triangular monitoring zone. The stations are located at coordinates P(2,5), Q(8,1), and R(4,9) on a coordinate grid where each unit represents 1 kilometer. What is the area of the triangular monitoring zone?

एक भूवैज्ञानिक सर्वेक्षण दल ने त्रिकोणीय मॉनिटरिंग ज़ोन बनाने वाले तीन अनुसंधान केंद्र स्थापित किए हैं। ये केंद्र एक निर्देशांक ग्रिड पर P(2,5), Q(8,1) और R(4,9) निर्देशांकों पर अवस्थित हैं, जहां प्रत्येक इकाई 1 किलोमीटर को दर्शाती है। त्रिकोणीय मॉनिटरिंग ज़ोन का क्षेत्रफल कितना है?

1 unit = 1 Km

RRB NTPC 2025-26 (Mar 2026) GL CBT-1

(a) 14 square kilometers
(b) 10 square kilometers
(c) 12 square kilometers
(d) 16 square kilometers

Answer: 4

Q. A triangular garden has vertices at coordinates A(2, 3), B(8, 11), and C(14, 7). A landscape designer wants to plant flowers covering exactly half of this triangular area. What area should be covered with flowers?

$\frac{1}{2} \times \frac{1}{2} |2\times4 + 8\times4 + 14(-3)|$

$= \frac{1}{4} |-72|$

$\Rightarrow 18$

एक त्रिभुजाकार बगीचे के शीर्ष निर्देशांक A(2, 3), B(8, 11), और C(14, 7) पर हैं। एक भूदृश्य डिज़ाइनर इस त्रिभुजाकार क्षेत्र के ठीक आधे हिस्से पर फूल लगाना चाहता है। फूलों से कितना क्षेत्र ढका होना चाहिए?

RRB Group D 2024-25 [Nov to feb 2026]

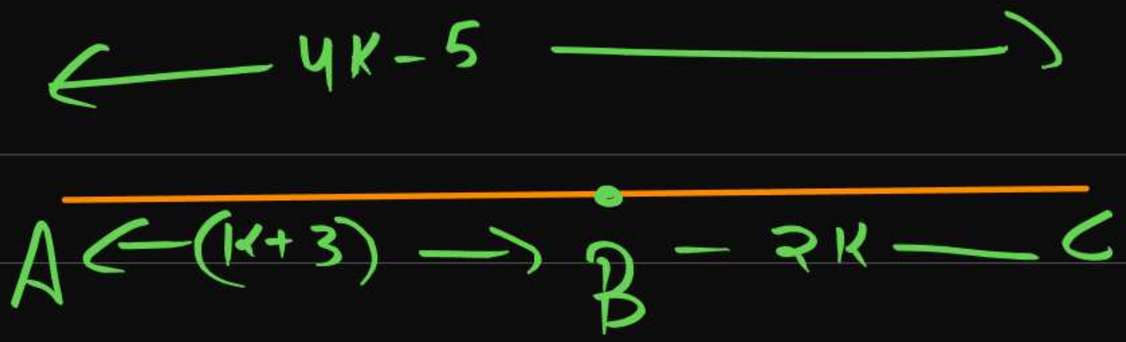
(a) 24 square units (b) 36 square units (c) 12 square units (d) 18 square units

Answer: 4

Q. If AB = k + 3, BC = 2k and AC = 4k – 5, then for what value of 'k' does B lie on AC?

यदि AB = k + 3, BC = 2k और AC = 4k – 5, तो ' k' के किस मान के लिए B, AC पर स्थित होगा ?

RRB Technician Grade III (2024)



$$4K - 5 = K + 3 + 2K$$

$$K = 8$$

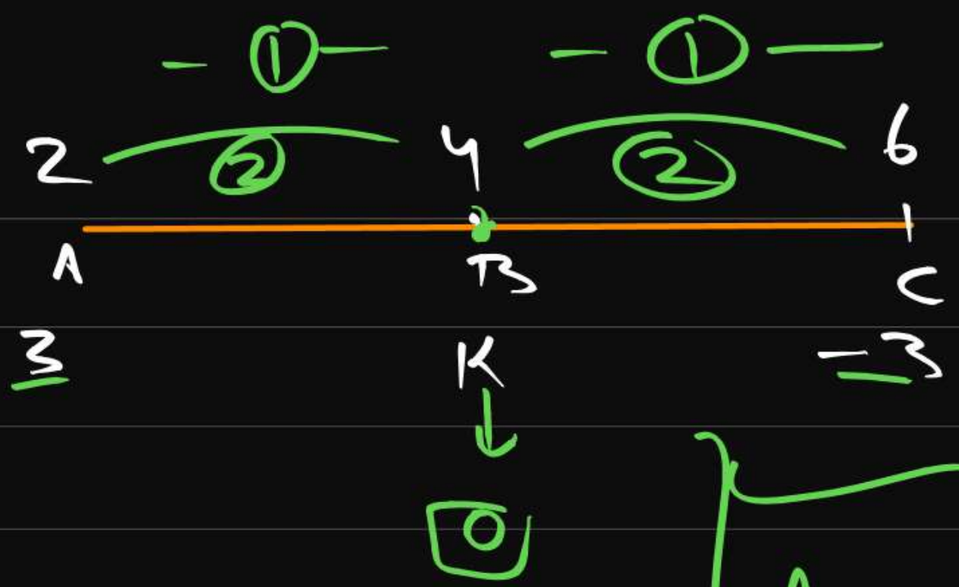
Good Questions

(a) 3 (b) 8 (c) 2 (d) 5

Answer: 2

Q-) Find the value of k if the points A(2, 3), B(4, k) and C(6, –3) are collinear.

k का मान ज्ञात करे यदि अंक A (2, 3), B (4, k) और C (6, –3) आपस में संरेख (एक ही रेखा पर) हैं



Slope method or area of triangle or section formula

$$A = 0 = \frac{1}{2}\left[2(k+3) + 4(-6) + 6(-3-k)\right]$$

$$4|k| = 0$$

$$\boxed{k = 0}$$

A) 0
B) 2
C) -1
D) 3

Q). Find the value of x if (3,4), (7,6) and (x,5) are co-linear.

यदि (3,4), (7,6) और (x, 5) सह-रैखिक हैं तो X का मान ज्ञात करें।

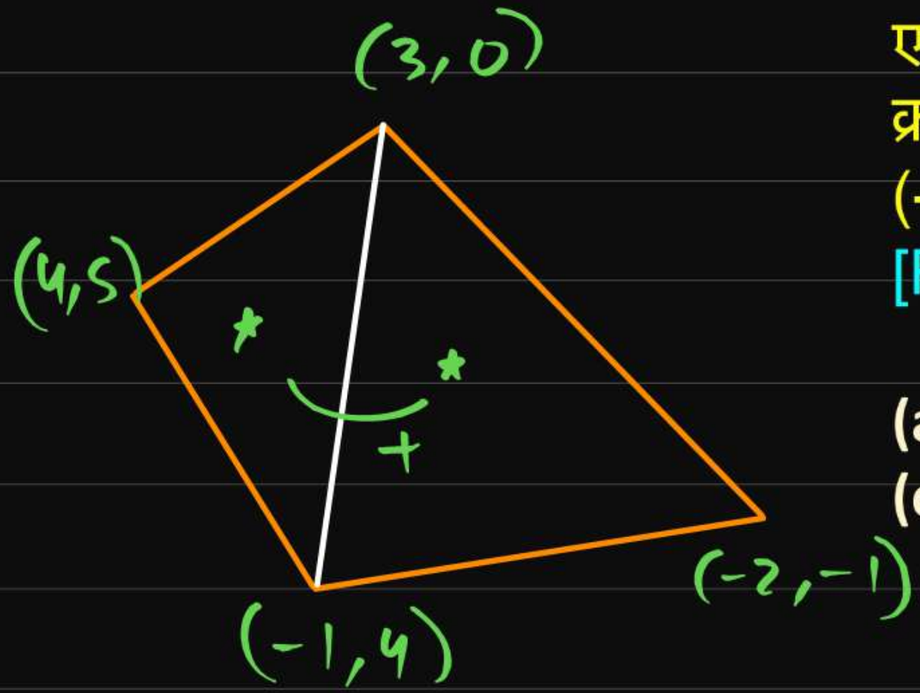
HW

(a) 3
(b) 6
(c) 5
(d) 4

Q. The area of a quadrilateral whose vertices are (3, 0), (4, 5), (-1, 4) and (- 2,- 1) taken in order, is:

एक चतुर्भुज का क्षेत्रफल जिसके शीर्ष क्रम में (3 , 0) , (4 , 5) , (- 1 , 4) और (-2 ,- 1) हैं , है :

[RRB Technician 2024 Grade-I]



(a) 30 sq.units (b) 48 sq.units
(c) 12 sq.units (d) 24 sq.units

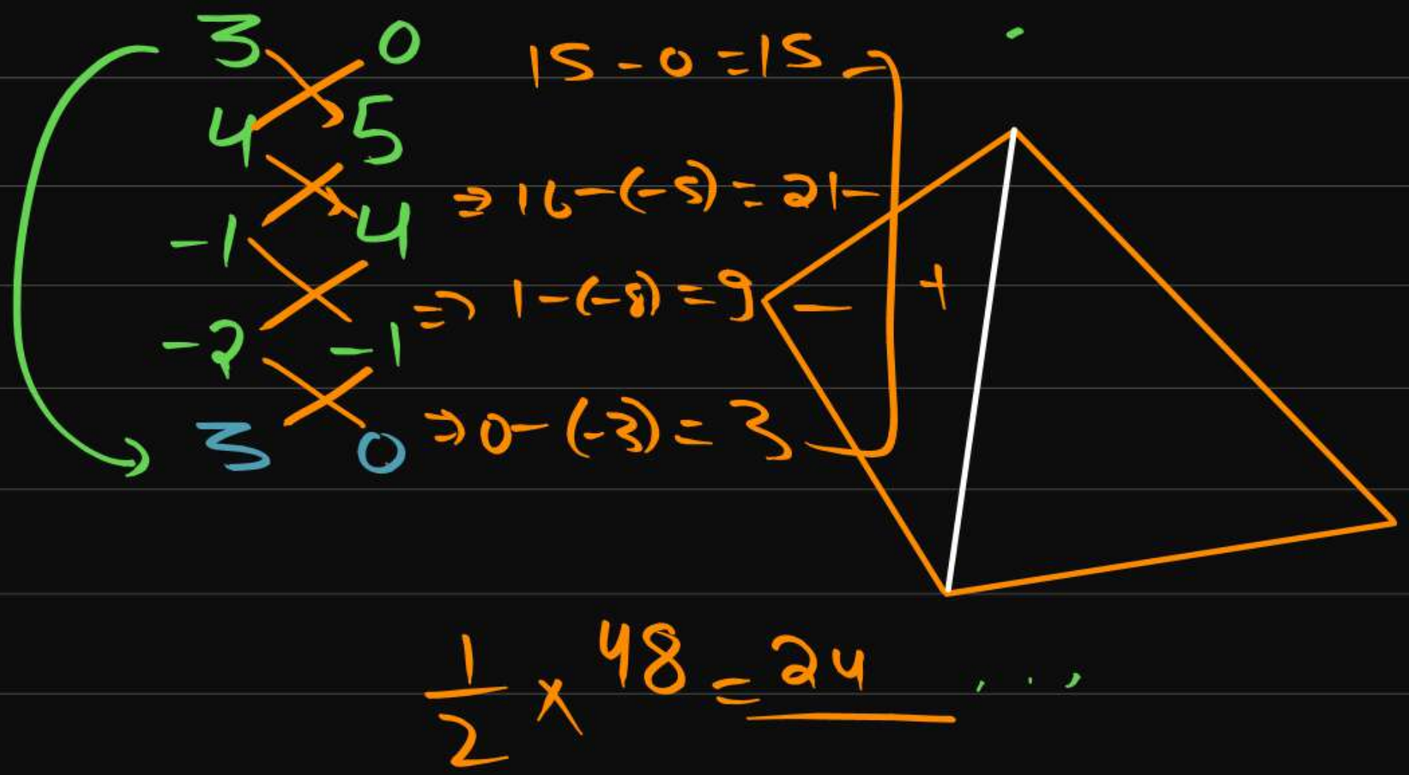
Answer: 4

Q. The area of a quadrilateral whose vertices are (3, 0), (4, 5), (-1, 4) and (- 2,- 1) taken in order, is:

एक चतुर्भुज का क्षेत्रफल जिसके शीर्ष क्रम में (3 , 0) , (4 , 5) , (- 1 , 4) और (-2 ,- 1) हैं , है :

[RRB Technician 2024 Grade-I]

(a) 30 sq.units (b) 48 sq.units
(c) 12 sq.units (d) 24 sq.units



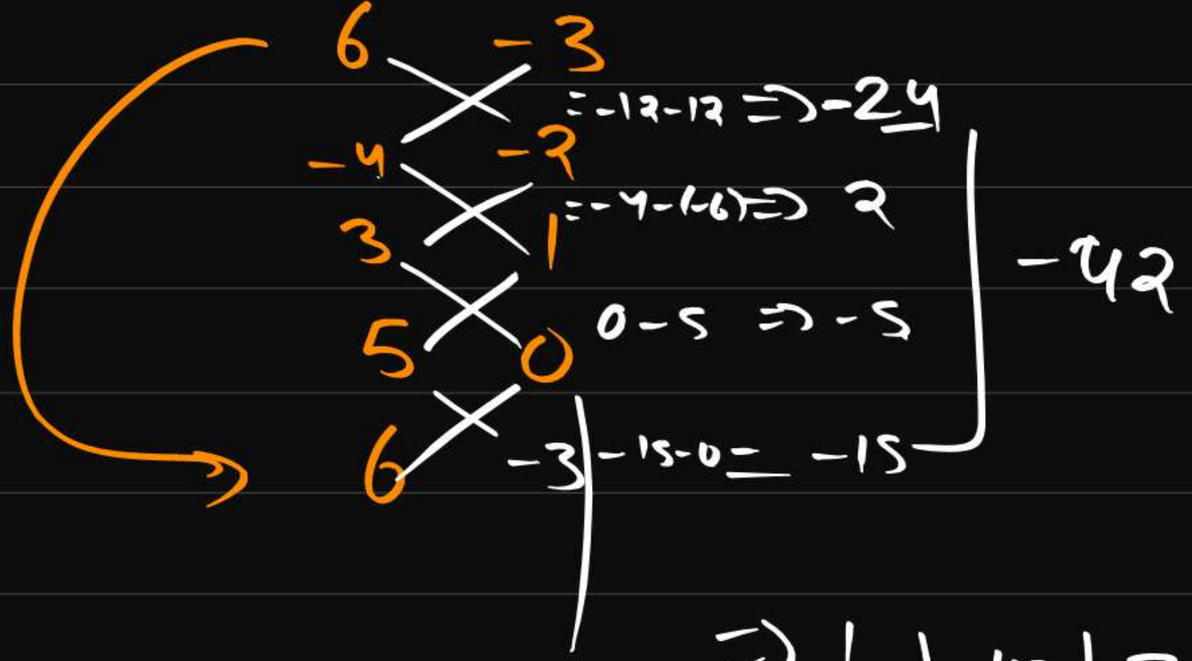
$\frac{1}{2} \times 48 = 24$ sq. units

Answer: 4

Q. Find the area of the quadrilateral ABCD, the coordinates of whose vertices are A(6,-3),B(-4,-2),C(3,1) and D(5,0)?

चतुर्भुज ABCD का क्षेत्रफल ज्ञात कीजिए जिसके शीर्षों के निर्देशांक A(6,-3),B(-4,-2),C(3,1) तथा D(5,0) हैं।

$-12-(+12)$



$\Rightarrow \frac{1}{2}|-42| = 21$

SSC MTS 2024

(a) 25 unit²
(b) 30 unit²
(c) 16 unit²
(d) 21 unit²
Ans: d

Q. A land plot is defined by points A(0,0), B(4,0), C(5,3), D(1,3). What is the area of the parallelogram representing this plot?

एक भूमि का भूखंड बिंदुओं A(0,0), B(4,0), C(5,3), D(1,3) द्वारा परिभाषित किया गया है। इस भूखंड को दर्शाने वाले समांतर चतुर्भुज का क्षेत्रफल क्या है?

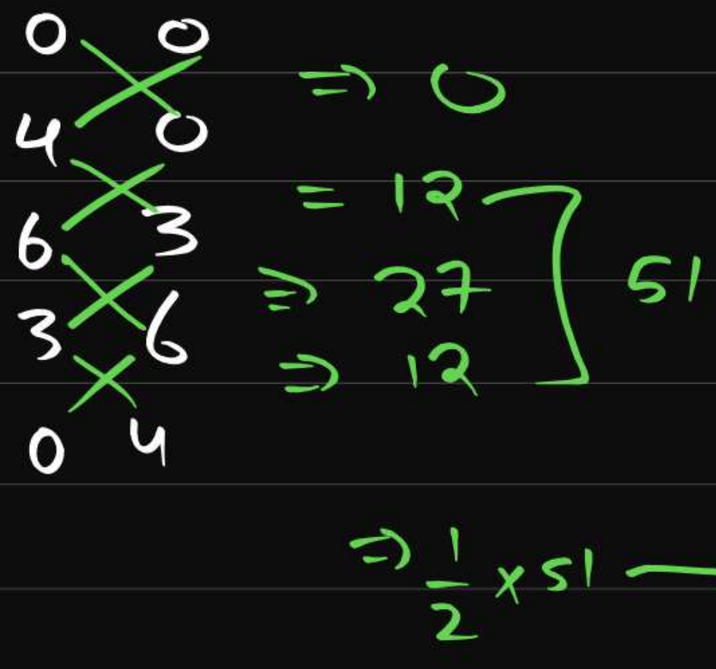
SSC CHSL 2025

(a) 12 (b) 8 (c) 14 (d) 10

Answer: 1

Q. The vertices of a pentagon are **A(0, 0), B(4, 0), C(6, 3), D(3, 6), and E(0, 4)**. Find the area of the pentagon.

एक पंचभुज के शीर्ष A(0, 0), B(4, 0), C(6, 3), D(3, 6) और E(0, 4) हैं। पंचभुज का क्षेत्रफल ज्ञात कीजिए।



(a) 22.5 sq. units
(b) 24 sq. units
(c) **25.5 sq. units**
(d) 27 sq. units

Answer: 1

Co-ordinate Geometry

निर्देशांक ज्यामिति

(Basic to high Level)

Latest Pattern

Co-ordinate Geometry

Part-4

Different Forms of Equation of a line

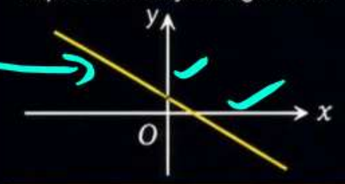
9 FORMS OF EQUATION OF A LINE

IN CO-ORDINATE GEOMETRY

1 GENERAL FORM

$$Ax + By + C = 0$$

Where A, B, C are real numbers with A and B not both zero. Represents any straight line.

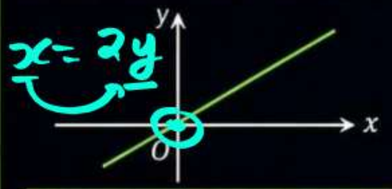


Most general form of a line.

2 LINE THROUGH ORIGIN

$$y = mx$$

Where m is the slope. (Line passes through origin $(0, 0)$).

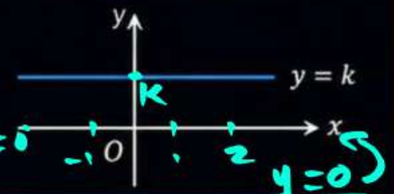


Special case of slope–intercept form with $c = 0$.

3 HORIZONTAL LINE

$$y = k$$

Where k is a constant. (Line parallel to x–axis).

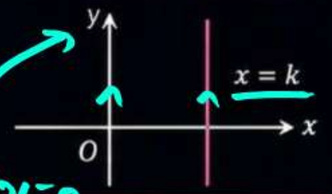


Slope $m = 0$.

4 VERTICAL LINE

$$x = k$$

Where k is a constant. (Line parallel to y–axis).

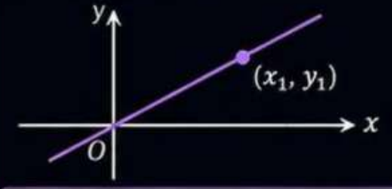


Slope is undefined.

5 POINT–SLOPE FORM

$$y - y_1 = m(x - x_1)$$

Where (x_1, y_1) is a point on the line and m is the slope.



Useful when one point on the line and slope are known.

6 TWO–POINT FORM

$$\frac{y - y_1}{y_2 - y_1} = \frac{x - x_1}{x_2 - x_1}$$

Where (x_1, y_1) and (x_2, y_2) are two distinct points on the line.

Useful when two points on the line are known.

7 INTERCEPT FORM

$$\frac{x}{a} + \frac{y}{b} = 1$$

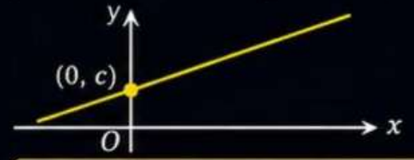
Where a and b are x–intercept and y–intercept respectively. (Line cuts axes at $(a, 0)$ and $(0, b)$).

Useful when intercepts on axes are known.

8 SLOPE–INTERCEPT FORM

$$y = mx + c$$

Where m is slope and c is the y–intercept (line cuts y–axis at $(0, c)$).

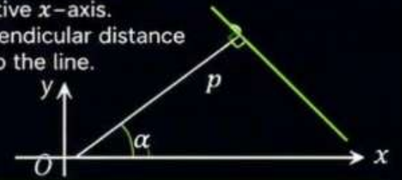


Most useful form when slope and y–intercept are known.

9 NORMAL FORM

$$x \cos\alpha + y \sin\alpha = p$$

Where α is the angle between the perpendicular from origin to the line and the positive x–axis. p is the perpendicular distance from origin to the line.



Useful in geometrical problems involving perpendicular distance.

IMPORTANT FACTS ABOUT LINES

SLOPE (m)

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

(for two points (x_1, y_1) and (x_2, y_2))

PARALLEL LINES

$$m_1 = m_2$$

Lines are parallel if their slopes are equal.

PERPENDICULAR LINES

$$m_1 m_2 = -1$$

Lines are perpendicular if product of slopes is -1.

ANGLE BETWEEN TWO LINES

$$\tan\theta = \left| \frac{m_2 - m_1}{1 + m_1 m_2} \right|$$

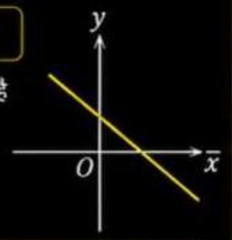
Where θ is the (acute) angle between the two lines.

रेखा के 9 विभिन्न रूप (Forms) | निर्देशांक ज्यामिति (Co-ordinate Geometry)

1 सामान्य रूप (General Form)

$$Ax + By + C = 0$$

जहाँ A, B, C वास्तविक संख्याएँ है और A तथा B दोनों शून्य नहीं हैं। यह किसी भी सीधी रेखा का सबसे सामान्य रूप है।

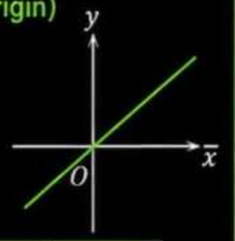


उदाहरण: $2x + 3y + 6 = 0$, $5x - 2y + 7 = 0$

2 मूल बिन्दु से गुजरने वाली रेखा (Line Through Origin)

$$y = mx$$

जहाँ m ढाल (slope) है। यह रेखा $(0, 0)$ बिन्दु से गुजरती है।

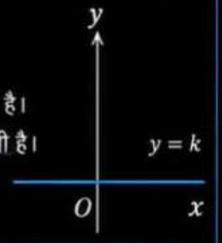


उदाहरण: $y = 2x$, $y = -\frac{1}{2}x$

3 क्षैतिज रेखा (Horizontal Line)

$$y = k$$

जहाँ k एक नियतांक (constant) है। यह रेखा x-अक्ष के समानांतर होती है।

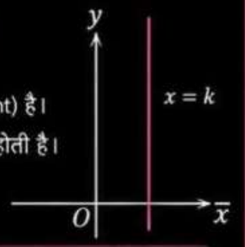


उदाहरण: $y = 4$, $y = -3$

4 लम्बवत रेखा (Vertical Line)

$$x = k$$

जहाँ k एक नियतांक (constant) है। यह रेखा y-अक्ष के समानांतर होती है।

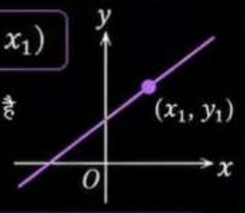


उदाहरण: $x = 3$, $x = -2$

5 बिन्दु-ढाल रूप (Point-Slope Form)

$$y - y_1 = m(x - x_1)$$

जहाँ (x_1, y_1) रेखा पर एक बिन्दु है और m ढाल (slope) है।

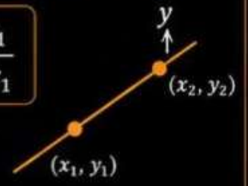


उदाहरण: $y - 2 = 3(x - 1)$

6 दो बिन्दु रूप (Two-Point Form)

$$\frac{y - y_1}{y_2 - y_1} = \frac{x - x_1}{x_2 - x_1}$$

जहाँ (x_1, y_1) और (x_2, y_2) रेखा पर दो भिन्न बिन्दु हैं।

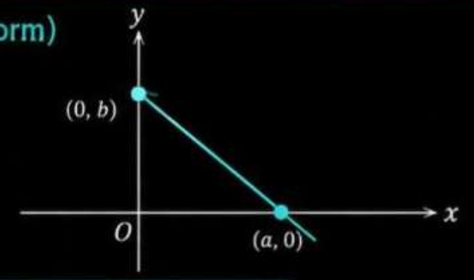


उदाहरण: $\frac{y-1}{4-1} = \frac{x-2}{5-2}$

7 अंतरछेद रूप (Intercept Form)

$$\frac{x}{a} + \frac{y}{b} = 1$$

जहाँ a और b क्रमशः x-अंतरछेद और y-अंतरछेद हैं। रेखा x-अक्ष को $(a, 0)$ और y-अक्ष को $(0, b)$ पर काटती है।

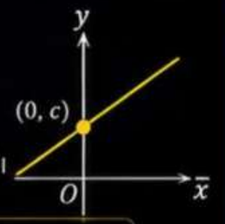


उदाहरण: $\frac{x}{5} + \frac{y}{3} = 1$ (x-अंतरछेद $= 5$, y-अंतरछेद $= 3$)

8 ढाल-अंतरछेद रूप (Slope-Intercept Form)

$$y = mx + c$$

जहाँ m ढाल (slope) है और c y-अंतरछेद (y-intercept) है। रेखा y-अक्ष को $(0, c)$ पर काटती है।

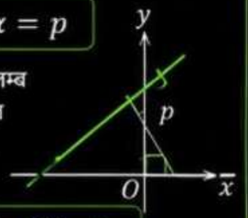


उदाहरण: $y = 2x + 3$, $y = -x + 1$

9 लंम्बवत दूरी (Normal Form)

$$x\cos\alpha + y\sin\alpha = p$$

जहाँ α वह कोण है जो रेखा का लम्ब मूल बिन्दु से बनाता है और p मूल बिन्दु से रेखा की लंबवत दूरी है।



उदाहरण: $x\cos 60° + y\sin 60° = 5$

★ नोट: इन रूपों का उपयोग प्रश्न के अनुसार सबसे सुविधाजनक रूप चुनकर करें। ★

महत्वपूर्ण तथ्य (Important Facts)

ढाल (Slope)

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

जहाँ (x_1, y_1) और (x_2, y_2) रेखा पर दो भिन्न बिन्दु हैं।

समान्तर रेखाएँ (Parallel Lines)

$$m_1 = m_2$$

दो रेखाएँ समान्तर हैं यदि उनकी ढालें बराबर हों।

लम्बवत रेखाएँ (Perpendicular Lines)

$$m_1 m_2 = -1$$

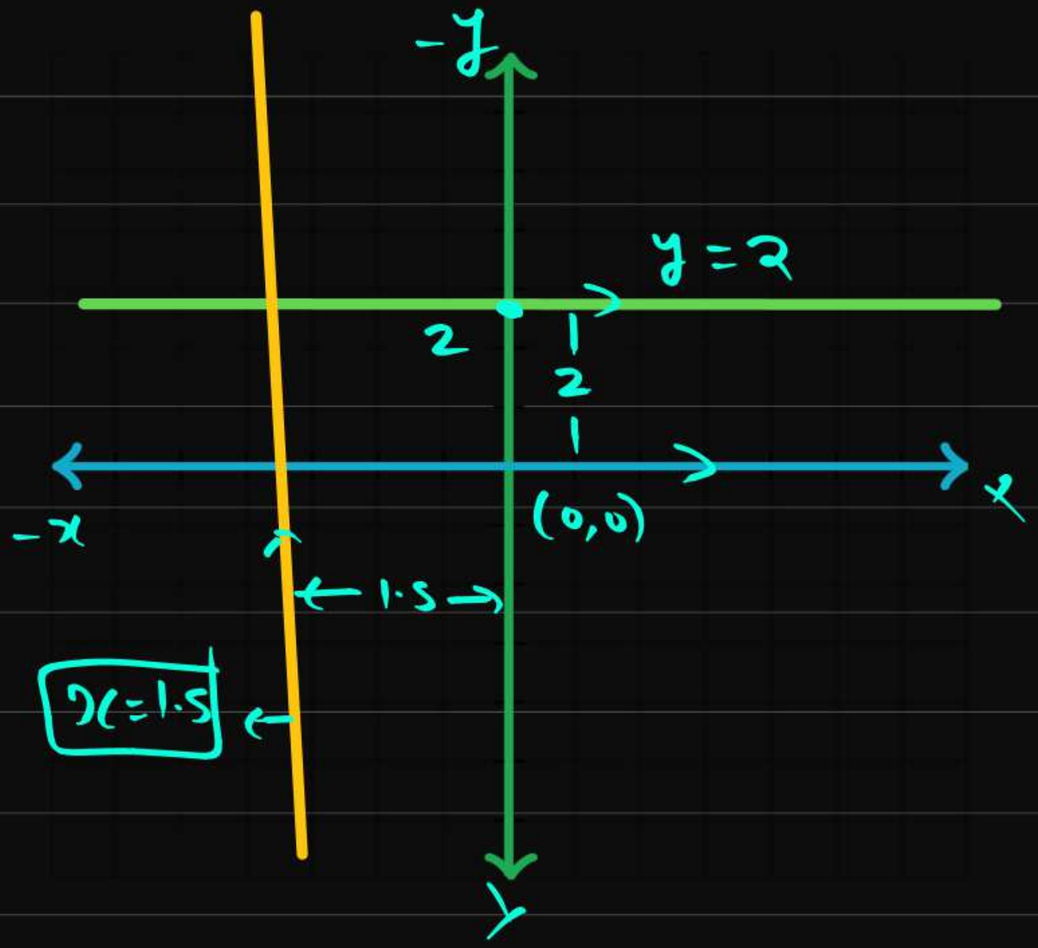
दो रेखाएँ लम्बवत हैं यदि ढालों का गुणनफल -1 हो।

दो रेखाओं के बीच कोण (θ) (Angle Between Two Lines)

$$\tan\theta = \left|\frac{m_2 - m_1}{1 + m_1 m_2}\right|$$

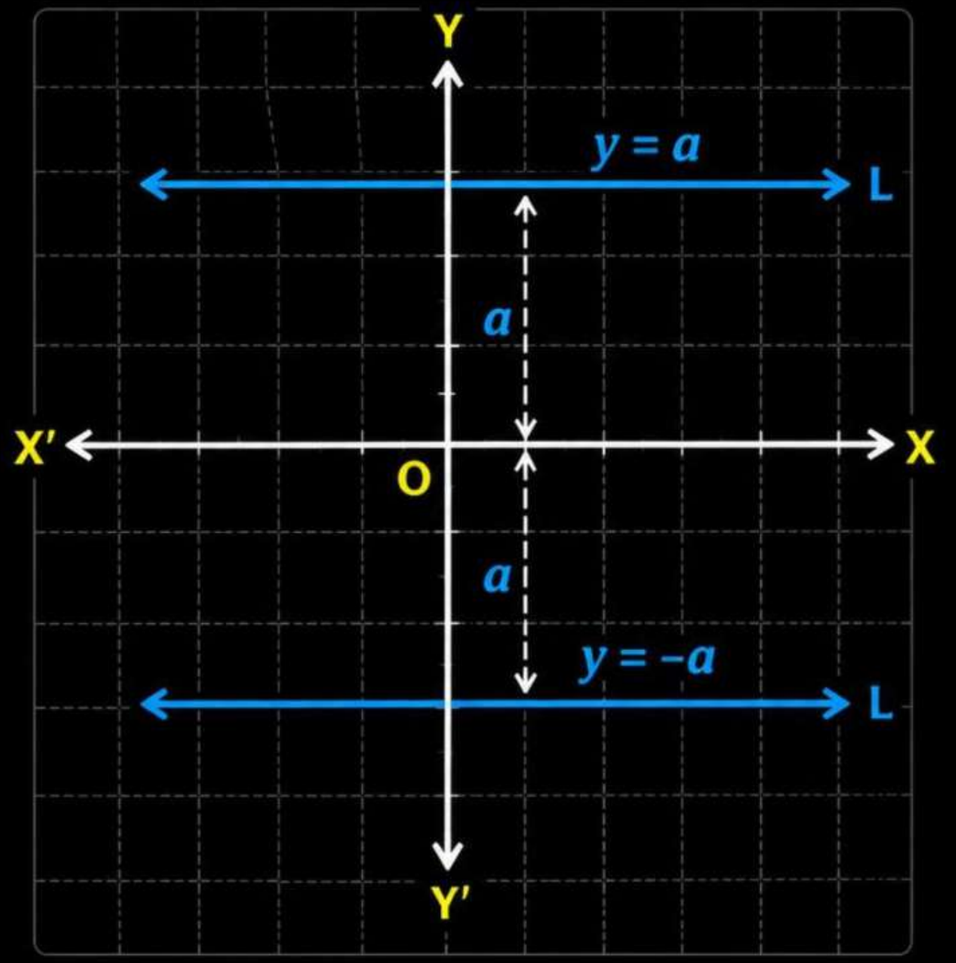
जहाँ θ दो रेखाओं के बीच तीव्र (acute) कोण है।

Horizontal and vertical lines
क्षैतिज और ऊर्ध्वाधर लाइनें

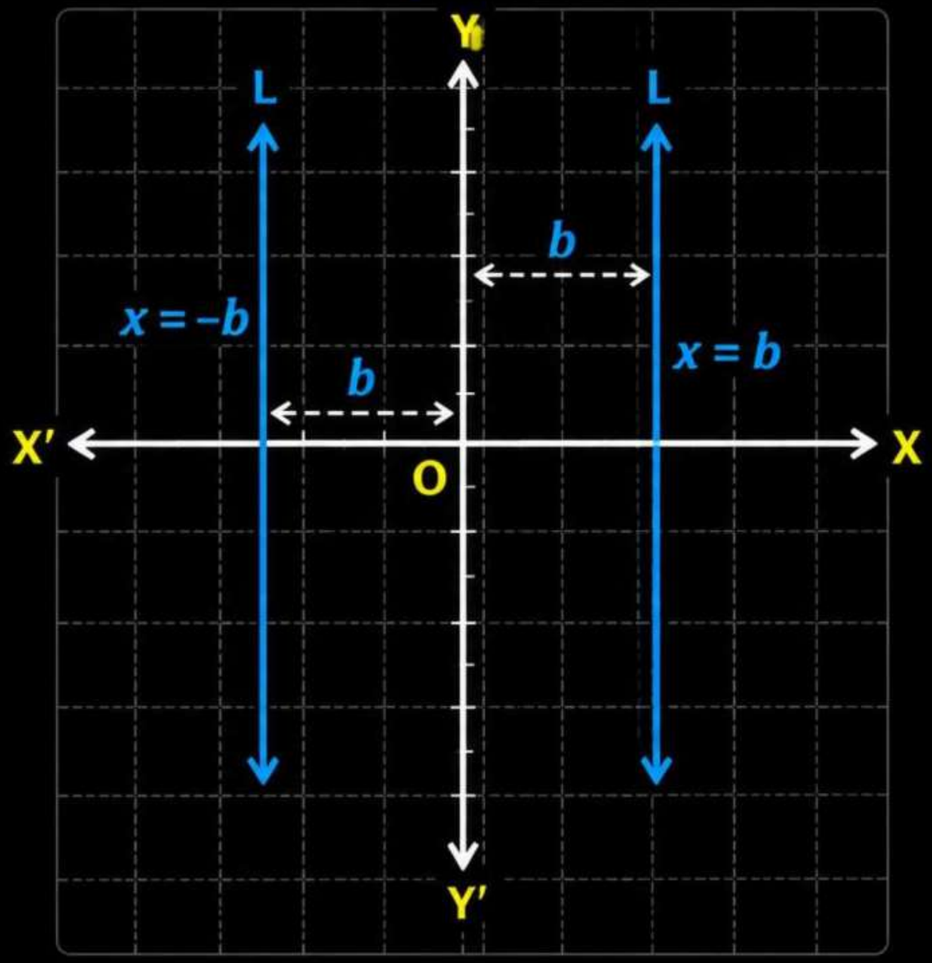


Horizontal and vertical lines

क्षैतिज और ऊर्ध्वाधर लाइनें



(a)

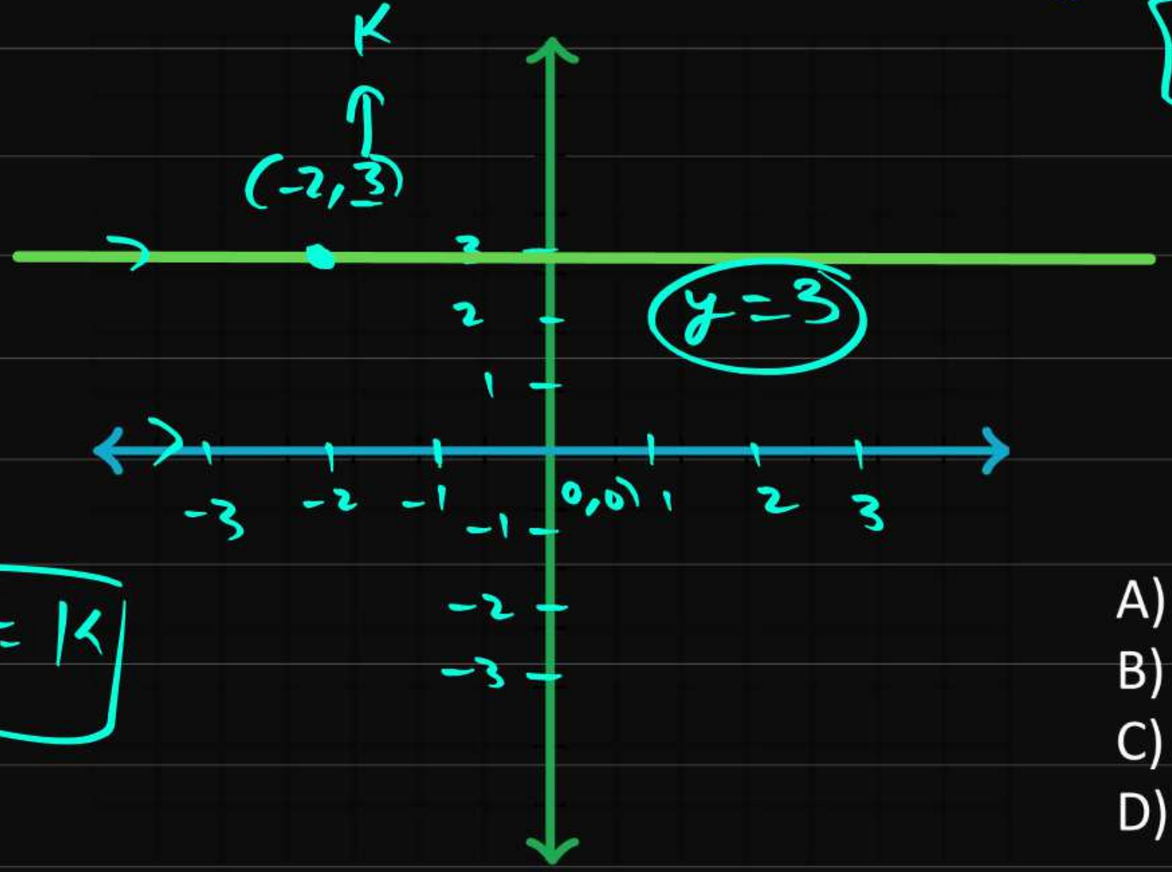


(b)

Q-) Find the equation of the line parallel to x axes and passing through (– 2, 3)

x अक्ष के समानांतर रेखा के समीकरण का पता लगाएं जो (- 2, 3) से गुजरती हो।

$y = 3$



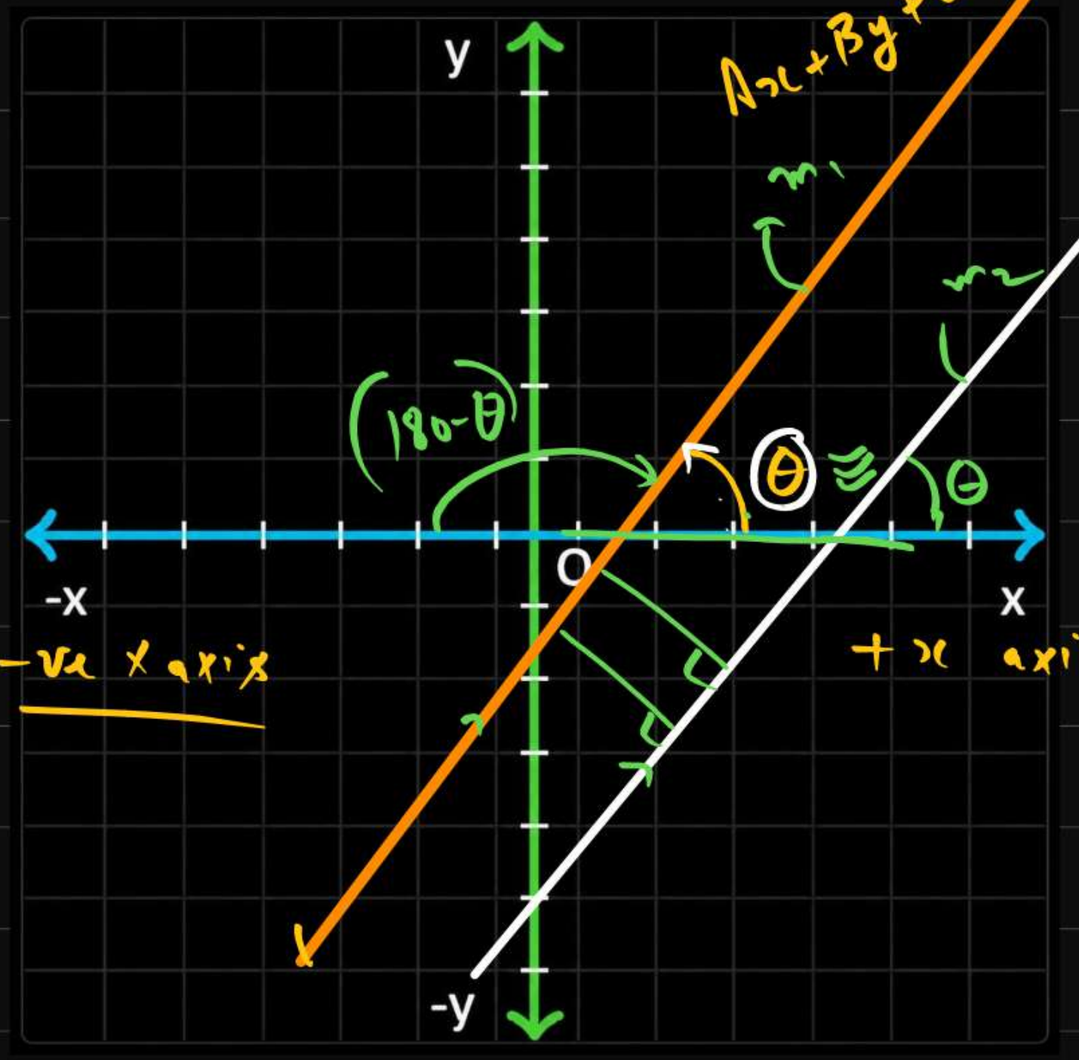
$y = k$

A) Y = 3
B) X +y =3
C) Y= -3
D) X = -2

|| to y axis & passes (-3, 8)

$x = 3$

Slope of a Line- रेखा की ढाल



Slope = $\tan\theta$

If lines

$m_1 = m_2$

SLOPE OF A LINE – रेखा की ढाल

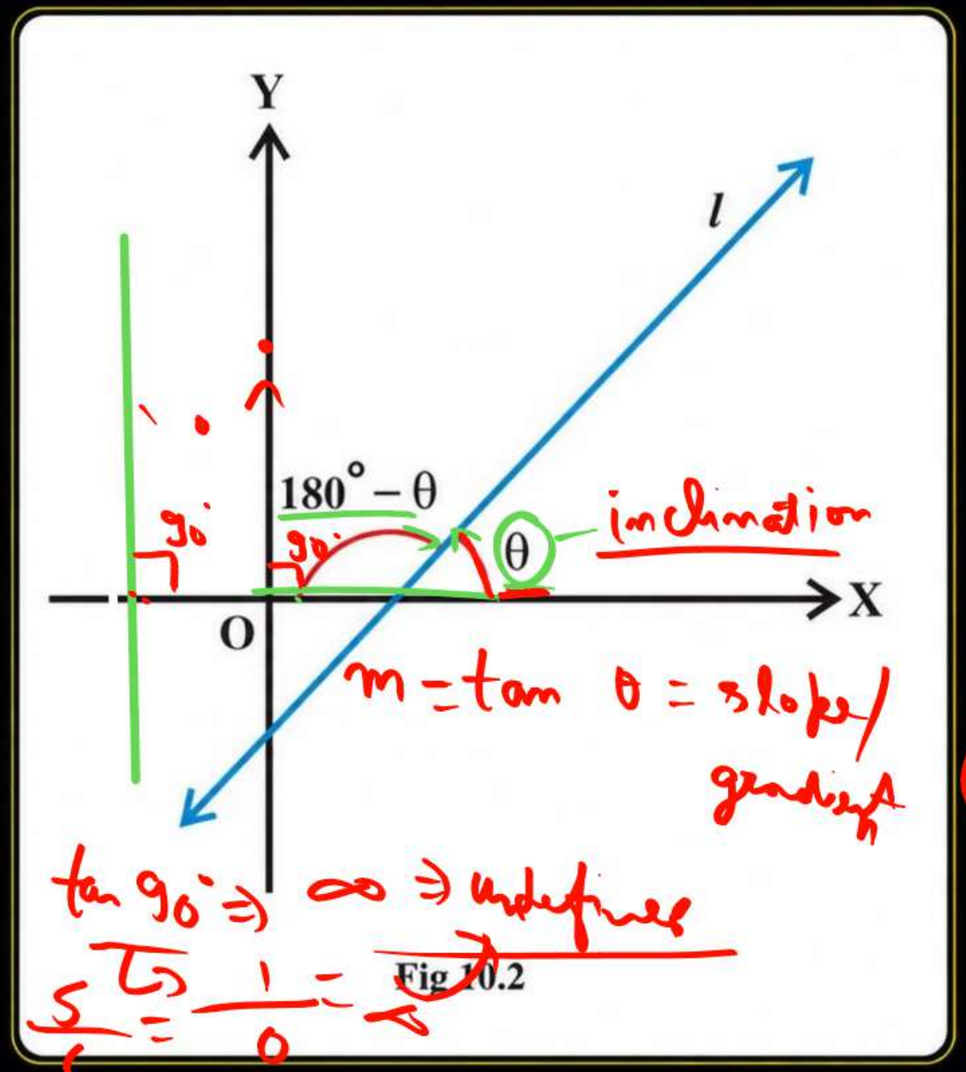


Fig 10.2

- A line in a coordinate plane forms two angles with the x-axis, which are supplementary (समपूरक).
  निर्देशांक तल में एक रेखा x-अक्ष के साथ दो कोण बनाती है, जो समपूरक (पूरक) होते हैं।
- If θ is the inclination of a line l, then tan θ is called the slope or gradient of the line l.
  यदि θ रेखा l का झुकाव कोण (inclination) है, तो tan θ को रेखा l की ढाल (slope) या प्रवणता (gradient) कहते हैं।
- The slope of a line whose inclination is 90° is not defined.
  जिस रेखा का झुकाव कोण 90° होता है, उसकी ढाल परिभाषित नहीं होती।
- The slope of a line is denoted by m.
  रेखा की ढाल को m द्वारा निरूपित (denote) किया जाता है।
- It may be observed that the slope of x-axis is zero and slope of y-axis is not defined.
  ध्यान दें कि x-अक्ष की ढाल शून्य (zero) होती है तथा y-अक्ष की ढाल परिभाषित नहीं होती।

Inclination of line – झुकाव कोण (θ)	Slope – ढाल (m)
$0° \le \theta < 180°$	$m = \tan\theta$

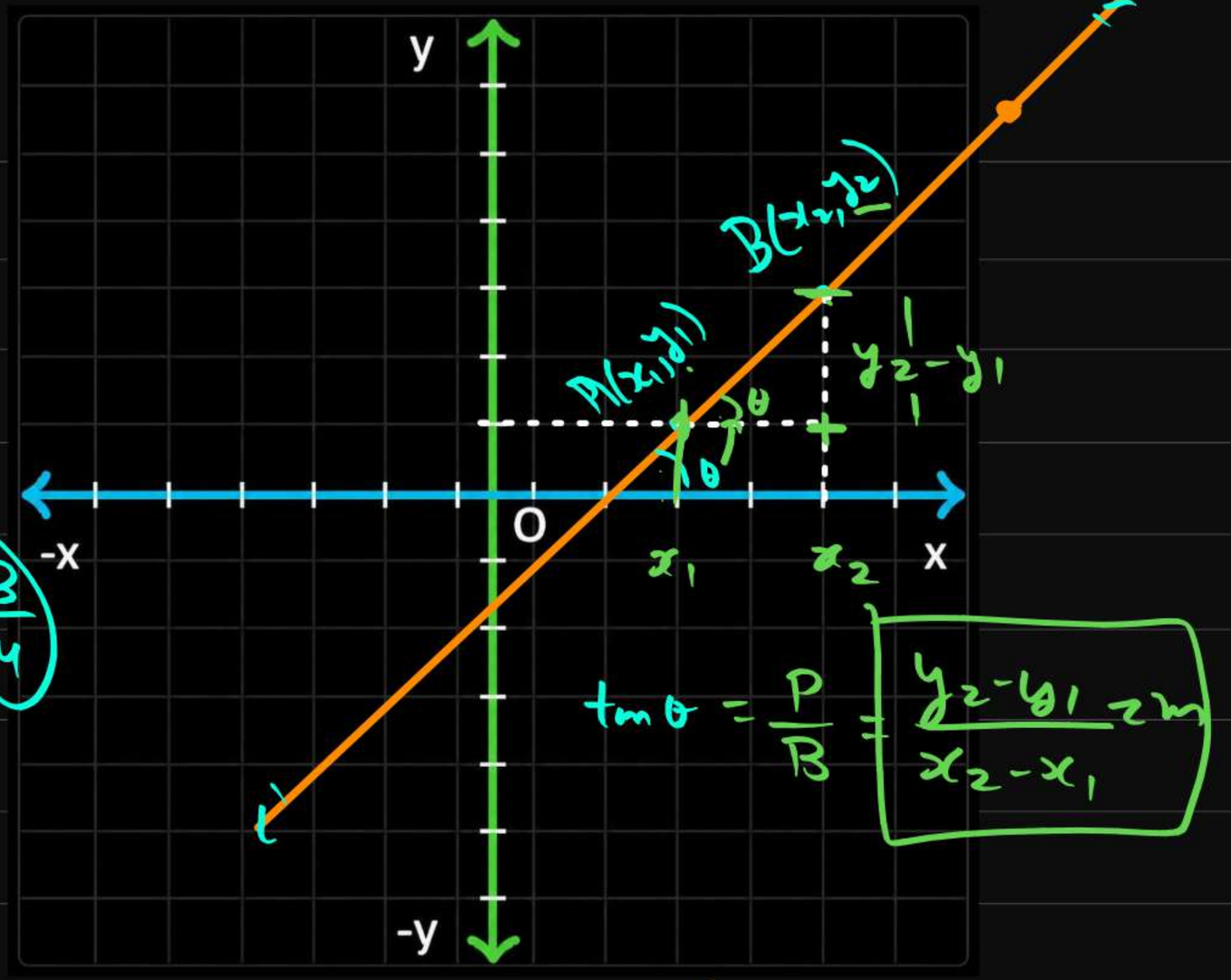
Slope of a line when coordinates of any two points on the line are given

रेखा का ढाल जब उस पर 2 बिंदु दिए हुवे हो

(6, 7)

(2, 4)

Slope $\Rightarrow m = \frac{7-4}{6-2} = \frac{3}{4}$



$$\tan\theta = \frac{P}{B} = \frac{y_2 - y_1}{x_2 - x_1} = m$$

Slope of a line when coordinates of any two points on the line are given

रेखा का ढाल जब उस पर 2 बिंदु दिए हुवे हो

$(6, 7)$

$(2, 4)$

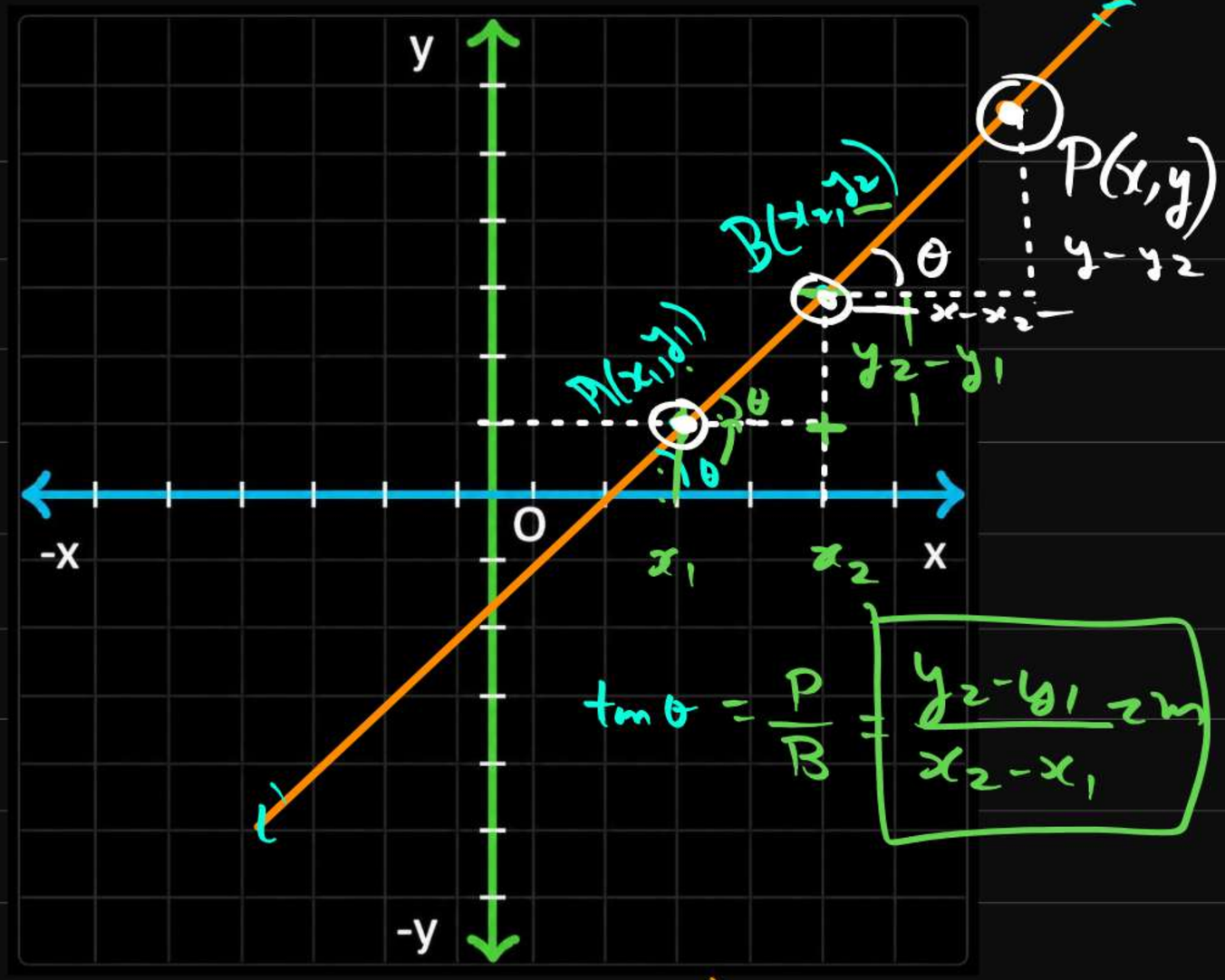
$$\frac{y - y_2}{x - x_2} = \frac{y_2 - y_1}{x_2 - x_1} \Rightarrow \frac{3}{4}$$

$7 \quad 4$

$6 \quad 2$

$$\frac{y - 7}{x - 6} = \frac{3}{4}$$

$$\therefore \quad 3x - 4y + 10 = 0$$



Slope of a line when coordinates of any two points on the line are given

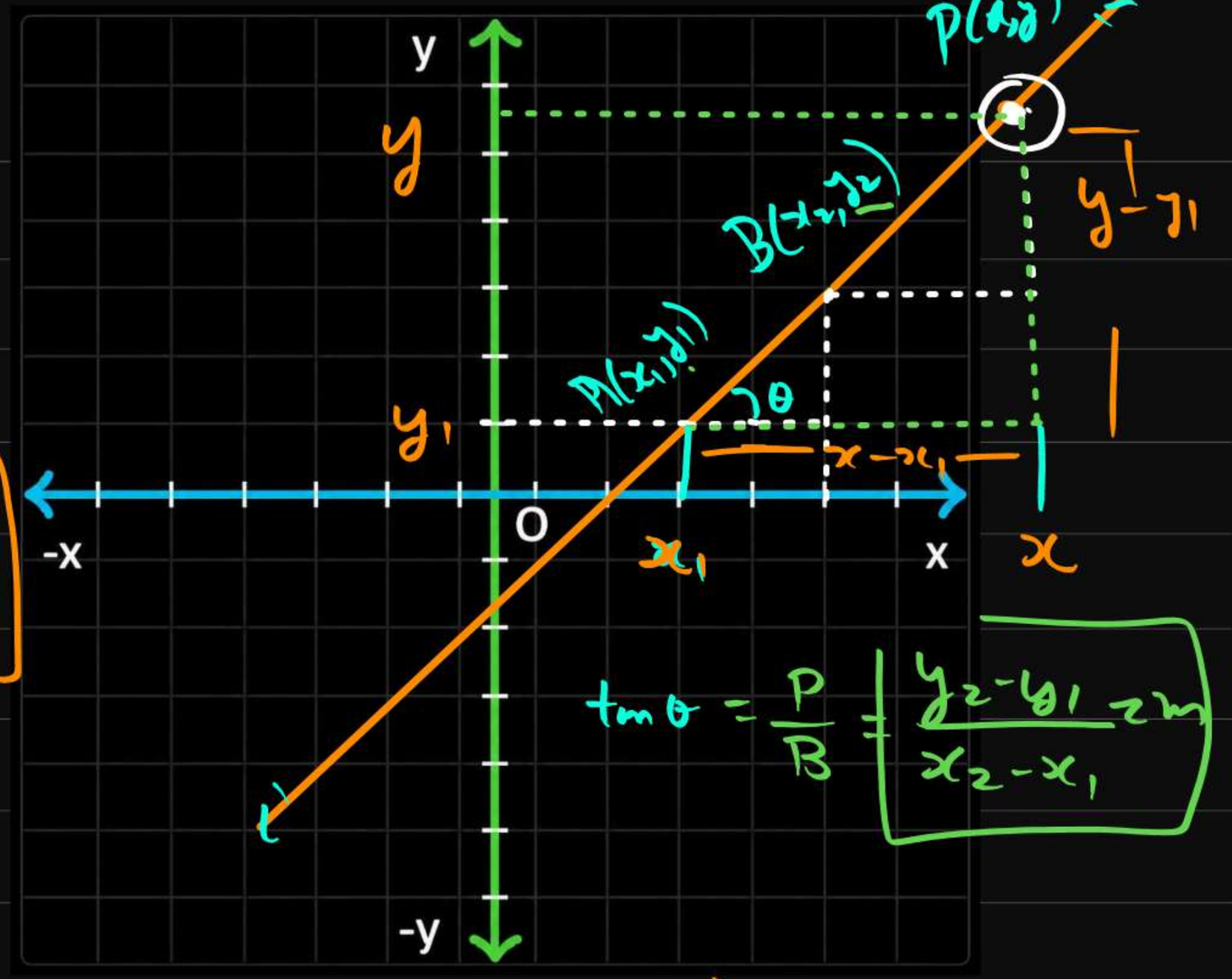
रेखा का ढाल जब उस पर 2 बिंदु दिए हुवे हो

$B \rightarrow (6, 7)$

$A \rightarrow (2, 4)$

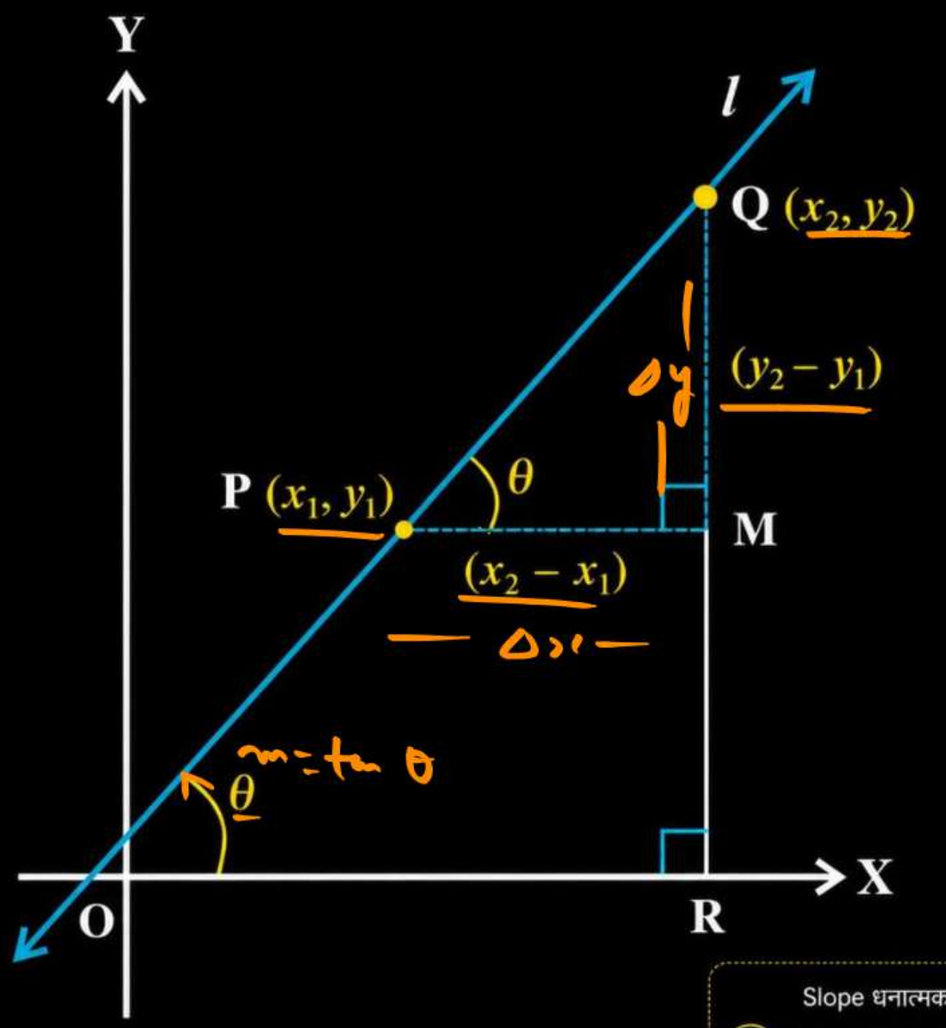
$$\frac{y - y_2}{x - x_2} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{y_1 - y_2}{x_1 - x_2}$$



$$\tan\theta = \frac{P}{B} = \frac{y_2 - y_1}{x_2 - x_1}$$

Slope of a Line (रेखा की ढाल)



Slope (m) / ढाल (m)

$$m = \tan\theta$$

जहाँ θ = रेखा द्वारा धनात्मक x-अक्ष के साथ बनाया गया कोण।

- दो बिंदुओं $P(x_1, y_1)$ और $Q(x_2, y_2)$ से रेखा की ढाल (m)

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

या

$$m = \frac{\Delta y}{\Delta x}$$

जहाँ $\Delta y = y_2 - y_1$, $\Delta x = x_2 - x_1$

- यदि रेखा की ढाल m हो, तो $m = \tan\theta$

- इस आकृति में,

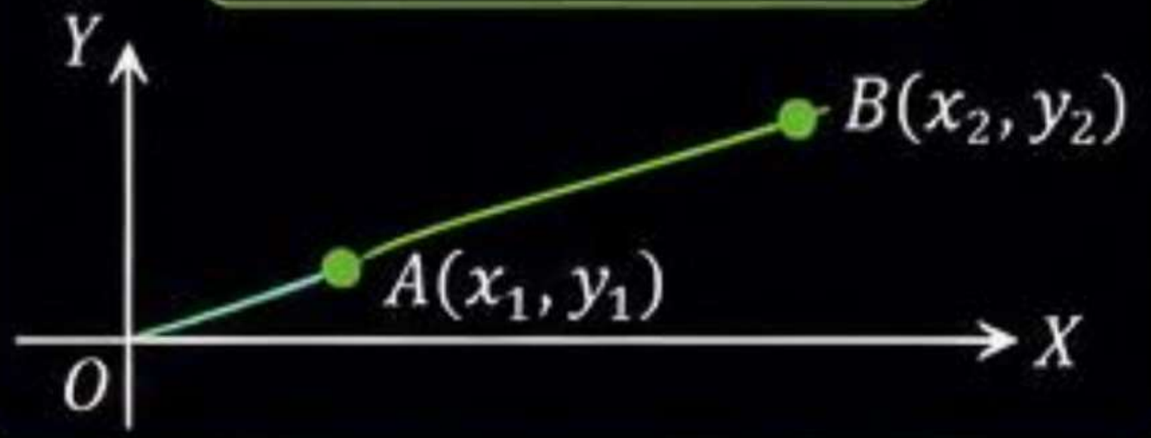
$PM = x_2 - x_1$ (क्षैतिज अंतर)
$QM = y_2 - y_1$ (ऊर्ध्वाधर अंतर)

Slope धनात्मक (Positive) ⇒ रेखा बाएँ से दाएँ ऊपर उठती है।
Slope ऋणात्मक (Negative) ⇒ रेखा बाएँ से दाएँ नीचे गिरती है।
Slope शून्य (Zero) ⇒ रेखा x-अक्ष के समानांतर होती है।
Slope अपरिभाषित (Not Defined) ⇒ रेखा y-अक्ष के समानांतर होती है।

2 SLOPE USING TWO POINTS
(दो बिंदुओं से ढाल)

If $A(x_1, y_1)$ and $B(x_2, y_2)$ lie on a line, then

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$



Point-slope form
बिंदु-ढाल रूप

Equation of a line when coordinates of any one point and slope of a line are given.

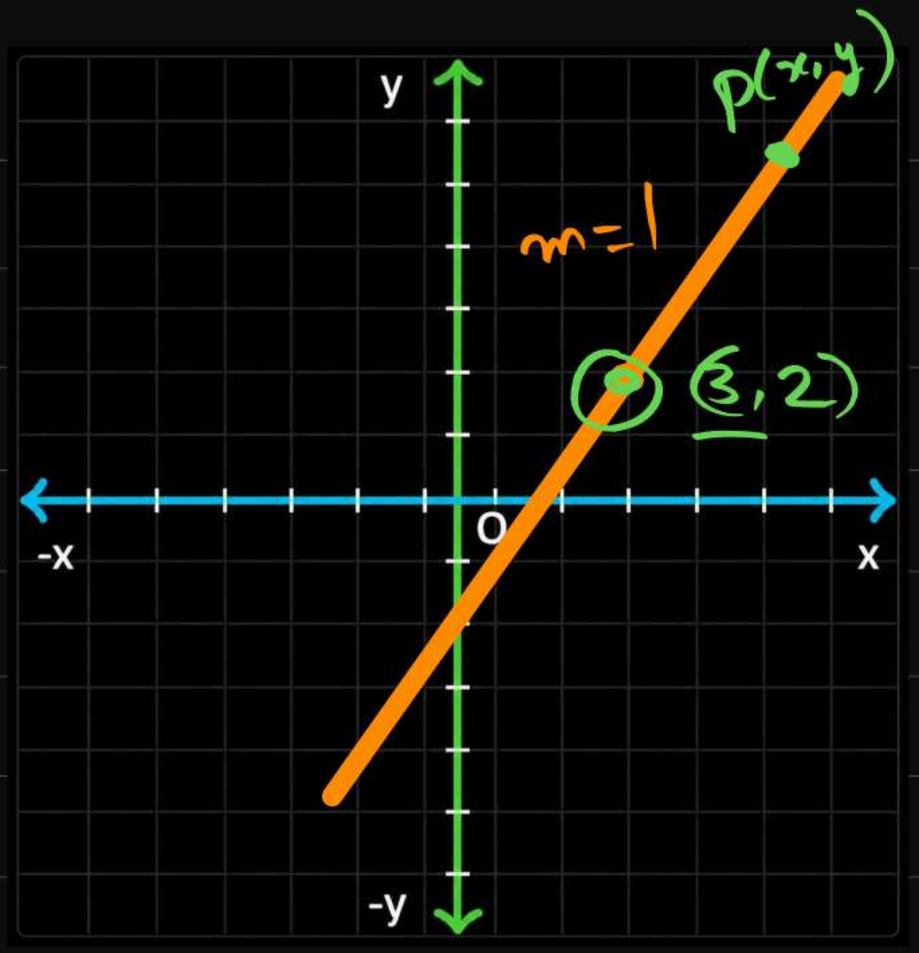
एक रेखा का समीकरण जब किसी एक बिंदु के निर्देशांक और एक रेखा का ढलान दिया जाता है।

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

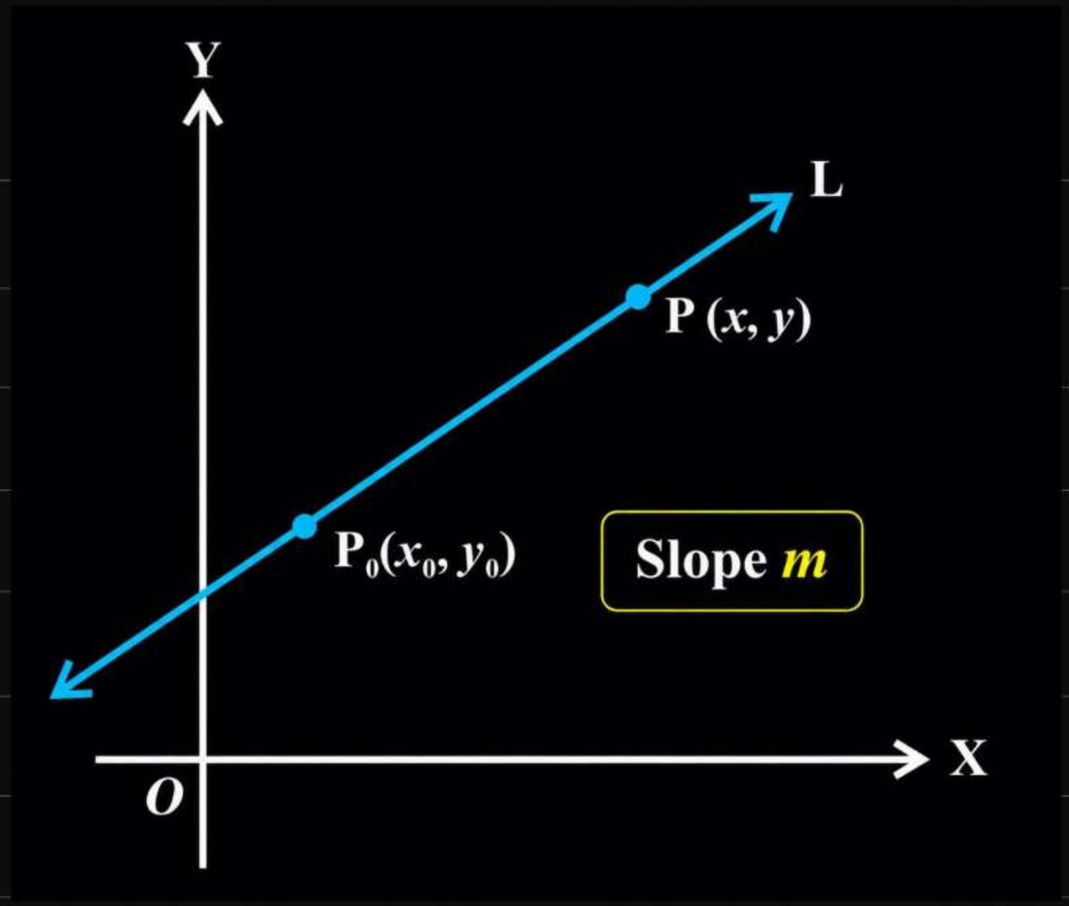
$$1 = \frac{y - 2}{x - 3}$$

$$x - y - 1 = 0$$

$$\boxed{x - y = 1}$$



Point-slope form
बिंदु-ढाल रूप



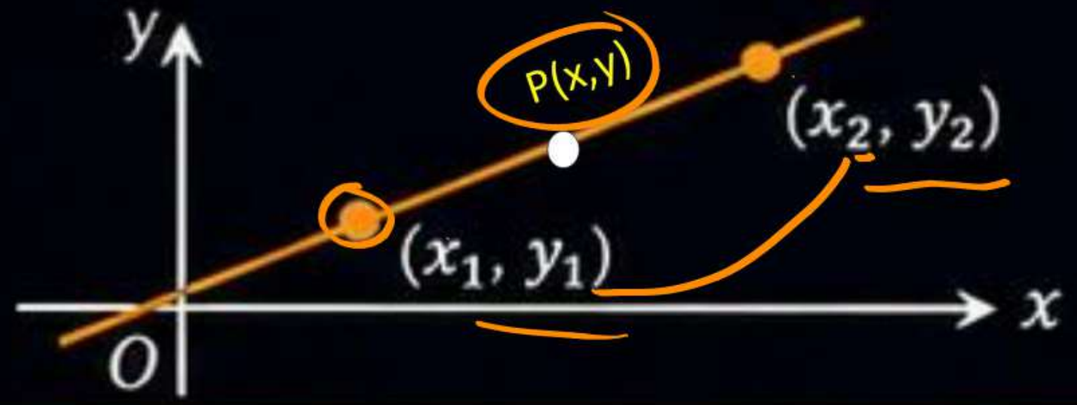
Suppose that $P_0 (x_0, y_0)$ is a fixed point on a non-vertical line L, whose slope is *m*. Let P (*x*, *y*) be an arbitrary point on L

मान लीजिए कि P_0 (x 0, y 0) एक गैर-ऊर्ध्वाधर रेखा L पर एक निश्चित बिंदु है, जिसका ढलान m है। मान लीजिए P (x, y) L पर एक मनमाना बिंदु है

6 TWO–POINT FORM

$$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$$

Where (x_1, y_1) and (x_2, y_2) are two distinct points on the line.



Useful when two points on the line are known.

Q. If a line intersects the x-axis at (6,0) and the y-axis at (0,−4), what is the slope of the line?

यदि एक रेखा x-अक्ष को (6,0) पर और y-अक्ष को (0,−4) पर काटती है, तो रेखा की प्रवणता क्या है?

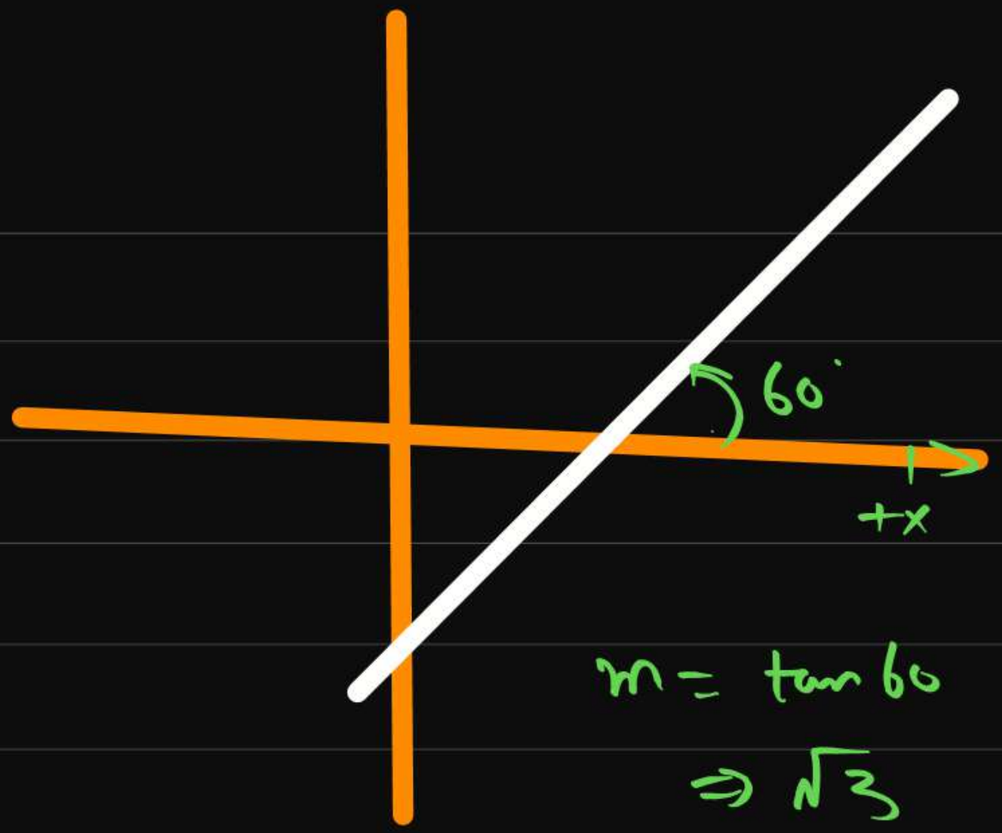
SSC CPO 2025

(a) 1/2 (b) 1/3 (c) 3/2 (d) 2/3

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$\Rightarrow \frac{-4 - (0)}{0 - (6)}$$

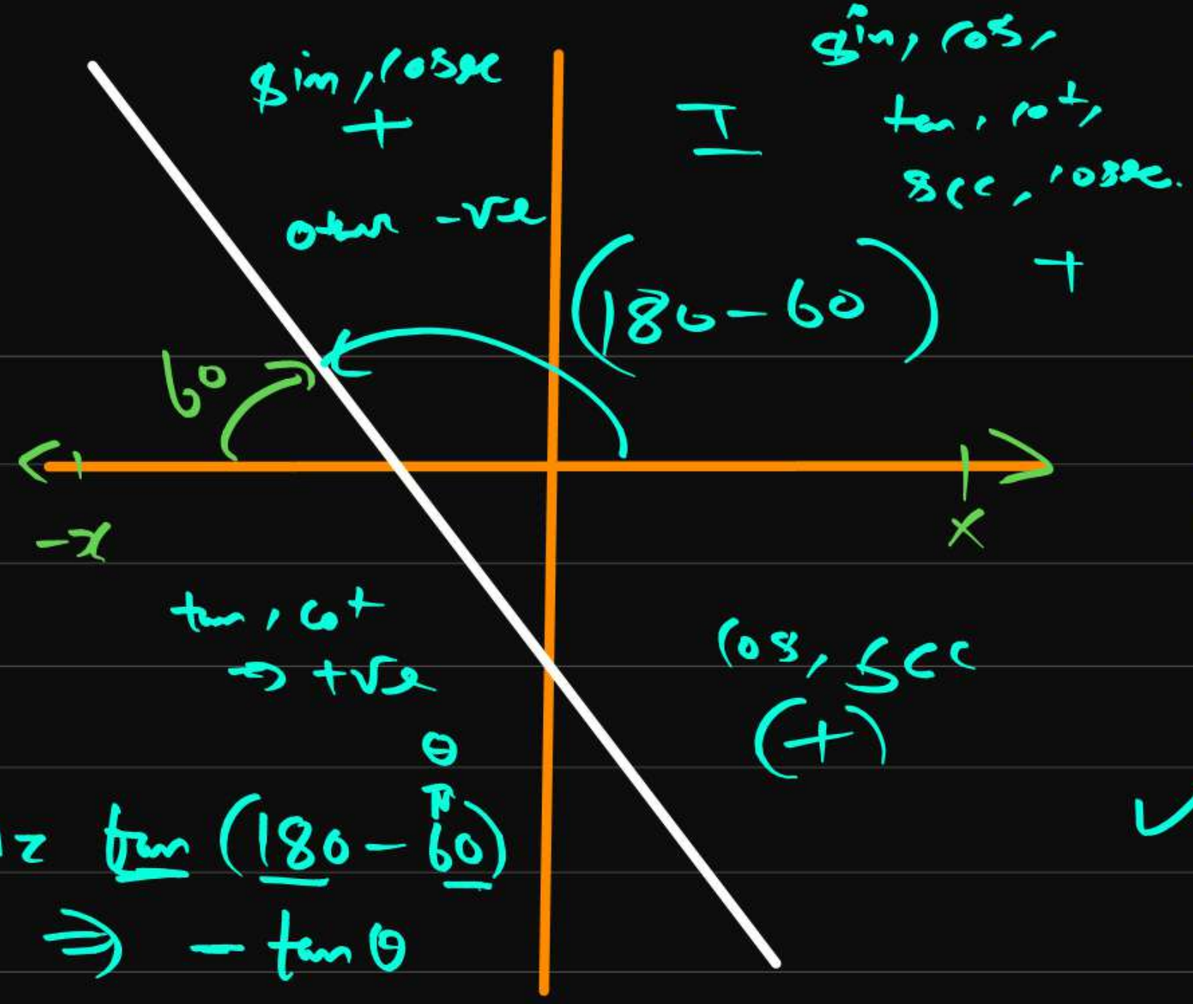
$$\Rightarrow \frac{-4}{-6} = \frac{2}{3}$$



Q. Find the slope of the line making inclination of 60° with the positive direction of *x*-axis.

धनात्मक x-अक्ष के साथ 60° का झुकाव बनाने वाली रेखा की ढाल ज्ञात कीजिए।

(a) 1/√3
(b) √3
(c) 1
(d) √2



$\text{Ans} \quad \tan(180-60)$

$\Rightarrow -\tan\theta$

$\Rightarrow -\tan 60$

$\Rightarrow -\sqrt{3}$

Q. Find the gradient of the line making inclination of 60° with the negative direction of *x*-axis.

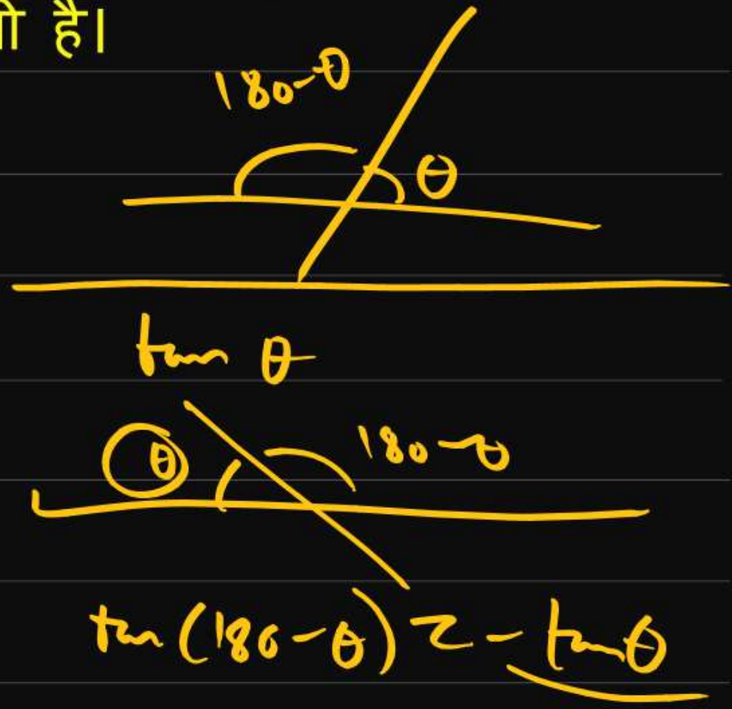
उस रेखा की ढलान (gradient) ज्ञात कीजिए जो x-अक्ष की ऋणात्मक दिशा के साथ 60° का झुकाव बनाती है।

(a) $-1/\sqrt{3}$

✓ (b) $-\sqrt{3}$

(c) $\sqrt{3}$

(d) $1/\sqrt{3}$



Q. Find angle of inclination of a line with slope 1.

1 ढलान वाली रेखा का झुकाव कोण ज्ञात कीजिए।

$m = \tan\theta = 1 = \tan 45^\circ$

$\theta = 45^\circ$

(a) 1 — 45° ✓

(b) $-\sqrt{3}$ — 60°

(c) $\sqrt{3}$ — 30°

(d) 1 — 80°

Q. What is the equation of line through (0,3) with slope -1?

ढाल -1 वाली तथा (0,3) से गुजरने वाली रेखा का समीकरण क्या है?

SSC CGL 2025

(a) $y = -x + 3$ (b) $y = x - 3$
(c) $y = -x - 3$ (d) $y = x + 3$

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$-1 = \frac{y - 3}{x - 0}$$

$$-x = y - 3$$

$$y = -x + 3$$

$$x + y = 3$$

Q. What is the equation of line through (0,3) with slope -1?

ढाल -1 वाली तथा (0,3) से गुजरने वाली रेखा का समीकरण क्या है?

SSC CGL 2025

(a) $y = -x + 3$ (b) $y = x - 3$

(c) $y = -x - 3$ (d) $y = x + 3$

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$-1 = \frac{y - 3}{x - 0}$$

$$-x = y - 3$$

$$x + y = 3$$

$$y = -x + 3$$

Q. If the graph of a linear equation passes through (0, −2) and has a slope −3, what is its equation?

यदि एक रैखिक समीकरण का ग्राफ (0, −2) से होकर गुजरता है और उसकी ढलान (slope) −3 है, तो उसका समीकरण क्या है?

SSC CHSL 2025

(a) $y = 3x - 2$ (b) $y = -3x - 2$
(c) $y = -3x + 2$ (d) $y = 3x + 2$

$$-3 = \frac{y+2}{x}$$

$$-2 - 3x = y$$

$$y = mx + c$$

m = slope

$$y = -3x + c$$

$m = \tan\theta = \tan(180-45) = -1 = \frac{y-4}{x-(-3)}$ (−tan 45°)

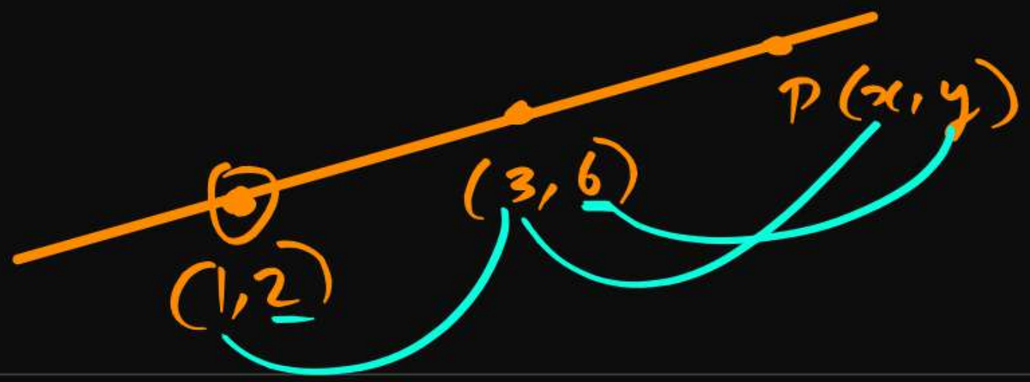
$-x+(-3) = y-4$

$x+y=1$

Q).Find the equation of the line when a line is making 45° angle with negative 'x' axis and passing through (-3,4).

उस रेखा का समीकरण ज्ञात करें जब वह रेखा ऋणात्मक 'x' अक्ष के साथ 45° का कोण बना रही हो और (-3,4) से गुजर रही हो।

(a) x-y=1
(b) x+y=1
(c) -3x+4y=0
(d) 4x-3y=0



Q. A line passes through the points (1, 2) and (3, 6). What is the equation of the line?

एक रेखा बिंदुओं (1, 2) और (3, 6) से होकर गुजरती है। रेखा का समीकरण क्या है?

SSC CHSL 2025

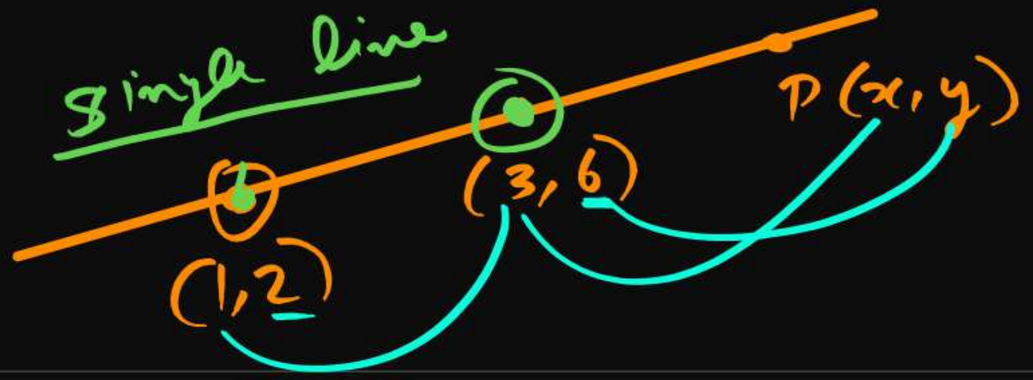
(a) y = 2x (b) y = x/3
(c) y = x/2 (d) y = 5x

$$m = \frac{6-2}{3-1} = \frac{y-6}{x-3}$$

$$2 = \frac{y-6}{x-3}$$

$$2x - y = 0$$

$$2x = y$$



$$m = \frac{6-2}{3-1} = \frac{y-6}{x-3}$$

$$2 = \frac{y-6}{x-3}$$

$$2x - y = 0$$

$$2x = y$$

Q. A line passes through the points (1, 2) and (3, 6). What is the equation of the line?

एक रेखा बिंदुओं (1, 2) और (3, 6) से होकर गुजरती है। रेखा का समीकरण क्या है?

Only 1 Eqⁿ can

SSC CHSL 2025

(a) y = 2x (b) y = x/3

(c) y = x/2 (d) y = 5x

Q. Find the equation of the line passing through (0, 0) and (5, 10).

(0, 0) और (5, 10) से होकर गुजरने वाली रेखा का समीकरण ज्ञात कीजिए।

SSC CPO 2025

(a) $y = 5x$ (b) $y = 3x$ (c) $y = 4x$ (d) $y = 2x$

$$\begin{matrix} 3x+y=6 \\ x-y+2=0 \end{matrix} + \quad x-2y=0$$

$$x+y=6$$

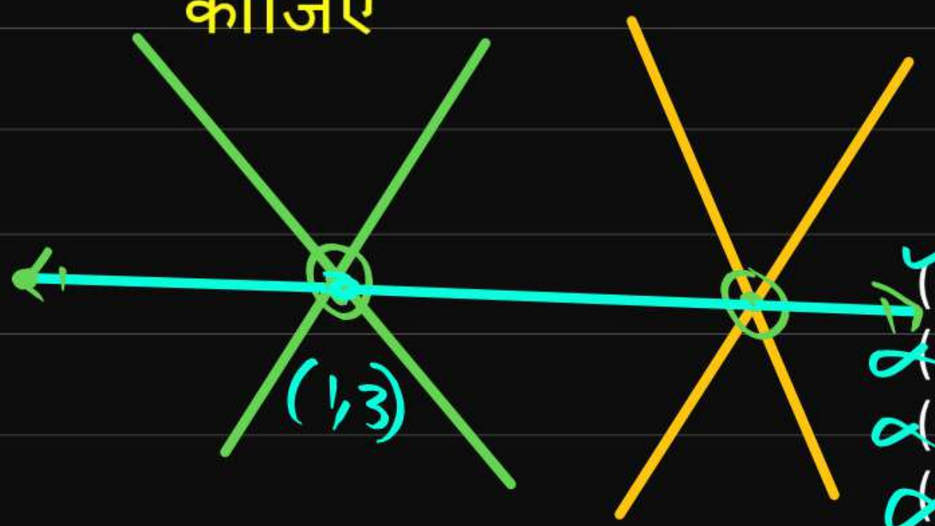
$4x = 4$

$x = 1$

$y = 3$

Q) . Find the equation of the line joining the point of intersection of $3x+y=6$ with $x-y+2=0$ and $x-2y=0$ with $x+y=6$.

$3x+y=6$ के साथ $x-y+2=0$ तथा $x-2y=0$ के साथ $x+y=6$ के प्रतिच्छेदन बिंदु से जुड़ने वाली रेखा का समीकरण ज्ञात कीजिए



(a) $x+3y-10=0$

(b) $2x-y+5=0$

(c) $3x-y+5=0$

(d) $x-3y+10=0$

Q) . Find the equation of the line joining the point of intersection of $3x+y=6$ with $x-y+2=0$ and $x-2y=0$ with $x+y=6$.

$3x+y=6$ के साथ $x-y+2=0$ तथा $x-2y=0$ के साथ $x+y=6$ के प्रतिच्छेदन बिंदु से जुड़ने वाली रेखा का समीकरण ज्ञात कीजिए

$3x+y=6$
$x-y+2=0$
$+$

$4x = 4$

$x = 1$

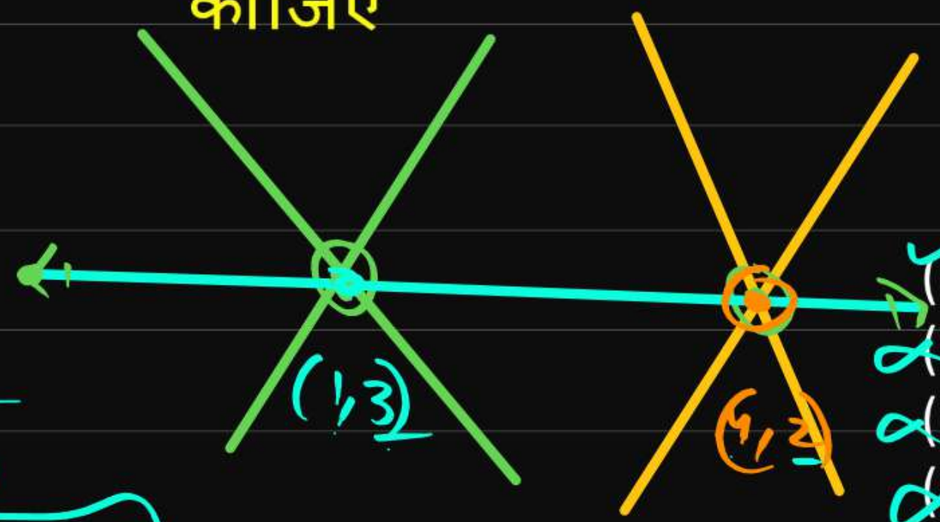
$y = 3$

$x-2y=0$
$x+y=6$

$3y = 6$

$y = 2$

$x = 4$



$\frac{y-3}{x-1} = \frac{1}{-3}$

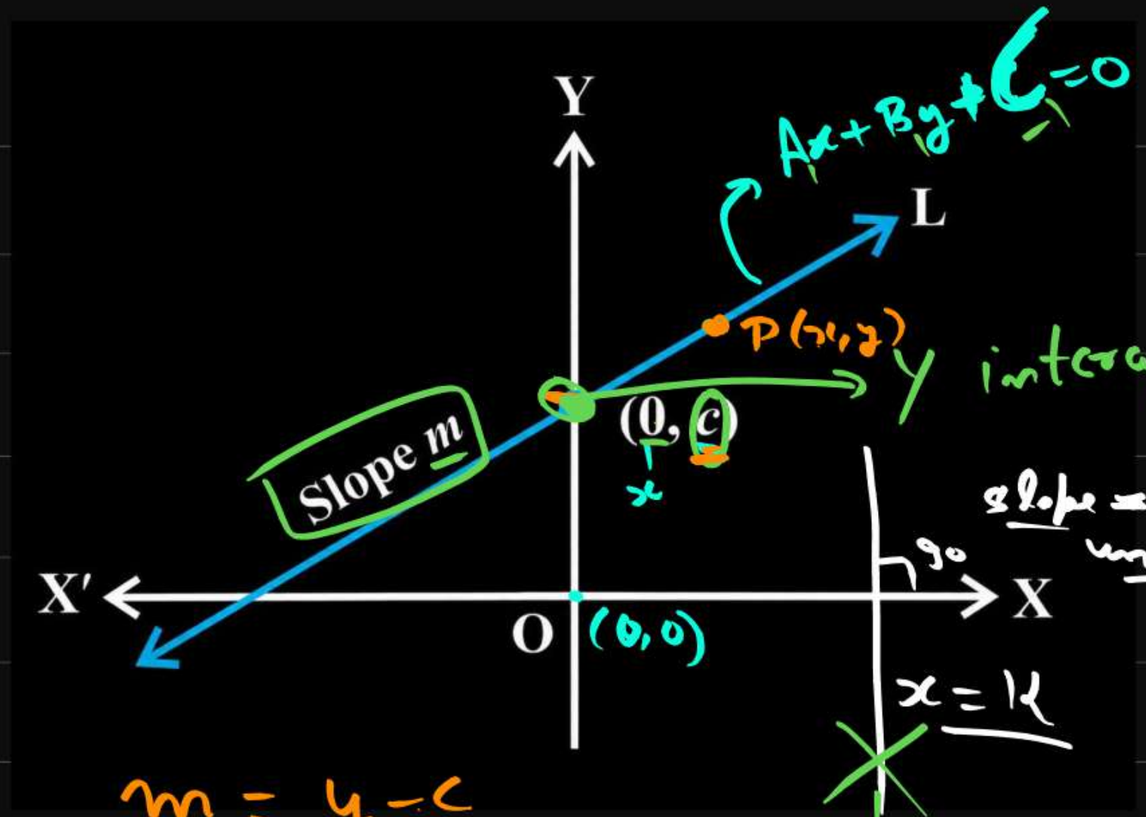
$\Rightarrow x+3y-10=0$

(a) $x+3y-10=0$
(b) $2x-y+5=0$
(c) $3x-y+5=0$
(d) $x-3y+10=0$

Y - intercept

Slope-intercept form
ढाल अंतःखंड रूप

Sometimes a line is known to us with its slope and an intercept on one of the axes. We will now find equations of such lines.
कभी-कभी एक रेखा हमें उसके ढलान और किसी एक अक्ष पर एक अवरोधन के साथ ज्ञात होती है। अब हम ऐसी रेखाओं के समीकरण ज्ञात करेंगे।



$Ax + By + C = 0$

slope = undefined

$m = \frac{y - c}{x - 0}$

$y = mx + c$

$y = 8x + 19$

Slope = ?

$y = mx + c$

$m = 8$

Y - intercept

Slope-intercept form
ढाल अंतःखंड रूप

Sometimes a line is known to us with its slope and an intercept on one of the axes. We will now find equations of such lines.
कभी-कभी एक रेखा हमें उसके ढलान और किसी एक अक्ष पर एक अवरोधन के साथ ज्ञात होती है। अब हम ऐसी रेखाओं के समीकरण ज्ञात करेंगे।

$A_1x + B_1y + C_1 = 0$

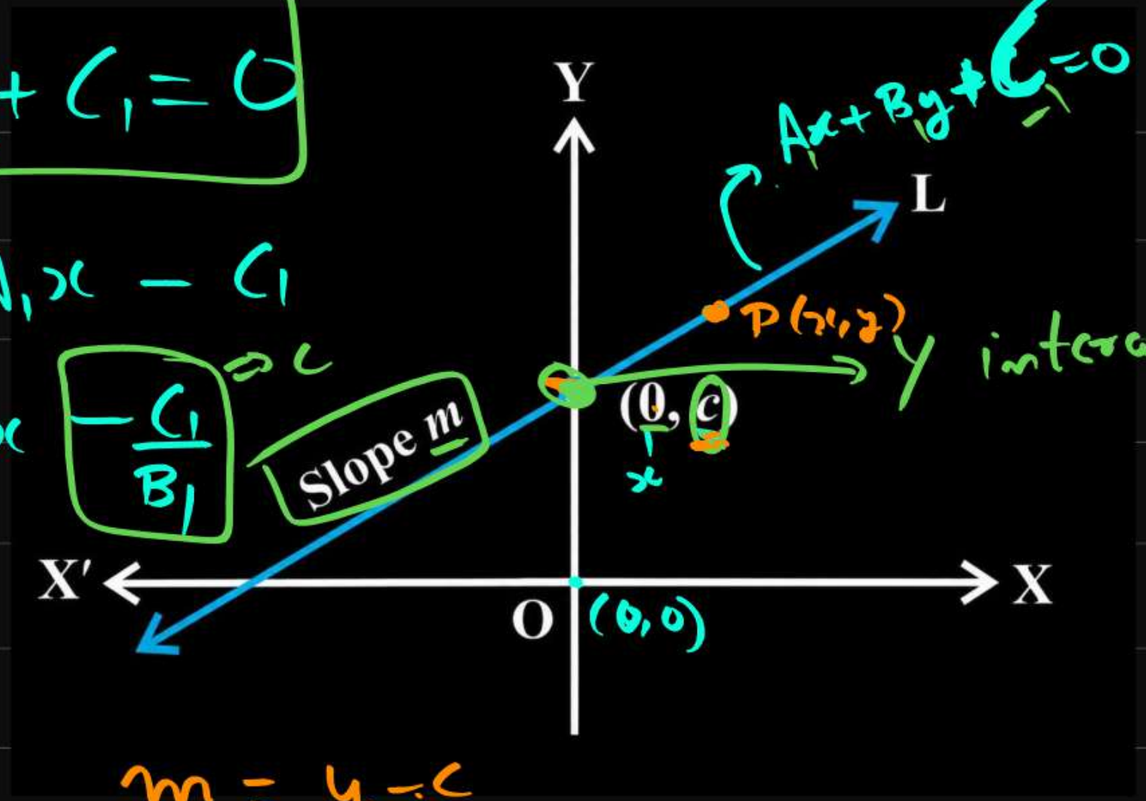
$B_1y = -A_1x - C_1$

$y = \left(\frac{-A_1}{B_1}\right)x \left(\frac{-C_1}{B_1}\right)$

$= c$

Slope m

m



$m = \frac{y-c}{x-0}$

$y = mx + c$

$2x + 3y + 8 = 0$

$3y = -2x - 8$

$y = \left(\frac{-2}{3}\right)x \left(\frac{-8}{3}\right)$

$m = \frac{-2}{3}$

$c = \frac{-8}{3}$

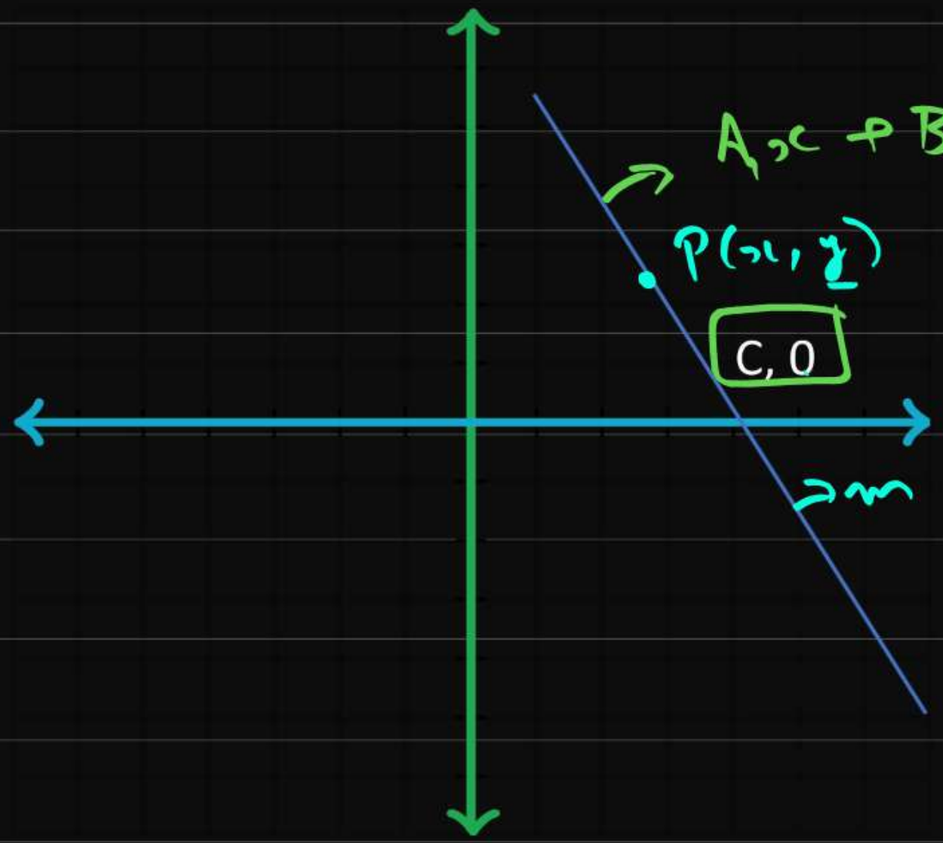
Slope of line if equation of a line is given

अगर किसी रेखा का समीकरण दिया गया हो, तो उसकी ढलान

$$Ax + By + C = 0$$

$$m = -\frac{A}{B}$$

X- intercept
(rarely asked)



$$m = \frac{y-0}{x-c}$$

$$x - c = \frac{y}{m}$$

$$\boxed{x = \frac{y}{m} + c}$$

Q. Write the equations of the lines for which tan θ = 1/2, where θ is the inclination of the line and:

(i) y-intercept is –3/2

(ii) x-intercept is 4.

प्रश्न। उन रेखाओं के समीकरण ज्ञात कीजिए जिनके लिए tan θ = 1/2 है, जहाँ θ रेखा का झुकाव कोण है तथा:

(i) y-अवरोधक = –3/2 है।

(ii) x-अवरोधक = 4 है।

(a) x – 2y – 3 = 0 ; x – 2y – 4 = 0

(b) x – 2y + 3 = 0 ; x – 2y + 4 = 0

(c) 2x – y – 3 = 0 ; 2x – y – 4 = 0

(d) x + 2y – 3 = 0 ; x + 2y – 4 = 0

1) $y = mx + c$

$$y = \frac{1}{2}x - \frac{3}{2}$$

$$2y = x - 3$$

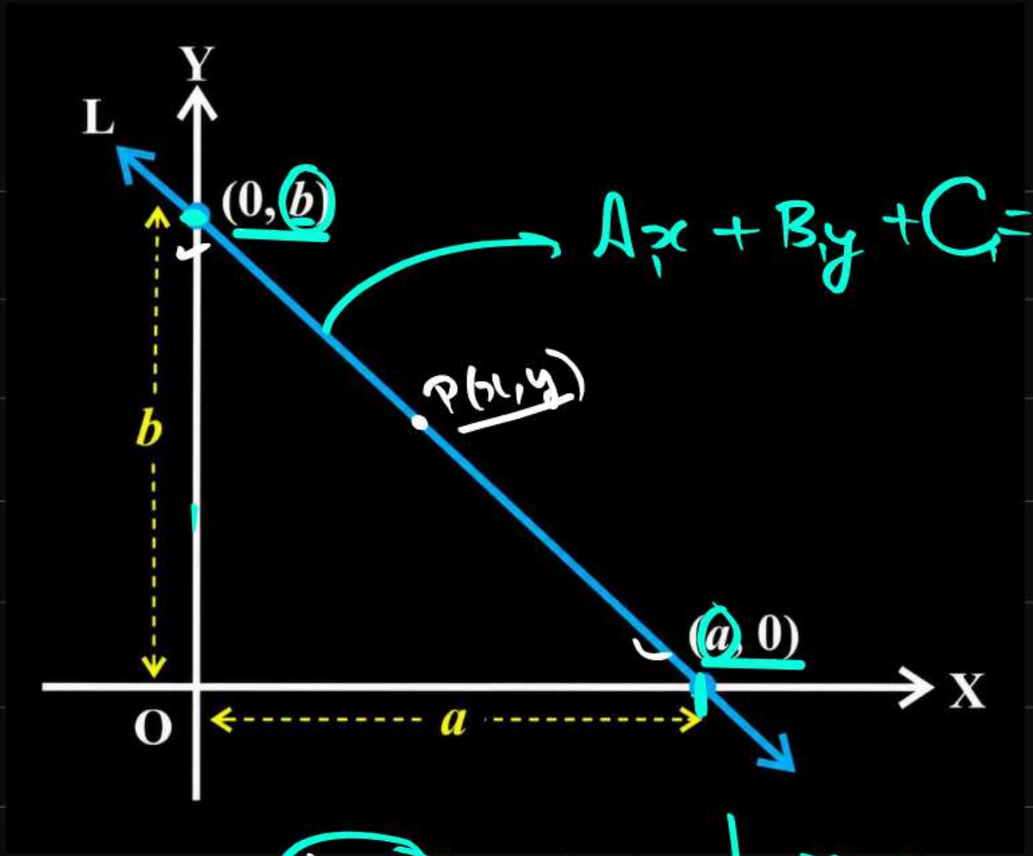
$$0 = x - 2y - 3$$

2) $x = \frac{y}{m} + c$

$$x = \frac{y}{1/2} + 4$$

$$x - 2y - 4 = 0$$

Intercept – form
अंतःखंड रूप



Suppose a line L makes x-intercept a and y-intercept b on the axes. Obviously L meets x-axis at the point (a, 0) and y-axis at the point (0, b)

मान लीजिए कि एक रेखा L अक्षों पर x-प्रतिच्छेद a और y-प्रतिच्छेद b बनाती है। स्पष्ट रूप से, L, x-अक्ष को बिंदु (a, 0) पर और y-अक्ष को बिंदु (0, b) पर मिलता है।

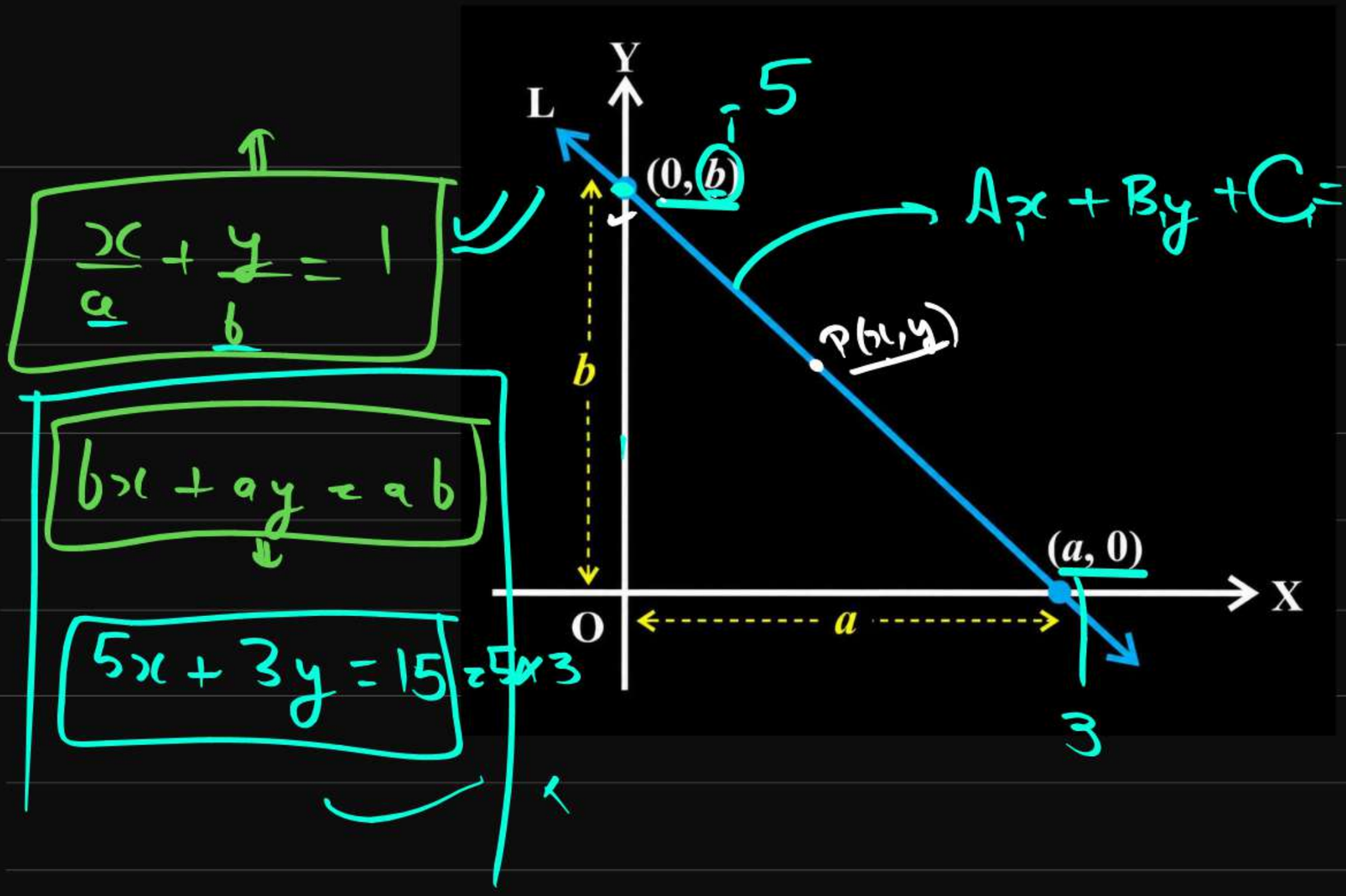
$$m = \frac{b-0}{0-a} = \frac{y-0}{x-a}$$

$$\frac{x-a}{-a} = \frac{y}{b}$$

$$-\frac{x}{a} + 1 = \frac{y}{b}$$

$$\boxed{\frac{x}{a} + \frac{y}{b} = 1}$$

Intercept – form
अंतःखंड रूप



Suppose a line L makes x-intercept a and y-intercept b on the axes. Obviously L meets x-axis at the point (a, 0) and y-axis at the point (0, b)

मान लीजिए कि एक रेखा L अक्षों पर x-प्रतिच्छेद a और y-प्रतिच्छेद b बनाती है। स्पष्ट रूप से, L, x-अक्ष को बिंदु (a, 0) पर और y-अक्ष को बिंदु (0, b) पर मिलता है।

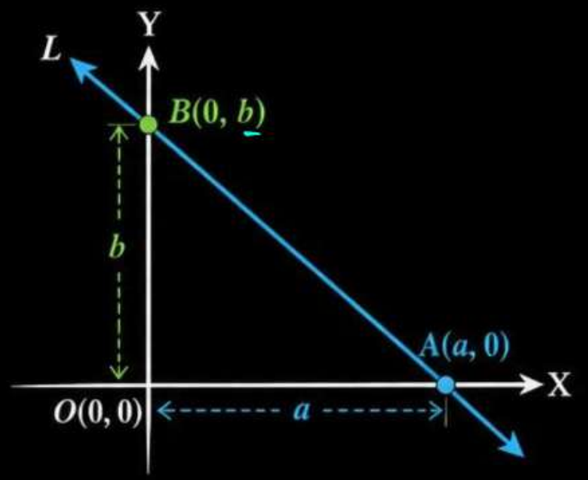
INTERCEPT FORM OF A LINE

रेखा का अंतःखंड रूप

1 CONCEPT / अवधारणा

A line L cuts the axes at:

- x-intercept = a at point $A(a, 0)$
- y-intercept = b at point $B(0, b)$



SLOPE OF LINE / ढाल

Slope $m = \dfrac{b-0}{0-a} = -\dfrac{b}{a}$

Thus, $m = -\dfrac{b}{a}$

2 DERIVATION USING TWO-POINT FORM
दो-बिंदु सूत्र द्वारा व्युत्पत्ति

The line passes through $A(a, 0)$ and $B(0, b)$.

Two-point form of a line:

$$\frac{y-y_1}{y_2-y_1} = \frac{x-x_1}{x_2-x_1}$$

$$\Rightarrow \frac{y-0}{b-0} = \frac{x-a}{0-a}$$

$$\Rightarrow \frac{y}{b} = \frac{x-a}{-a}$$

$$\Rightarrow \frac{y}{b} = -\frac{x}{a} + 1$$

दोनों ओर 1 जोड़ने पर

$$\frac{x}{a} + \frac{y}{b} = 1$$

इसलिए, रेखा का **अंतःखंड रूप**

$$\frac{x}{a} + \frac{y}{b} = 1$$

3 DERIVATION USING SLOPE INTERCEPT FORM
ढाल-अवरोधक रूप द्वारा व्युत्पत्ति

From the figure, slope $m = -\dfrac{b}{a}$

Using slope–intercept form $y = mx + c$

Here, $c = b$ (y-intercept)

$$y = -\frac{b}{a}x + b$$

$$\Rightarrow \frac{y}{b} = -\frac{x}{a} + 1$$

$$\Rightarrow \frac{x}{a} + \frac{y}{b} = 1$$

4 KEY POINTS / मुख्य बिंदु

- ★ a = x-intercept (where line meets x-axis)
- ★ b = y-intercept (where line meets y-axis)
- ★ Intercept form: $\dfrac{x}{a} + \dfrac{y}{b} = 1$
- ★ Slope of line: $m = -\dfrac{b}{a}$

Special Cases / विशेष स्थितियाँ

1. अगर रेखा मूल बिंदु (0,0) से गुजरती है, तो $a = \infty$ या $b = \infty$
2. अगर रेखा x-अक्ष के समानांतर है, तो $b = \infty$
3. अगर रेखा y-अक्ष के समानांतर है, तो $a = \infty$

Q. Find the x-intercept of $3x - 6y = 12$.

$3x - 6y = 12$ का x-अंतःखंड ज्ञात करें।

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(a) 3 (b) 4 (c) 5 (d) –2

$$\frac{3x}{12} - \frac{6y}{12} = 1$$

$$\frac{x}{4} + \frac{y}{(-2)} = 1$$

$a = 4$ → x-intercept

$b = -2$ → y intercept

Q. What is the slope in the equation $y = 4x - 2$?

समीकरण $y = 4x - 2$ में ढलान क्या है ?

$y = mx + c$

SSC CPO 2025

(a) 0 (b) 1 (c) 4 (d) 2

Q. Find slope of line $7x + 4y = 8$.

रेखा $7x + 4y = 8$ का ढाल ज्ञात कीजिए।

SSC CGL 2025

(a) -7/4 (b) 7/4 (c) -4/7 (d) 4/7

$$y = mx + c$$

$$7x + 4y = 8$$

$$y = \frac{-7}{4}x + \frac{8}{4}$$

m

$$Ax + By + C = 0$$

$$7x + 4y - 8 = 0$$

$$m = -\frac{7}{4}$$

Q. If $y = mx + 1$ passes through $(2,5)$, find m.

यदि $y = mx + 1$, $(2,5)$ से होकर गुजरता है, तो m ज्ञात कीजिए।

SSC CGL 2025

(a) 2 (b) 1 (c) 0 (d) 3

$y = mx + 1$

$5 = 2m + 1$

$4 = 2m$

$\boxed{m = 2}$

Q. If line passing through (3, −1) with slope 4; find y-intercept.

यदि (3, −1) से होकर गुजरने वाली रेखा की ढलान 4 है; तो y-अक्ष पर अंतःखंड ज्ञात कीजिए।

SSC CGL 2025

(a) −13 (b) 12 (c) 14 (d) −14

$x = \frac{y}{m} + c$

$3 = \frac{-1}{4} + c$

$3 + \frac{1}{4} = c$

$\frac{13}{4} = c$

x - intercept.

1

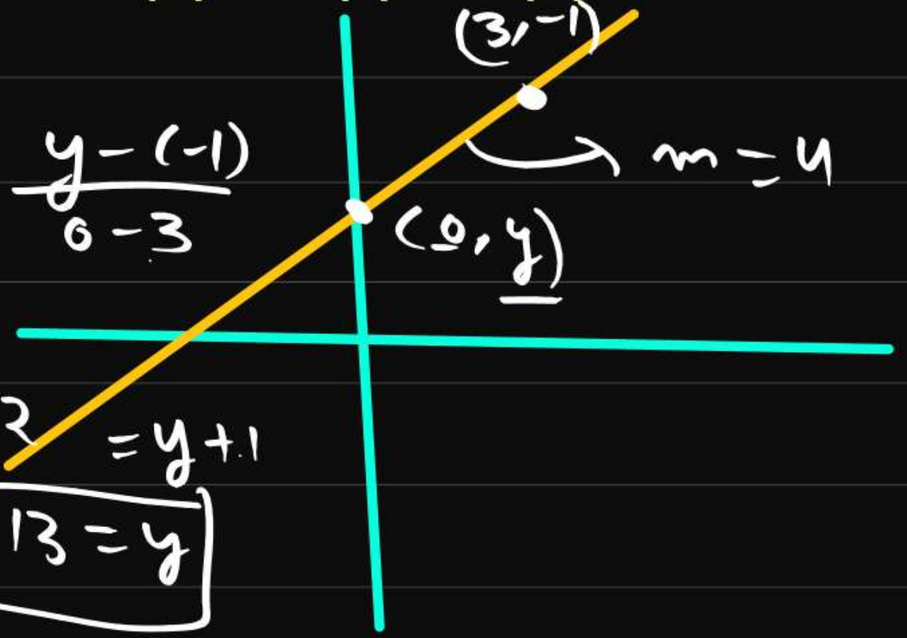
y intercept

$y = mx + c$

$-1 = 4(3) + c$

$-13 = c$

2



$4 = \frac{y-(-1)}{0-3}$

$-12 = y + 1$

$-13 = y$

Q. If $y = mx + c$ passes through (2, 5) and (4, 9), find m.

यदि $y = mx + c$ (2, 5) और (4, 9) से होकर गुजरता है, तो m का मान ज्ञात कीजिए।

$$m = \frac{9-5}{4-2} = \frac{4}{2} = 2$$

SSC CPO 2025

(a) 4 (b) 2 (c) 3 (d) 5

Q. If a line has x-intercept 4 and y-intercept -2, what is its equation in standard form?

यदि एक रेखा का x-अंतःखण्ड 4 और y-अंतःखण्ड -2 है, तो इसका समीकरण मानक रूप में क्या है?

SSC CPO 2025

(a) x−2y=4 (b) -x−2y=4 (c) -x+2y=4
(d) x+2y=4

1)

$bx + ay = ab$

$-2 \quad 4$

$-2x + 4y = -2 \times 4$

$x - 2y = 4$

2)

$\frac{x}{a} + \frac{y}{b} = 1 \; ; \; \frac{x}{4} + \frac{y}{-2} = 1$

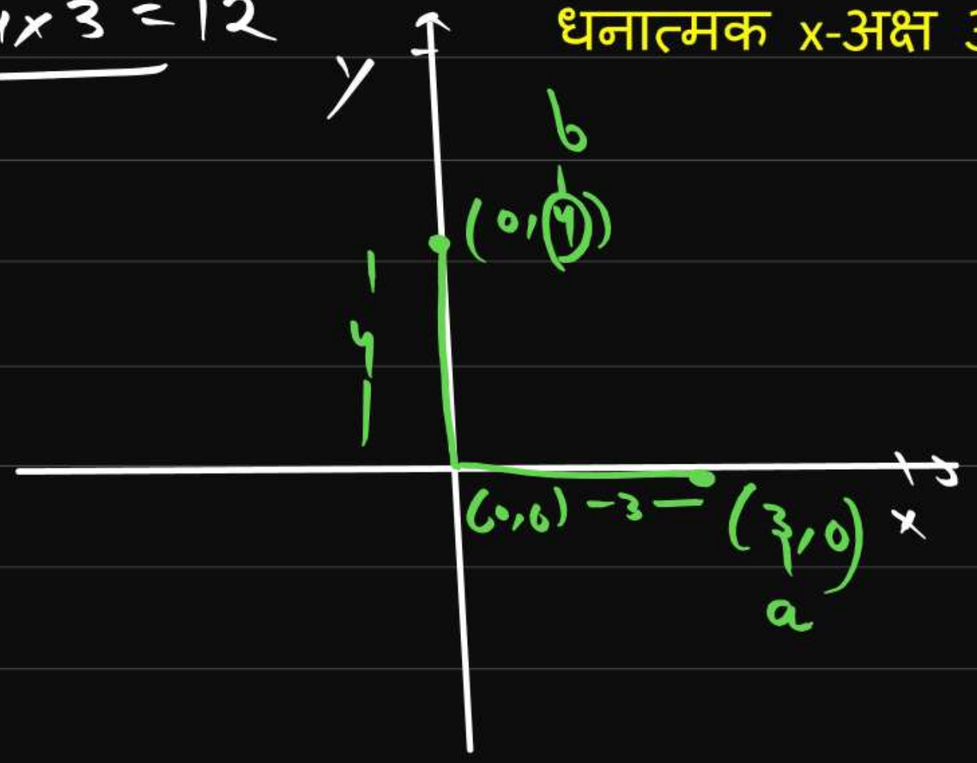
$x - 2y = 4$

Q) Find the equation of the line which intersects at 4 and 3 units length from origin form the positive x-axis and y-axis respectively.

उस रेखा का समीकरण ज्ञात कीजिए जो मूल बिन्दु से 4 और 3 इकाई लंबाई पर प्रतिच्छेद करती है, क्रमशः धनात्मक x-अक्ष और y-अक्ष बनाती है।

$bx + ay = ab$

$4x + 3y = 4 \times 3 = 12$



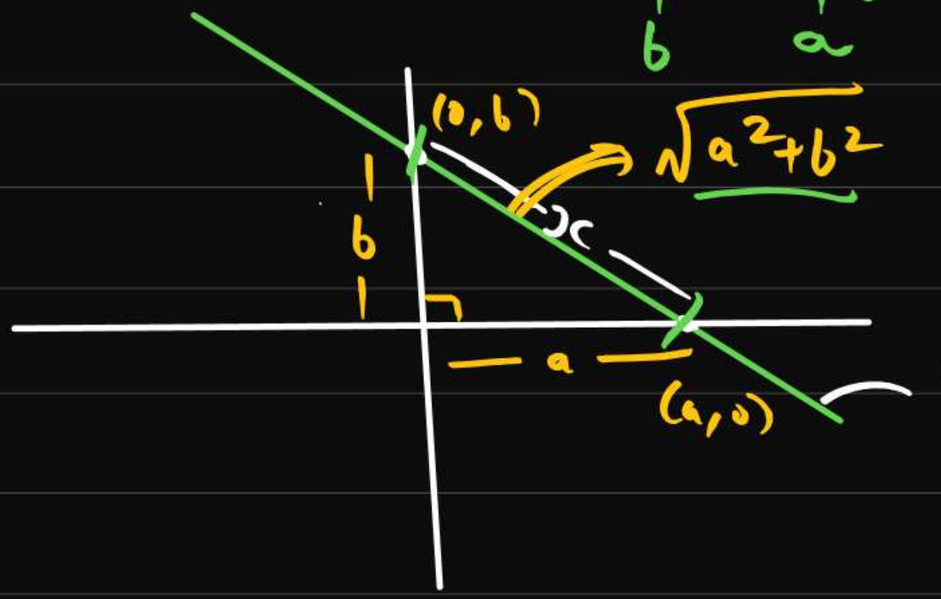
(a) 4x+3y=12
(b) 4x-3y=12
(c) 3x+4y=12
(d) 3x-4y=12

Q. The shortest length of the intercept between the x-axis and y-axis of the equation $6x + 8y - 48 = 0$ is:

समीकरण $6x + 8y - 48 = 0$ के x-अक्ष और y-अक्ष के बीच अवरोधन की सबसे छोटी लंबाई कितनी है?

$6x + 8y = 48$

b a $a \times b$



$$\frac{6x}{48} + \frac{8y}{48} = 1$$

$$\frac{x}{8} + \frac{y}{6} = 1$$

a b

$l = \sqrt{a^2+b^2} = \sqrt{8^2+6^2} = \sqrt{100} = 10$

SSC CHSL 2024 T-1

(a) 6 इकाई
(b) 12 इकाई
(c) 10 इकाई
(d) 14 इकाई

$6x + 10y + ab$

$5 \quad 12 \quad 5 \times 12$

Q. The length of the portion of the straight line $5x + 12y = 60$ intercepted between the axes is:

सरल रेखा $5x + 12y = 60$ के अक्षों के बीच अंतःखंडित (intercepted) भाग की लंबाई ज्ञात कीजिए।

$$\frac{5x}{60} + \frac{12y}{60} = 1$$

$$\frac{x}{12} + \frac{y}{5} = 60$$

a b

$\sqrt{a^2+b^2} \Rightarrow 13$

$\Rightarrow \sqrt{12^2+5^2}$

(a) 13
(b) 5
(c) 12
(d) 17

SSC CHSL 2024 T-1

Q. If the graph of the equation 20x + 21y = 420 cuts the coordinate axes at P and Q, then what is the measure of the length (in units) of PQ?

यदि समीकरण 20x + 21y = 420 का ग्राफ निर्देशांक अक्षों को P और Q पर काटता है, तो PQ की लंबाई की माप (इकाइयों में) क्या है?

$20 \times 21 = 420$

$\downarrow \quad \downarrow$

$b \quad a$

$\sqrt{20^2 + 21^2}$

$\Rightarrow 29$

SSC CHSL 2024 T-1

(a) 35
(b) 25
(c) 40
(d) 29

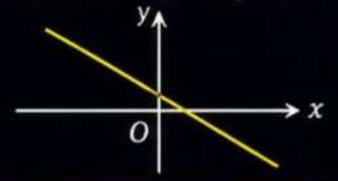
9 FORMS OF EQUATION OF A LINE

IN CO-ORDINATE GEOMETRY

1 GENERAL FORM

$$Ax + By + C = 0$$

Where A, B, C are real numbers with A and B not both zero. Represents any straight line.

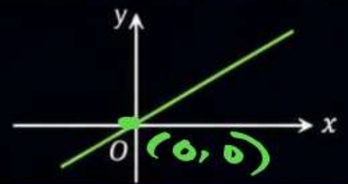


Most general form of a line.

2 LINE THROUGH ORIGIN

$$y = mx$$

Where m is the slope. (Line passes through origin $(0, 0)$).

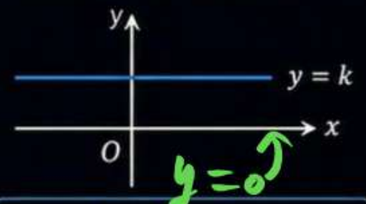


Special case of slope–intercept form with $c = 0$.

3 HORIZONTAL LINE

$$y = k$$

Where k is a constant. (Line parallel to x–axis).



Slope $m = 0$.

4 VERTICAL LINE

$$x = k$$

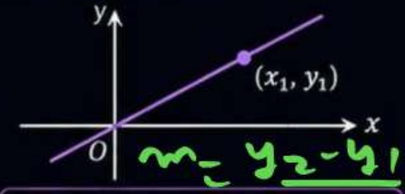
Where k is a constant. (Line parallel to y–axis).

Slope is undefined.

5 POINT–SLOPE FORM

$$y - y_1 = m(x - x_1)$$

Where (x_1, y_1) is a point on the line and m is the slope.

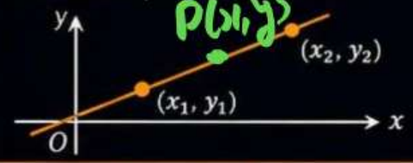


Useful when one point on the line and slope are known.

6 TWO–POINT FORM

$$\frac{y - y_1}{y_2 - y_1} = \frac{x - x_1}{x_2 - x_1}$$

Where (x_1, y_1) and (x_2, y_2) are two distinct points on the line.

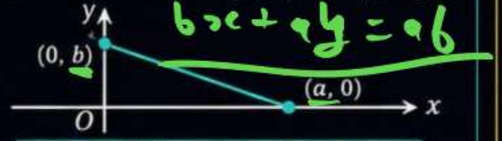


Useful when two points on the line are known.

7 INTERCEPT FORM

$$\frac{x}{a} + \frac{y}{b} = 1$$

Where a and b are x–intercept and y–intercept respectively. (Line cuts axes at $(a, 0)$ and $(0, b)$).



Useful when intercepts on axes are known.

8 SLOPE–INTERCEPT FORM

$$y = mx + c$$

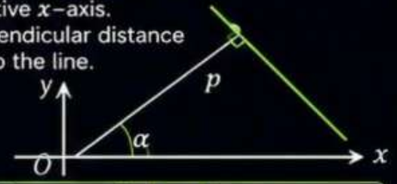
Where m is slope and c is the y–intercept (line cuts y–axis at $(0, c)$).

Most useful form when slope and y–intercept are known.

9 NORMAL FORM

$$x \cos\alpha + y \sin\alpha = p$$

Where α is the angle between the perpendicular from origin to the line and the positive x–axis. p is the perpendicular distance from origin to the line.



Useful in geometrical problems involving perpendicular distance.

IMPORTANT FACTS ABOUT LINES

SLOPE (m)

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

(for two points (x_1, y_1) and (x_2, y_2))

PARALLEL LINES

$$m_1 = m_2$$

Lines are parallel if their slopes are equal.

PERPENDICULAR LINES

$$m_1 m_2 = -1$$

Lines are perpendicular if product of slopes is -1.

ANGLE BETWEEN TWO LINES

$$\tan\theta = \left| \frac{m_2 - m_1}{1 + m_1 m_2} \right|$$

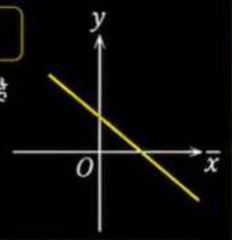
Where θ is the (acute) angle between the two lines.

रेखा के 9 विभिन्न रूप (Forms) | निर्देशांक ज्यामिति (Co-ordinate Geometry)

1 सामान्य रूप (General Form)

$$Ax + By + C = 0$$

जहाँ A, B, C वास्तविक संख्याएँ है और A तथा B दोनों शून्य नहीं हैं। यह किसी भी सीधी रेखा का सबसे सामान्य रूप है।

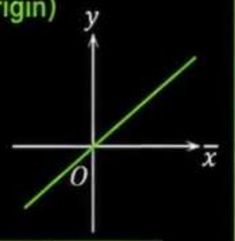


उदाहरण: $2x + 3y + 6 = 0$, $5x - 2y + 7 = 0$

2 मूल बिन्दु से गुजरने वाली रेखा (Line Through Origin)

$$y = mx$$

जहाँ m ढाल (slope) है। यह रेखा $(0, 0)$ बिन्दु से गुजरती है।

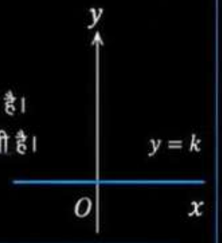


उदाहरण: $y = 2x$, $y = -\frac{1}{2}x$

3 क्षैतिज रेखा (Horizontal Line)

$$y = k$$

जहाँ k एक नियतांक (constant) है। यह रेखा x-अक्ष के समानांतर होती है।

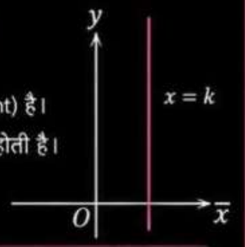


उदाहरण: $y = 4$, $y = -3$

4 लम्बवत रेखा (Vertical Line)

$$x = k$$

जहाँ k एक नियतांक (constant) है। यह रेखा y-अक्ष के समानांतर होती है।

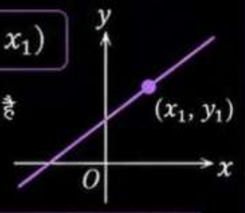


उदाहरण: $x = 3$, $x = -2$

5 बिन्दु-ढाल रूप (Point-Slope Form)

$$y - y_1 = m(x - x_1)$$

जहाँ (x_1, y_1) रेखा पर एक बिन्दु है और m ढाल (slope) है।

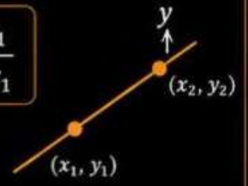


उदाहरण: $y - 2 = 3(x - 1)$

6 दो बिन्दु रूप (Two-Point Form)

$$\frac{y - y_1}{y_2 - y_1} = \frac{x - x_1}{x_2 - x_1}$$

जहाँ (x_1, y_1) और (x_2, y_2) रेखा पर दो भिन्न बिन्दु हैं।

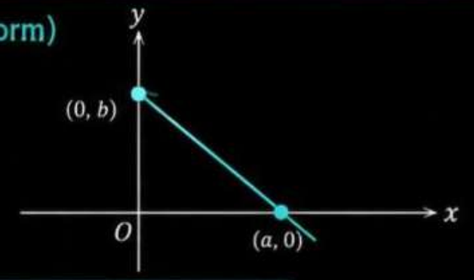


उदाहरण: $\frac{y-1}{4-1} = \frac{x-2}{5-2}$

7 अंतरछेद रूप (Intercept Form)

$$\frac{x}{a} + \frac{y}{b} = 1$$

जहाँ a और b क्रमशः x-अंतरछेद और y-अंतरछेद हैं। रेखा x-अक्ष को $(a, 0)$ और y-अक्ष को $(0, b)$ पर काटती है।

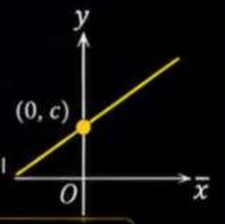


उदाहरण: $\frac{x}{5} + \frac{y}{3} = 1$ (x-अंतरछेद $= 5$, y-अंतरछेद $= 3$)

8 ढाल-अंतरछेद रूप (Slope-Intercept Form)

$$y = mx + c$$

जहाँ m ढाल (slope) है और c y-अंतरछेद (y-intercept) है। रेखा y-अक्ष को $(0, c)$ पर काटती है।

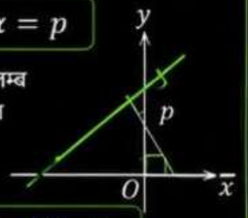


उदाहरण: $y = 2x + 3$, $y = -x + 1$

9 लंम्बवत दूरी (Normal Form)

$$x\cos\alpha + y\sin\alpha = p$$

जहाँ α वह कोण है जो रेखा का लम्ब मूल बिन्दु से बनाता है और p मूल बिन्दु से रेखा की लंबवत दूरी है।



उदाहरण: $x\cos 60° + y\sin 60° = 5$

★ नोट: इन रूपों का उपयोग प्रश्न के अनुसार सबसे सुविधाजनक रूप चुनकर करें। ★

महत्वपूर्ण तथ्य (Important Facts)

ढाल (Slope)

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

जहाँ (x_1, y_1) और (x_2, y_2) रेखा पर दो भिन्न बिन्दु हैं।

समान्तर रेखाएँ (Parallel Lines)

$$m_1 = m_2$$

दो रेखाएँ समान्तर हैं यदि उनकी ढालें बराबर हों।

लम्बवत रेखाएँ (Perpendicular Lines)

$$m_1 m_2 = -1$$

दो रेखाएँ लम्बवत हैं यदि ढालों का गुणनफल -1 हो।

दो रेखाओं के बीच कोण (θ) (Angle Between Two Lines)

$$\tan\theta = \left|\frac{m_2 - m_1}{1 + m_1 m_2}\right|$$

जहाँ θ दो रेखाओं के बीच तीव्र (acute) कोण है।

Co-ordinate Geometry

निर्देशांक ज्यामिति

(Basic to high Level)

Latest Pattern

Part-5

Parallel, perpendicular & intersecting Lines.

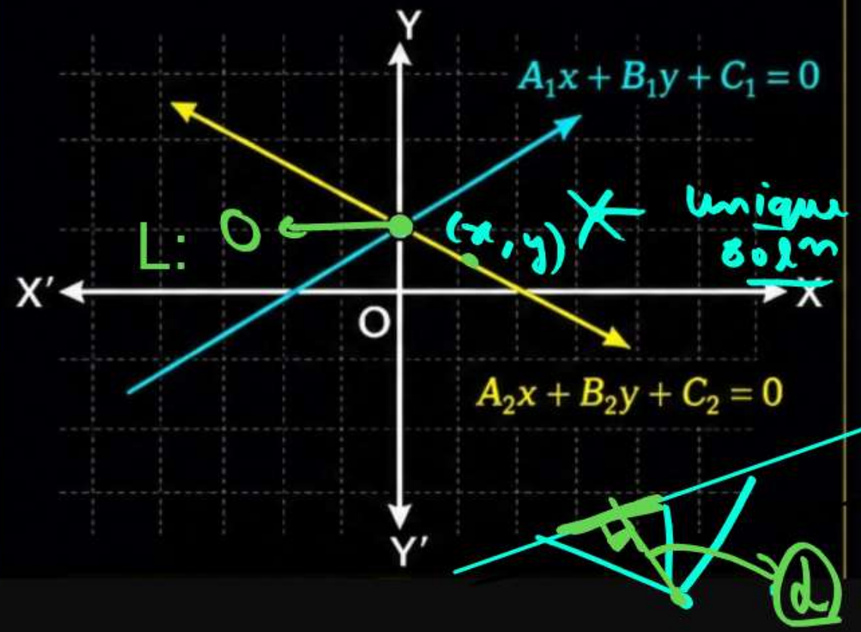
Distance between point and a line and parallel lines.

Intersecting, Parallel and Co-incident Lines

$A_1x + B_1y + C_1 = 0$ $A_2x + B_2y + C_2 = 0$

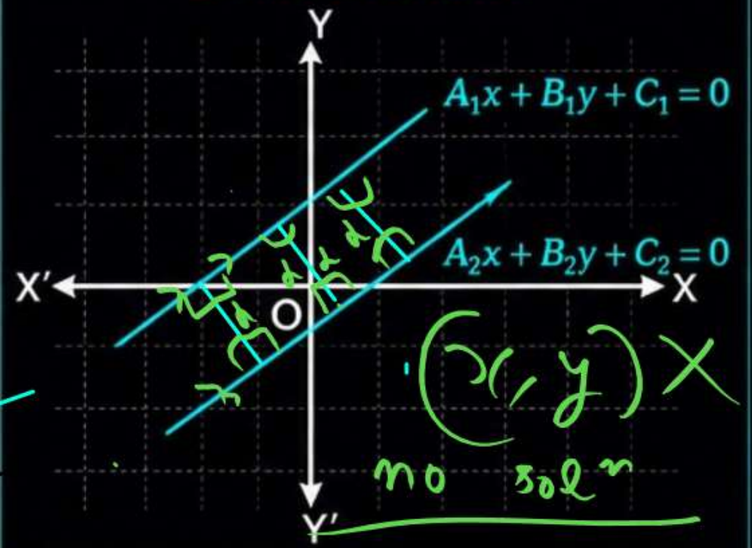
1. INTERSECTING LINES

Lines intersect at one point.



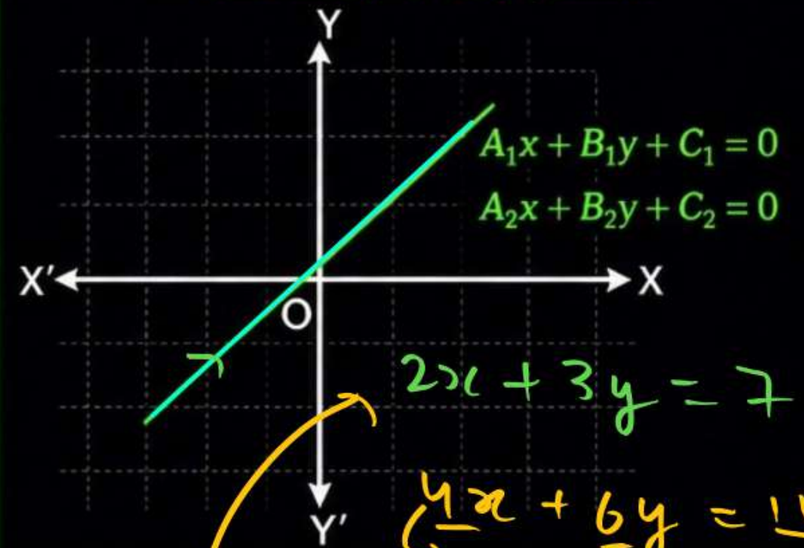
2. PARALLEL LINES

Lines are parallel and do not intersect.



3. CO-INCIDENT LINES

Lines coincide and have infinitely many points.



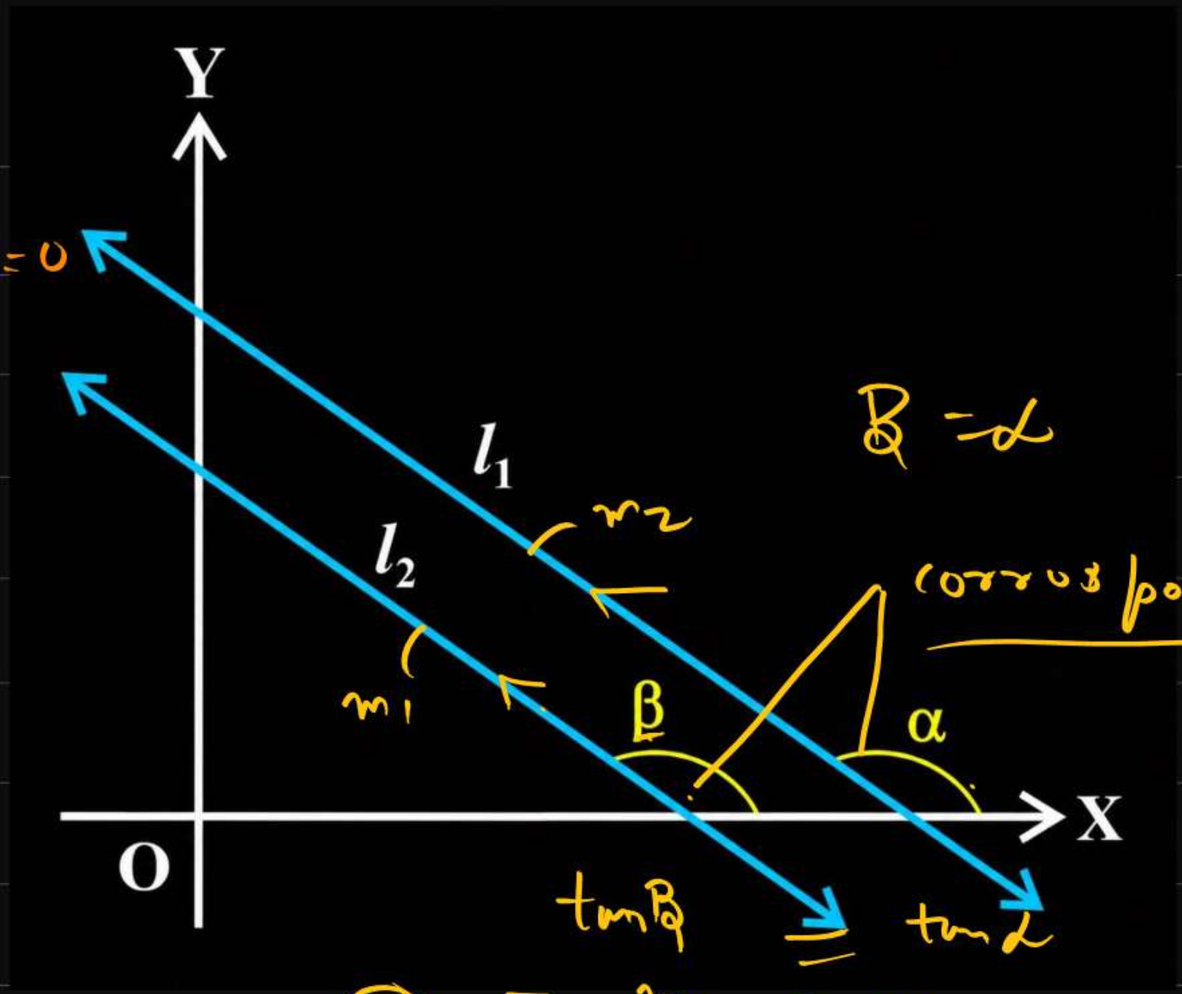
Parallel Lines समानांतर रेखाएं

$A_1x + B_1y + C_1 = 0 \quad A_2x + B_2y + C_2 = 0$

$$m = \frac{-A_1}{B_1} = \frac{-A_2}{B_2}$$

$$\frac{A_1}{B_1} = \frac{A_2}{B_2}$$

$$\frac{A_1}{A_2} = \frac{B_1}{B_2}$$

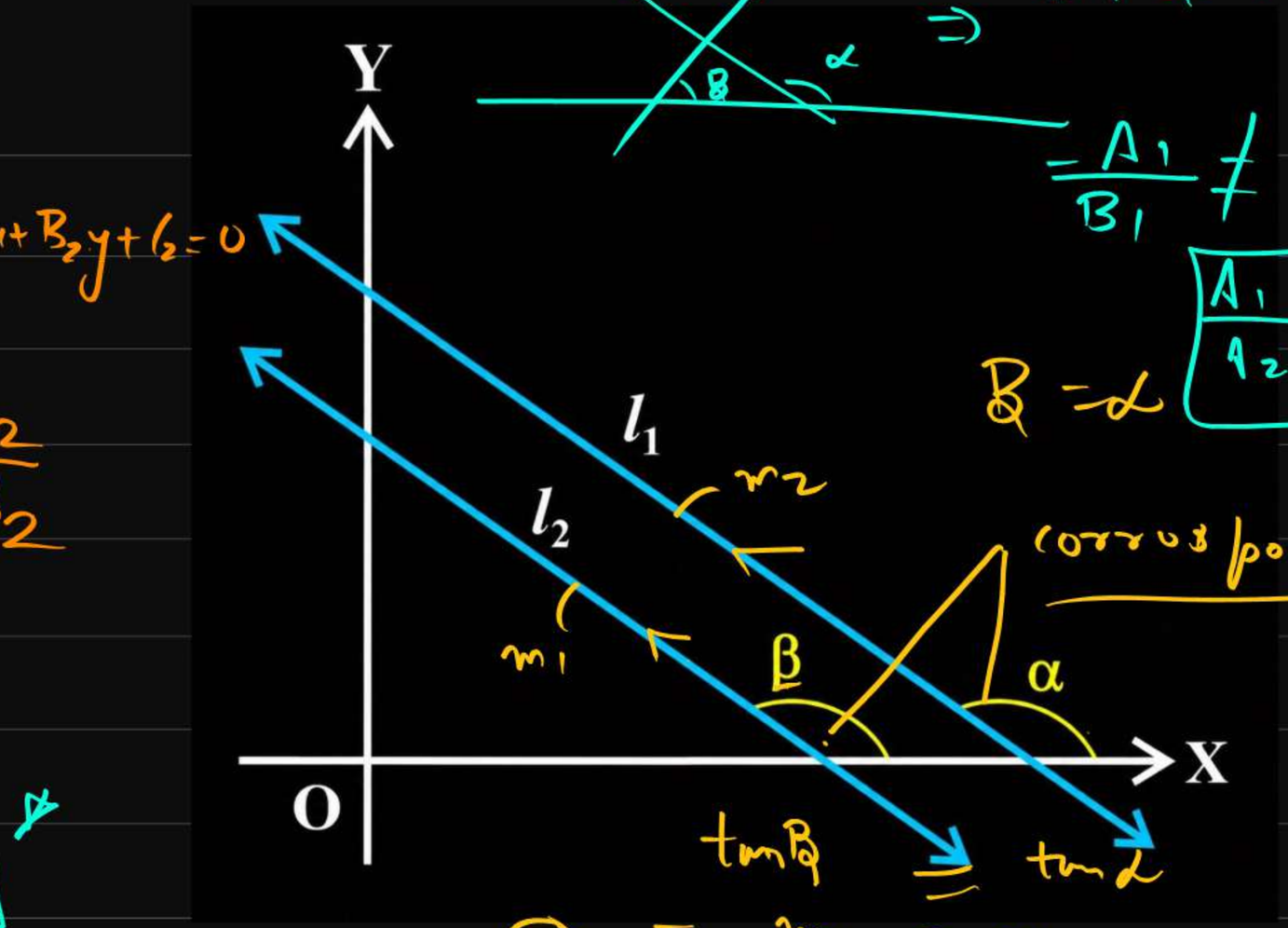


Parallel Lines समानांतर रेखाएं

Intersecting Lines

$m_1 \neq m_2$

$\Rightarrow$



$\theta = \alpha$

$\frac{-A_1}{B_1} \neq \frac{-A_2}{B_2}$

$\frac{A_1}{A_2} \neq \frac{B_1}{B_2}$

(1) $A_1x + B_1y + C_1 = 0$ $A_2x + B_2y + C_2 = 0$

$m = \frac{-A_1}{B_1} = \frac{-A_2}{B_2}$

$\frac{A_1}{B_1} = \frac{A_2}{B_2}$

$\frac{A_1}{A_2} = \frac{B_1}{B_2}$ ✗

Conditions for parallelism and perpendicularity of lines in terms of their slopes

2 रेखाओं के समानांतर और परस्पर लंब होने का प्रतिबंध

If the line L1 is parallel to L2 then their inclinations are equal, i.e.,
$\alpha = \beta$, and hence, $\tan\alpha = \tan\beta$ Therefore $m_1 = m_2$, i.e., their slopes are equal.

अगर रेखा L1, L2 के समानांतर है, तो उनके झुकाव बराबर होंगे, यानी $\alpha = \beta$, और इसलिए $\tan\alpha = \tan\beta$ होगा। इसलिए $m_1 = m_2$, यानी उनके ढलान बराबर होंगे।

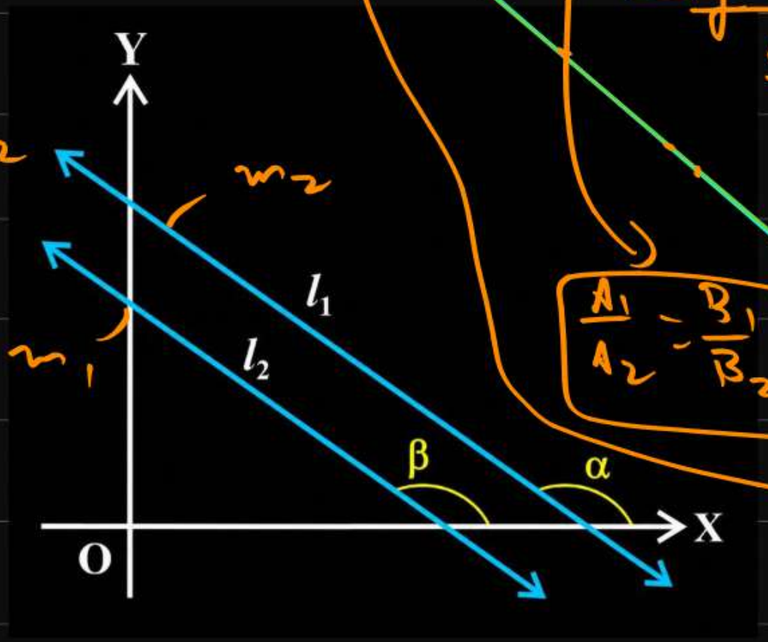
$$\text{Slope} = \tan\alpha = \tan\beta = m$$

Unique Parallel Lines:

1) No solution $\Rightarrow (x, y)$ ✗
2) No common point

$$\frac{A_1}{A_2} = \frac{B_1}{B_2} \neq \frac{C_1}{C_2}$$

$m = m_1 = m_2$



$A_1x + B_1y + C_1 = 0$
$A_2x + B_2y + C_2 = 0$

Infinitely many solutions

$$\frac{A_1}{A_2} = \frac{B_1}{B_2} = \frac{C_1}{C_2}$$

Conditions for parallelism and perpendicularity of lines in terms of their slopes
2 रेखाओं के समानांतर और परस्पर लंब होने का प्रतिबंध

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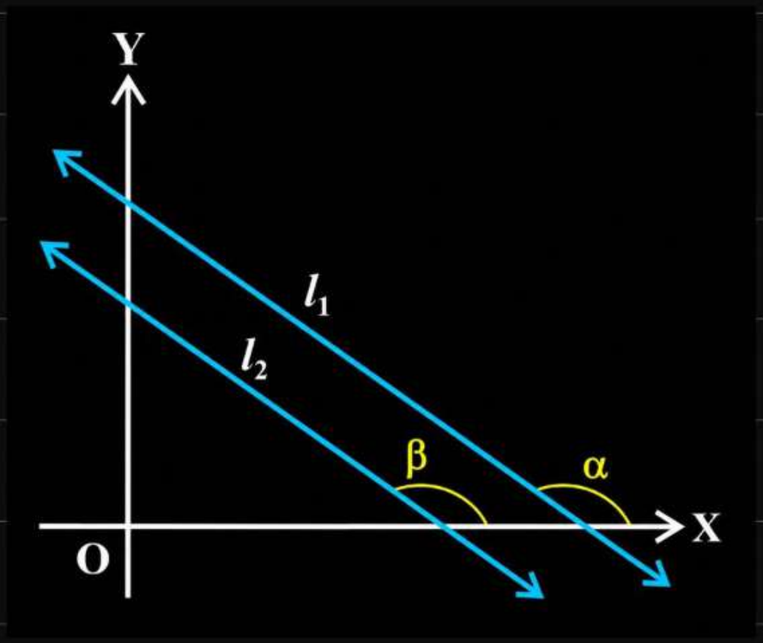
अगर रेखा L1, L2 के समानांतर है, तो उनके झुकाव बराबर होंगे, यानी $\alpha = \beta$, और इसलिए $\tan\alpha = \tan\beta$ होगा। इसलिए $m1 = m2$, यानी उनके ढलान बराबर होंगे।

$$\text{Slope} = \tan\alpha = \tan\beta = m$$

Slope of lines in general Equation form
सामान्य समीकरण रूप में रेखाओं की ढलान

$$A_1x + B_1y + C_1 = 0 \qquad A_2x + B_2y + C_2 = 0$$

$$m = -\frac{A_1}{B_1} = -\frac{A_2}{B_2} \quad \Big| \quad \frac{A_1}{A_2} = \frac{B_1}{B_2}$$



Conditions for parallelism and perpendicularity of lines in terms of their slopes

2 रेखाओं के समानांतर और परस्पर लंब होने का प्रतिबंध

Conversion from 1 perpendicular line to other

$\forall\ k \neq 7$

A_1 B_1 $C_1 = -2$

$3x + 5y = 2$

$6x + 10y = -1$

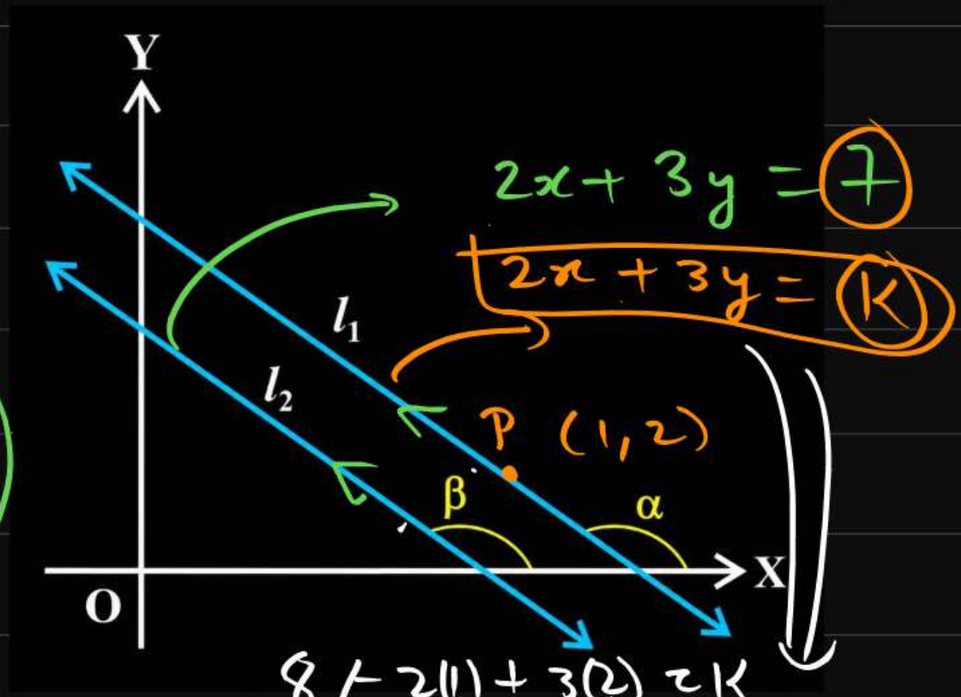
A_2 B_2 $C_2 = +1$

$\frac{3}{6} = \frac{5}{10} \neq \frac{-2}{1}$

$\frac{1}{2} = \frac{1}{2} \neq \frac{-2}{1}$

$2[3x+5y] = -1$

$3x+5y = \frac{-1}{2}$



Direct Conversion for equation of parallel line

$$A_1x + B_1y + C_1 = 0$$

Find the equation of the line that is parallel to this line.

उस रेखा का समीकरण ज्ञात कीजिए जो इस रेखा के समांतर है।

$Ax + By + C_1 = 0$

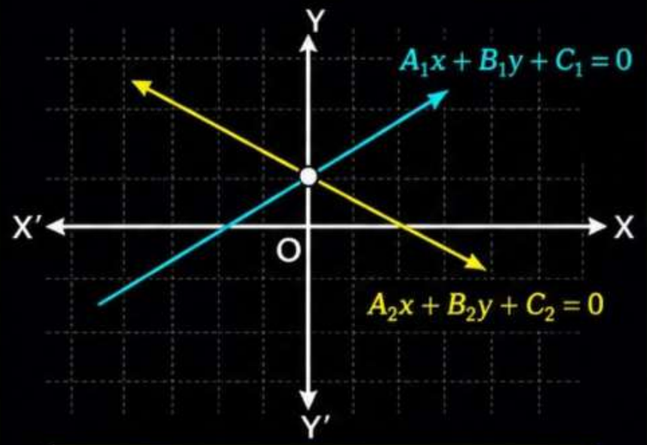
$\parallel$ ($Ax + By + C_2 = 0$) $\neq$

Intersecting, Parallel and Co-incident Lines

$A_1x + B_1y + C_1 = 0$ $A_2x + B_2y + C_2 = 0$

1. INTERSECTING LINES

Lines intersect at one point.



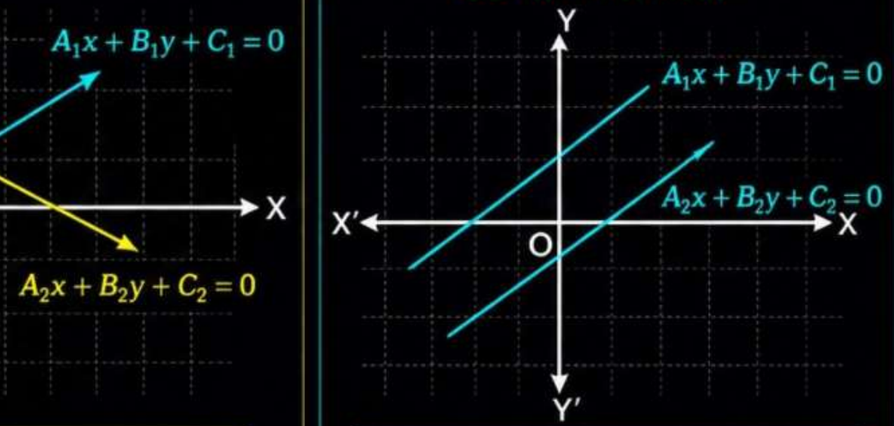
CONDITION

$$\frac{A_1}{A_2} \neq \frac{B_1}{B_2}$$

The ratios of coefficients of x and y are not equal.

2. PARALLEL LINES

Lines are parallel and do not intersect.



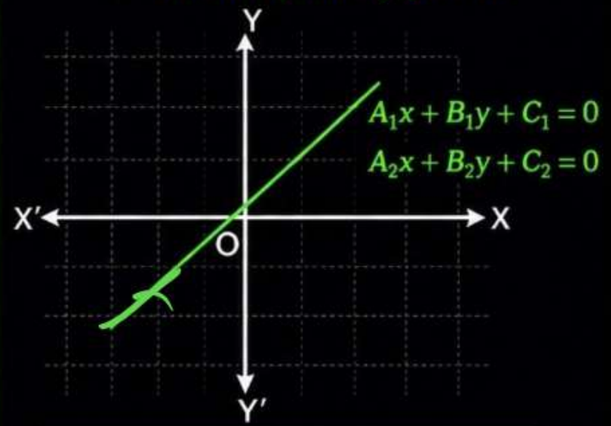
CONDITION

$$\frac{A_1}{A_2} = \frac{B_1}{B_2} \neq \frac{C_1}{C_2}$$

The ratios of coefficients of x and y are equal but the ratio of constants is not equal.

3. CO-INCIDENT LINES

Lines coincide and have infinitely many points.



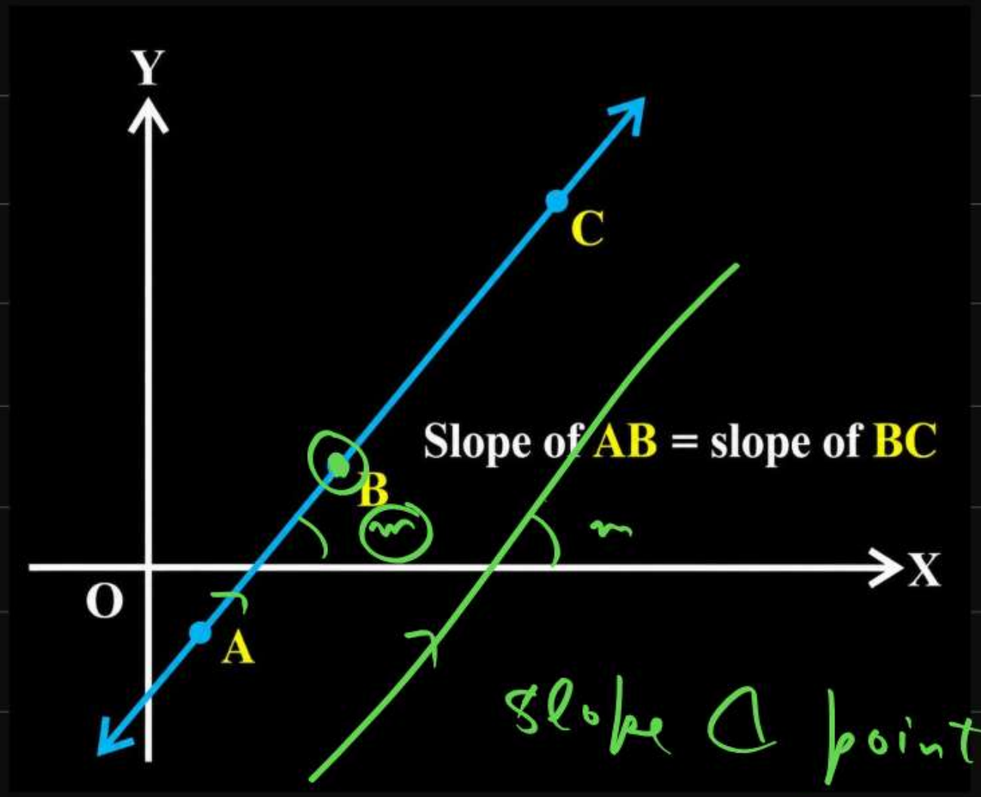
CONDITION

$$\frac{A_1}{A_2} = \frac{B_1}{B_2} = \frac{C_1}{C_2}$$

The ratios of coefficients of x, y and constants are all equal.

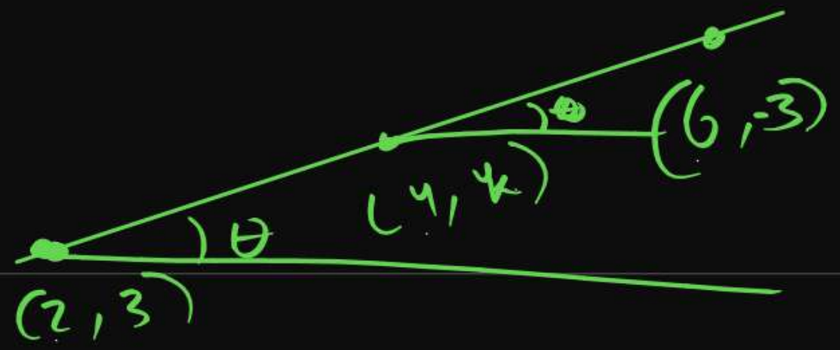
NOTE: If $A_1 = A_2$, $B_1 = B_2$ and $C_1 = C_2$, then both the equations represent the same line.

Collinearity of three points
तीन बिंदुओं की संरेखता



We know that slopes of two parallel lines are equal. If two lines having the same slope pass through a common point, then two lines will coincide.

हम जानते हैं कि दो समानांतर रेखाओं के ढाल समान हैं। यदि एक ही ढाल वाली दो रेखाएं एक सामान्य बिंदु से गुजरती हैं, तो दो रेखाएं तो संपाति (coincident) होती है



Q-) Find the value of k if the points A(2, 3), B(4, k) and C(6, –3) are collinear.

K का मान ज्ञात करें यदि अंक A (2, 3), B (4, k) और C (6, –3) आपस में संरेख (एक ही रेखा पर) हैं

Slope method or area of triangle or section formula

$$\tan\theta = m = \frac{K-3}{4-2} = \frac{-3-K}{6-4}$$

$$2K = 0$$

$$\boxed{K = 0}$$

A) 0
B) 2
C) -1
D) 3

Q. If the lines given by $3x + 2ky = 2$ and $3x + 5y = -1$ are parallel, then the value of k is:

यदि $3x + 2ky = 2$ और $3x + 5y = -1$ द्वारा दी गई रेखाएँ समानांतर हैं, तो k का मान है:

SSC CPO 2025

(a) (b) (c) (d)

$$\frac{A_1}{A_2} = \frac{B_1}{B_2}$$

$$\frac{3}{3} = \frac{2K}{5}$$

$$K = \frac{5}{2}$$

Q. Two lines are planned such that the equations representing their paths are: $(k-4)x + 2y = 6$ and $2x + y = 4$. If the lines are to be constructed parallel to each other, what value of k will ensure that?

$Ax + By + C = 0$

$\frac{k-4}{2} = \frac{2}{1}$

$k = 8$

दो रेखाएँ इस प्रकार खींची गई हैं कि उनके पथों को निरूपित करने वाले समीकरण : $(k-4)x + 2y = 6$ और $2x + y = 4$ हैं। यदि रेखाओं की एक दूसरे के समानांतर रचना की जानी है, तो k का कौन-सा मान यह सुनिश्चित करेगा?

RRB Group D 2024-25 [Nov to feb 2026]

(a) 6 (b) 7 (c) 8 (d) 10

$Ax + By + C = 0$

$m = -\frac{A}{B}$

$m = \frac{-3}{2}$

Q. Which of these lines is parallel to $3x + 2y = 5$?

इनमें से कौन सी रेखा $3x + 2y = 5$ के समांतर है ?

$3x + 2y = k$

SSC CGL 2025

(a) $3x + 2y = 7$ (b) $2x + 3y = 5$

(c) $3x - 2y = 5$ (d) $x + y = 5$

$-3/2$ $-2/3$

$\frac{-3}{-2} = \frac{3}{2}$ α $\frac{-1}{1}$ k

$\times 2$

Parallel

$6x + 4y = 14$

$3x + 2y = 17$

$3x + 2y = 91$

$m = -\frac{6}{4} = -\frac{3}{2}$

Q. Which line is not parallel to the line $6x - 2y = 10$?

कौन सी रेखा, रेखा $6x - 2y = 10$ के समानांतर नहीं है?

$m = \frac{-(6)}{(-2)} = 3$

$3x - y = 5$

SSC CPO 2025

(a) $3x - y = 4$

(b) $y = 3x - 1$ $\quad 3x - y = 1$

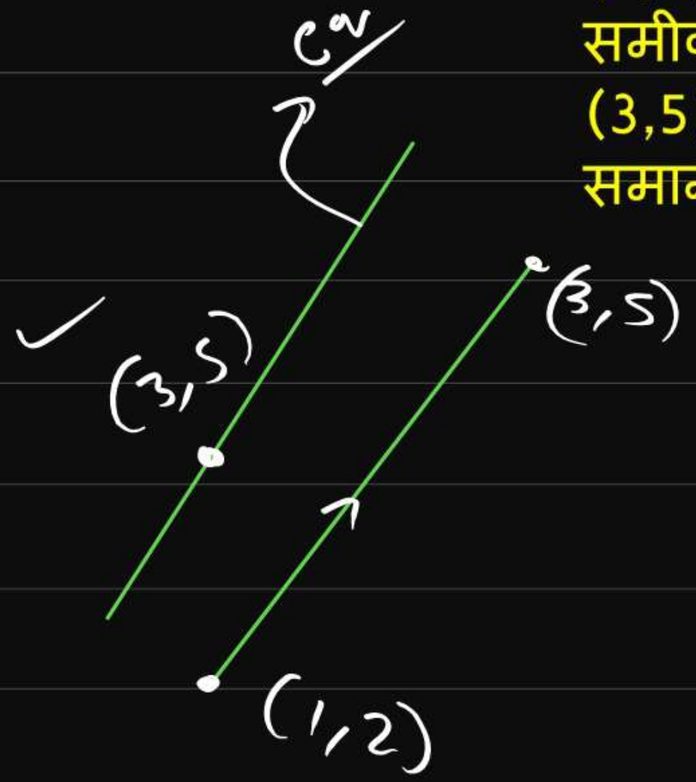
(c) $y = 4x + 5$ $\quad 4x - y = -5$

(d) $12x - 4y = 9$

$3x - y = \frac{9}{4}$

Q). Find the equation of a line passing through (3,5) and parallel to the line passing through (1,2) and (3,5).

(3,5) से गुजरने वाली एक रेखा का समीकरण ज्ञात कीजिए जो (1,2) और (3,5) से गुजरने वाली रेखा के समानांतर है ।



(a) $3x+2y=1$

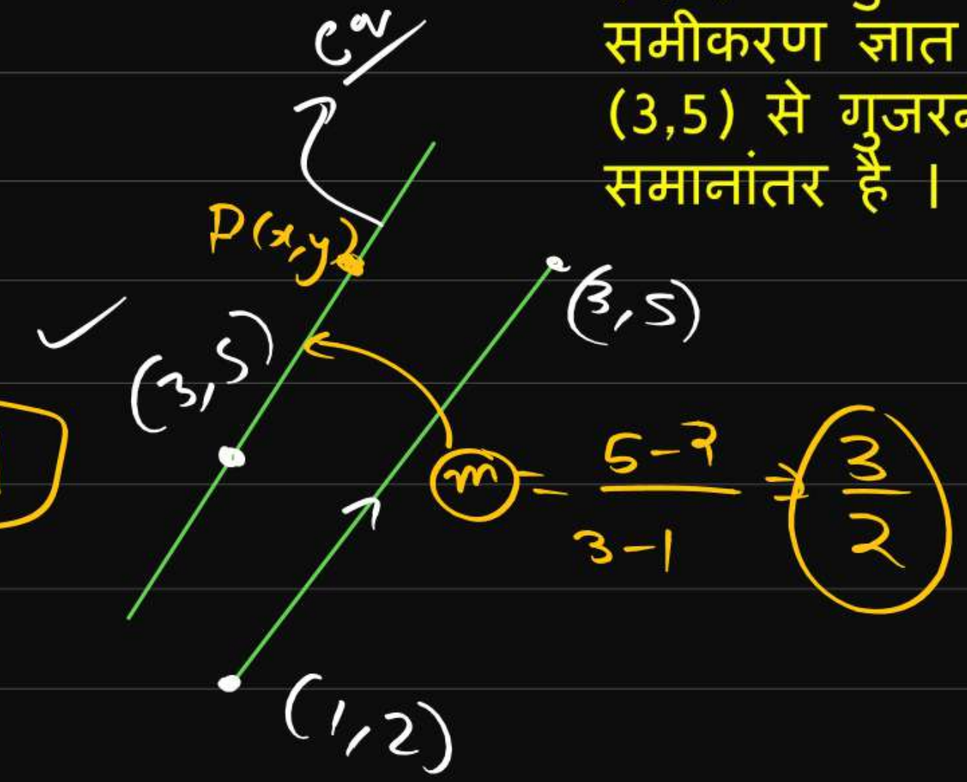
(b) $3x-2y=-1$

(c) $3x-2y=19$

(d) $2x-3y=-1$

Q). Find the equation of a line passing through (3,5) and parallel to the line passing through (1,2) and (3,5).

(3,5) से गुजरने वाली एक रेखा का समीकरण ज्ञात कीजिए जो (1,2) और (3,5) से गुजरने वाली रेखा के समानांतर है ।



$$\frac{3}{2} = \frac{y-5}{x-3}$$

$$3x-2y = -10+9 = -1$$

(a) $3x+2y=1$

(b) $3x-2y=-1$

(c) $3x-2y=19$

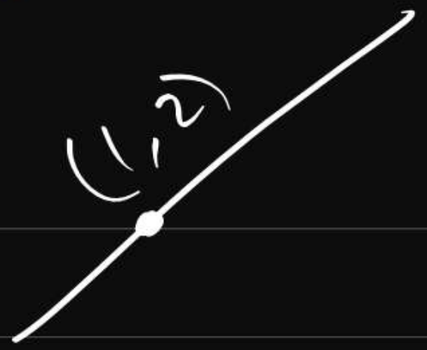
(d) $2x-3y=-1$

Q. Find the equation of a line that is parallel to y = 3x - 5 and passes through the point (1, 2).

उस रेखा का समीकरण ज्ञात कीजिए जो y = 3x - 5 के समानांतर है और बिंदु (1, 2) से होकर गुजरती है।

$Ax + By + C_1 = 0$
$+ C_2 = 0$. **SSC CHSL 2025**

(a) y = 3x - 1 (b) y = 3x + 1
(c) y = 3x - 2 (d) y = 3x + 2



$y = mx + c$

$y = 3x - 5$

m

$3 = \frac{y-2}{x-1}$

$y = 3x + k$

$(1,2)$

$k = -1$

$y = 3x - 1$

$y - 3x + 5 = 0$

$y - 3x + k = 0 \Rightarrow$

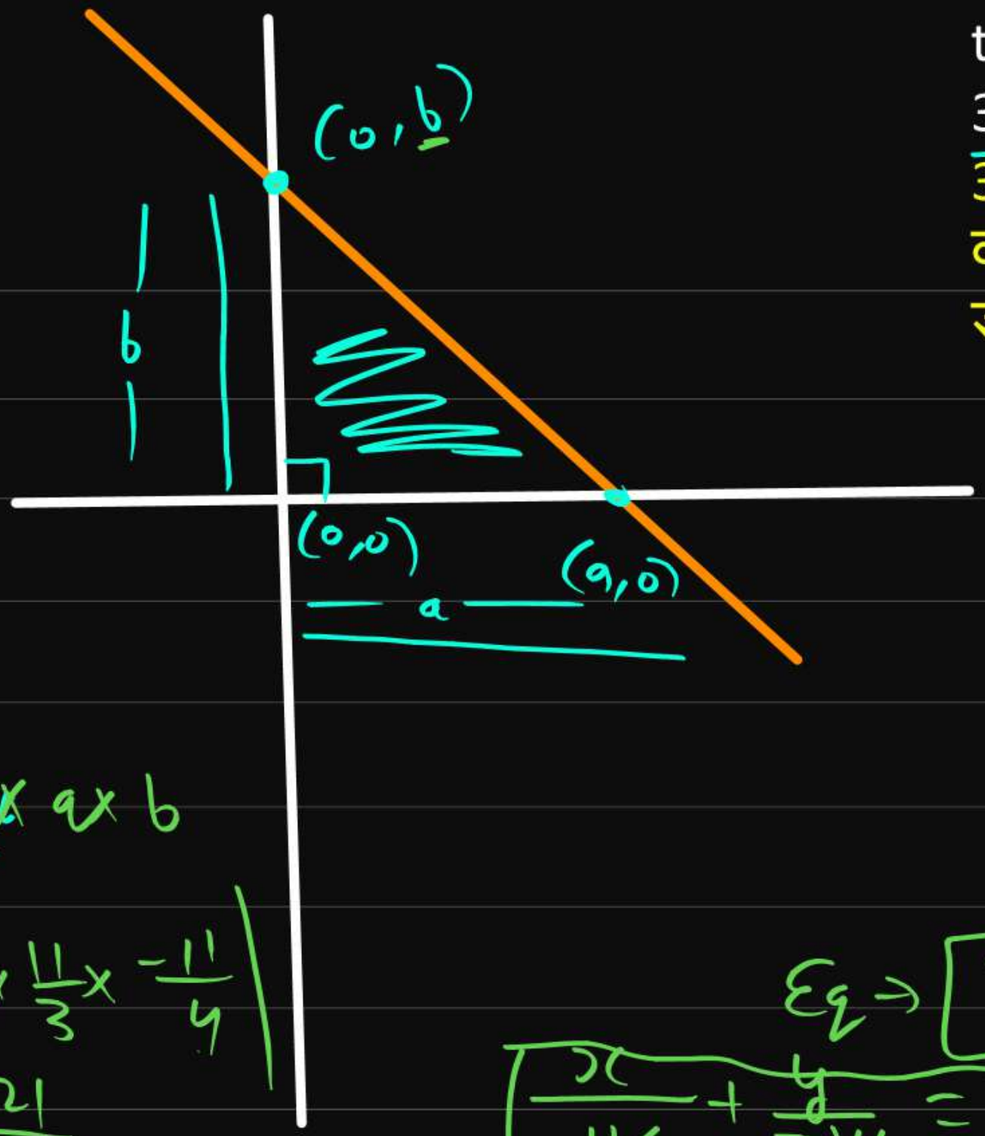
$k = 1$

$y - 3x + 1 = 0$; $y = 3x - 1$

Q) Find the area between the axis and the line passing through (1,-2) parallel to 3x-4y=5.

3x-4y=5 के समांतर (1,-2) से गुजरने वाली रेखा और अक्षों के बीच का क्षेत्रफल ज्ञात कीजिए।

(good question)



$3x - 4y = K$

(1) (-2)

$3 + 8 = 11 = K$

Eq → $3x - 4y = 11$

$\frac{x}{11/3} + \frac{y}{-11/4} = 1$

Area $\Delta = \frac{1}{2} \times a \times b$

$\Rightarrow \left|\frac{1}{2} \times \frac{11}{3} \times \frac{-11}{4}\right|$

$\Rightarrow \frac{121}{24}$

(a) 112/25
(b) 114/25
(c) 121/24
(d) 73/12

Q. What will be the value of K for which the system of equations 2x + 2y − 5 = 0 and 6x + ky + 7 = 0 has no solution?

parallel

K का वह मान क्या होगा जिसके लिए समीकरणों के निकाय 2x + 2y − 5 = 0 और 6x + ky + 7 = 0 का कोई हल नहीं है?

$\frac{2}{6} = \frac{2}{K}$

$K = 6$

SSC CPO 2025

(a) 2 (b) 3 (c) 4 (d) 6

Q. The graphs of two linear equations, x + 2y = 15 and 4x + 8y = 13 will be:

दो रैखिक समीकरण, x + 2y = 15 और 4x + 8y = 13 के ग्राफ _____ होंगे।

SSC CHSL 2024 T-1

(a) intersect at one point
एक बिंदु पर प्रतिच्छेदित

(b) intersect at two points
दो बिंदुओं पर प्रतिच्छेदित

(c) co-incident
संपाती

(d) parallel
समांतर

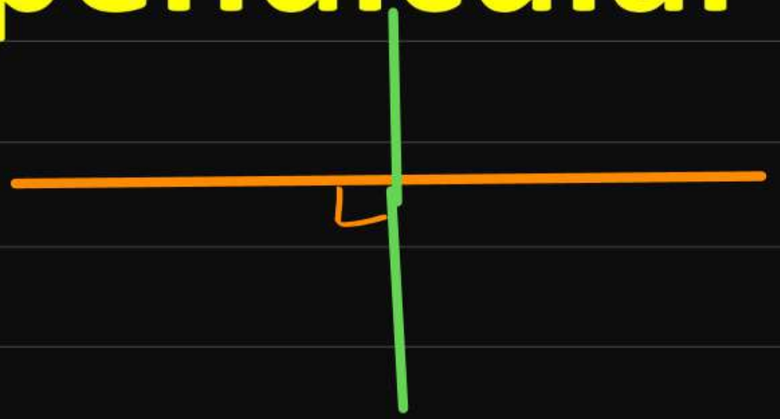
$$1x + 2y - 15 = 0$$
$$4x + 8y - 13 = 0$$

$$\frac{1}{4} \quad \frac{2}{8} \quad \frac{-15}{-13}$$

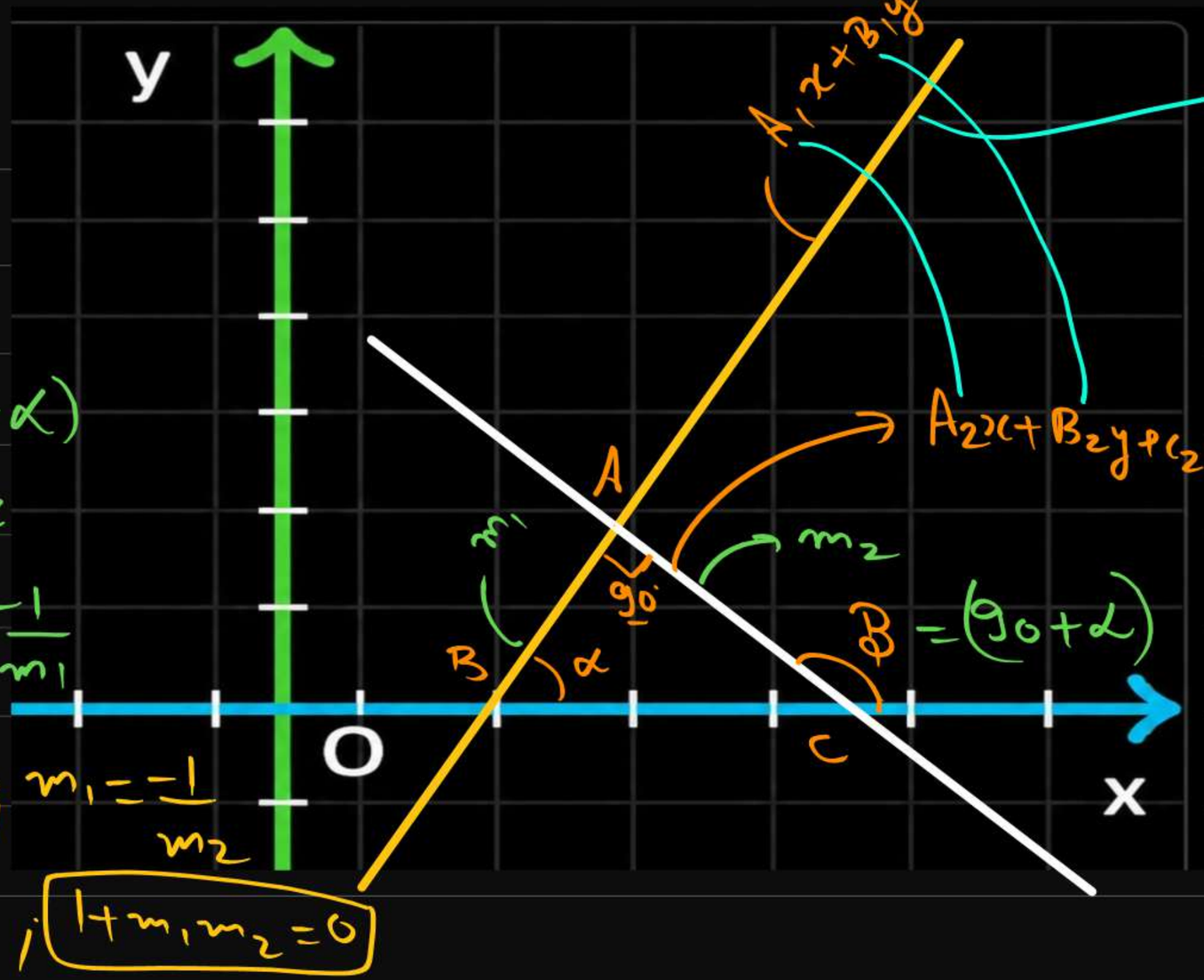
$$\frac{1}{4} = \frac{1}{4} \neq \frac{15}{13}$$

$$\frac{A_1}{A_2} = \frac{B_1}{B_2} \neq \frac{C_1}{C_2}$$

Perpendicular lines



Perpendicular lines लम्बवत रेखायें



$m_1 = \tan\alpha$

$m_2 = \tan\beta$

$= \tan(90^\circ + \alpha)$

$\Rightarrow -\cot\alpha$

$\Rightarrow \frac{-1}{\tan\alpha} = \frac{-1}{m_1}$

$m_2 = \frac{-1}{m_1}$; $m_1 = \frac{-1}{m_2}$

$m_1 m_2 = -1$; $1 + m_1 m_2 = 0$

$m_1 = \frac{-A_1}{B_1}$

$m_2 = \frac{-A_2}{B_2}$

$\frac{+A_1}{B_1} = \frac{+1}{(-A_2)/B_2}$

$\frac{A_1}{B_1} = \frac{-B_2}{A_2}$

$A_1A_2 = -B_1B_2$

$A_1A_2 + B_1B_2 = 0$

If the lines L1 and L2 are perpendicular

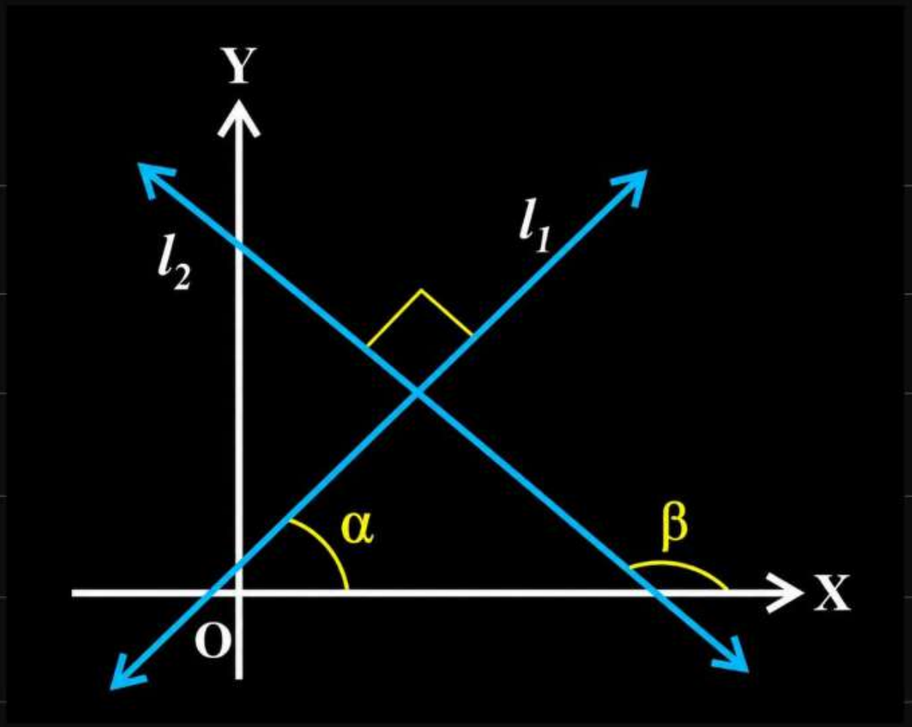
Relation in Slopes

$m_1 = \tan(\alpha)$

$m_2 = \tan\beta = \tan(\alpha + 90°)$

$m_2 = -\cot\alpha$

$m_2 = -1/m_1$



If the lines L1 and L2 are perpendicular

Relation in Equations form

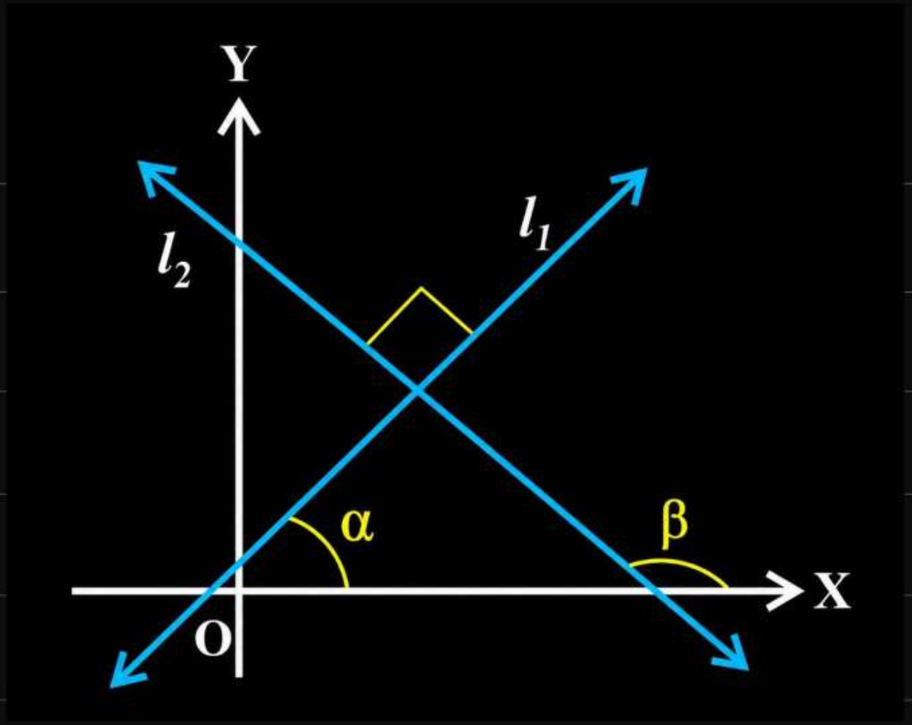
$m_2 = -1/m_1$

$m_1 = -\frac{1}{m_2}$; $1 + m_1 m_2 = 0$

$A_1x + B_1y + C_1 = 0$ $\quad$ $A_2x + B_2y + C_2 = 0$

$-\frac{A_1}{B_1} = \frac{-1}{(-A_2/B_2)}$

$A_1A_2 + B_1B_2 = 0$



$A_1A_2 + B_1B_2 = 0$

Q. Are these lines $y = 2x + 3$ and $y = -(1/2)x - 1$ perpendicular?

एक 8-भुजीय बहुभुज के 7 आंतरिक कोणों के मापों का योग 1,000° है। 8वें बाह्य कोण का माप क्या है?

SSC CGL 2025

(a) Yes
(b) No
(c) Only if they intersect at origin
(d) Cannot be determined

A_1 B_1

$2x - 1y + 3 = 0$

$\frac{1}{2}x + 1y + 1 = 0$

A_2 B_2

$A_1A_2 + B_1B_2$

$2 \times \frac{1}{2} + -1 \times 1 = 0$

Lines are perpendicular

Q. Are these lines $y = 2x + 3$ and $y = -(1/2)x - 1$ perpendicular?

एक 8-भुजीय बहुभुज के 7 आंतरिक कोणों के मापों का योग 1,000° है। 8वें बाह्य कोण का माप क्या है?

$y = mx + c$

SSC CGL 2025

(a) Yes
(b) No
(c) Only if they intersect at origin
(d) Cannot be determined

$1 + m_1 m_2 = 0$ (⊥)

$1 + 2 \times \left(-\frac{1}{2}\right) = 0$

$A_1 \quad B_1$

$2x - 1y + 3 = 0$

$\frac{1}{2}x + 1y + 1 = 0$

$A_2 \quad B_2$

$A_1 A_2 + B_1 B_2$

$2 \times \frac{1}{2} + -1 \times 1 = 0$

Lines are perpendicular

Q. What is the slope of line perpendicular to $y = -3x + 7$?

$y = -3x + 7$ के लंबवत रेखा का ढाल (slope) क्या होगा?

SSC CGL 2025

(a) -1/3 (b) 3 (c) 1/3 (d) -3

$m_2 = \frac{-1}{m_1} \Rightarrow \frac{-1}{-3} = \frac{1}{3}$

Q. The lines $2x - 3y = 6$ and $kx + y = 4$ are perpendicular. What is the value of k?

रेखाएँ $2x - 3y = 6$ और $kx + y = 4$ लंबवत हैं। k का मान क्या है?

SSC CPO 2025

(a) 1/2 (b) 3/2 (c) 2 (d) 1

$$A_1A_2 + B_1B_2 = 0$$

$$2k + -3 \times 1 = 0$$

$$k = \frac{3}{2}$$

Q-) Line through the points (–2, 6) and (4, 8) is perpendicular to the line through the points (8, 12) and (*x*, 24). Find the value of *x*.

बिंदुओं (-2, 6) और (4, 8) को मिलाने वाली रेखा, (8, 12) और (x, 24) बिंदुओं को मिलाने वाली रेखा पर लंब है। x का मान ज्ञात करें।

$$m_1 = \frac{8-6}{4-(-2)} = \frac{2}{6} = \frac{1}{3}$$

$$m_2 = -\frac{1}{m_1} \Rightarrow -\frac{1}{1/3} \Rightarrow -3$$

$$-3 = \frac{24-12}{x-8}$$

$$-x + 8 = 4$$

$$x = 4$$

A) 3
B) 2
C) -2
D) 4

Q. Which of the following represents a line perpendicular to the line $4x + 5y = 20$?

निम्नलिखित में से कौन सी रेखा, रेखा $4x + 5y = 20$ के लंबवत है?

SSC CHSL 2025

(a) $5x - 7y = 8$ (b) $4x + 5y = 12$

(c) $y = 5x/4 - 3$ (d) $y = -4x/5 + 1$

$m_1 = \frac{-4}{5}$

$m_2 = \frac{-1}{-4/5}$

$\Rightarrow \frac{5}{4}$

$m = \frac{5}{4}$ $\quad$ $m = -\frac{4}{5}$

a) $20 - 35 \neq 0$ $\quad$ b) $16 + 25 \neq 0$

c)

Direct Conversion for equation of perpendicular line

$$-\frac{A_1}{B_1} = \frac{-1}{-A/B_2}$$

1) $$A_1x + B_1y + C_1 = 0$$

2) $$\therefore x + B_2 y \therefore C_2 = 0$$

Find the equation of the line that is perpendicular to this line.

उस रेखा का समीकरण ज्ञात कीजिए जो इस रेखा पर लंब है।

$$A_1A_2 + B_1B_2 = 0$$

$$A_1A_2 = -B_1B_2$$

$$-\frac{A_1}{B_1} = \frac{B_2}{A_2}$$

$$\frac{B_2}{-A_1} = \frac{A_2}{B_1} = K$$

$$B_2 = -A_1K$$

$$A_2 = B_1K$$

$$(K B_1 x + K(-A_1) y + C_2 = 0) \frac{1}{K}$$

2) $$B_1x - A_1y + \left(\frac{C_2}{K}\right) = 0$$

$$\downarrow$$

$$C$$

Direct Conversion for equation of perpendicular line

$$-\frac{A_1}{B_1} = \frac{-1}{-A/B_2}$$

1) $A_1x + B_1y + C_1 = 0$

$x \to -1$

$B_1x - A_1y + C = 0$

↳ Req. $\perp$ line

Find the equation of the line that is perpendicular to this line.

उस रेखा का समीकरण ज्ञात कीजिए जो इस रेखा पर लंब है।

$B_1x - A_1y + \left(\frac{C_2}{K}\right) = 0$

$\frac{C_2}{K} \to C$

Q. If the equation of a line is 3x - 5y = 15, what is the equation of the line perpendicular to it that passes through the point (3, 2)?

यदि एक रेखा का समीकरण 3x - 5y = 15 है , तो उस रेखा का समीकरण क्या है जो इसके लंबवत है और बिंदु (3, 2) से होकर गुजरती है ?

SSC CPO 2025

(a) y = (5/3)x + 7 (b) y = -(5/3)x + 7
(c) y = (5/3)x - 7 (d) y = -(3/5)x + 7

$3x - 5y - 15 = 0$

$\times -1$

$-5x + -3y + C = 0$

$5x + 3y = C$ → passes through (3, 2)

$3 \quad 2$

$21 = C$

$5x + 3y = 21 \Rightarrow y = -\frac{5x}{3} + 7$

Q. If the equation of a line is 3x - 5y = 15, what is the equation of the line perpendicular to it that passes through the point (3, 2)?

$3x - 5y = 15$

Slope of line $= \frac{B}{A}$

$\Rightarrow \frac{-5}{3}$

यदि एक रेखा का समीकरण 3x - 5y = 15 है , तो उस रेखा का समीकरण क्या है जो इसके लंबवत है और बिंदु (3, 2) से होकर गुजरती है ?

$y = mx + C$

SSC CPO 2025

(a) y = (5/3)x + 7 (b) y = -(5/3)x + 7

(c) y = (5/3)x - 7 (d) y = -(3/5)x + 7

$$Ax + By + C = 0$$

$m_1 \Rightarrow -\frac{A}{B}$

↳ Perpendicular Line

$m_2 = \frac{-1}{-A/B} \Rightarrow \frac{B}{A}$

Q. Find the equation of the line that passes through the point (2, 5) and is perpendicular to the line y = 3x + 4.

$y = 3x + y$

$1 \text{ slope} = \frac{-1}{3}$

उस रेखा का समीकरण ज्ञात कीजिए जो बिंदु (2, 5) से होकर गुजरती है और रेखा y = 3x + 4 के लंबवत है।

SSC CHSL 2025

(a) y = -x/3 + 6 (b) y = -3x + 5

(c) y = -6x - 15 (d) y = -x/3 + 17/3

$y = \frac{-2}{3} + \frac{17}{3} = 5$

(2, 5)

Q. A line passes through the point (2, −1) and is perpendicular to the line $y = (-2/5)x + 3$.
What is its equation?

एक रेखा बिंदु (2, −1) से होकर गुजरती है और रेखा $y = (-2/5)x + 3$ पर लंबवत है।
इसका समीकरण क्या है?

SSC CPO 2025

(a) $y = 2x - 6$
(b) $y = (3/2)x - 6$
(c) $y = (5/2)x - 6$
(d) $y = (5/2)x - 2$

$m_2 = -\frac{5}{2}$

$y = \frac{5}{2}x + k$

$-1 = \frac{5}{2} \times 2 + k$

$k = -6$

$y = \frac{5}{2}x - 6$

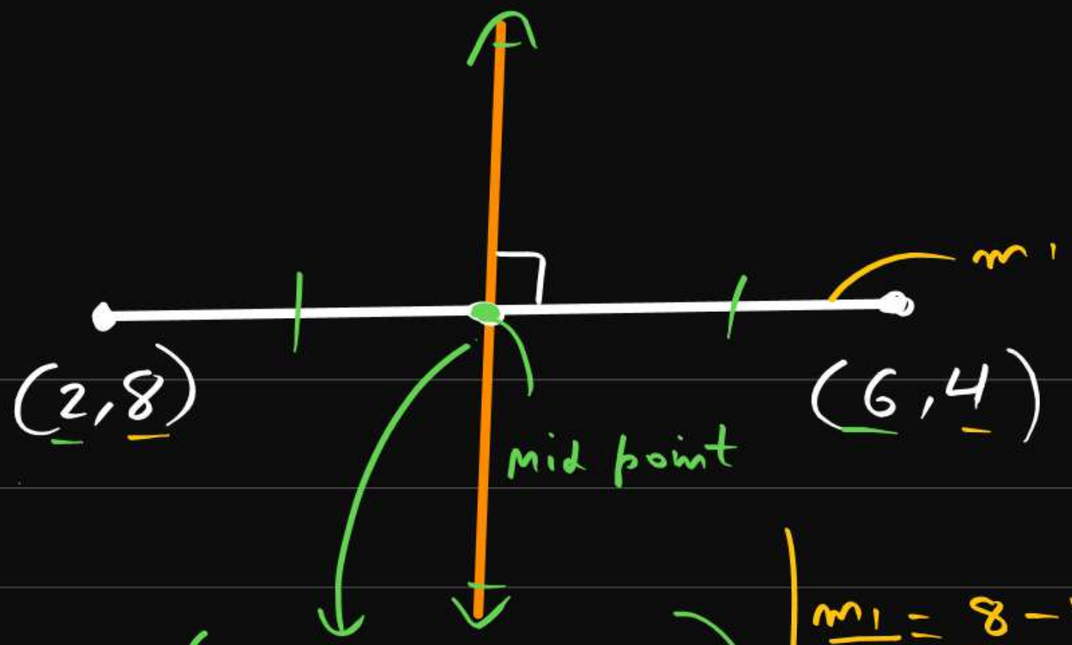
Q. What is the equation of the perpendicular bisector of the line segment connecting the points **(2, 8) and (6, 4)?**

बिंदुओं **(2, 8) और (6, 4)** को जोड़ने वाले रेखाखंड के लंब समद्विभाजक का समीकरण क्या है?

SSC CGL 2025

(a) $y = -x - 10$ (b) $y = x - 2$

(c) $y = -x + 10$ (d) $y = x + 2$



$\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$

$\left(\frac{2+6}{2}, \frac{8+4}{2}\right)$

$(4, 6)$

$m_1 = \frac{8-4}{2-6}$

$\Rightarrow \frac{4}{-4} = -1$

$m_2 = +1$

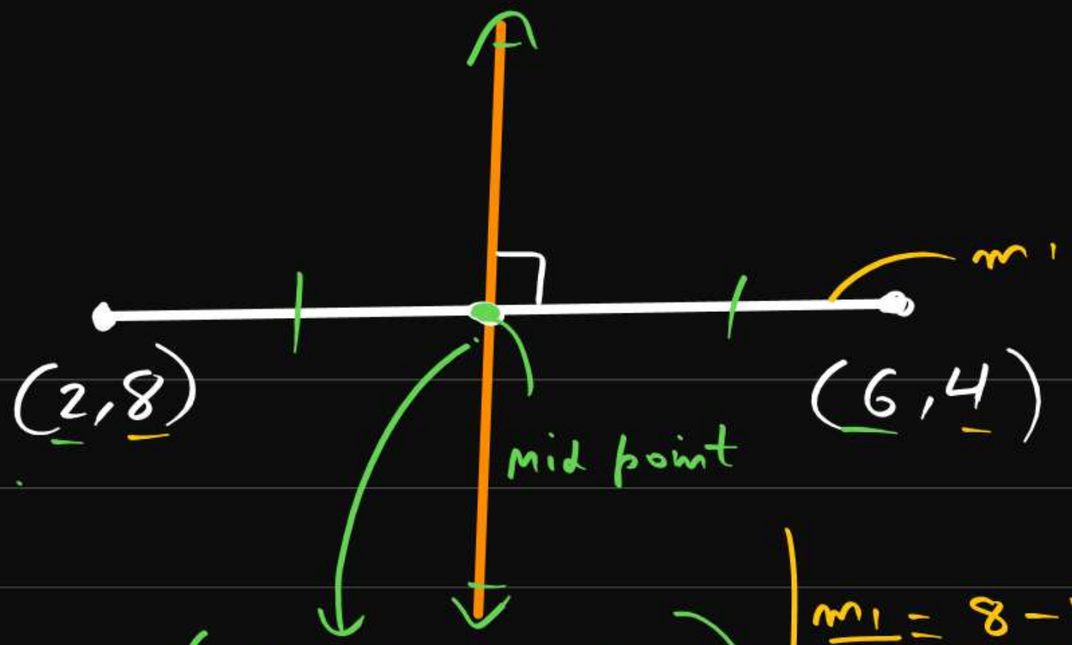
$y = mx + c$

Q. What is the equation of the perpendicular bisector of the line segment connecting the points **(2, 8) and (6, 4)?**

बिंदुओं **(2, 8) और (6, 4)** को जोड़ने वाले रेखाखंड के लंब समद्विभाजक का समीकरण क्या है?

SSC CGL 2025

(a) $y = -x - 10$ **(b)** $y = x - 2$
(c) $y = -x + 10$ **(d)** $y = x + 2$



$\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$

$\left(\frac{2+6}{2}, \frac{8+4}{2}\right)$

$(4, 6)$

$m_1 = \frac{8-4}{2-6}$

$\Rightarrow \frac{4}{-4} = -1$

$m_1 \times m_2 = -1$

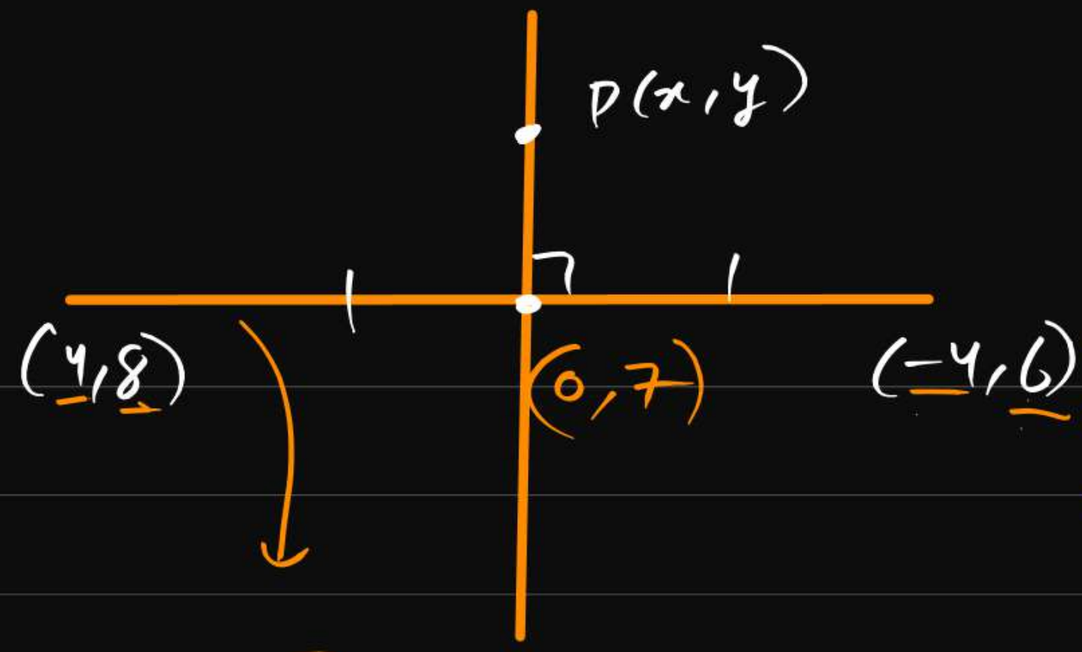
$m_2 = +1$

$y = mx + c$;

$y = x + c$

$6 = 4 + c$; $c = 2$

$y = x + 2$



Q) Find the equation of the perpendicular bisector of PQ, where P and Q are the points (4,8) and (-4,6) respectively.

PQ के लंबवत द्विभाजक का समीकरण ज्ञात करें, जहां P और Q क्रमशः बिंदु (4,8) और (-4,6) हैं।

$m_1 = \frac{2}{8} = +\frac{1}{4}$

$m_2 = -4 = \frac{y-7}{x-0}$

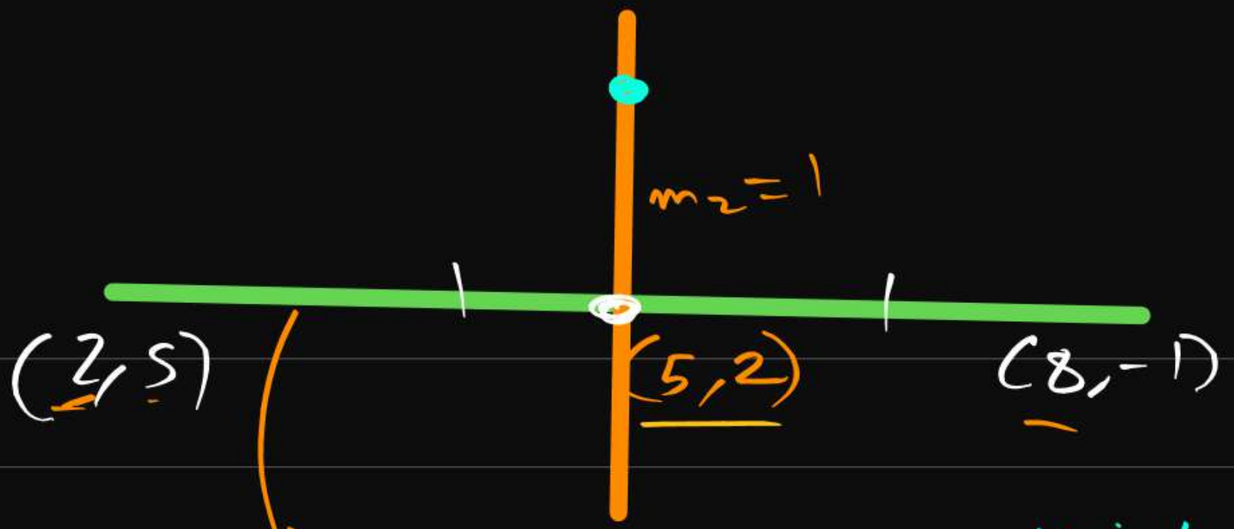
$y + 4x - 7 = 0$

(a) $3x+y-7=0$

(b) $3x+2y-5=0$

(c) $4x+y-7=0$

(d) $4x+2y+5=0$



$$m_1 = \frac{6}{-6} = -1$$

$$m_2 = +1$$

y intercept

$$y = m_2 x + c$$

$$5 = 1(2) + c$$

$$3 = c$$

Q. . A line L is the perpendicular bisector of the line segment connecting points A(2, 5) and B(8, -1). What is the y-intercept of line L?

. एक रेखा L, बिंदुओं A(2, 5) और B(8, -1) को जोड़ने वाले रेखाखंड का लंब समद्विभाजक है। रेखा L का y-अंतःखंड क्या है?

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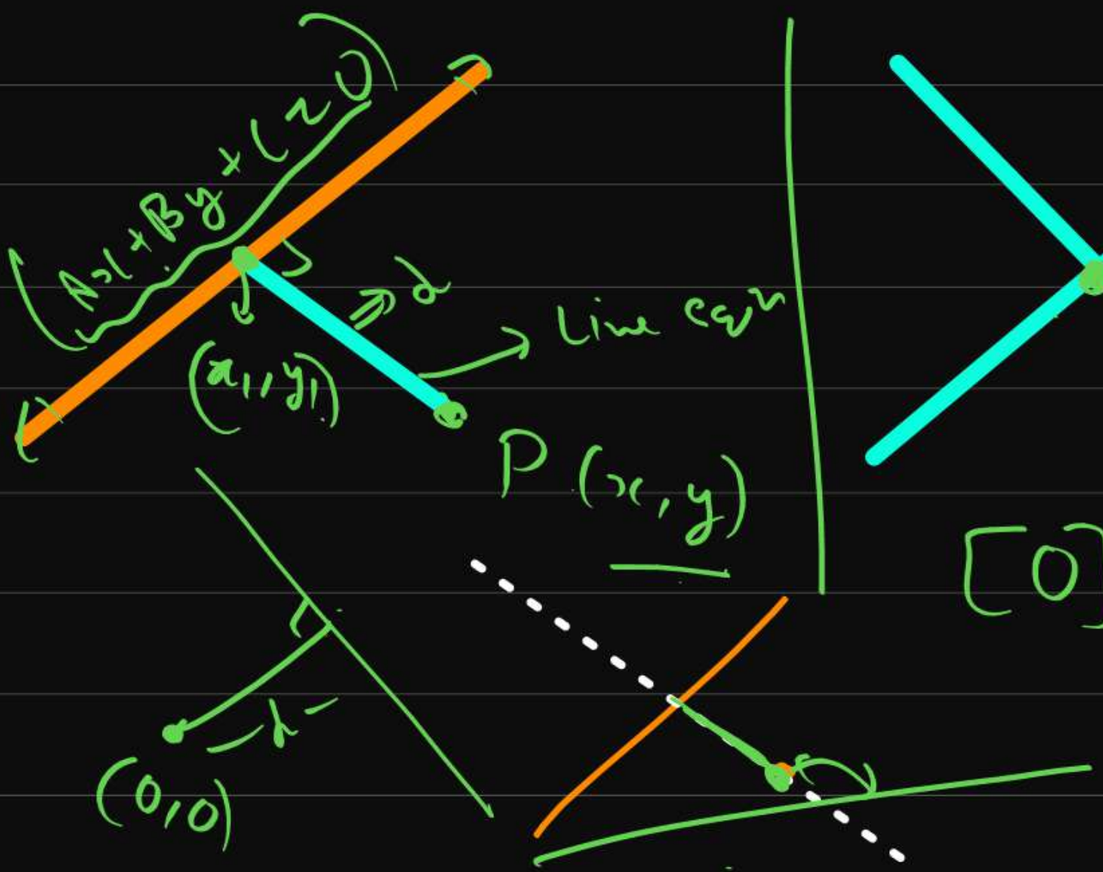
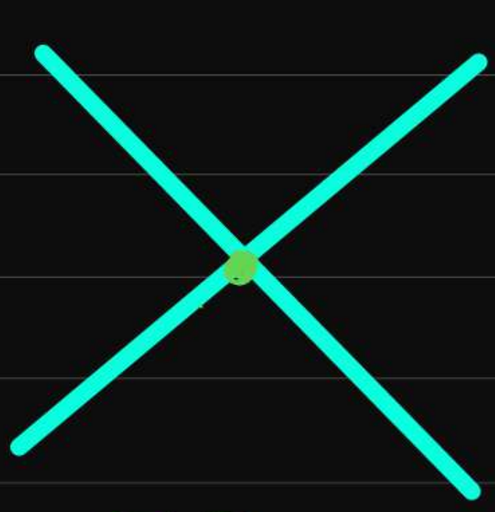
(a) –3 (b) 2 (c) 3 (d) 5

Distance of a point from a line
एक रेखा से एक बिंदु की दूरी



Distance between Intersecting lines
प्रतिच्छेदी रेखाओं के बीच की दूरी



[0]

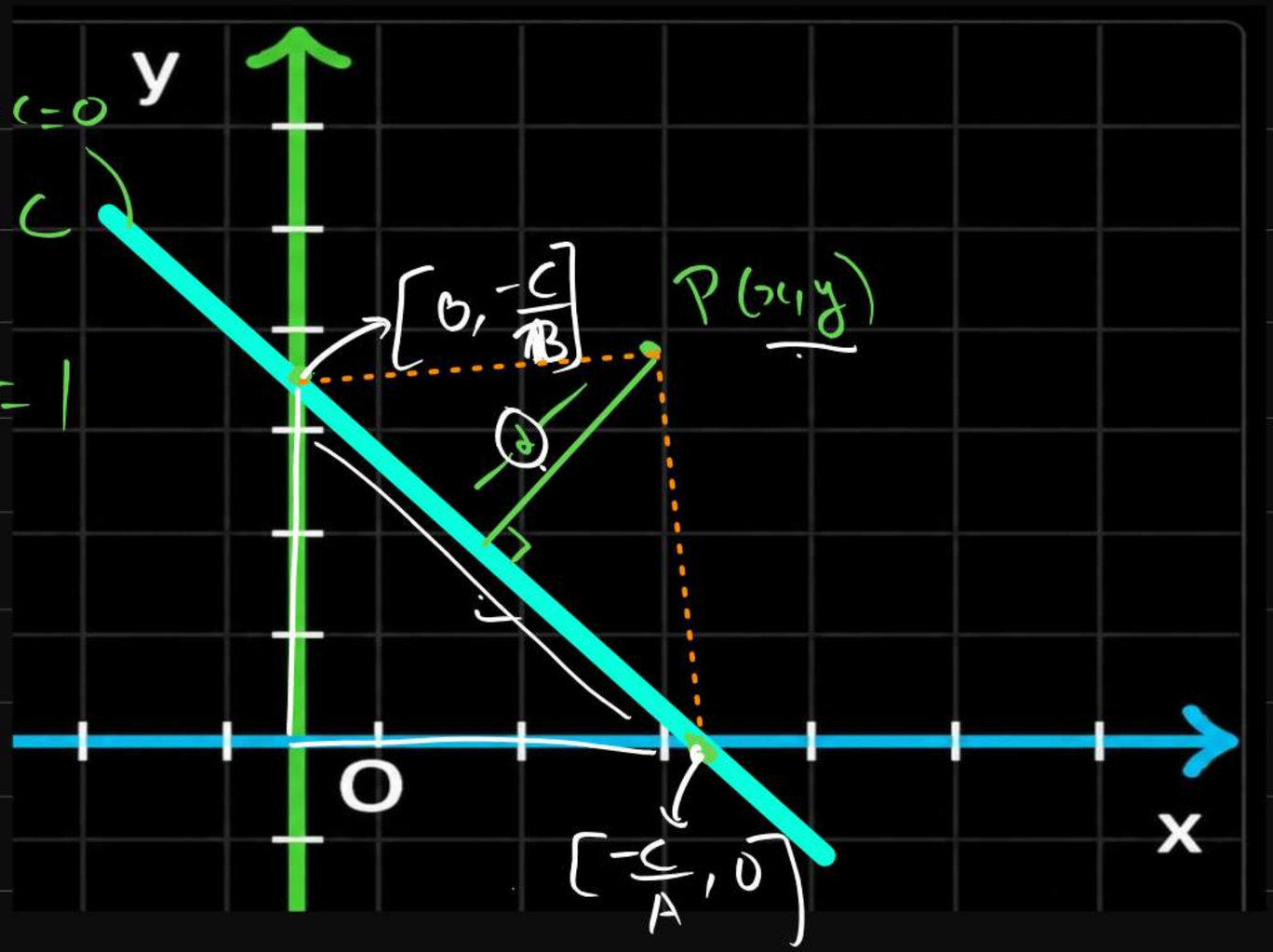
Distance between parallel lines
समानांतर रेखाओं के बीच की दूरी

Distance of a Point From a Line
एक पंक्ति से एक बिंदु की दूरी

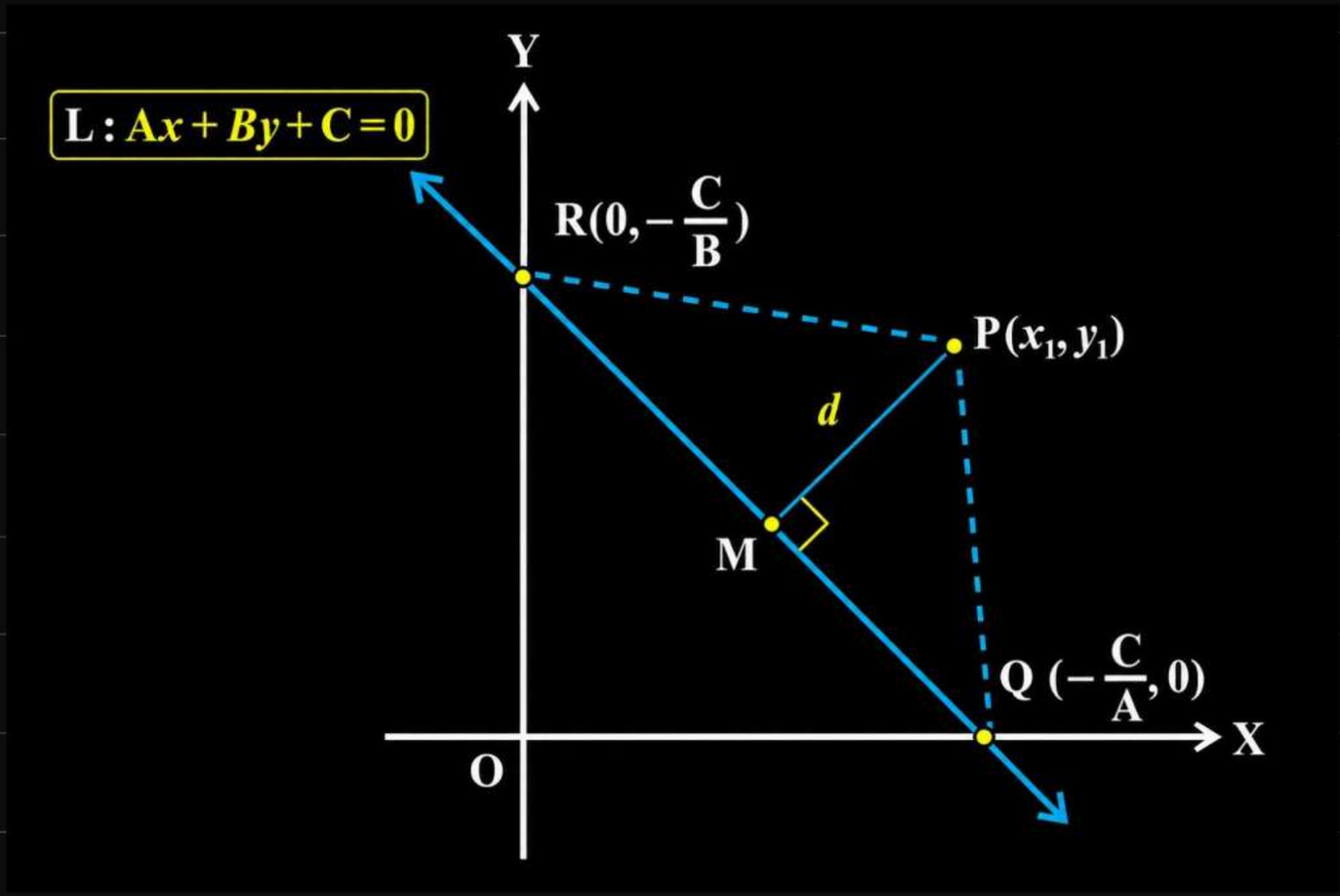
$Ax + By + C = 0$

$Ax + By = -C$

$$\frac{x}{\left(\frac{-C}{A}\right)} + \frac{y}{\left(\frac{-C}{B}\right)} = 1$$



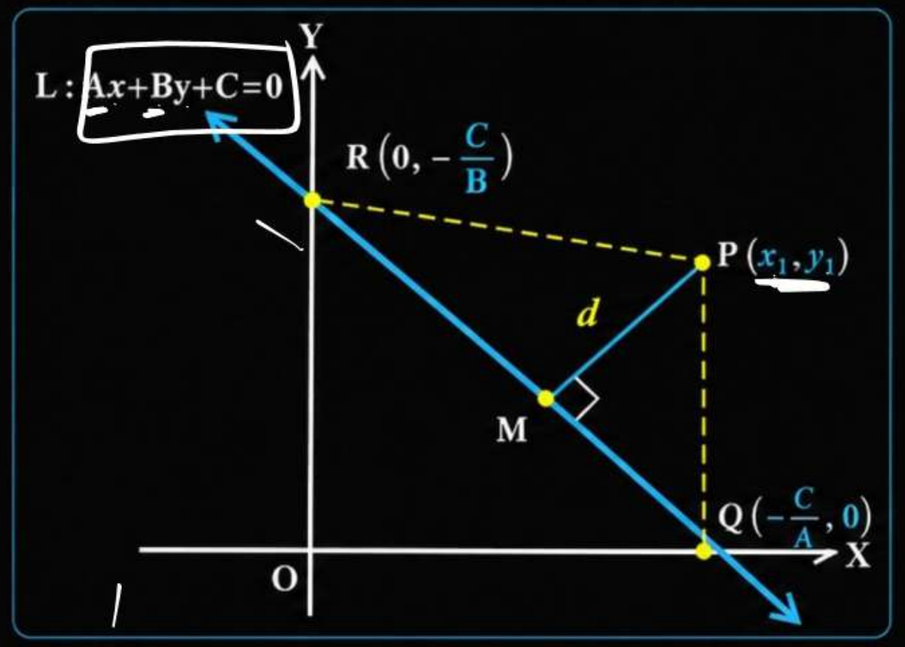
Distance of a Point From a Line
एक पंक्ति से एक बिंदु की दूरी



Distance of a Point From a Line

एक पंक्ति से एक बिंदु की दूरी

Let the line be : $Ax + By + C = 0$



- R is y-intercept : $R(0, -C/B)$
- Q is x-intercept : $Q(-C/A, 0)$
- P is any point : $P(x_1, y_1)$
- M is foot of perpendicular from P on the line.
- PM = d = perpendicular distance (required)

(1) Area of triangle ΔPQR

$$\text{area}(\Delta PQR) = \frac{1}{2} \cdot PM \cdot QR \Rightarrow PM = \frac{2 \cdot \text{area}(\Delta PQR)}{QR} \quad \ldots(1)$$

QR = distance between $R\left(0, -\frac{C}{B}\right)$ and $Q\left(-\frac{C}{A}, 0\right)$

$$= \sqrt{\left(0 + \frac{C}{A}\right)^2 + \left(\frac{C}{B} - 0\right)^2} = \frac{|C|}{|AB|}\sqrt{A^2 + B^2}$$

(2) Area of triangle ΔPQR

$$\text{area}(\Delta PQR) = \frac{1}{2}\left|x_1\left(0 + \frac{C}{B}\right) + \left(-\frac{C}{A}\right)\left(\frac{C}{B} = y_1\right) + 0(y_1 - 0)\right|$$

$$= \frac{1}{2}\left|x_1\frac{C}{B} + y_1\frac{C}{A} + \frac{C^2}{AB}\right|$$

$$= \frac{1}{2}\left|\frac{C}{AB}\right| |Ax_1 + By_1 + C|$$

$$2 \cdot \text{area}(\Delta PQR) = \frac{|C|}{|AB|}|Ax_1 + By_1 + C|$$

(3) Substitute in (1)

$$PM = \frac{2 \cdot \text{area}(\Delta PQR)}{QR} = \frac{\frac{|C|}{|AB|}|Ax_1 + By_1 + C|}{\frac{|C|}{|AB|}\sqrt{A^2 + B^2}} \Rightarrow PM = \frac{|Ax_1 + By_1 + C|}{\sqrt{A^2 + B^2}}$$

(4) Result (Perpendicular distance)

$$d = PM = \frac{|Ax_1 + By_1 + C|}{\sqrt{A^2 + B^2}}$$

Hence, the perpendicular distance (d) of the line $Ax + By + C = 0$ from the point (x_1, y_1) is

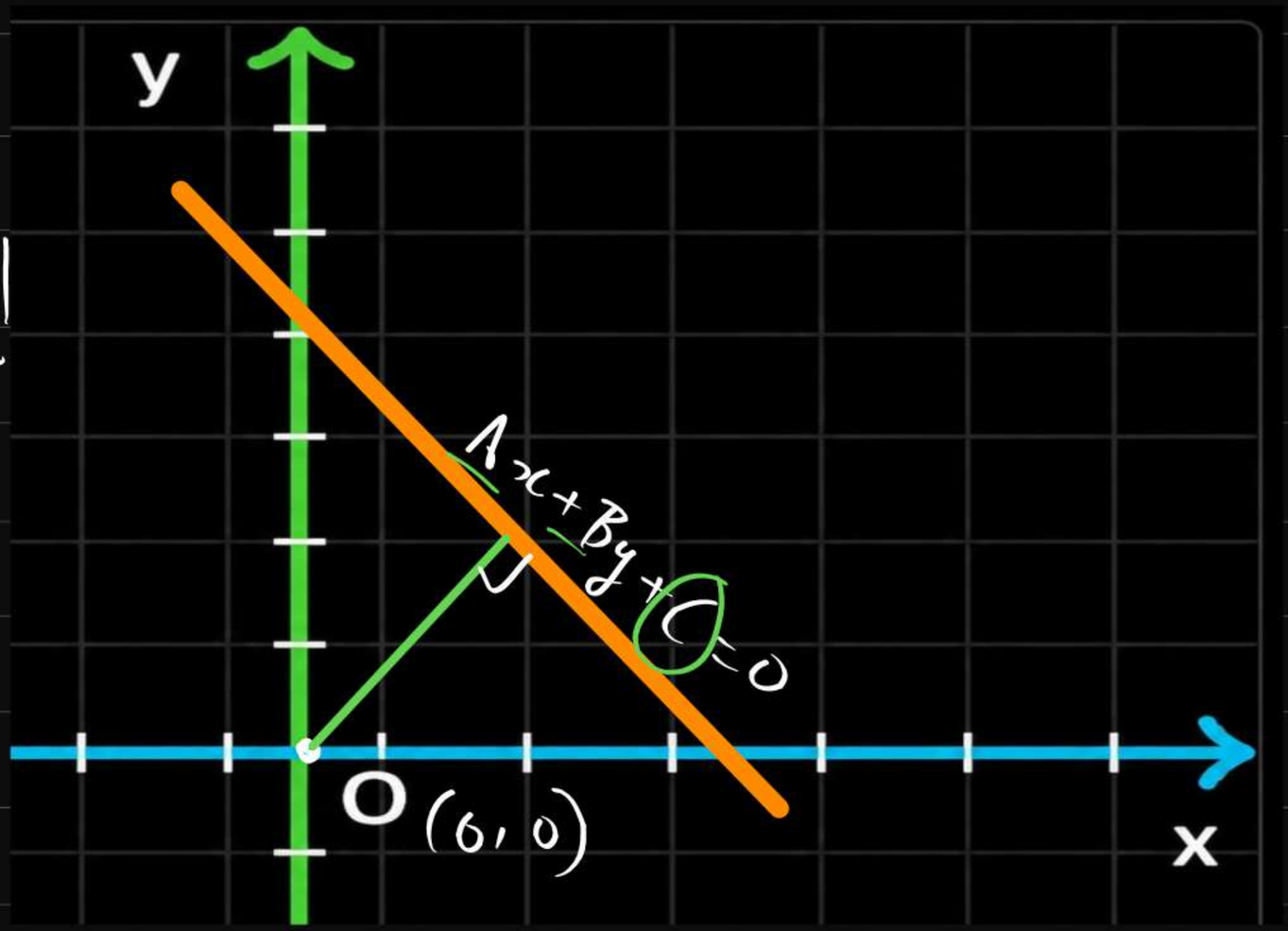
$$d = \frac{|Ax_1 + By_1 + C|}{\sqrt{A^2 + B^2}}$$

Key Points : • Works for any line $Ax + By + C = 0$ • (x_1, y_1) can be anywhere in the plane • Always take absolute value

Distance of a line from origin
मूल से एक पंक्ति की दूरी

$$d = \frac{|Ax_1 + By_1 + C|}{\sqrt{A^2 + B^2}}$$

$$d \Rightarrow \frac{|C|}{\sqrt{A^2 + B^2}}$$



Distance of a line from origin
मूल से एक पंक्ति की दूरी

$$d = \frac{|C|}{\sqrt{A^2 + B^2}}$$

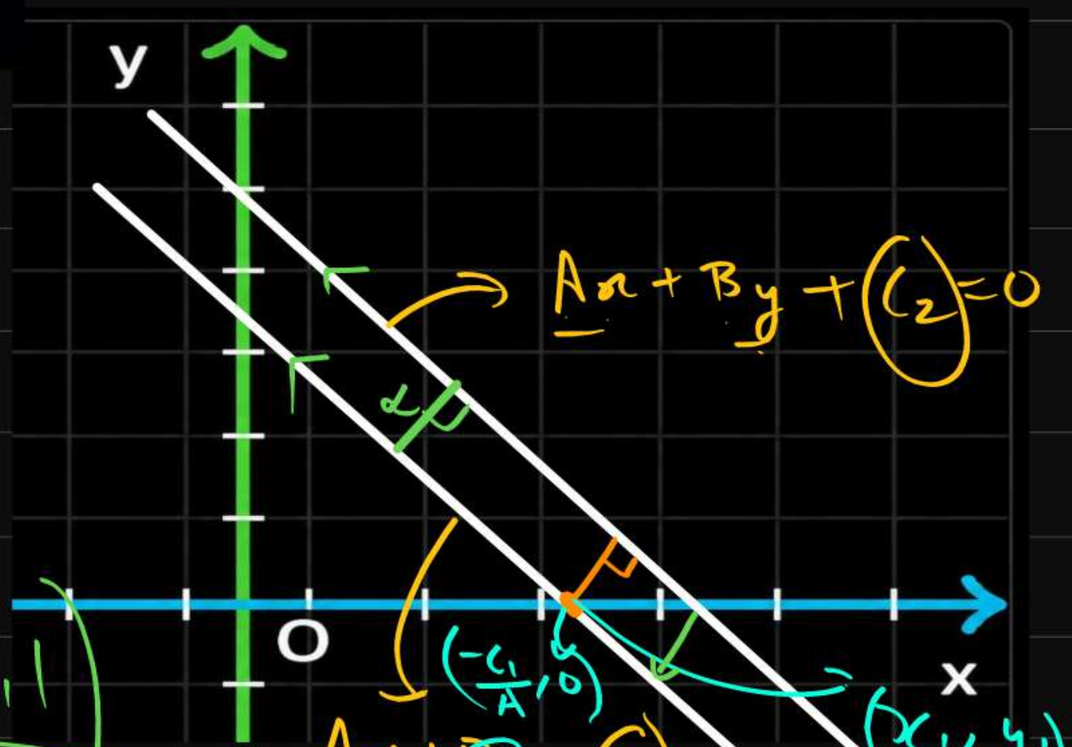
Distance between two parallel lines
दो समानांतर रेखाओं के बीच की दूरी

$$Ax + By + C_1 = 0 \qquad Ax + By + C_2 = 0$$

$$d = \frac{|C_1 - C_2|}{\sqrt{A^2 + B^2}}$$

$$d = \frac{|Ax_1 + By_1 + C_2|}{\sqrt{A^2 + B^2}}$$

$$\Rightarrow \frac{\left|Ax - \frac{C_1}{A} + B(0) + C_2\right|}{\sqrt{A^2 + B^2}} \Rightarrow \frac{|C_2 - C_1|}{\sqrt{A^2 + B^2}}$$



$\because y = 0 \Rightarrow x = -\frac{C_1}{A}$

Distance between two parallel lines
दो समानांतर रेखाओं के बीच की दूरी

Our parallel lines are:-

Distance between two parallel lines
दो समानांतर रेखाओं के बीच की दूरी

$$d = \frac{|\ C_1 - C_2\ |}{\sqrt{A^2 + B^2}}$$

$Ax + By + C = 0$

$$d = PM = \frac{|Ax_1 + By_1 + C|}{\sqrt{A^2 + B^2}}$$

Q. Find the distance of the point P(2,5) from the line 6x + 8y = 12.

बिंदु P(2,5) की रेखा 6x +8y=12 से दुरी ज्ञात करें।

$P(2, 5)$

x_1, y_1

$6x + 8y + (-12) = 0$

A B C

$$d = \frac{|6 \times (2) + 8(5) + (-12)|}{\sqrt{6^2 + 8^2}} = ?$$

$$\Rightarrow \frac{40}{10} = 4$$

(a) 2
(b) 5
(c) 4
(d) 8

$$d = PM = \frac{|Ax_1 + By_1 + C|}{\sqrt{A^2 + B^2}}$$

Q-) Find the distance of the point (3, – 5) from the line $3x - 4y - 26 = 0$

बिंदु (3, – 5) की समीकरण $3x - 4y - 26 = 0$ से दुरी ज्ञात करें

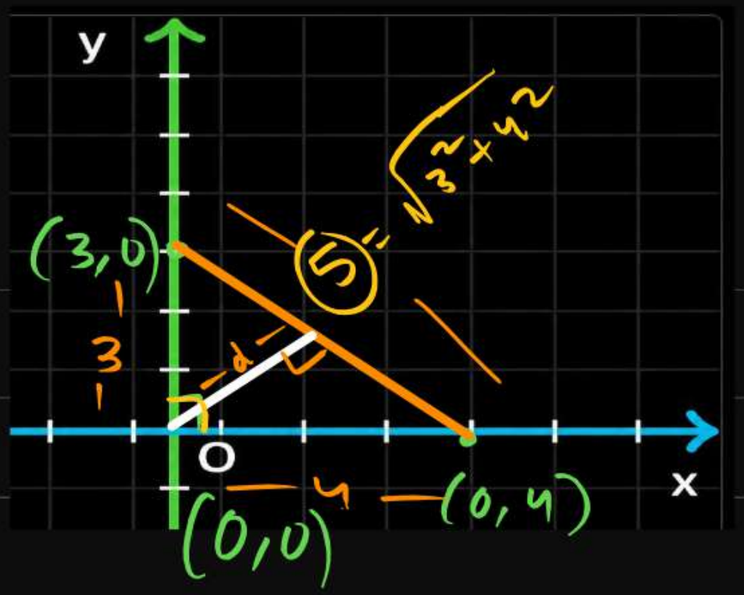
$$d = \frac{|3(3) - 4(-5) - 26|}{\sqrt{3^2 + (-4)^2}}$$

$$\Rightarrow \boxed{\frac{3}{5}}$$

(a) 3/5 unit
(b) 4/5 unit
(c) 5/3 unit
(d) 6/5 unit

Two Methods

$$d = \frac{|C|}{\sqrt{A^2 + B^2}}$$



$\frac{1}{2} \times 4 \times 3 = \frac{1}{2} \times 5 \times d$

$d = \frac{12}{5} = 2.4$

Q. What is the minimum distance from the origin (0,0) to a point on the line segment connecting the points A(3, 0) and B(0, 4)?

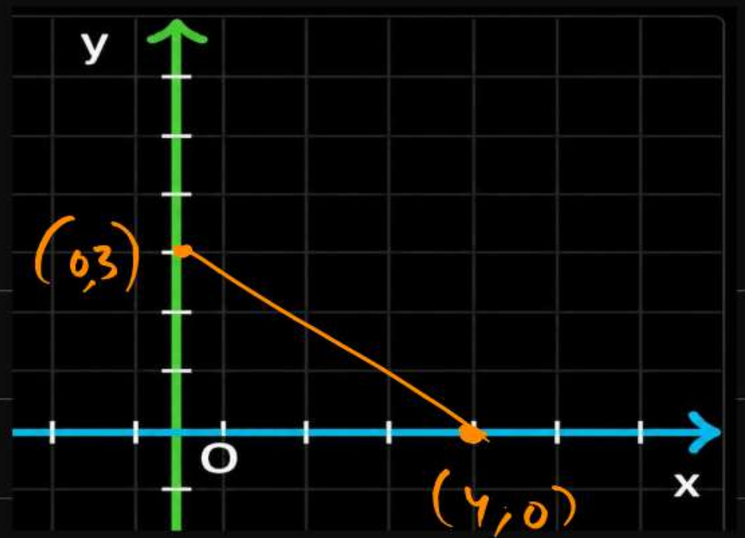
मूलबिंदु (0,0) से बिंदुओं A(3, 0) और B(0, 4) को जोड़ने वाले रेखाखंड पर एक बिंदु तक की न्यूनतम दूरी क्या है?

SSC CGL 2025

(a) (a) 2.4 (b) (b) 3.2
(c) (c) 4.6 (d) (d) 6.8

Two Methods

$$d = \frac{|C|}{\sqrt{A^2 + B^2}}$$



Q. What is the minimum distance from the origin (0,0) to a point on the line segment connecting the points A(3, 0) and B(0, 4)?

मूलबिंदु (0,0) से बिंदुओं A(3, 0) और B(0, 4) को जोड़ने वाले रेखाखंड पर एक बिंदु तक की न्यूनतम दूरी क्या है?

SSC CGL 2025

(a) (a) 2.4 (b) (b) 3.2
(c) (c) 4.6 (d) (d) 6.8

$$\frac{x}{4} + \frac{y}{3} = 1$$

$$3x + 4y = 12$$

$$(3)x + (4)y + (-12) = 0$$

A B C

$$d = \frac{|-12|}{\sqrt{3^2+4^2}} = \frac{12}{5} = 2.4$$

$$d = \frac{|C_1 - C_2|}{\sqrt{A^2 + B^2}}$$

Use this formula only after conversion in the particular form

A B C_1

$8x + 6y + 15 = 0$

$(16x + 12y - 18 = 0) \times \frac{1}{2}$

$8x + 6y - 9 = 0$

C_2

$$d = \frac{|15 - (-9)|}{\sqrt{8^2 + 6^2}} \Rightarrow \frac{24}{10} = 2.4$$

Q) Find the distance between two parallel lines 8x+6y+15=0 and 16x+12y-18=0.

दो समांतर रेखाओं 8x+6y+15=0 और 16x+12y-18=0 के बीच की दुरी ज्ञात करें।

(a) 2.4
(b) 3
(c) 2.1
(d) 4.2

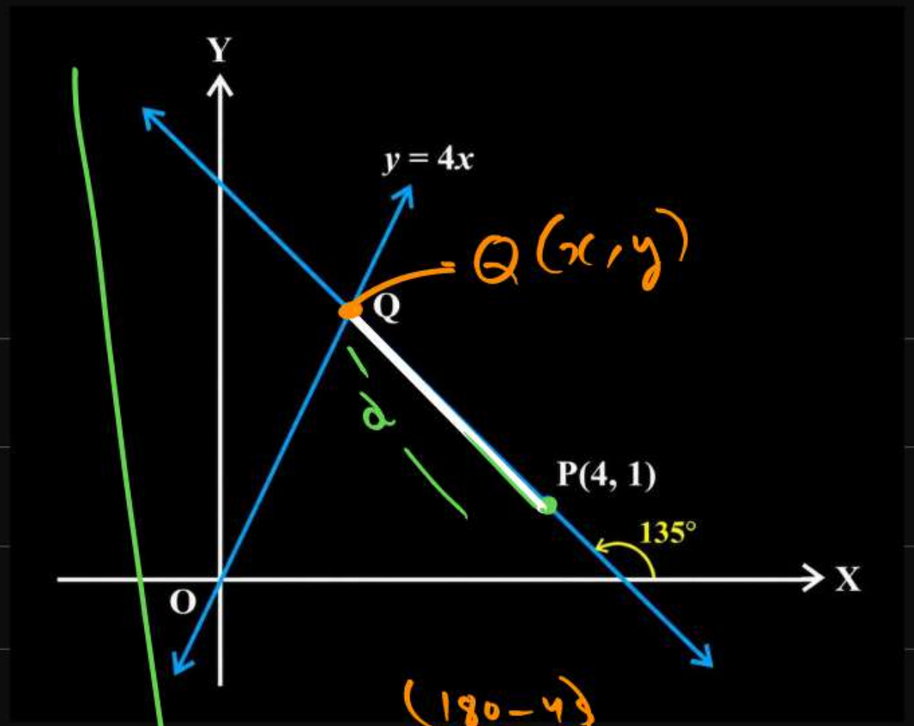
Q-) Find the distance of the line $4x - y = 0$ from the point P (4, 1) measured along the line making an angle of 135° with the positive *x*-axis.

बिंदु P (4, 1) से रेखा 4x - y = 0 की दूरी उस रेखा के अनुदिश ज्ञात कीजिये जो धन x- अक्ष के साथ 135 □ का कोण बनती हो|

A) $2\sqrt{3}$
B) $3\sqrt{2}$
C) 9
D) 4

Q-) Find the distance of the line $4x - y = 0$ from the point P (4, 1) measured along the line making an angle of 135° with the positive x-axis.

बिंदु P (4, 1) से रेखा 4x - y = 0 की दूरी उस रेखा के अनुदिश ज्ञात कीजिये जो धन x- अक्ष के साथ 135 ° का कोण बनती हो|



$y = 4x$

$x + y = 5$

$4x$

$x = 1$

$y = 4$

$(180 - 45)$

$\tan(135) = \frac{y-1}{x-4} = -(\tan 45)$

$4 - x = y - 1$

$x + y = 5$

(1, 4) — d — (4, 1)

$d = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2} = \sqrt{(4-1)^2+(1-4)^2}$

$= 3\sqrt{2}$

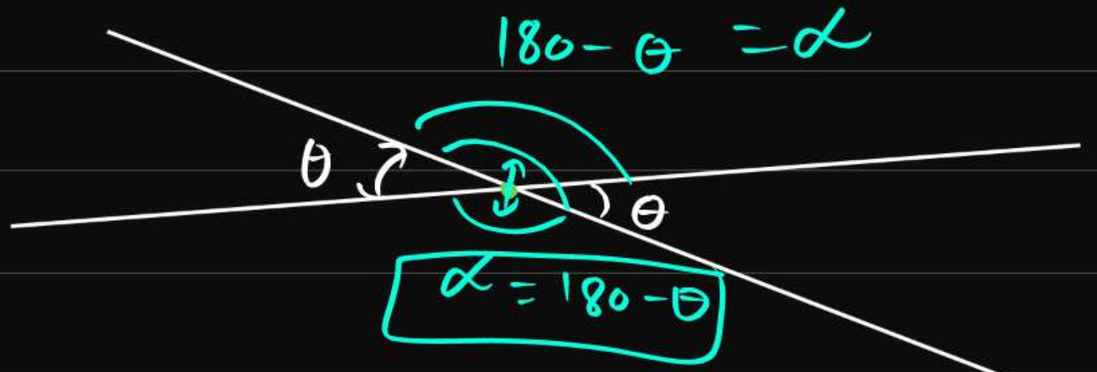
A) $2\sqrt{3}$

B) $3\sqrt{2}$

C) 9

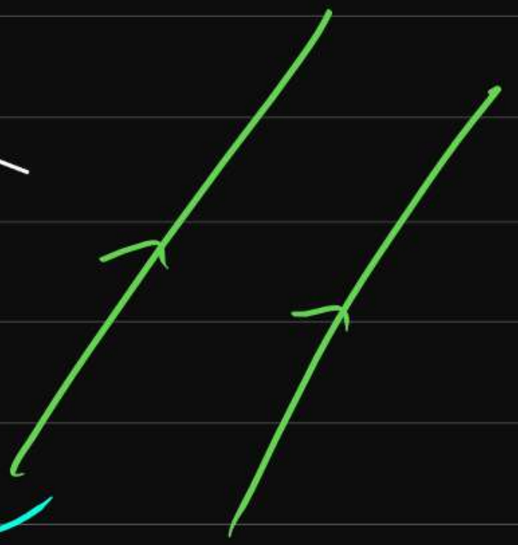
D) 4

Angle between two lines
दो रेखाओं के बीच का कोण

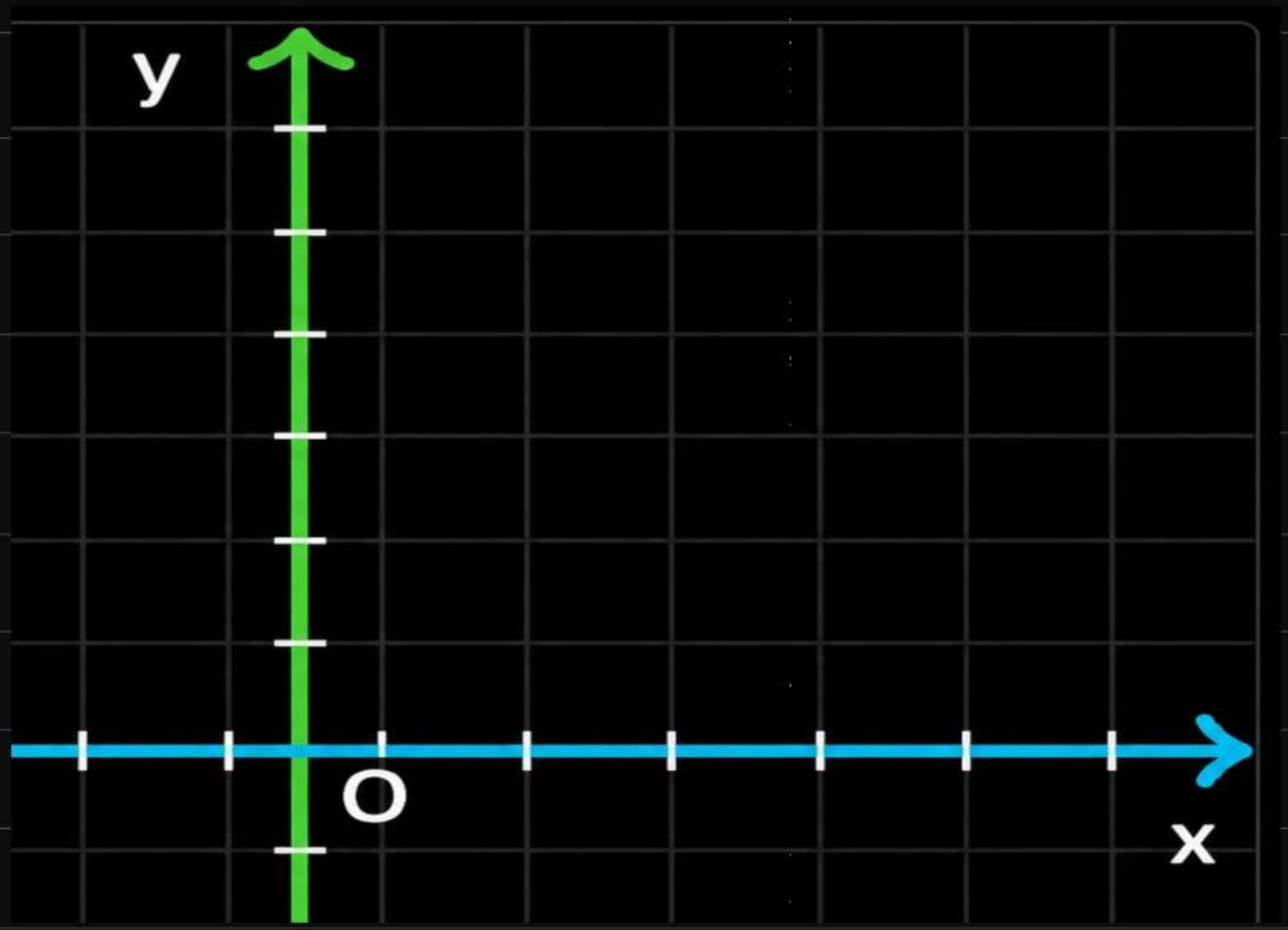


45° ✓

180 - 45 = 135° ✓



Angle between two lines
दो रेखाओं के बीच का कोण



$$\tan(A-B) = \frac{\tan A - \tan B}{1+\tan A \tan B}$$

Angle between two lines

दो रेखाओं के बीच का कोण

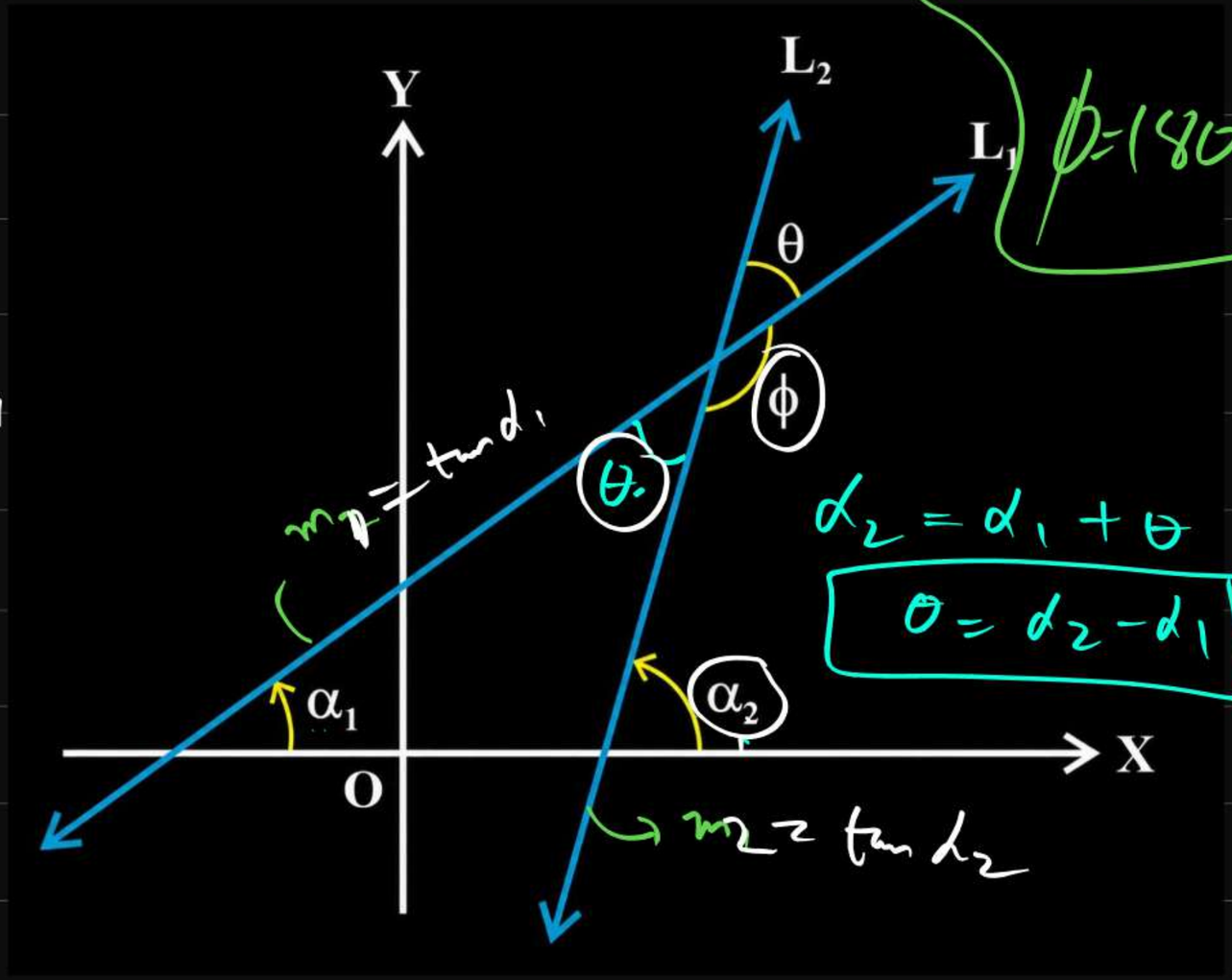
$$0 < \theta < 90$$

$$\phi = 180 - \theta$$

$$\tan\theta = \tan(\alpha_2 - \alpha_1)$$

$$\Rightarrow \frac{\tan\alpha_2 - \tan\alpha_1}{1+\tan\alpha_2 \tan\alpha_1}$$

$$\boxed{\tan\theta \Rightarrow \left|\frac{m_2 - m_1}{1+m_1 m_2}\right|}$$



$$\alpha_2 = \alpha_1 + \theta$$

$$\theta = \alpha_2 - \alpha_1$$

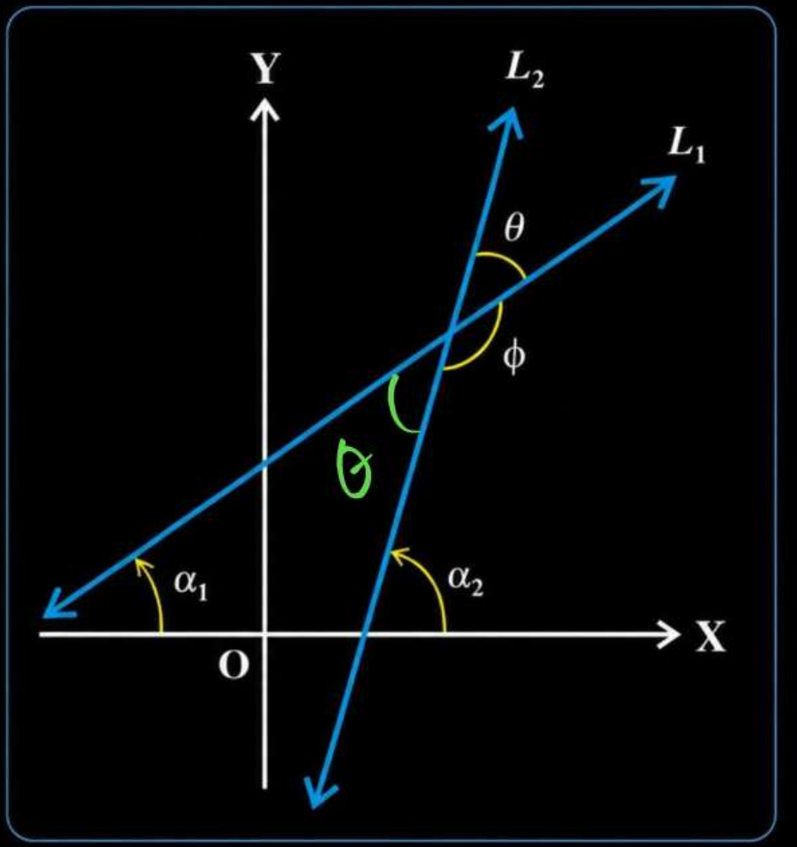
Angle Between Two Lines
दो रेखाओं के बीच का कोण

Let the lines

$L_1 : y = m_1x + c_1$ (slope m_1, angle of inclination α_1)

$L_2 : y = m_2x + c_2$ (slope m_2, angle of inclination α_2)

Let θ be the acute angle between the two lines.



1. Relation between θ and α_1, α_2

- Since L_1 and L_2 intersect each other,

  $\theta = \alpha_2 - \alpha_1$ and $\alpha_1, \alpha_2 \neq 90°$

- Therefore,

$$\tan\theta = \tan(\alpha_2 - \alpha_1) = \frac{\tan\alpha_2 - \tan\alpha_1}{1 + \tan\alpha_1 \tan\alpha_2} = \frac{m_2 - m_1}{1 + m_1m_2} \quad (\text{as } 1 + m_1m_2 \neq 0)$$

- Also, $\phi = 180° - \theta$ (supplementary angle)

$$\tan\phi = \tan(180° - \theta) = -\tan\theta = -\frac{m_2 - m_1}{1 + m_1m_2} \quad (\text{as } 1 + m_1m_2 \neq 0)$$

2. Final Result

$$\tan\theta = \left|\frac{m_2 - m_1}{1 + m_1m_2}\right|, \text{ as } 1 + m_1m_2 \neq 0$$

Here θ is the acute angle between the two lines. (i.e., $0° \leq \theta < 90°$)

★ **Note :** If $1 + m_1m_2 = 0$ (i.e., $m_1m_2 = -1$), the lines are perpendicular and $\theta = 90°$.

θ, α

Q) If θ is the angle between the two lines y = (7 – 4√3)x + 5 and y = (7 + 4√3)x – 9, then find the value of tan θ.

यदि θ दो रेखाओं y = (7 – 4√3)x + 5 और y = (7 + 4√3)x – 9 के बीच का कोण है, तो tan θ का मान ज्ञात कीजिए।

(a) √3,- √3
(b) 1,-1
(c) 4√3,-4 √3
(d) 0

$y = mx + c$

$y = (7 - 4\sqrt{3})x + 5$. $y = (7 + 4\sqrt{3})x - 9$

m_1 m_2

$\tan(180 - \theta) = -\tan\theta$

$\tan\theta = \left|\frac{m_2 - m_1}{1 + m_1 m_2}\right|$

$7^2 - (4\sqrt{3})^2$

$49 - 48$

$\Rightarrow 1$

$\Rightarrow \left|\frac{(7+4\sqrt{3}) - (7-4\sqrt{3})}{1 + (7+4\sqrt{3})(7-4\sqrt{3})}\right|$

$\Rightarrow \left|\frac{8\sqrt{3}}{1+1}\right| \Rightarrow |4\sqrt{3}| = 4\sqrt{3}$

$+4\sqrt{3}$

$-4\sqrt{3}$

Q-) Find the angle between the lines $y - \sqrt{3}x - 5 = 0$ and $\sqrt{3}y + x + 6 = 0$.

रेखाओं $y - \sqrt{3}x - 5 = 0$ और $\sqrt{3}y + x + 6 = 0$ के बीच का कोण ज्ञात कीजिए।

$y - \sqrt{3}x - 5 = 0 \qquad \sqrt{3}y + x + 6 = 0.$

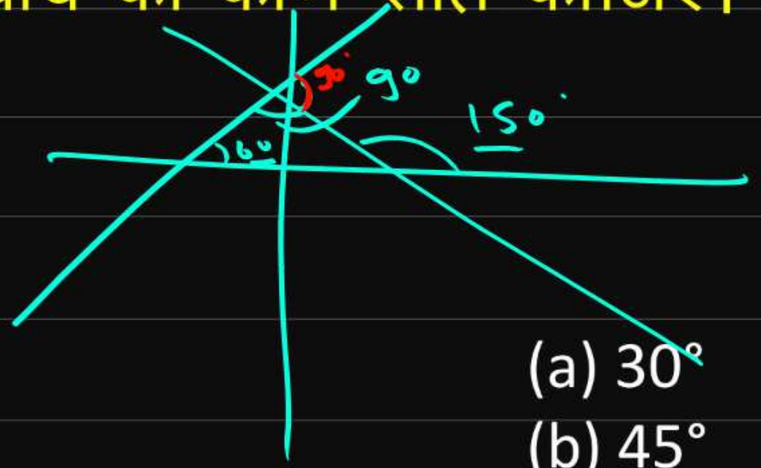
$y = \sqrt{3}x + 5 \qquad y = \frac{-1}{\sqrt{3}}x - \frac{6}{\sqrt{3}}$

$m_1 = \tan 60^\circ \qquad m_2 \Rightarrow \tan 150^\circ$

$$\tan\theta = \left|\frac{\sqrt{3} - \left(\frac{-1}{\sqrt{3}}\right)}{1 + \sqrt{3} \times \left(\frac{-1}{\sqrt{3}}\right)}\right| \Rightarrow$$

$0 = 1 + m_1 m_2 \Rightarrow$ Perpendicular lines

$\therefore \theta = 90^\circ$

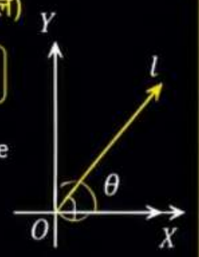


(a) 30°
(b) 45°
(c) 60°
(d) 90°

ALL FORMULAS RELATED TO SLOPE OF A LINE रेखा की ढाल से संबंधित सभी सूत्र

1 SLOPE USING ANGLE OF INCLINATION (झुकाव कोण से ढाल)

$$m = \tan\theta$$



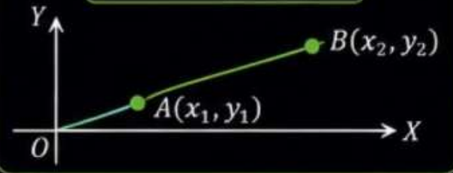
θ = angle made by the line with positive x-axis, measured anticlockwise.

θ = रेखा द्वारा धनात्मक x-अक्ष के साथ बनाया गया कोण (वामावर्ती मापा गया)।

2 SLOPE USING TWO POINTS (दो बिंदुओं से ढाल)

If $A(x_1, y_1)$ and $B(x_2, y_2)$ lie on a line, then

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$



3 SLOPE FROM GENERAL EQUATION (सामान्य समीकरण से ढाल)

For $Ax + By + C = 0$

$$m = -\frac{A}{B}$$

(Note: $B \neq 0$, otherwise slope is not defined.)

4 SLOPE FROM $y = mx + c$ ($y = mx + c$ से ढाल)

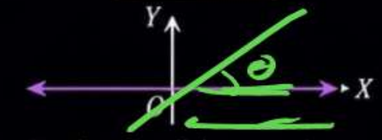
$$y = mx + c$$

Here, coefficient of x is the slope.

$$\text{Slope } (m) = m$$

5 SLOPE OF X-AXIS (x-अक्ष की ढाल)

$$m = 0$$



Any line parallel to x-axis has slope 0.

x-अक्ष के समानांतर किसी भी रेखा की ढाल 0 होती है।

6 SLOPE OF Y-AXIS (y-अक्ष की ढाल)

$$m = \text{Not Defined}$$

Any line parallel to y-axis has slope not defined.

y-अक्ष के समानांतर किसी भी रेखा की ढाल परिभाषित नहीं होती।

7 PARALLEL LINES (समांतर रेखाएं)

$$m_1 = m_2$$

If two lines are parallel, their slopes are equal.

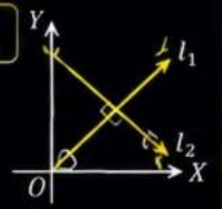
यदि दो रेखाएं समांतर हों, तो उनकी ढाल समान होती है।

8 PERPENDICULAR LINES (लम्बवत् रेखाएं)

$$m_1 m_2 = -1$$

or

$$m_2 = \frac{-1}{m_1}$$



If two lines are perpendicular, product of their slopes $= -1$.

यदि दो रेखाएं लम्बवत् हों, तो उनकी ढालों का गुणनफल -1 होता है।

9 ANGLE BETWEEN TWO LINES (दो रेखाओं के बीच कोण)

If slopes are m_1 and m_2, then the angle θ between them is

$$\tan\theta = \left|\frac{m_2 - m_1}{1 + m_1 m_2}\right|$$

θ = acute angle between the two lines.

θ = दोनों रेखाओं के बीच न्यून (तीव्र) कोण।

10 DIRECTED ANGLE BETWEEN TWO LINES (निर्देशित कोण)

For directed angle θ,

$$\tan\theta = \frac{m_2 - m_1}{1 + m_1 m_2}$$

Sign of θ depends on direction.

θ का चिह्न दिशा पर निर्भर करता है।

$$A_1A_2 + B_1B_2 = 0$$

Equation of a tangent to a circle if centre of a circle

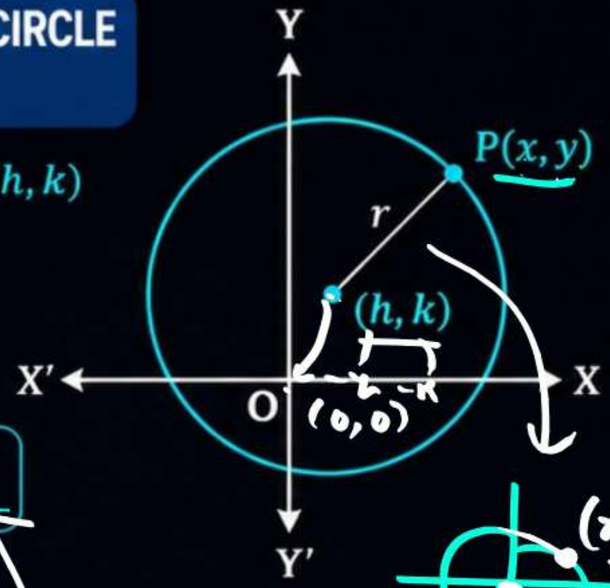
Latest Questions

1. STANDARD EQUATION OF A CIRCLE
वृत्त का मानक समीकरण

The equation of a circle with centre (h, k) and radius r is

जिस वृत्त का केंद्र (h, k) तथा त्रिज्या r हो, उसका समीकरण है

$$(x-h)^2 + (y-k)^2 = r^2$$



Where | जहाँ

(h, k)	= Coordinates of centre	(h, k)	= केंद्र के निर्देशांक
r	= Radius of the circle	r	= वृत्त की त्रिज्या

KEY POINTS | मुख्य बातें

- If centre is $(0, 0)$, equation becomes $x^2 + y^2 = r^2$
- All points on the circle are at a distance r from the centre.

- यदि केंद्र $(0, 0)$ हो, तो समीकरण होगा $x^2 + y^2 = r^2$
- वृत्त पर स्थित सभी बिंदु केंद्र से r दूरी पर होते हैं।

1. STANDARD EQUATION OF SPHERE
गोले का मानक समीकरण

The equation of a sphere with centre (h, k, l) and radius r is

जिस गोले का केंद्र (h, k, l) और त्रिज्या r हो, उसका समीकरण है

$$(x-h)^2+(y-k)^2+(z-l)^2=r^2$$

$r^2 = (x-h)^2+(y-k)^2+(z-l)^2$

Where | जहाँ

(h, k, l)	=	Coordinates of centre	(h, k, l)	=	केंद्र के निर्देशांक
r	=	Radius of the sphere	r	=	गोले की त्रिज्या

KEY POINTS | मुख्य बातें

- If centre is $(0, 0, 0)$, equation becomes

$$x^2+y^2+z^2=r^2$$

- All points on the sphere are at a distance r from the centre.

- यदि केंद्र $(0, 0, 0)$ हो, तो समीकरण होगा

$$x^2+y^2+z^2=r^2$$

- गोले पर स्थित सभी बिंदु केंद्र से r दूरी पर होते हैं।

If the lines L1 and L2 are perpendicular

Relation in Slopes

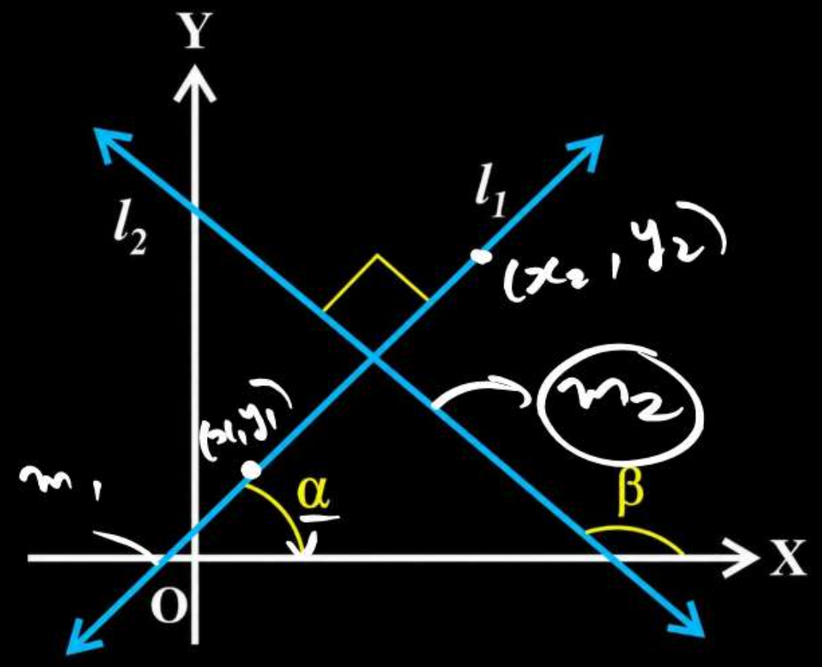
$m_1 = \tan(\alpha)$

$m_2 = \tan\beta = \tan(\alpha + 90°)$

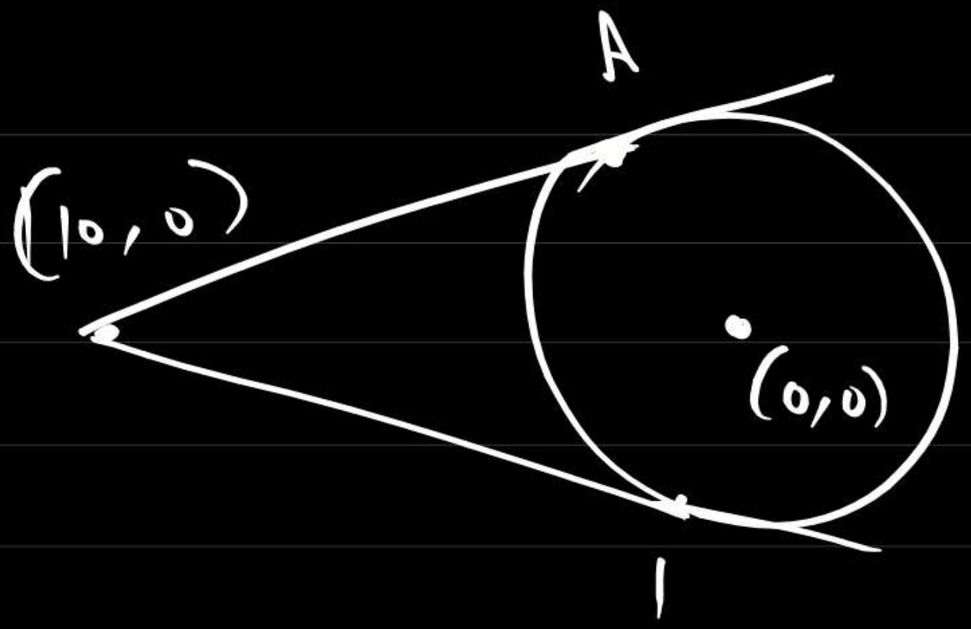
$m_2 = -\cot\alpha$

$m_2 = -1/m_1$

$m_1 m_2 = -1$



$m_1 = \tan\alpha = \dfrac{y_2 - y_1}{x_2 - x_1}$



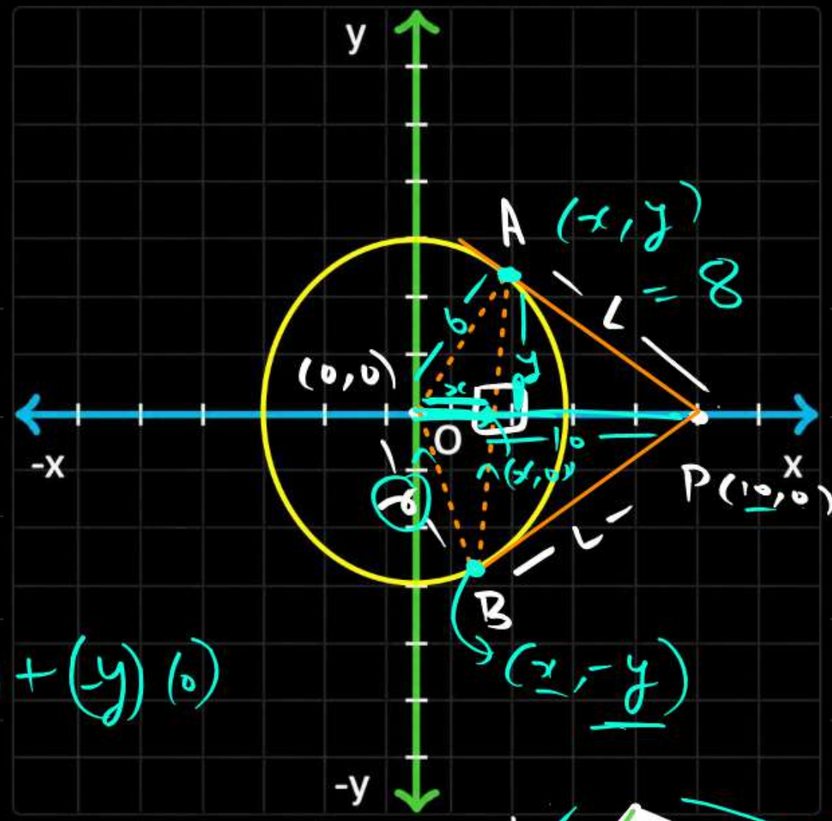
Q. A satellite control station is located at point P(10, 0). It needs to establish two signal beams that just graze the edge of a circular satellite dish centered at the origin O(0, 0) with a radius of 6 units. These beams will touch the dish at exactly two points — call them A and B. Find the coordinates of the points of contact A and B of the tangents from P(10, 0) to the circle.

एक उपग्रह नियंत्रण स्टेशन बिंदु P(10, 0) पर स्थित है। इसे दो सिग्नल किरणें स्थापित करनी हैं, जो मूल बिंदु O(0, 0) पर केंद्रित तथा 6 इकाई त्रिज्या वाली एक वृत्ताकार सैटेलाइट डिश के किनारे को स्पर्श करती हुई गुजरती हैं। ये किरणें डिश को ठीक दो बिंदुओं पर स्पर्श करेंगी — इन्हें A और B कहते हैं। बिंदु P(10, 0) से वृत्त पर खींची गई स्पर्श रेखाओं के स्पर्श बिंदु A और B के निर्देशांक ज्ञात कीजिए।

Delhi Police Head Constable Ministerial 2025

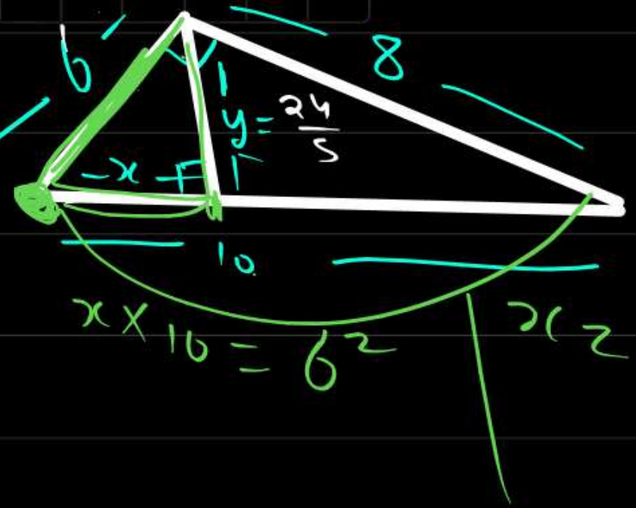
25/7, -17/7
18/5, -24/5
19/2, -23/2
13/4, -17/4

Q. A satellite control station is located at point P(10, 0). It needs to establish two signal beams that just graze the edge of a circular satellite dish centered at the origin O(0, 0) with a radius of 6 units. These beams will touch the dish at exactly two points — call them A and B. Find the coordinates of the points of contact A and B of the tangents from P(10, 0) to the circle.



$6^2 = x(10) + (y)(0)$

$\frac{18}{5} = \frac{36}{10} = x$



$\frac{1}{2} \times 6 \times 8 = \frac{1}{2} \times 10 \times y$

$\frac{24}{5} = y$

$x \times 10 = 6^2$

$x = \frac{18}{5}$

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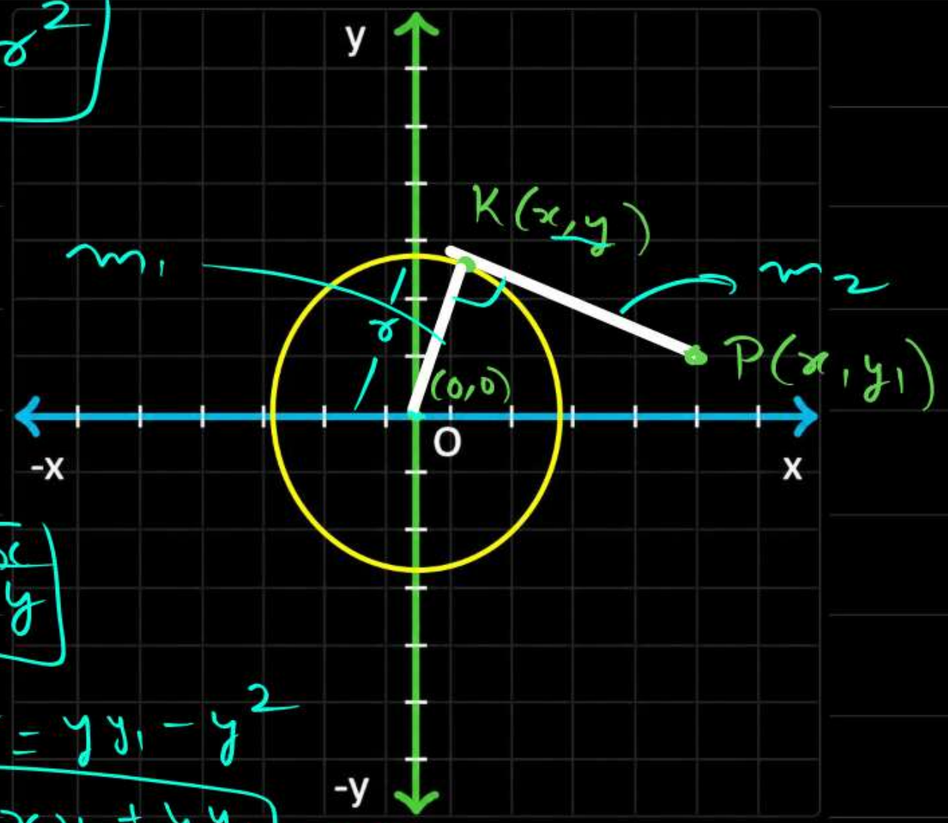
25/7, -17/7
18/5, -24/5 ✓
19/2, -23/2
13/4, -17/4

Equation of a tangent to a circle if centre of a circle is at origin

$x^2+y^2=r^2$ ⇓ Circle ; $xx_1+yy_1=r^2$ Tangent



$$m_1=\frac{y-0}{x-0}=\frac{y}{x}$$

$$m_2=\frac{-1}{m_1}=\frac{-1}{y/x}=\boxed{\frac{-x}{y}}$$

$$\frac{-x}{y}=\frac{y_1-y}{x_1-x}\ ;\ -xx_1+x^2=yy_1-y^2$$

$$\boxed{x^2+y^2=xx_1+yy_1}=r^2$$

1. Circle with centre at origin

For the circle

$$\boxed{x^2 + y^2 = r^2}$$

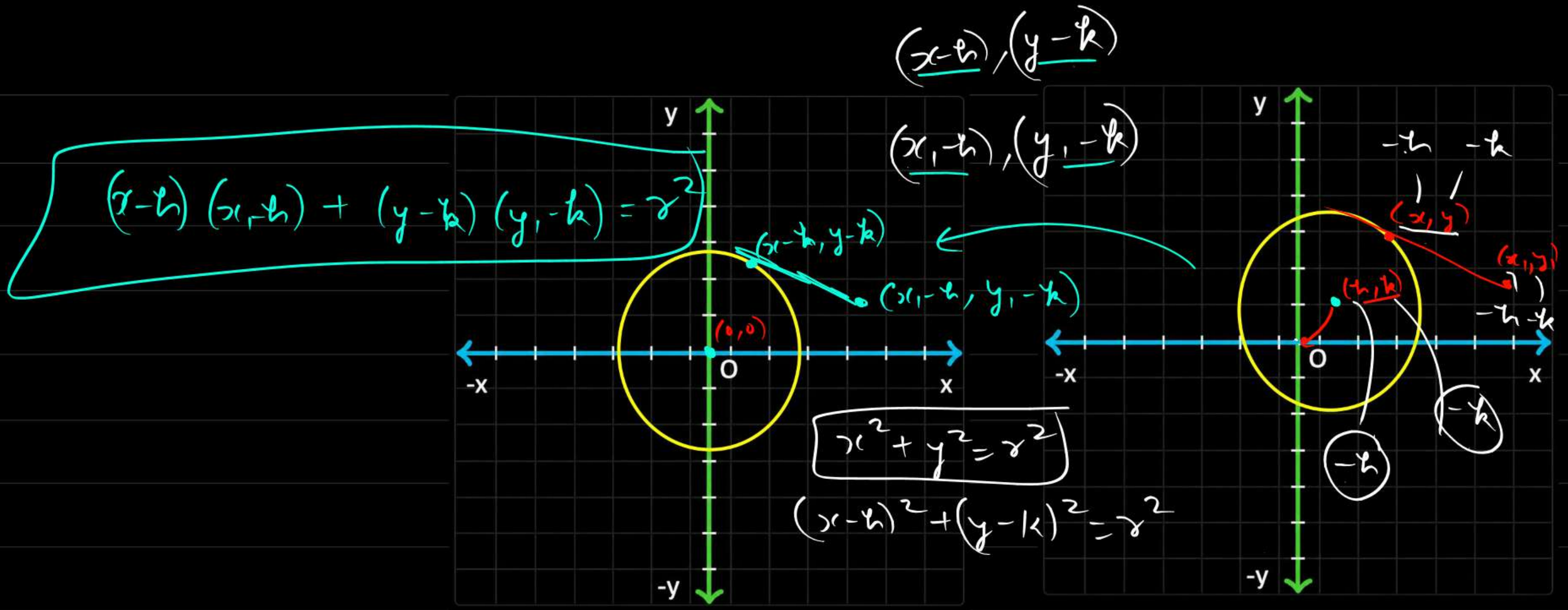
the equation of tangent at a point $P(x_1, y_1)$ on the circle is:

$$\boxed{xx_1 + yy_1 = r^2}$$

Shortcut: Replace $x^2 \to xx_1$ and $y^2 \to yy_1$.

Let the point on the circle be (x,y) and the external point be (x1,y1)
Use slope method to prove

Equation of a tangent to a circle, if centre is not at origin.



3. Circle with centre (h, k)

For

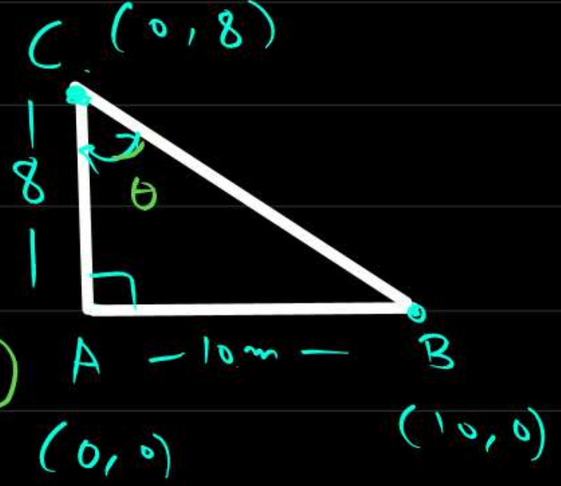
$$(x-h)^2 + (y-k)^2 = r^2$$

tangent at $P(x_1, y_1)$:

$$\boxed{(x-h)(x_1-h) + (y-k)(y_1-k) = r^2}$$

Q. An engineer is designing a solar panel mount in the shape of triangle ABC, where point A is at the base of a building, point B is 10 meters east of A, on the ground, and point C is a point on the rooftop directly above A, 8 meters high. To improve sunlight exposure, the engineer wants to calculate the angle at point C, formed between the rooftop and the inclined panel from B to C. Find the angle at point C (i.e., angle BCA) in triangle ABC, where A is (0, 0), B is (10, 0), and C is (0, 8).

एक इंजीनियर त्रिभुज ABC के आकार में एक सोलर पैनल माउंट डिज़ाइन कर रहा है, जहाँ बिंदु A भवन के आधार पर है, बिंदु B, A से 10 मीटर पूर्व दिशा में भूमि पर स्थित है, और बिंदु C छत पर A के ठीक ऊपर 8 मीटर की ऊँचाई पर स्थित है। सूर्य के प्रकाश के बेहतर उपयोग के लिए, इंजीनियर बिंदु C पर बनने वाले उस कोण की गणना करना चाहता है, जो छत और B से C तक जाने वाले झुके हुए पैनल के बीच बनता है। दिए गए निर्देशांकों A(0, 0), B(10, 0) और C(0, 8) के अनुसार, त्रिभुज ABC में कोण BCA ज्ञात कीजिए।



$\tan\theta = \frac{10}{8}$

$= \frac{5}{4} = 1.25$

$\theta = \tan^{-1}(1.25)$

Delhi Police Head Constable Ministerial 2025

tan	0°	30°	45°	60°	90°
	0	$\frac{1}{1.73}$	1	1.73	∞

1.25

(a) 51.3°

(b) 35.4°

(c) 40°

(d) 61.3°

Co-ordinate Geometry

निर्देशांक ज्यामिति

(Basic to high Level)

Latest Pattern

Part-6

(Reflection of a point.)
&
Different Centres of a triangle.

Summary
सारांश

- Slope (m) of a non-vertical line passing through the points (x_1, y_1) and (x_2, y_2) is given by

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{y_1 - y_2}{x_1 - x_2}, \quad x_1 \neq x_2.$$

- दो बिंदुओं (x_1, y_1) और (x_2, y_2) से गुजरने वाली गैर-ऊर्ध्वाधर रेखा का ढाल (m) निम्न प्रकार दिया जाता है।

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{y_1 - y_2}{x_1 - x_2}, \quad x_1 \neq x_2.$$

- If a line makes an angle $á$ with the positive direction of x-axis, then the slope of the line is given by

$$m = \tan á, \quad á \neq 90°.$$

- यदि कोई रेखा x-अक्ष की धनात्मक दिशा के साथ $á$ कोण बनाती है, तो रेखा का ढाल निम्न प्रकार दिया जाता है।

$$m = \tan á, \quad á \neq 90°.$$

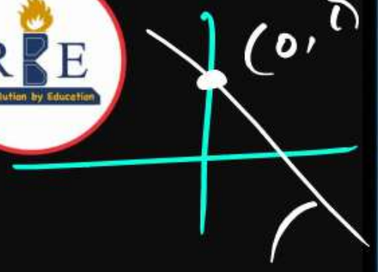
- Slope of horizontal line is zero and slope of vertical line is undefined.

Horizontal line	$\Rightarrow$	$m = 0$
Vertical line	$\Rightarrow$	m = undefined

- क्षैतिज रेखा का ढाल शून्य होता है तथा ऊर्ध्वाधर रेखा का ढाल अपरिभाषित होता है।

क्षैतिज रेखा	$\Rightarrow$	$m = 0$
ऊर्ध्वाधर रेखा	$\Rightarrow$	m = अपरिभाषित

ENGLISH	हिंदी
An acute angle (say θ) between lines L_1 and L_2 with slopes m_1 and m_2 is given by $\tan\theta = \left\lvert \frac{m_2 - m_1}{1 + m_1 m_2} \right\rvert$, $1 + m_1 m_2 \neq 0$.	दो रेखाओं L_1 और L_2 जिनके ढाल m_1 और m_2 हैं, के बीच का न्यून कोण (मान लें θ) निम्न द्वारा दिया जाता है $\tan\theta = \left\lvert \frac{m_2 - m_1}{1 + m_1 m_2} \right\rvert$, $1 + m_1 m_2 \neq 0$.
Two lines are *parallel* if and only if their slopes are equal.	दो रेखाएँ *समान्तर (parallel)* तभी और केवल तभी होंगी जब उनके ढाल *बराबर* (equal) हों।
Two lines are *perpendicular* if and only if product of their slopes is -1.	दो रेखाएँ *लम्बवत् (perpendicular)* तभी और केवल तभी होंगी जब उनके ढालों का गुणनफल -1 हो।
Three points A, B and C are collinear, if and only if slope of AB = slope of BC.	तीन बिंदु A, B और C *सहरेखीय (collinear)* तभी और केवल तभी होंगे जब AB का ढाल = BC का ढाल हो।
Equation of the *horizontal line* having distance a from the x-axis is either $y = a$ or $y = -a$.	x-अक्ष से दूरी a रखने वाली *क्षैतिज रेखा (horizontal line)* का समीकरण या तो $y = a$ या $y = -a$ होता है।
Equation of the *vertical line* having distance b from the y-axis is either $x = b$ or $x = -b$.	y-अक्ष से दूरी b रखने वाली *ऊर्ध्वाधर रेखा (vertical line)* का समीकरण या तो $x = b$ या $x = -b$ होता है।
The point (x, y) lies on the line with slope m and through the fixed point (x_0, y_0), if and only if its coordinates satisfy the equation $y - y_0 = m(x - x_0)$.	बिंदु (x, y) उस रेखा पर स्थित है जिसका ढाल m है और जो स्थिर बिंदु (x_0, y_0) से होकर जाती है, तभी और केवल तभी जब इसके निर्देशांक समीकरण $y - y_0 = m(x - x_0)$ को संतुष्ट करें।
Equation of the line passing through the points (x_1, y_1) and (x_2, y_2) is given by $y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$.	बिंदुओं (x_1, y_1) और (x_2, y_2) से होकर जाने वाली रेखा का समीकरण निम्न द्वारा दिया जाता है $y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$.



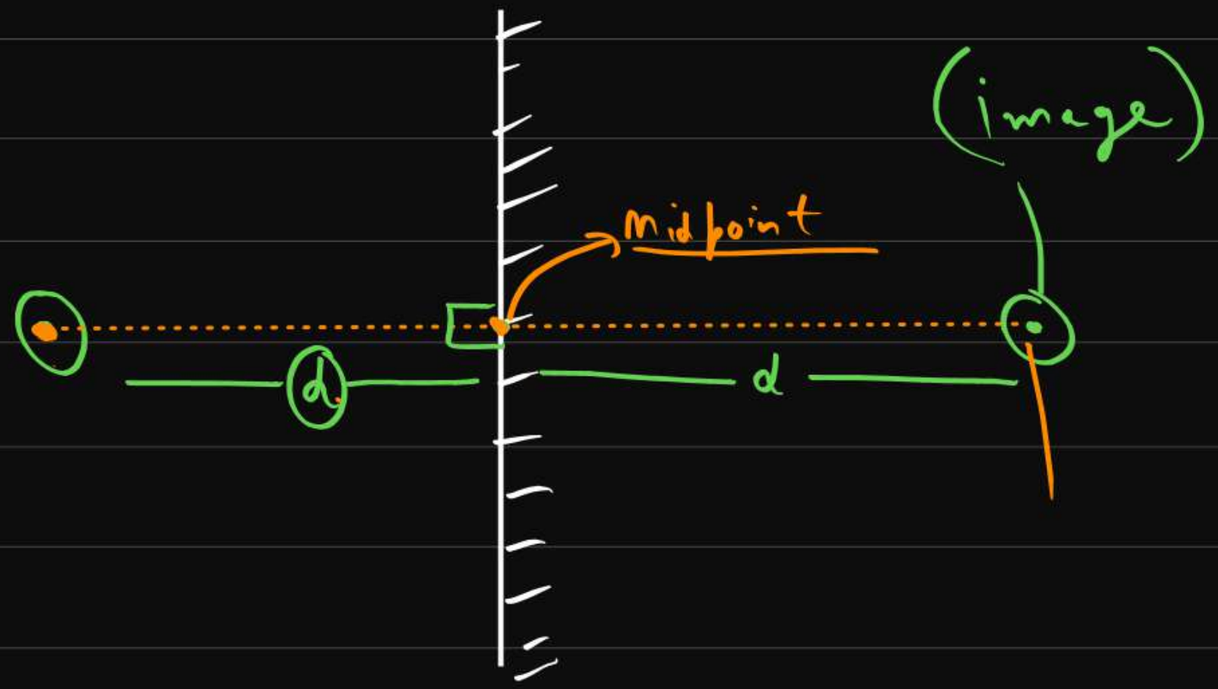
◆ The point (x, y) on the line with slope m and y-intercept c lies on the line if and only if $y = mx + c$.	◆ बिंदु (x, y) जो ढाल m तथा y-अवरोध c वाली रेखा पर स्थित हो, उसी स्थिति में उस रेखा पर होगा जब और केवल जब $y = mx + c$.
◆ If a line with slope m makes x-intercept d. Then equation of the line is $y = m(x - d)$.	◆ यदि ढाल m वाली रेखा x-अवरोध d बनाती है, तब रेखा का समीकरण है । $y = m(x - d)$.
◆ Equation of a line making intercepts a and b on the x-and y-axis, respectively, is $\frac{x}{a} + \frac{y}{b} = 1$.	◆ जो रेखा x-अक्ष तथा y-अक्ष पर क्रमशः a और b अवरोध बनाती है, उसका समीकरण है $\frac{x}{a} + \frac{y}{b} = 1$.
◆ The equation of the line having normal distance from origin p and angle between normal and the positive x-axis ω is given by $x \cos\omega + y \sin\omega = p$.	◆ मूल बिंदु से लंबवत दूरी p तथा लंब (नॉर्मल) और धनात्मक x-अक्ष के बीच कोण ω वाली रेखा का समीकरण है $x \cos\omega + y \sin\omega = p$.
◆ Any equation of the form $Ax + By + C = 0$, with A and B are not zero, simultaneously, is called the *general linear equation* or *general equation of a line.*	◆ रूप $Ax + By + C = 0$ का कोई भी समीकरण, जहाँ A और B एक साथ शून्य न हों, समीकालीन (simultaneously), *सामान्य रेखीय समीकरण* या *रेखा का सामान्य समीकरण* कहलाता है।
◆ The perpendicular distance (d) of a line $Ax + By + C = 0$ from a point (x_1, y_1) is given by $d = \frac{\lvert Ax_1 + By_1 + C\rvert}{\sqrt{A^2 + B^2}}$.	◆ बिंदु (x_1, y_1) से रेखा $Ax + By + C = 0$ की लंबवत दूरी (d) है $d = \frac{\lvert Ax_1 + By_1 + C\rvert}{\sqrt{A^2 + B^2}}$.
◆ Distance between the parallel lines $Ax + By + C_1 = 0$ and $Ax + By + C_2 = 0$, is given by $d = \frac{\lvert C_1 - C_2\rvert}{\sqrt{A^2 + B^2}}$.	◆ समांतर रेखाओं $Ax + By + C_1 = 0$ और $Ax + By + C_2 = 0$ के बीच की दूरी (d) है $d = \frac{\lvert C_1 - C_2\rvert}{\sqrt{A^2 + B^2}}$.

Reflection of point

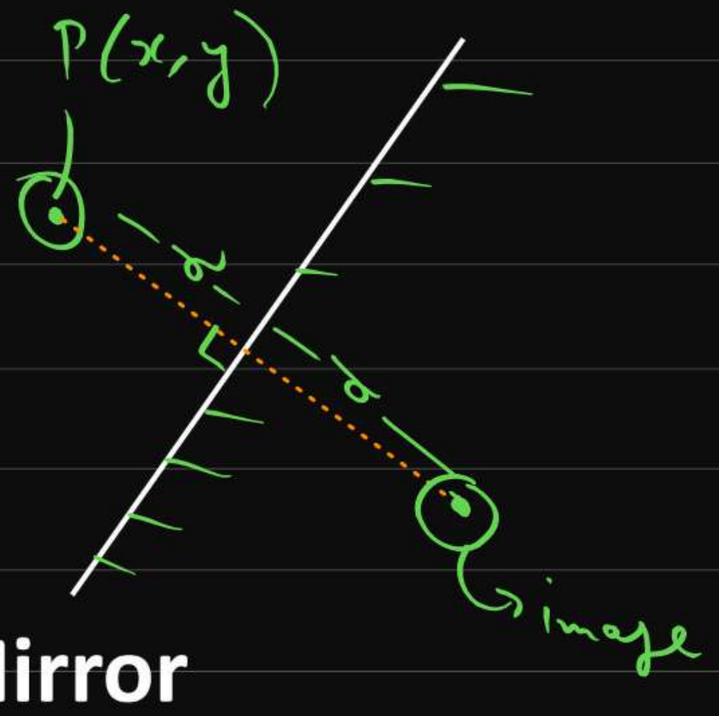
बिंदु का प्रतिबिंब

Reflection of point

बिंदु का प्रतिबिंब

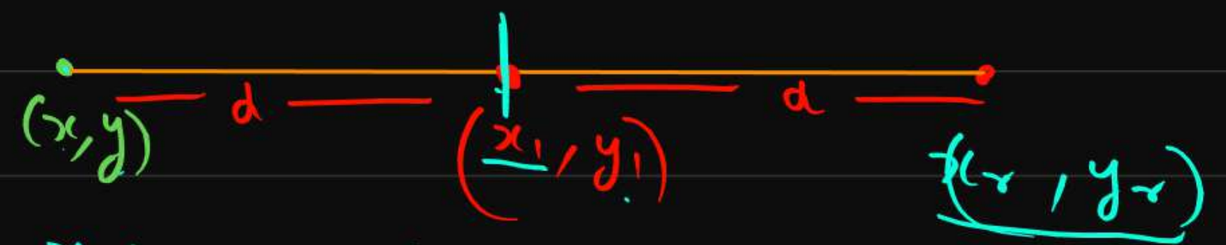


Reflection of point
बिंदु का प्रतिबिंब



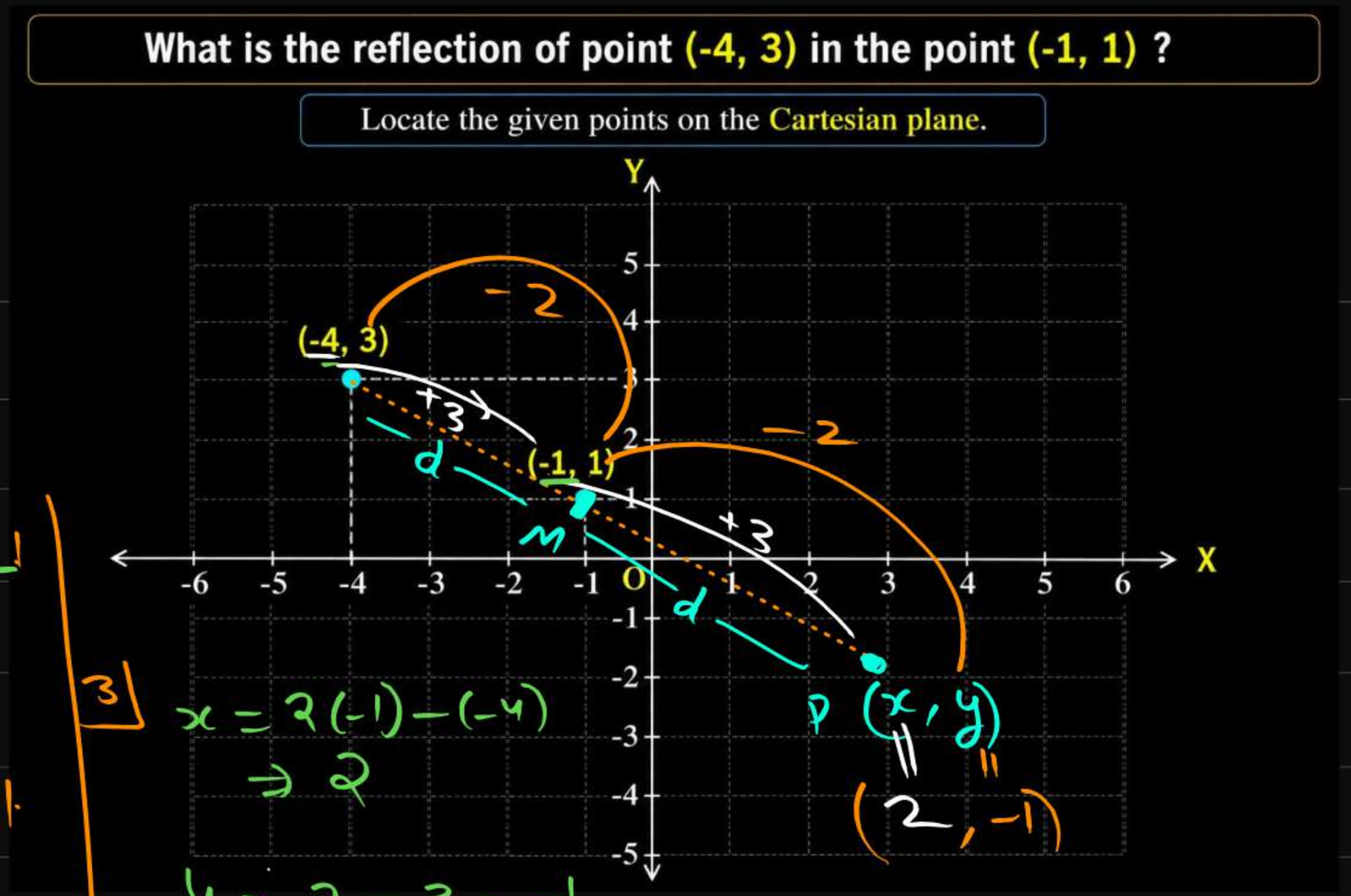
Reflection of point
बिंदु का प्रतिबिंब

Across a point



$$\frac{x + x_2}{2} = x_1$$

$$x_2 = 2x_1 - x \qquad y_2 = 2y_1 - y$$



2] $\frac{x+(-4)}{2} = -1$

$x = 2$

$\frac{y+3}{2} = 1$

$y = -1$

3] $x = 2(-1) - (-4)$

$\Rightarrow 2$

$y = 2(1) - 3 = -1$

$(2, 1)$

Q. What is the reflection of point (-4, 3) in the point (-1,1) ?

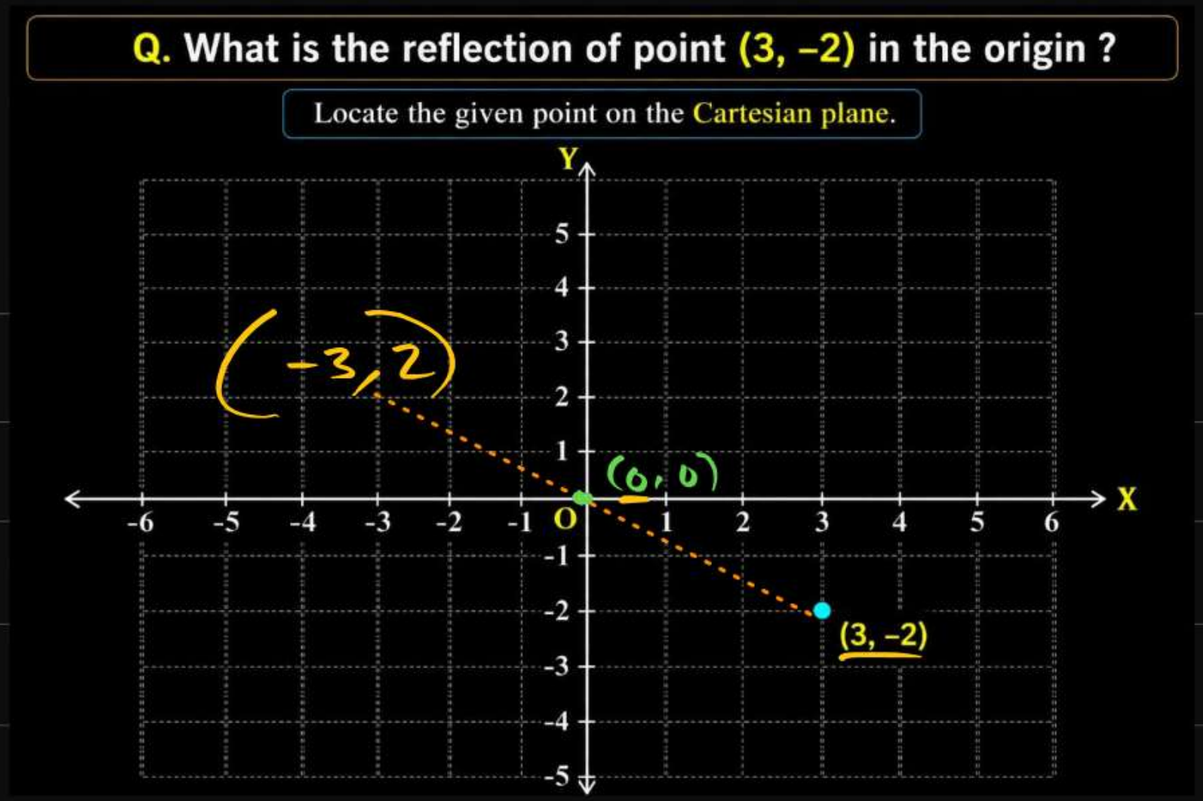
(-1,1) बिंदु में बिंदु (-4, 3) का प्रतिबिम्ब क्या है?

$-2 + 4 = 2$

$2 - 3 = -1$

(A) (2,-1)
(B) (2, −2)
(C) (3, 12)
(D) (3, -1)

Q. What is the reflection of point (3, –2) in the origin ?



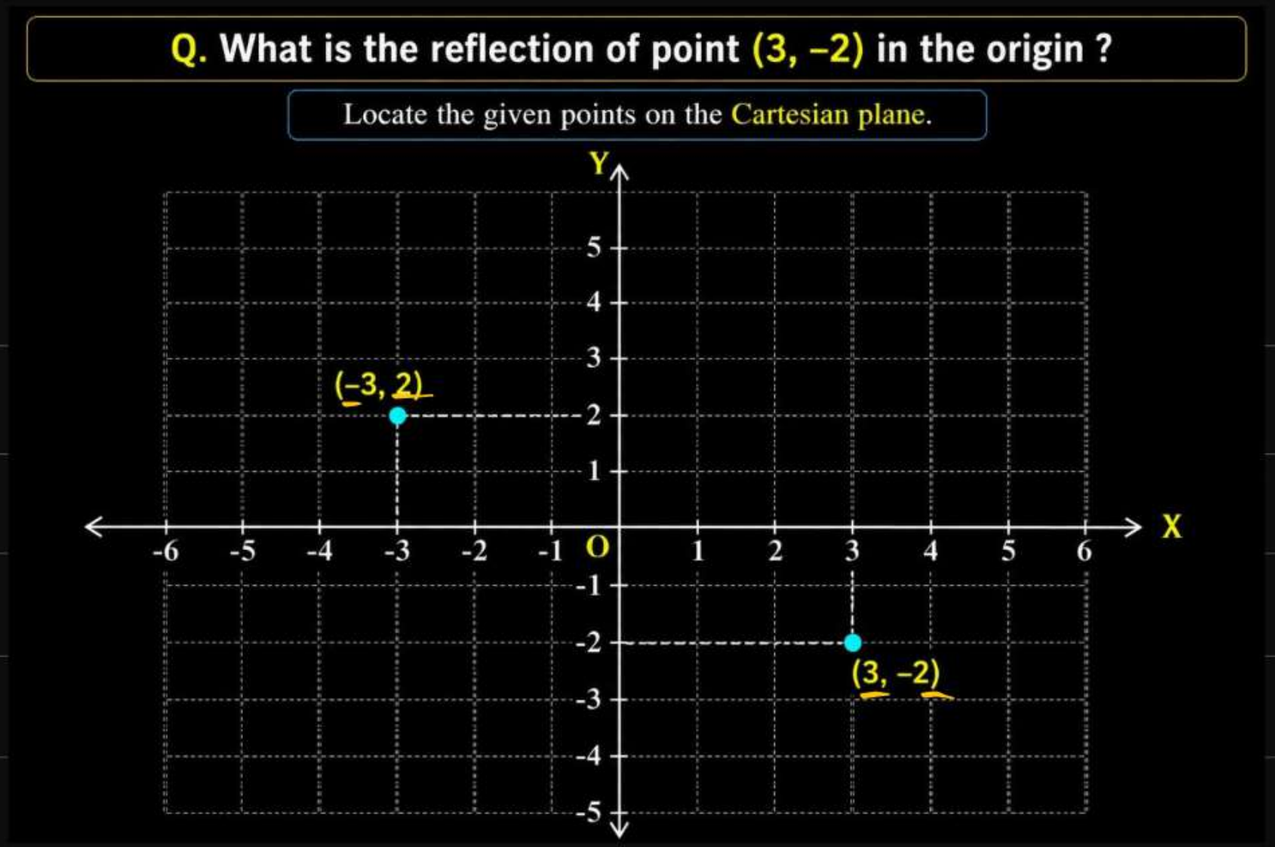
$x_r = 2(0) - (-3) = -3$

$y_r = 2(0) - (-2) = +2$

Q. What is the reflection of point (3, –2) in the origin ?
मूल बिंदु में बिंदु (3, –2) का प्रतिबिम्ब क्या है?

$(-3, +2)$

(A) (–3, 2)
(B) (–3, –2)
(C) (3, 2)
(D) (–2, –3)



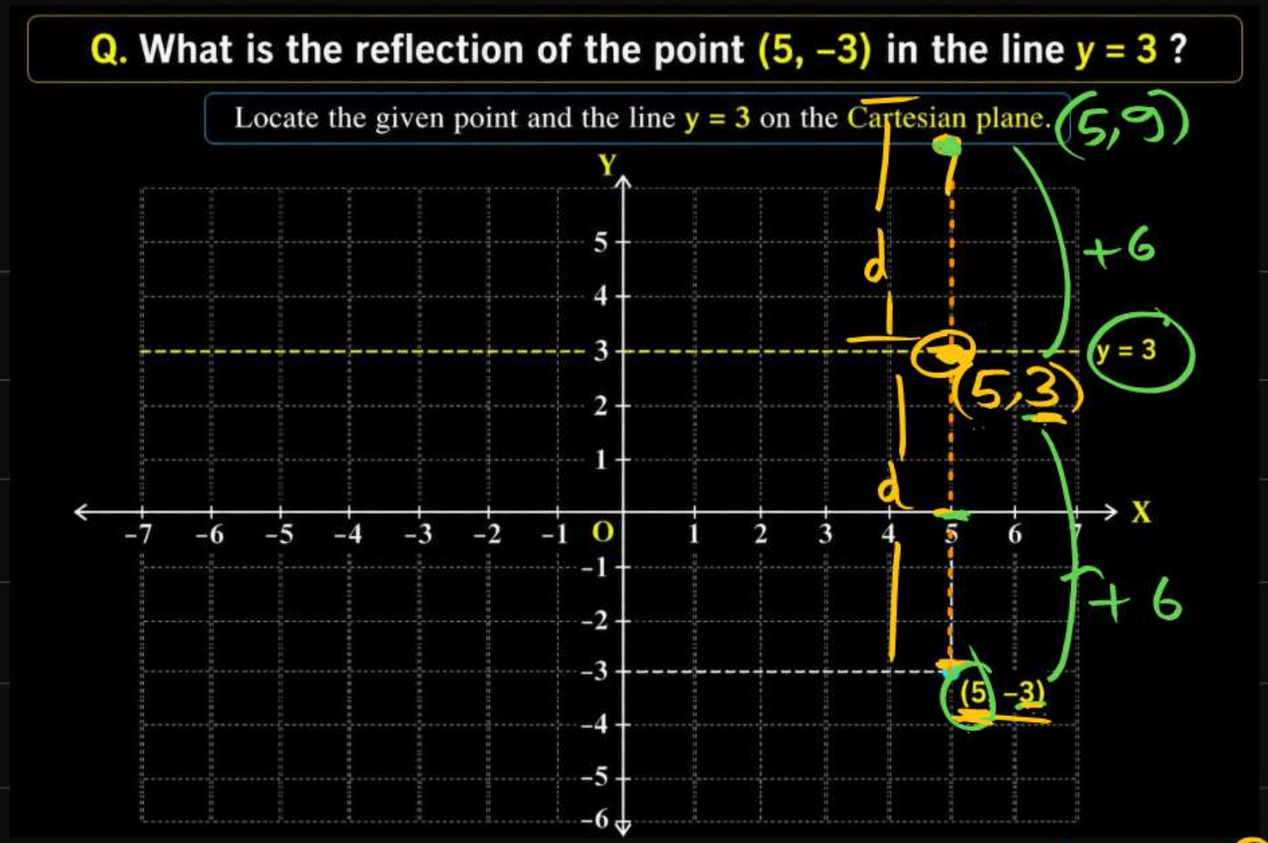
Q. What is the reflection of point (3, –2) in the origin ?
मूल बिंदु में बिंदु (3, –2) का प्रतिबिम्ब क्या है?

Sign inverse

(A) (–3, 2)
(B) (–3, –2)
(C) (3, 2)
(D) (–2, –3)

Q. What is the reflection of the point (5,-3) in the line y = 3?
रेखा y = 3 में निर्देशांक बिंदु (5, -3) का प्रतिबिंब क्या है?

SSC CGL 2019 Tier-2



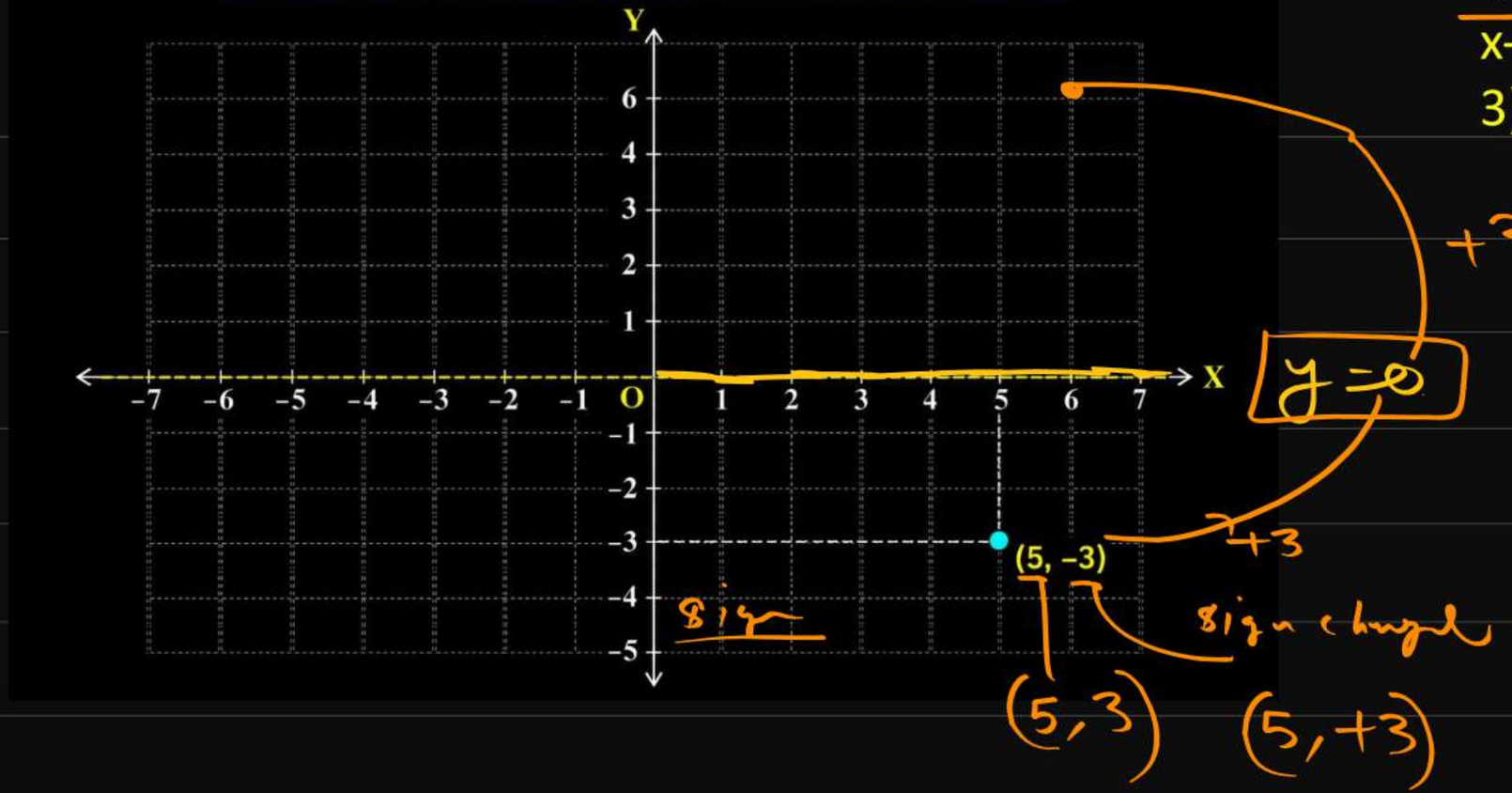
$x = 5$

$y = 9 \Rightarrow y_r = 2(3) - (-3) \Rightarrow 9$

(a) (5,-6)
(b) (-5,3)
(c) (5,9)
(d) (5,3)

Q. What is the reflection of the point (5, –3) across the x axis?

Locate the given point and the x axis on the Cartesian plane.



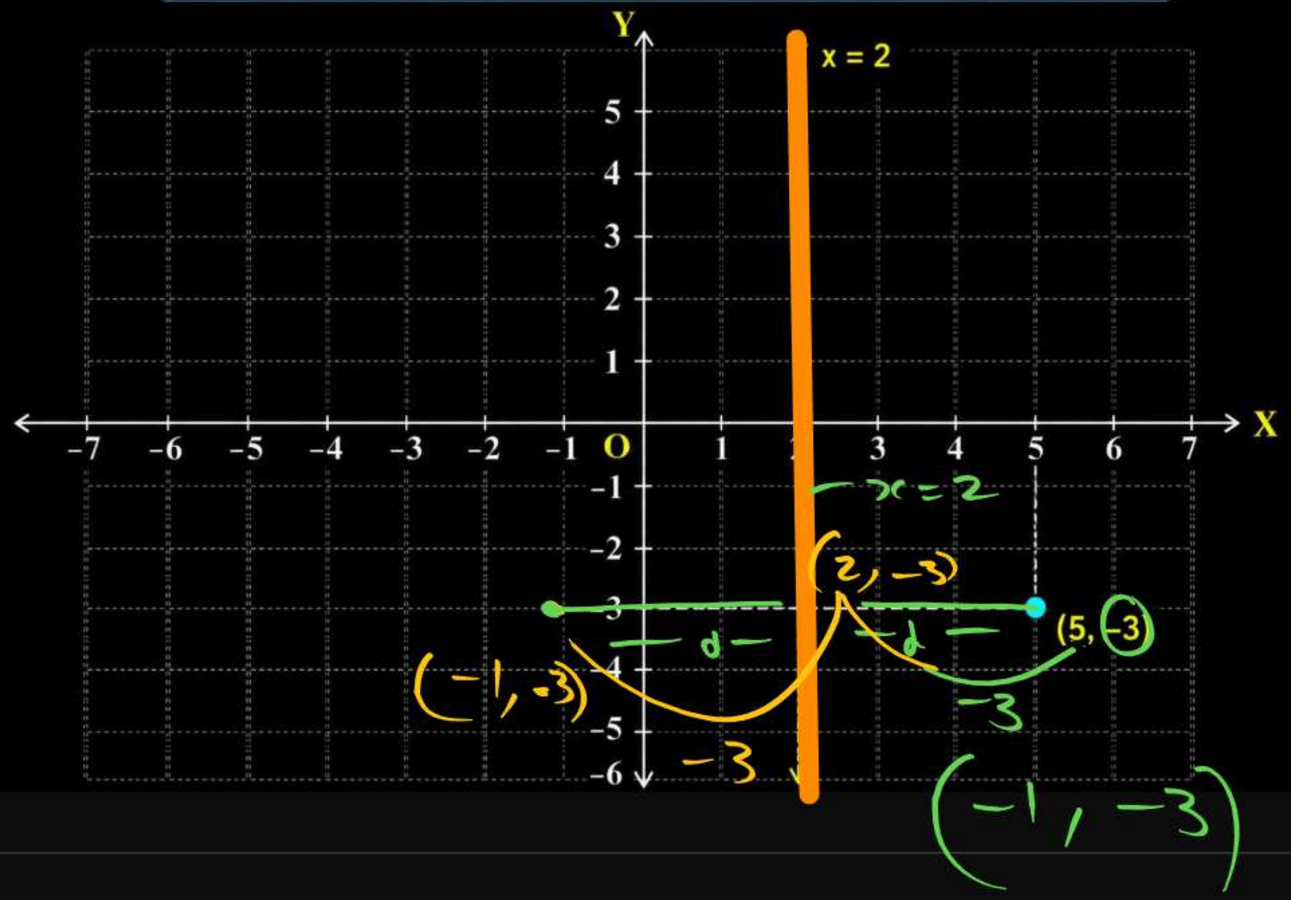
Q. What is the reflection of the point (5,-3) across the x axis?

x-अक्ष में निर्देशांक बिंदु (5, -3) का प्रतिबिंब क्या है?

(a) (5,-6)
(b) (5,3)
(c) (5,9)
(d) (5,3)

Q. What is the reflection of the point (5, –3) in the line x = 2?

Locate the given point and the line x = 2 on the Cartesian plane.



Q. What is the reflection of the point (5,-3) in the line x = 2?

रेखा x = 2 में निर्देशांक बिंदु (5, -3) का प्रतिबिंब क्या है?

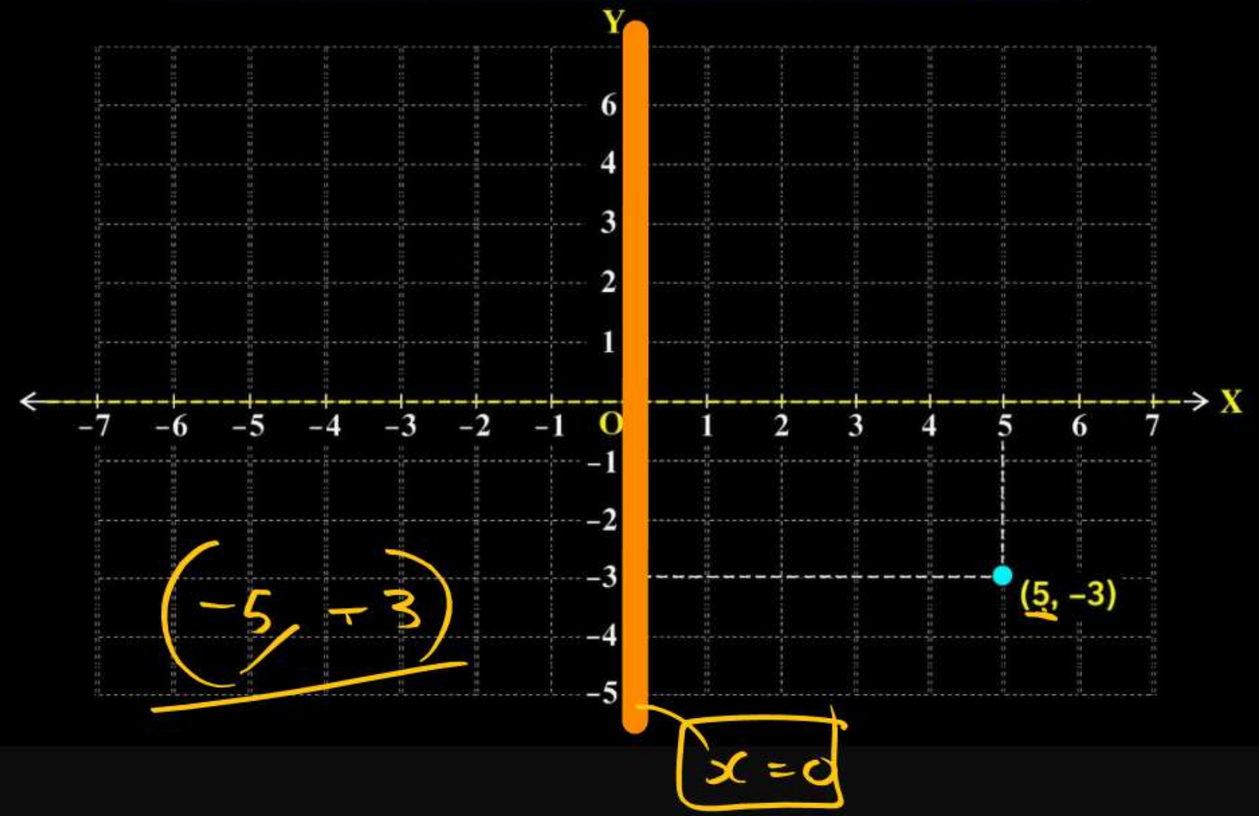
$x_2 = 2 \times (2) - 5$

$\Rightarrow (-1)$

(a) (5,-6)
(b) (-5,3)
(c) (-1,-3)
(d) (5,3)

Q. What is the reflection of the point (5, –3) across the x axis?

Locate the given point and the x axis on the Cartesian plane.

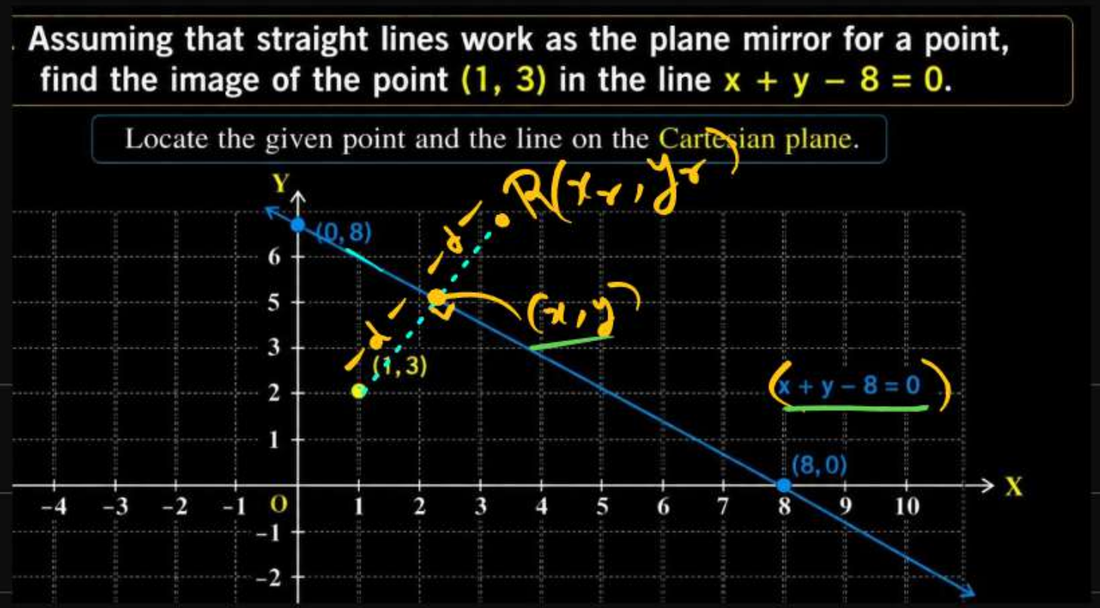


Q. What is the reflection of the point (5,-3) in the y axis?

y-अक्ष x = 3 में निर्देशांक बिंदु (5, -3) का प्रतिबिंब क्या है?

x = 0

(a) (2,-6)
(b) (-1,3)
(c) (-5,-3)
(d) (5,4)



Q. Assuming that straight lines work as the plane mirror for a point, find the image of the point (1, 3) in the line x + y- 8 = 0.

यह मानते हुए कि सीधी रेखाएँ बिंदु के लिए समतल दर्पण के रूप में काम करती हैं, (1, 3) बिंदु का x + y- 8 = 0. रेखा में प्रतिबिंब ज्ञात कीजिये।

$Ax + By + C = 0$

1 line → $Bx - Ay + C_2 = 0$

(1) $x + 1y = 8$

$x - y = K$

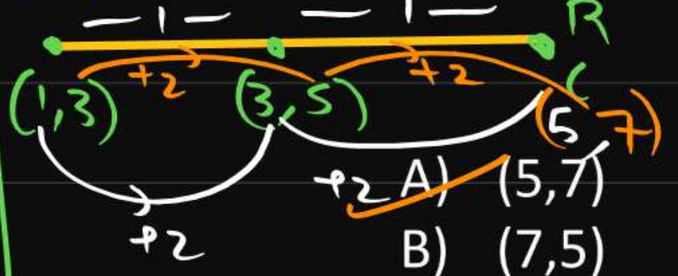
$1 - 3 = -2$

$x - y = -2$

$x + y = 8$

$x = 3$

$y = 5$

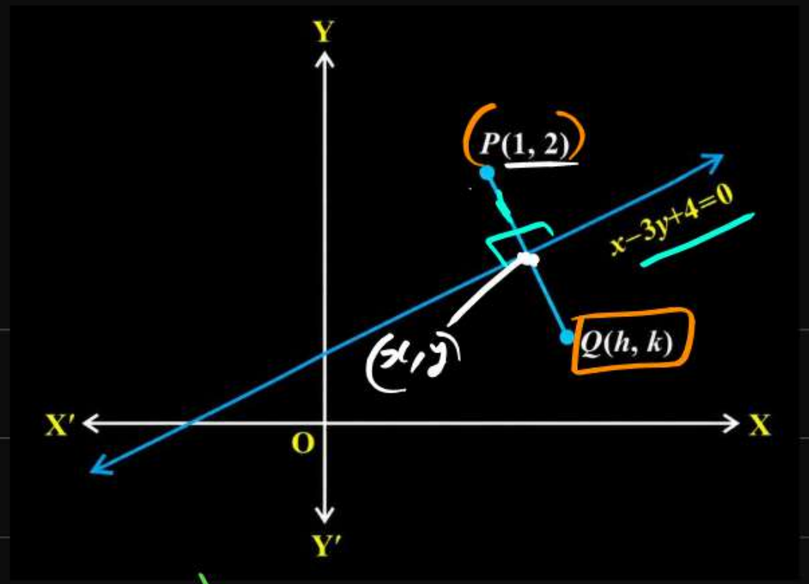


A) (5,7)

B) (7,5)

C) (3,5)

D) (5,3)



Q-) Assuming that straight lines work as the plane mirror for a point, find the image of the point (1, 2) in the line $x - 3y + 4 = 0$.

यह मानते हुए कि सीधी रेखाएँ बिंदु के लिए समतल दर्पण के रूप में काम करती हैं, (1, 2) बिंदु का $x - 3y + 4 = 0$ रेखा में प्रतिबिंब ज्ञात कीजिये।

A) (1/5,7/5)
B) (6/5, 7/5) ✓
C) (4/5, 11/5)
D) (11/10, 17/10)

$x - 3y + 4 = 0$

$-3x \quad -1y = K$

1 line ⇒ $3x + y = -K$ $(1, 2)$

$3 + 2 = 5 = -K$

$3x + y = 5$ ... (i)

$(x - 3y = -4) \times 3$

$10y = 17$

$y = \frac{17}{10}$; $x = \frac{11}{10}$

$h = 2 \times \frac{11}{10} - 1 = \frac{6}{5}$

$K = 2 \times \frac{17}{10} - 2 = 7/5$

Incircle, circumcircle, orthocenter Centroid

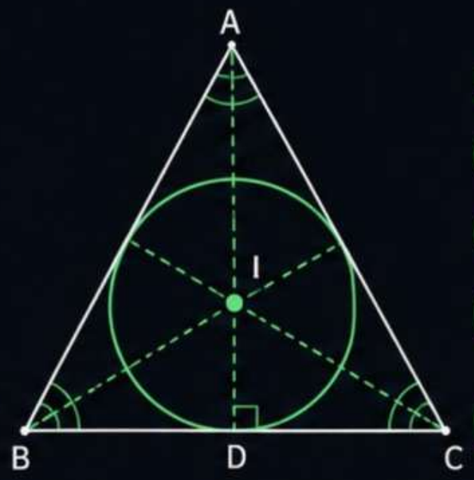
अंतर्वृत्त, परिवृत्त, लंबकेंद्र, केंद्रक

INCIRCLE

The circle that touches all three sides of a triangle from the inside.

CONSTRUCTION

1. Draw the angle bisectors of any two angles of the triangle.
2. Mark their intersection point I (incenter).
3. From I, drop a perpendicular to any side. Let the foot be D.
4. With center I and radius ID, draw the circle. It will touch all three sides.



IMPORTANCE / KEY PROPERTIES

- I is equidistant from all three sides.
- The incircle touches each side at exactly one point.
- Area of triangle = $r \times s$ where r = inradius, s = semiperimeter.
- The incircle is unique for every triangle.

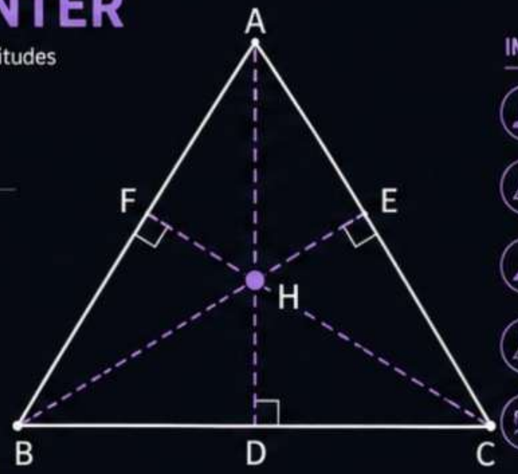
I is called the incenter and $r = ID$ is the inradius.

ORTHOCENTER

The point where all three altitudes of a triangle intersect.

CONSTRUCTION

1. From vertex A, draw an altitude to side BC.
2. From vertex B, draw an altitude to side AC.
3. The intersection of these two altitudes is H (orthocenter).
4. (The third altitude from C will also pass through H.)



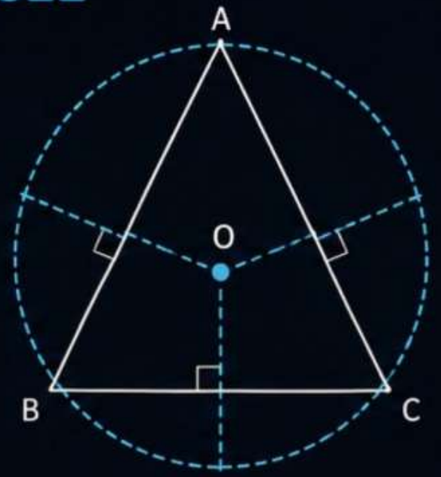
IMPORTANCE / KEY PROPERTIES

- H is the intersection point of the three altitudes.
- In an acute triangle, H is inside the triangle.
- In a right triangle, H is at the right-angled vertex.
- In an obtuse triangle, H is outside the triangle.
- The orthocenter need not lie inside the triangle.

H is called the orthocenter.

CIRCUMCIRCLE

The circle that passes through all three vertices of a triangle.



CONSTRUCTION

1. Draw the perpendicular bisectors of any two sides of the triangle.
2. Mark their intersection point O (circumcenter).
3. With center O and radius OA (or OB or OC), draw the circle.

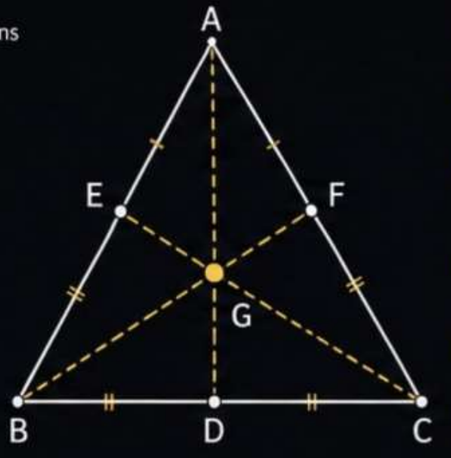
IMPORTANCE / KEY PROPERTIES

- O is equidistant from all three vertices.
- The circumcircle passes through A, B and C.
- Radius $R = OA = OB = OC$.
- For an acute triangle, O is inside the triangle.
- For a right triangle, O is the midpoint of the hypotenuse.
- For an obtuse triangle, O is outside the triangle.

O is called the circumcenter and R is the circumradius.

CENTROID

The point where all three medians of a triangle intersect.



CONSTRUCTION

1. Find the midpoint of side BC. Let it be D.
2. Draw the median AD.
3. Find the midpoint of side AC. Let it be E.
4. Draw the median BE.
5. Their intersection point is G (centroid). The third median from C also passes through G.

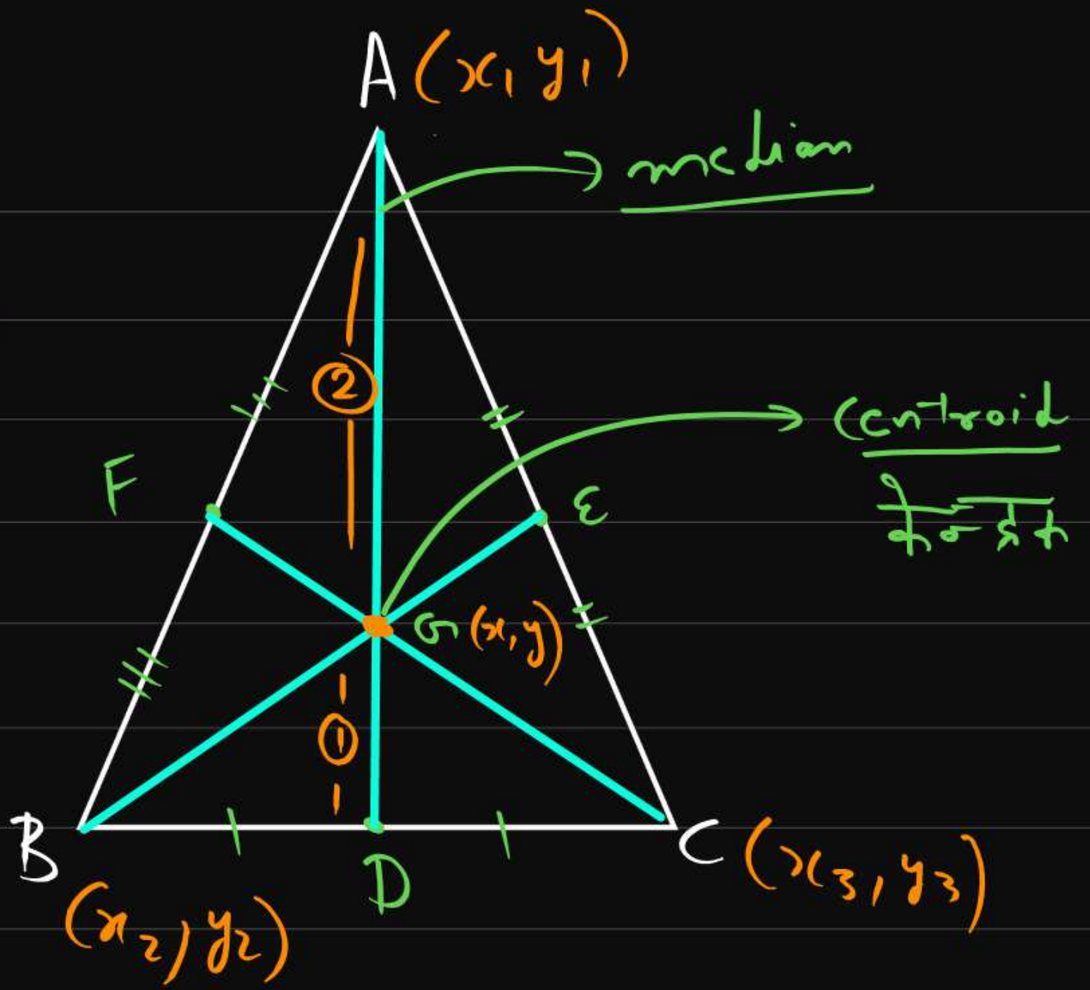
IMPORTANCE / KEY PROPERTIES

- G divides each median in the ratio 2:1.
- $AG = \frac{2}{3}AD, \quad GD = \frac{1}{3}AD$ (similarly for the other medians).
- The centroid is always inside the triangle.
- It is the center of gravity of the triangle (balance point).
- The centroid is unique for every triangle.

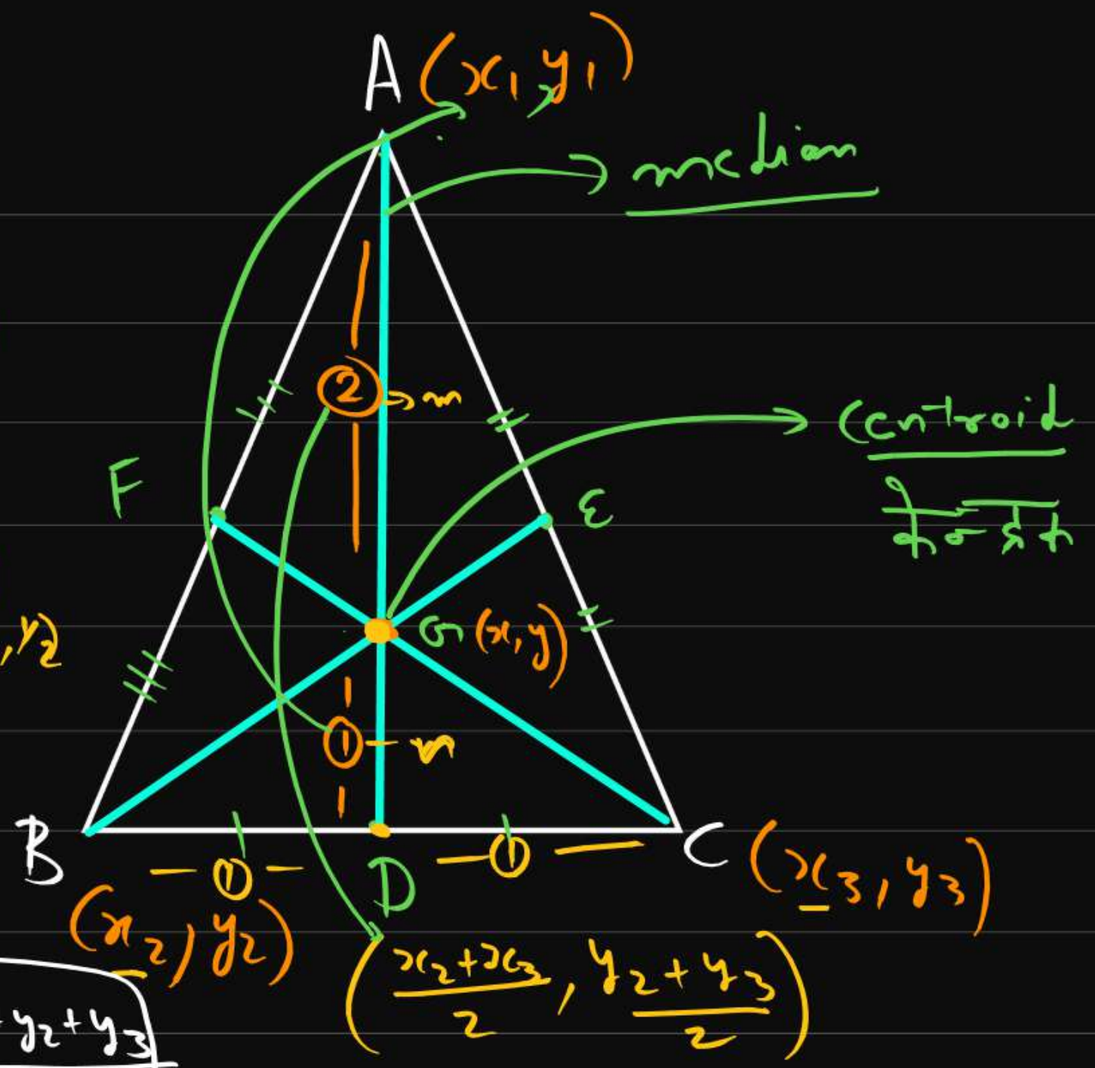
G is called the centroid.

Centroid केन्द्रक

$$G(\bar{x}, \bar{y}) = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

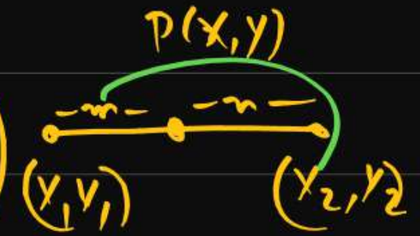


Centroid केन्द्रक



$$G(x, y) = \left(\frac{x_1+x_2+x_3}{3}, \frac{y_1+y_2+y_3}{3}\right)$$

$$\left(X = \frac{m x_2 + n x_1}{m+n}\right)$$



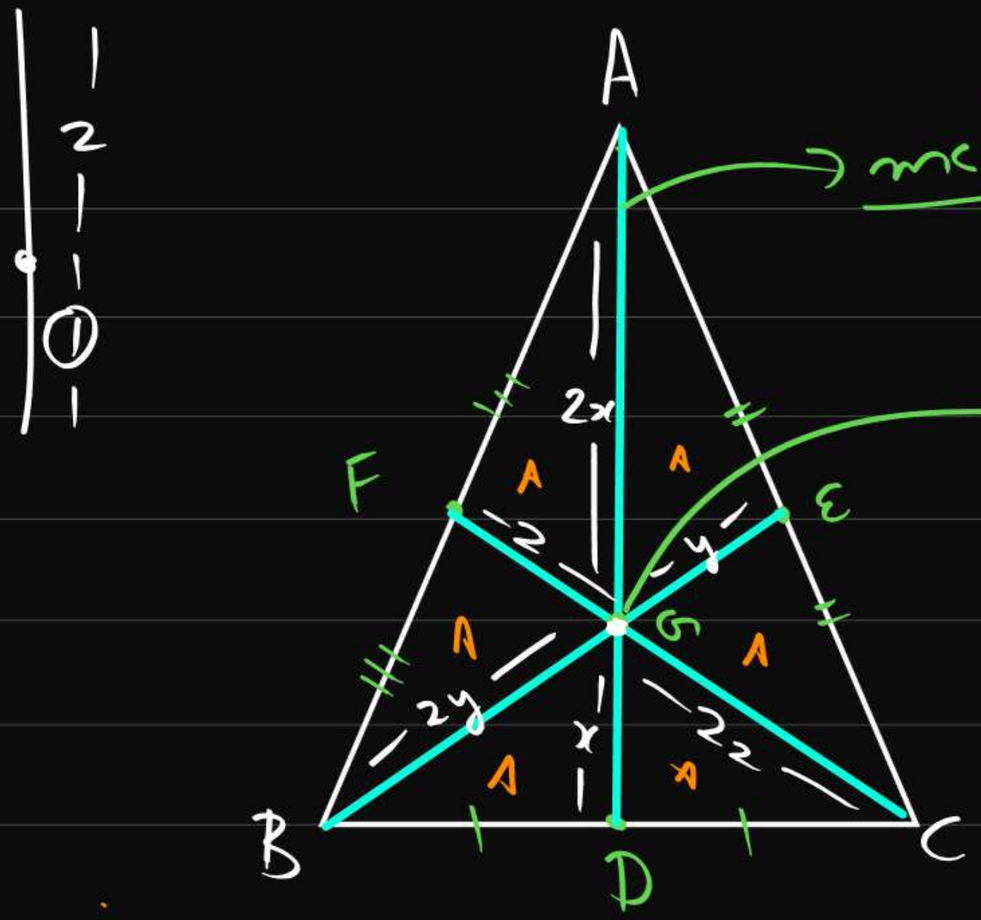
$$x = \frac{2\left[\frac{x_2+x_3}{2}\right] + 1(x_1)}{2+1}$$

$$\Rightarrow \frac{x_1+x_2+x_3}{3}$$

Simil.

$$y = \frac{y_1+y_2+y_3}{2}$$

Centroid केन्द्रक



Centroid केन्द्रक

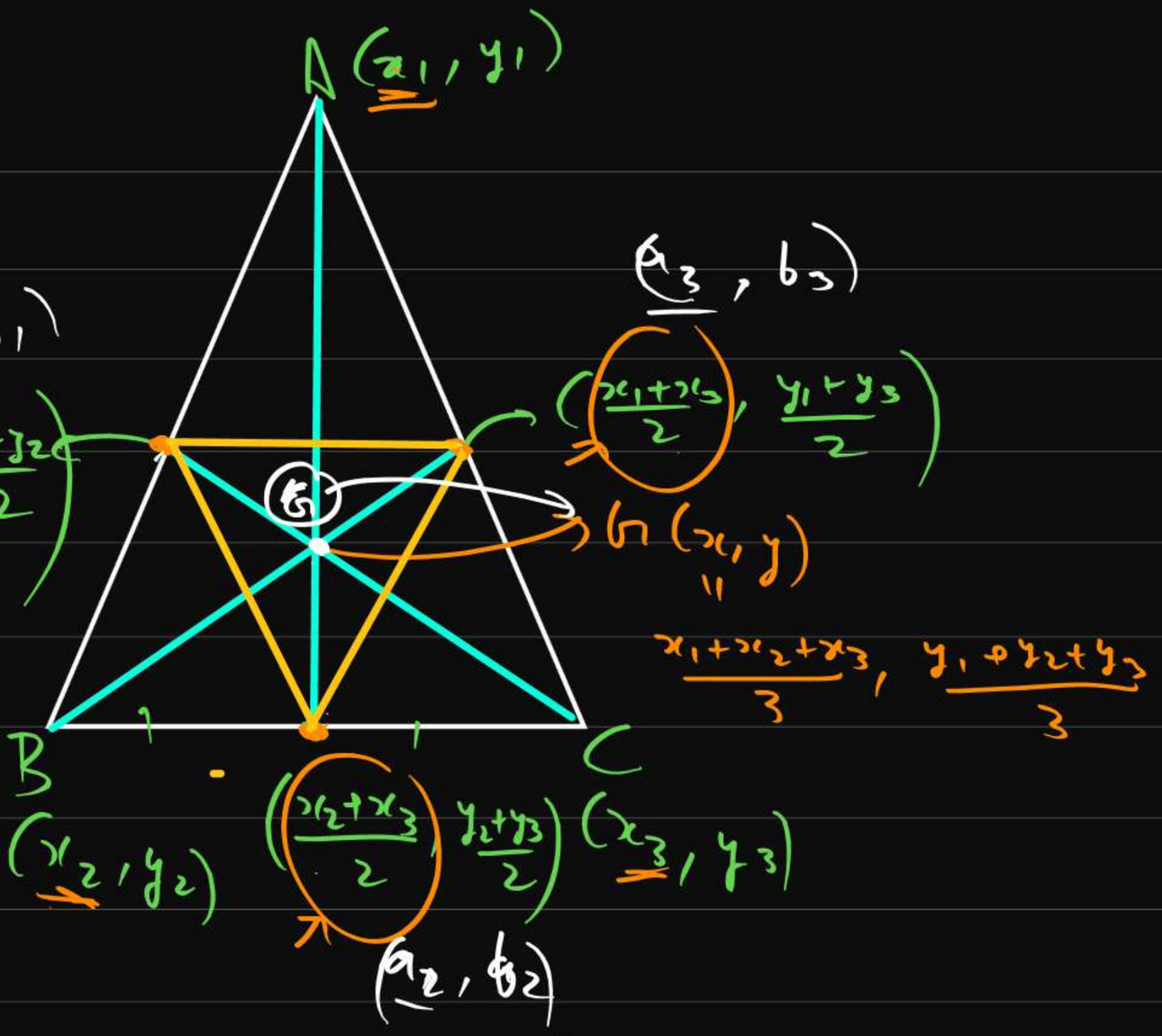
$a_1 \quad a_2 \quad a_3$

$1 \quad 1 \quad 1$

$\frac{(x_1+x_2)}{2} + \frac{x_2+x_3}{2} \neq \frac{x_1+x_3}{2}$

$\Rightarrow (x_1 + x_2 + x_3)$

$$\frac{a_1+a_2+a_3}{3} = \frac{x_1+x_2+x_3}{3} = x$$

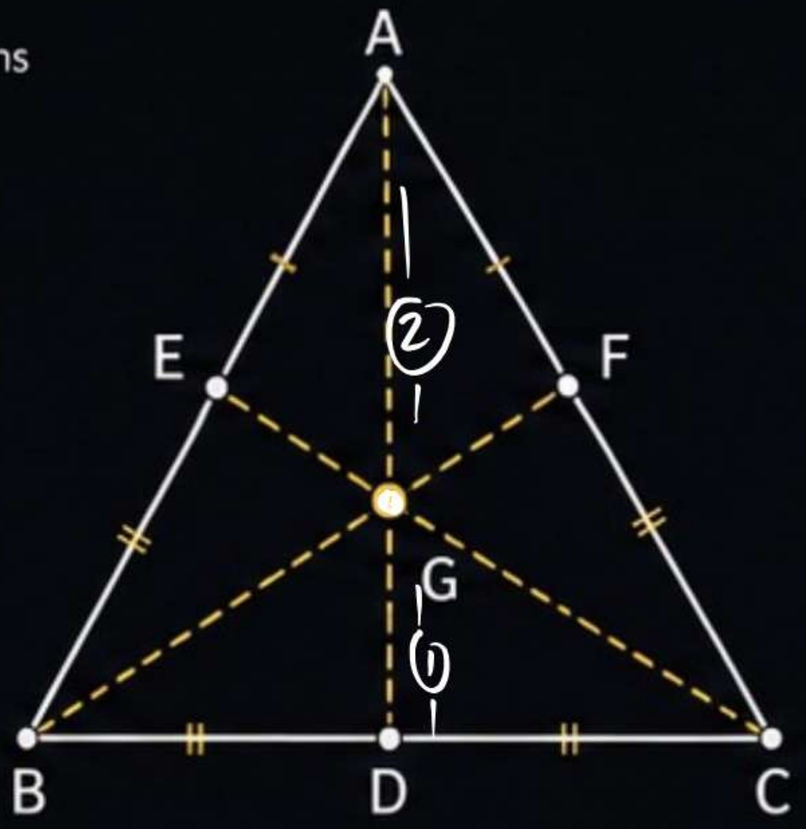


CENTROID

The point where all three medians of a triangle intersect.

CONSTRUCTION

1. Find the midpoint of side BC. Let it be D.
2. Draw the median AD.
3. Find the midpoint of side AC. Let it be E.
4. Draw the median BE.
5. Their intersection point is G (centroid). The third median from C also passes through G.



G is called the centroid.

IMPORTANCE / KEY PROPERTIES

- G divides each median in the ratio 2:1.
- $AG = \frac{2}{3}AD, \quad GD = \frac{1}{3}AD$ (similarly for the other medians).
- The centroid is always inside the triangle.
- It is the center of gravity of the triangle (balance point).
- The centroid is unique for every triangle.

Q. A triangle has vertices A(1,2), B(5,2), and C(3,8). What is the y-coordinate of its centroid?

$$y = \frac{y_1 + y_2 + y_3}{3}$$

$$\Rightarrow \frac{2+2+8}{3} = \frac{12}{3} = 4$$

एक त्रिभुज के शीर्ष A(1,2), B(5,2) और C(3,8) हैं। इसके केंद्रक का y-निर्देशांक क्या है?

SSC CHSL 2025

(a) 2 (b) 5 (c) 6 (d) 4

Q. Find the coordinates of the centroid for a triangle with vertices located at the points (2, 1), (5, 4), and (8, 1).

एक त्रिभुज के केन्द्रक के निर्देशांक ज्ञात करें जिसके शीर्ष (2, 1), (5, 4) और (8, 1) हैं।

$$x = \frac{2+5+8}{3} = \frac{15}{3} = 5$$

$$y = \frac{1+4+1}{3} = \frac{6}{3} = 2$$

SSC CPO 2025

(a) (5, 2) (b) (6, 2) (c) (5, 3) (d) (6, 3)

Q. A triangle has vertices at A(0,0), B(4,0), and C(2,6). What is the centroid of the triangle?

$\frac{6}{3}, \frac{6}{3}$

2, 2

एक त्रिभुज के शीर्ष A(0,0), B(4,0) और C(2,6) हैं। त्रिभुज का केंद्रक क्या है?

SSC CHSL 2025

(a) (2, 2) (b) (3, 2) (c) (2, 3) (d) (4, 3)

Q. If the centroid of a triangle is (5,5) and two vertices are (2,3) and (4,7), the third vertex is:

(x_3, y_3)

यदि एक त्रिभुज का केंद्रक (5,5) है और उसके दो शीर्ष (2,3) और (4,7) हैं, तो तीसरा शीर्ष है:

$$5 = \frac{2+4+x_3}{3}$$

$$15 = 5 \times 3 = 2+4+x_3$$

$$x_3 = 9$$

$$y = 15 - 3 - 7 = 5$$

SSC CPO 2025

(a) (5,9) (b) (9,5) (c) (7,5) (d) (5,7)

Q. In a triangle with vertices P(x, y), Q(5, -1), and R(2, 6), the Centroid is at G(4, 2). What are the coordinates of vertex P?

$x = 12 - (7) = 5$

$y = 6 - [-1+6] = 1$

एक त्रिभुज में जिसके शीर्ष P(x, y), Q(5, -1) और R(2, 6) हैं, केंद्रक G(4, 2) पर है। शीर्ष P के निर्देशांक क्या हैं?

SSC CGL 2025

(a) (5, 1) (b) (5, -1) (c) (4, 1) (d) (4, 2)

Q. A (x, 5), B (3, – 4) and C (– 6, y) are the three vertices of a triangle ABC. If (3.5, 4.5) is the centroid of the triangle, find the point (x + 3, y – 4).

A (x, 5), B (3, - 4) और C (- 6, y) त्रिभुज ABC के तीन शीर्ष हैं। यदि (3.5, 4.5) त्रिभुज का केन्द्रक है , तो बिंदु (x + 3, y - 4) ज्ञात कीजिए।

[NTPC Graduate level CBT-I 2025]

(a) (16.5, 8.5) (b) (13.5, 12.5)
(c) (8.5, 16.5) (d) (15.5, 9.5)

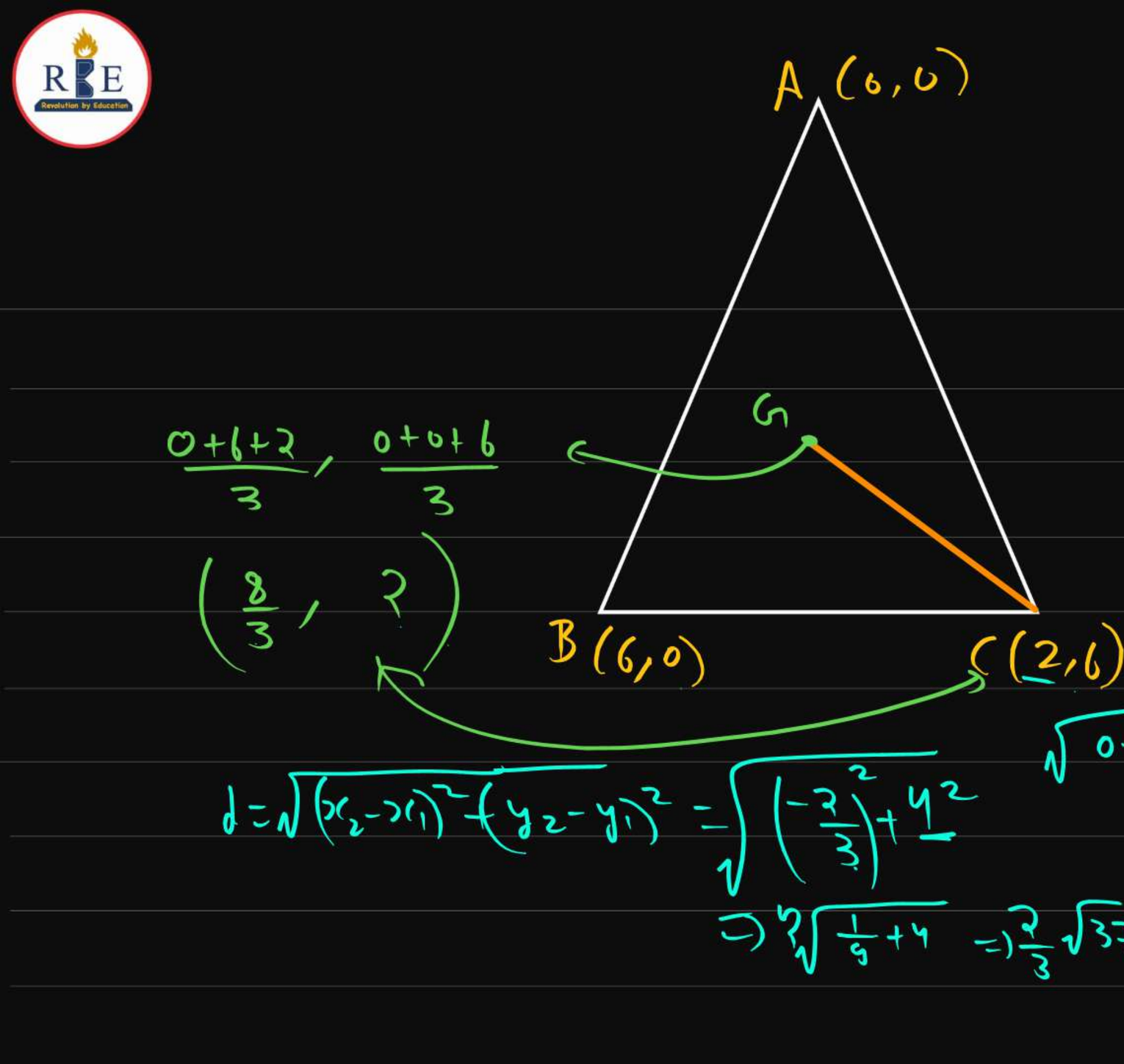
$x = \frac{3.5 \times 3 - (3 + -6)}{}$

$= 13.5$

$y = 13.5 - [5 - 4]$

$\Rightarrow 12.5$

$(16.5, 8.5)$

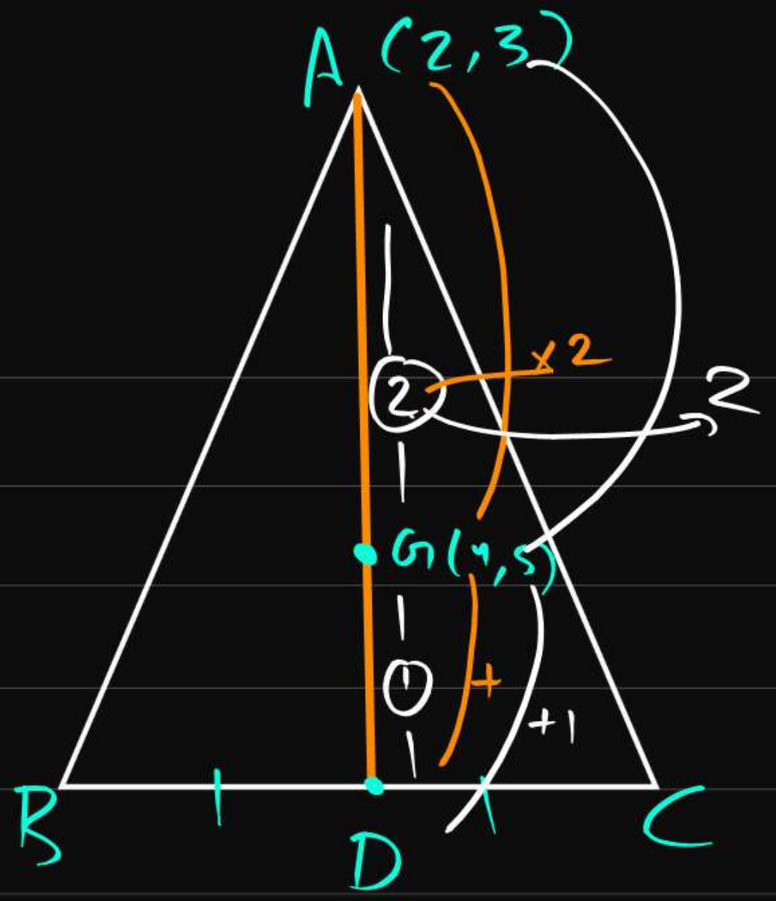


Q. In a triangular steel frame, points A(0,0), B(6,0), and C(2,6) form a triangle. A vertical load acts at the centroid. What is the distance from vertex C to the centroid?

एक त्रिभुजाकार स्टील फ्रेम में, बिंदु A(0,0), B(6,0) और C(2,6) एक त्रिभुज बनाते हैं। एक ऊर्ध्वाधर भार केंद्रक पर कार्य करता है। शीर्ष C से केंद्रक तक की दूरी क्या है?

SSC CHSL 2025

(a) √13 units (b) √16.4 units (c) √12 units (d) √14 units



Q. Triangle ABC has its centroid at point G(4, 5), and vertex A is located at (2, 3). If point D is the midpoint of side BC, what are the coordinates of D?

त्रिभुज ABC का केंद्रक G(4, 5) पर है, और शीर्ष A (2, 3) पर स्थित है। यदि बिंदु D भुजा BC का मध्यबिंदु है, तो D के निर्देशांक क्या होंगे?

SSC CGL 2025

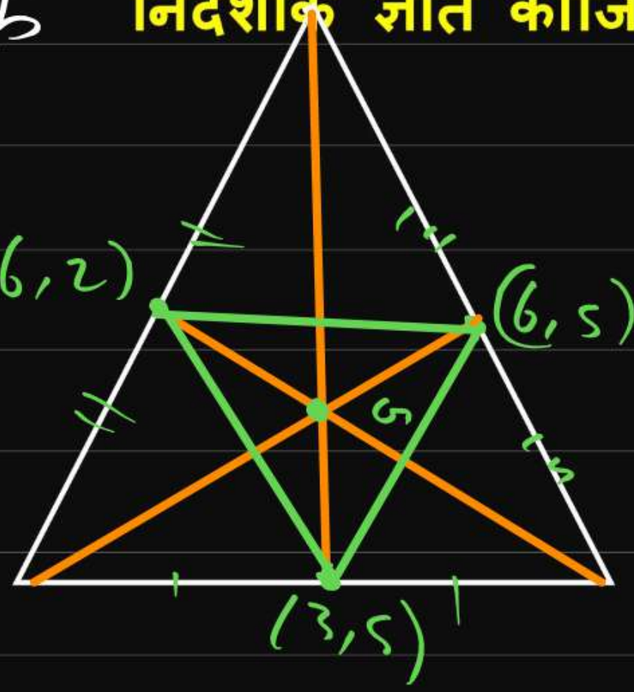
(a) (5, 6) (b) (5, 8) (c) (6, 7) (d) (6, 8)

Q. The coordinates of the midpoints of the sides of a triangle are (6,2), (3,5) and (6,5). What are the coordinates of the centroid of the triangle?

किसी त्रिभुज की भुजाओं के मध्यबिंदुओं के निर्देशांक (6,2), (3,5) और (6,5) हैं। त्रिभुज के केंद्रक के निर्देशांक ज्ञात कीजिए।

$$x = \frac{6+3+6}{3} = \frac{15}{3} = 5$$

$$y = \frac{2+5+5}{3} \Rightarrow \frac{12}{3} = 4$$



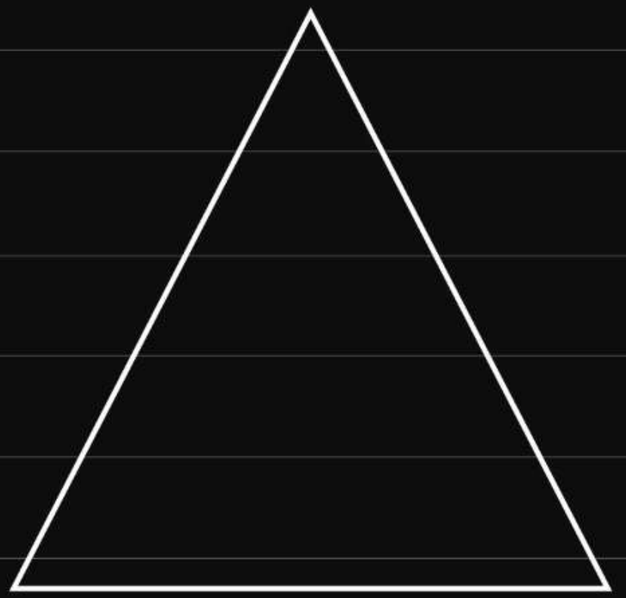
(a) (5, 4)
(b) (4, 5)
(c) (5, 5)
(d) (4, 4)

Q. The coordinates of the midpoints of the sides of a triangle are (5,3), (4,4) and (4,5). What will be the centroid coordinates of this triangle?

त्रिभुज की भुजाओं के मध्य बिन्दुओं के निर्देशांक (5,3), (4,4) तथा (4,5) है | इस त्रिभुज के केन्द्रक के निर्देशांक क्या होंगे?

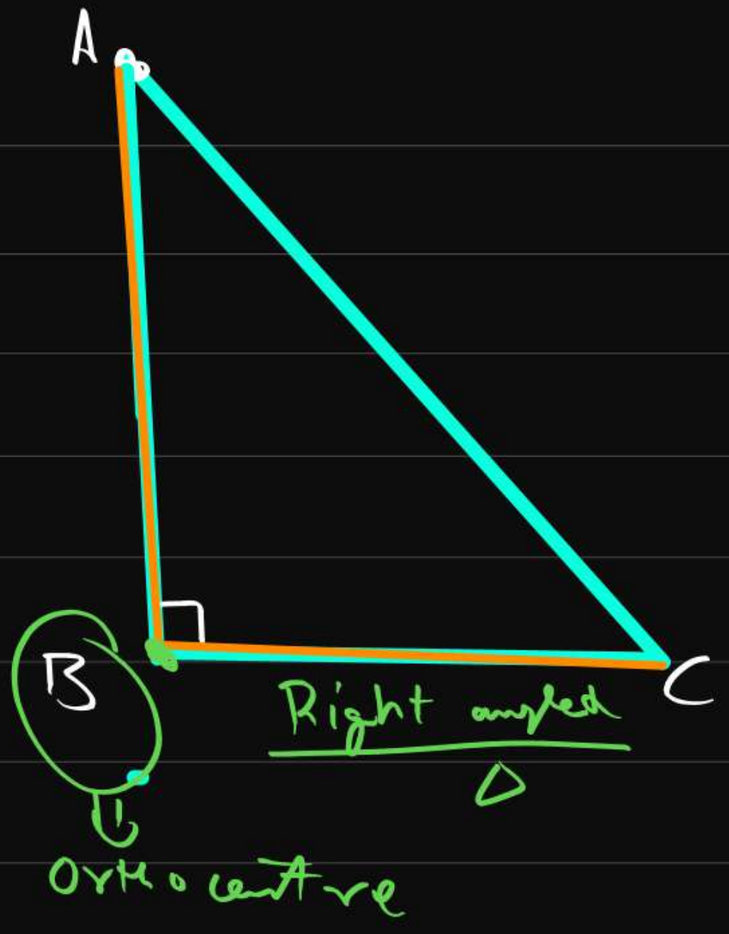
$$\frac{5+4+4}{3}, \frac{3+4+5}{3}$$

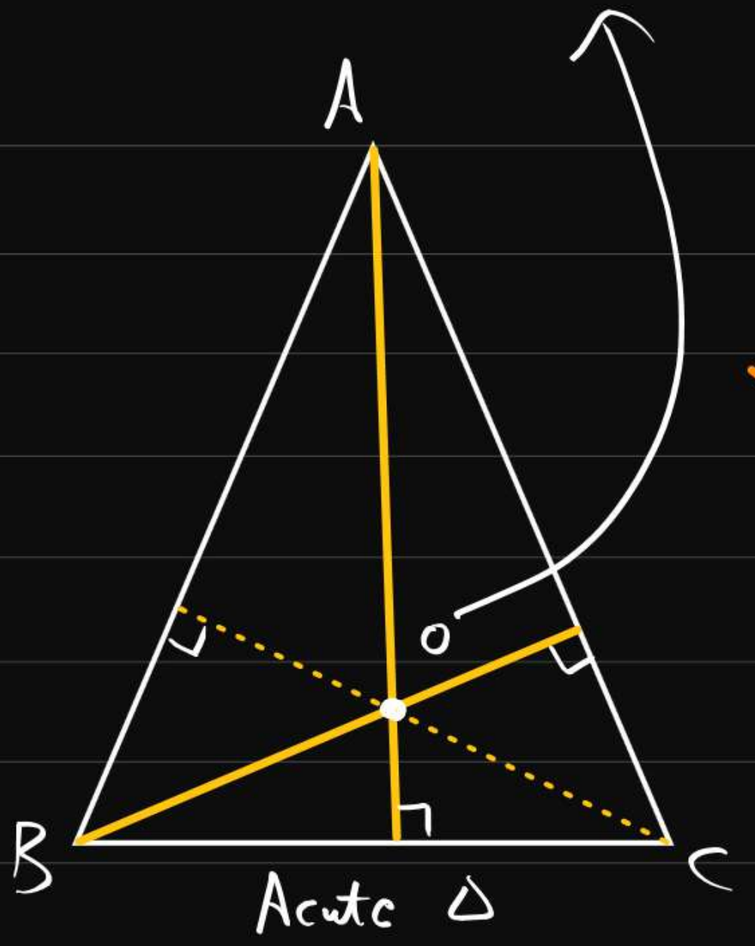
$$\frac{13}{3}, 4$$

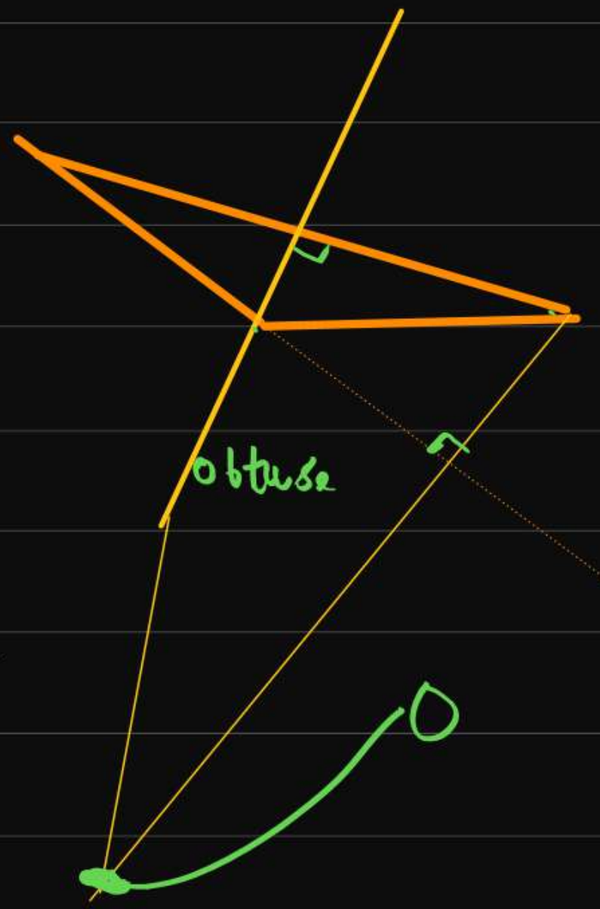


(a) (13/3,4)
(b) (13/2,6)
(c) (5/3,4)
(d) (10/3,4)

Orthocenter लंबकेंद्र





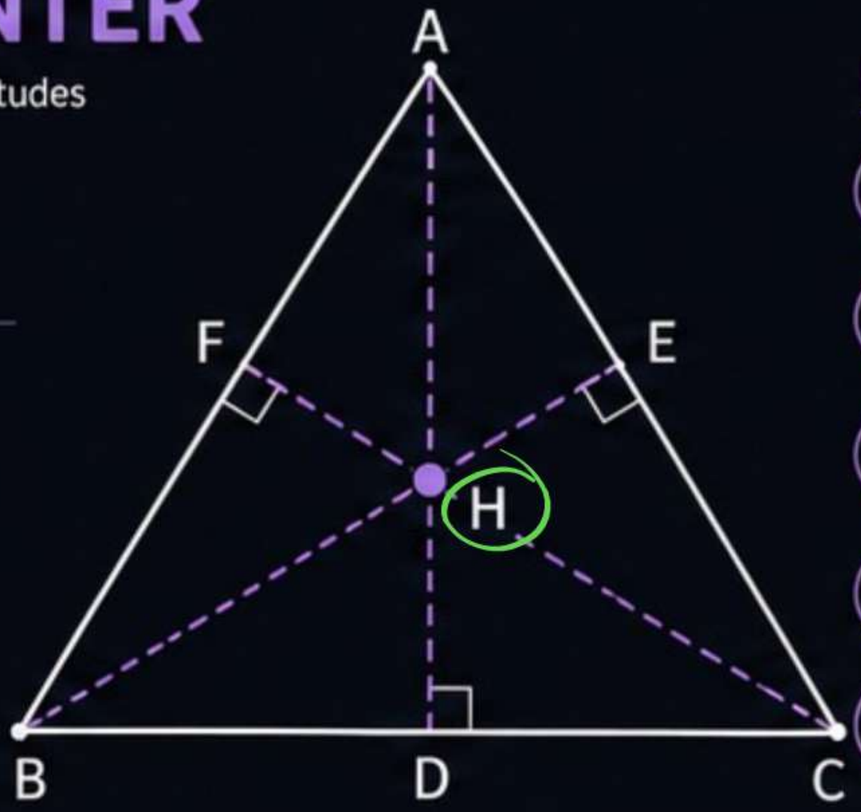


ORTHOCENTER

The point where all three altitudes of a triangle intersect.

CONSTRUCTION

1. From vertex A, draw an altitude to side BC.
2. From vertex B, draw an altitude to side AC.
3. The intersection of these two altitudes is H (orthocenter).
4. (The third altitude from C will also pass through H.)



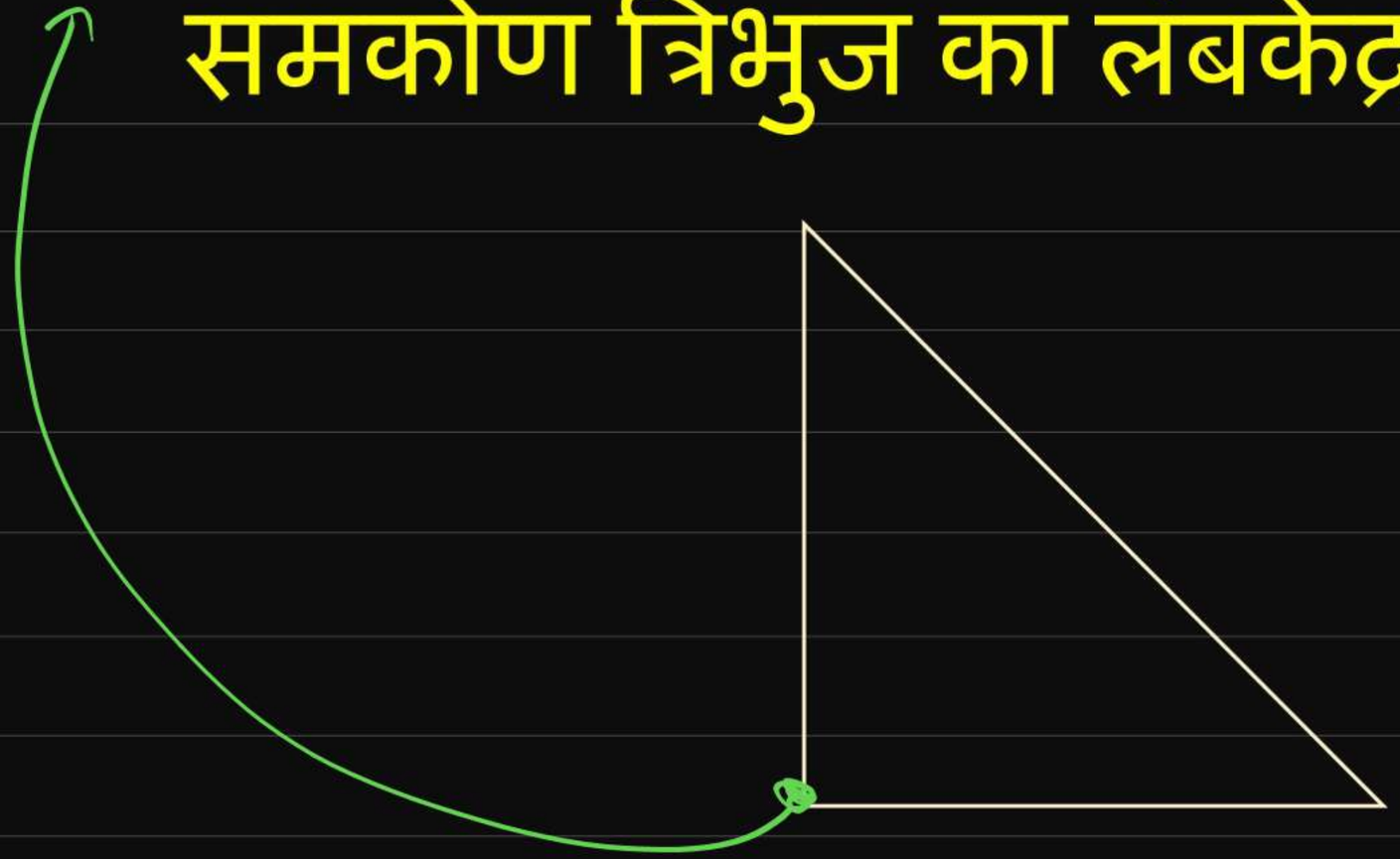
H is called the orthocenter.

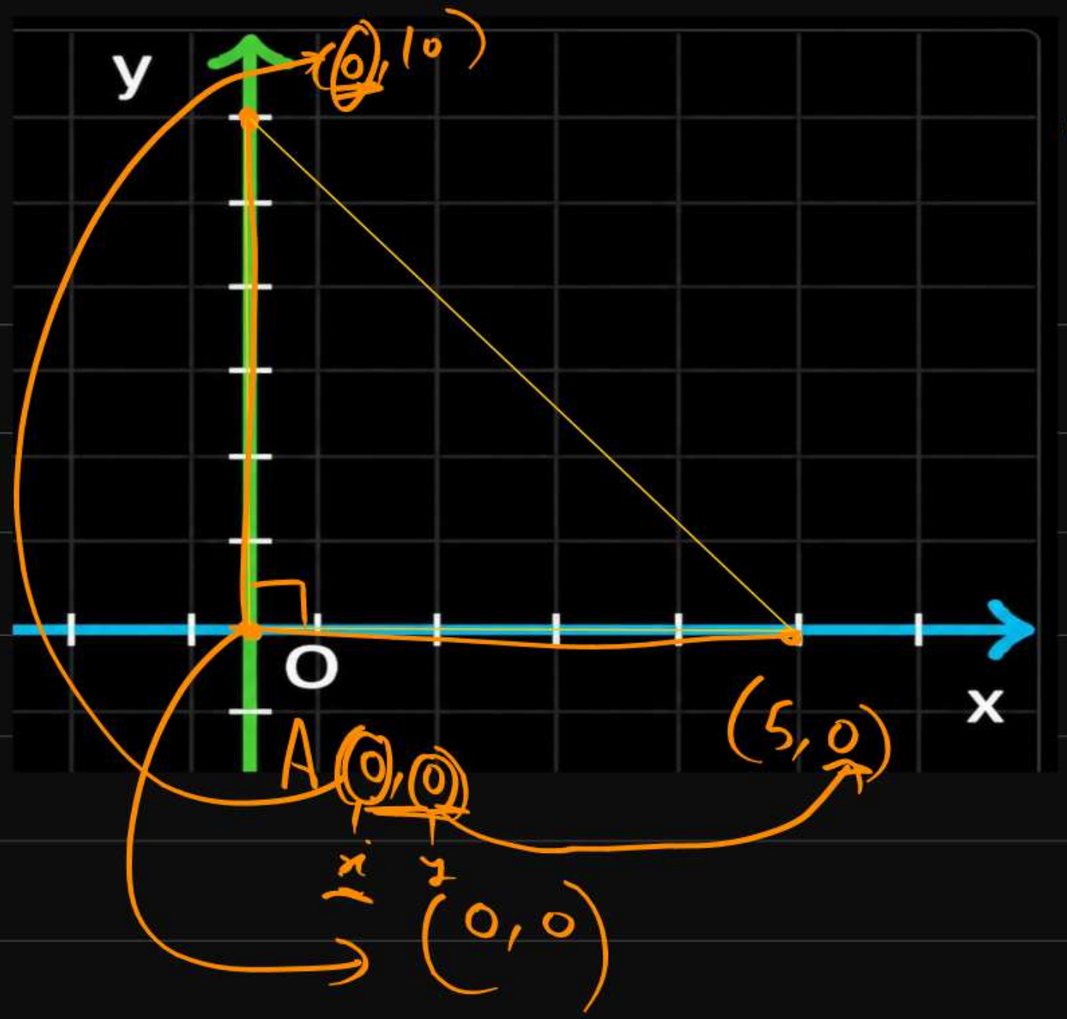
IMPORTANCE / KEY PROPERTIES

- H is the intersection point of the three altitudes.
- In an acute triangle, H is inside the triangle.
- In a right triangle, H is at the right-angled vertex.
- In an obtuse triangle, H is outside the triangle.
- The orthocenter need not lie inside the triangle.

Orthocenter of right angled triangle

समकोण त्रिभुज का लंबकेंद्र



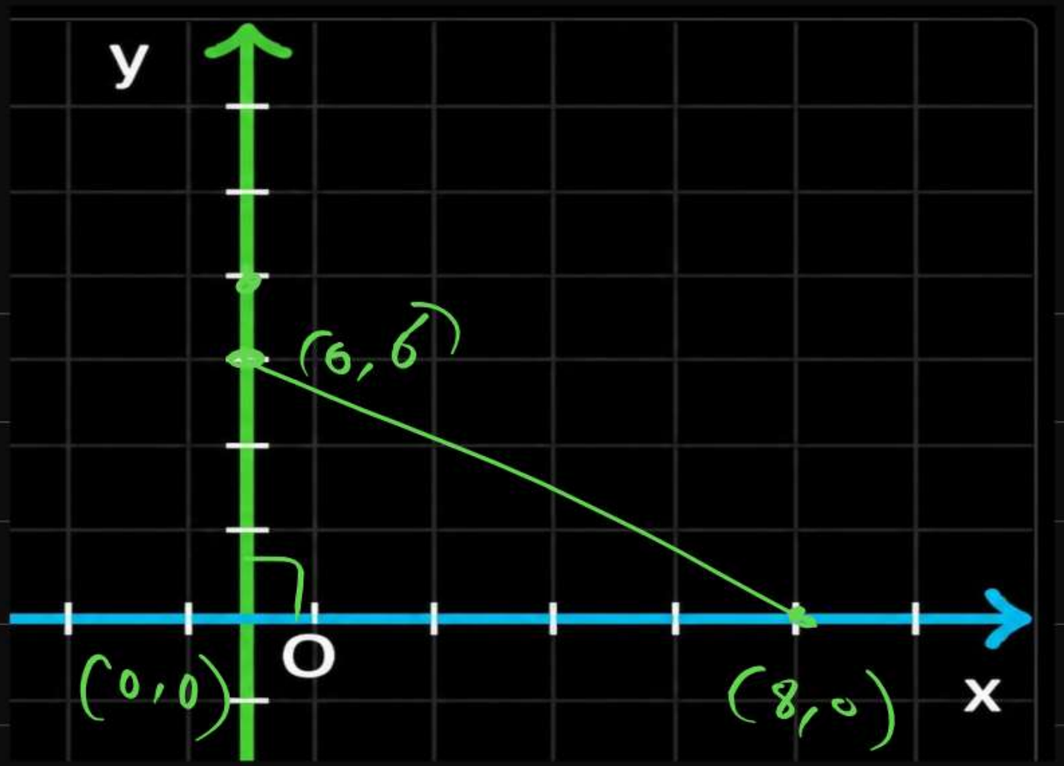


Q. For a triangle with vertices A(0, 0), B(5, 0), and C(0, 10), where is its Orthocenter located?

एक त्रिभुज जिसके शीर्ष A(0, 0), B(5, 0) और C(0, 10) हैं, उसका लंबकेंद्र कहाँ स्थित है?

SSC CPO 2025

(a) (0, 0) (b) (0, 5)
(c) (5, 0) (d) (5, 10)

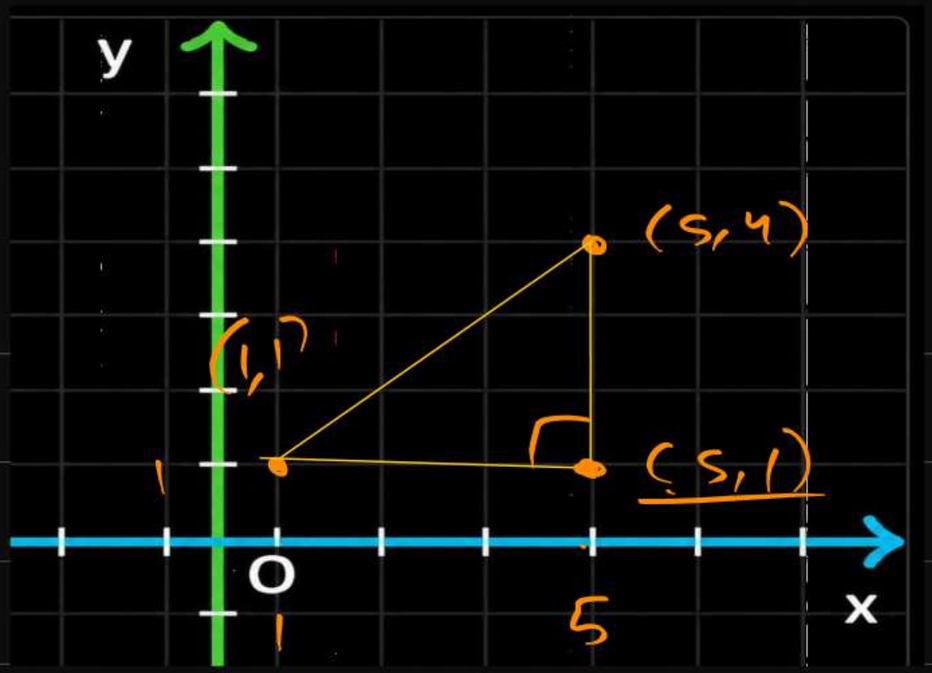


Q. The coordinates of the vertices of a triangle are A(0,0), B(8,0), and C(0,6). What is the location of the orthocenter?

एक त्रिभुज के शीर्षों के निर्देशांक A(0,0), B(8,0) और C(0,6) हैं। लंब केंद्र का स्थान क्या है?

SSC CHSL 2025

(a) (0, 0) (b) (4, 3) (c) (8, 0) (d) (0, 6)



Q) Find the coordinates of the orthocentre of a triangle whose vertices are (1,1), (5,1) and (5,4).
त्रिभुज की लंबकेन्द्र के निर्देशांक ज्ञात कीजिए जिनके शीर्षांक (1,1), (5,1) and (5,4) हैं।

(a) (2,3)
(b) (5,1)
(c) (2,4)
(d) (1,0)

$m_2 M_2 = -1$

Q) Find the coordinates of the ortho-centre of a triangle whose vertices are (7,5), (1,-3) and (-7,3).

त्रिभुज की लंबकेन्द्र के निर्देशांक ज्ञात कीजिए जिनके शीर्षांक (7,5), (1, -3) और (-7,3) हैं।

$AC = m_2 = \frac{5-3}{7-(-7)} = \frac{2}{14} = \left(\frac{1}{7}\right)$

$M_2 = -7 = \frac{y-(-3)}{x-1}$

$y+3 = -7x+7$

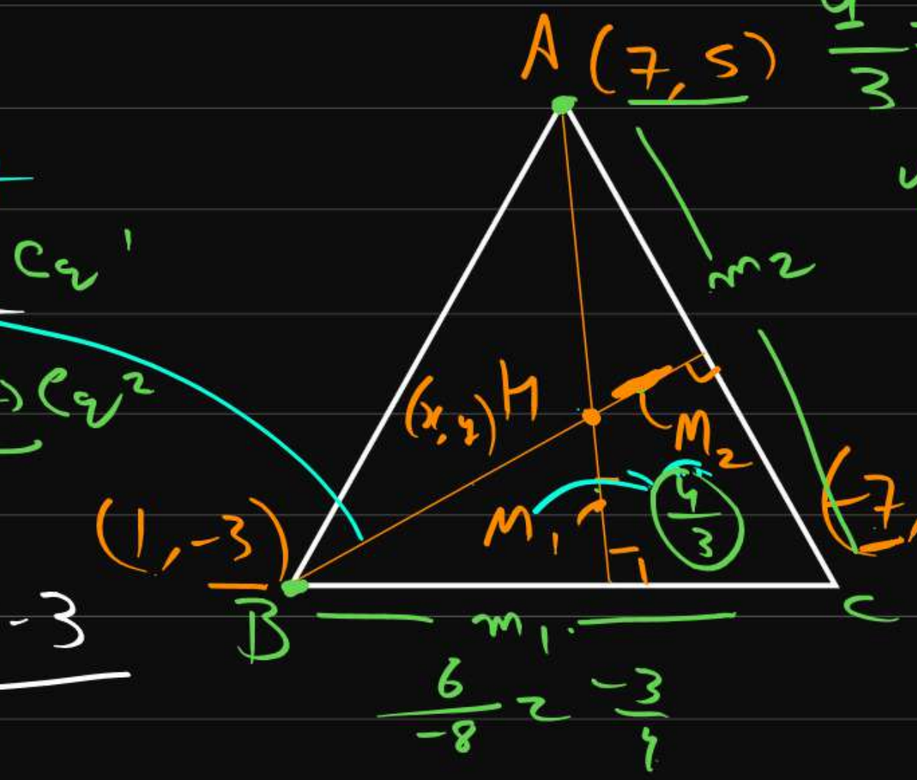
$y+7x = 4$ $\times 3$ C_1

$\times 2$

$-3y+4x = 13 \rightarrow C_2$

$25x = 25$

$x = 1 ; y = -3$



$\frac{4}{3} = \frac{y-5}{x-7}$

$4x-3y = 13$

A) (1,3)

B) (1,-3)

C) (3,-1)

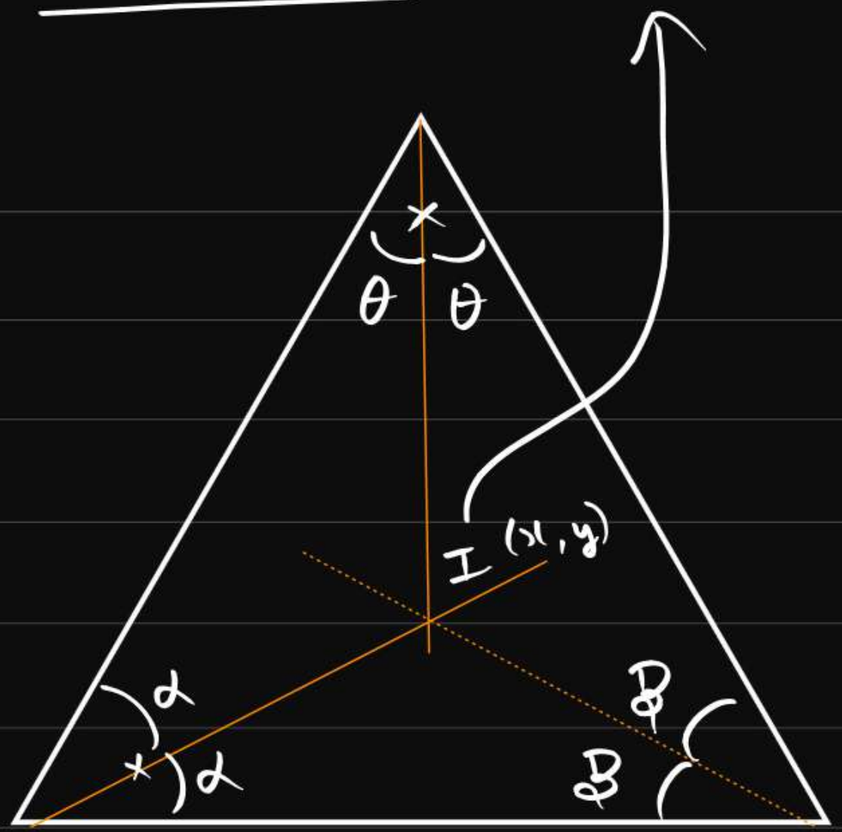
D) (4,1)

Q) Find the coordinates of the orthocentre of a triangle whose vertices are (-2,1), (2,3) and (5,4).
त्रिभुज की लंबकेन्द्र के निर्देशांक ज्ञात कीजिए जिनके शीर्षांक (-2,1), (2,3) and (5,4) हैं।

HW

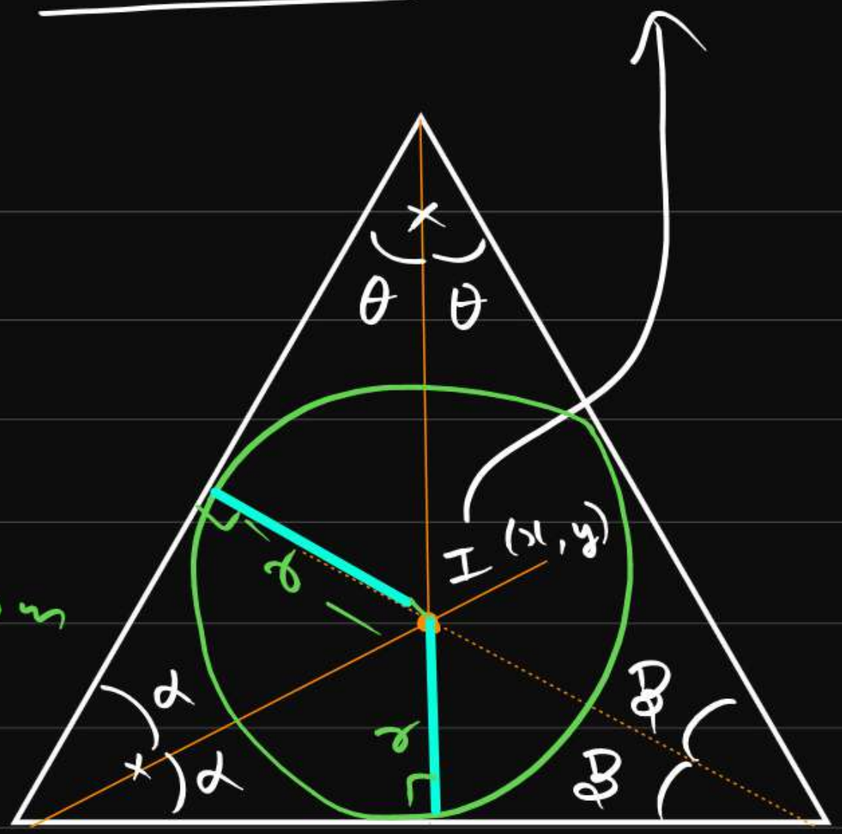
(a) (-2,3)
(b) (-19,52)
(c) (3,-2)
(d) (52,-19)

Incentre अन्तःकेन्द्र



Incentre अन्तःकेन्द्र

All sides are equidistant from incentre

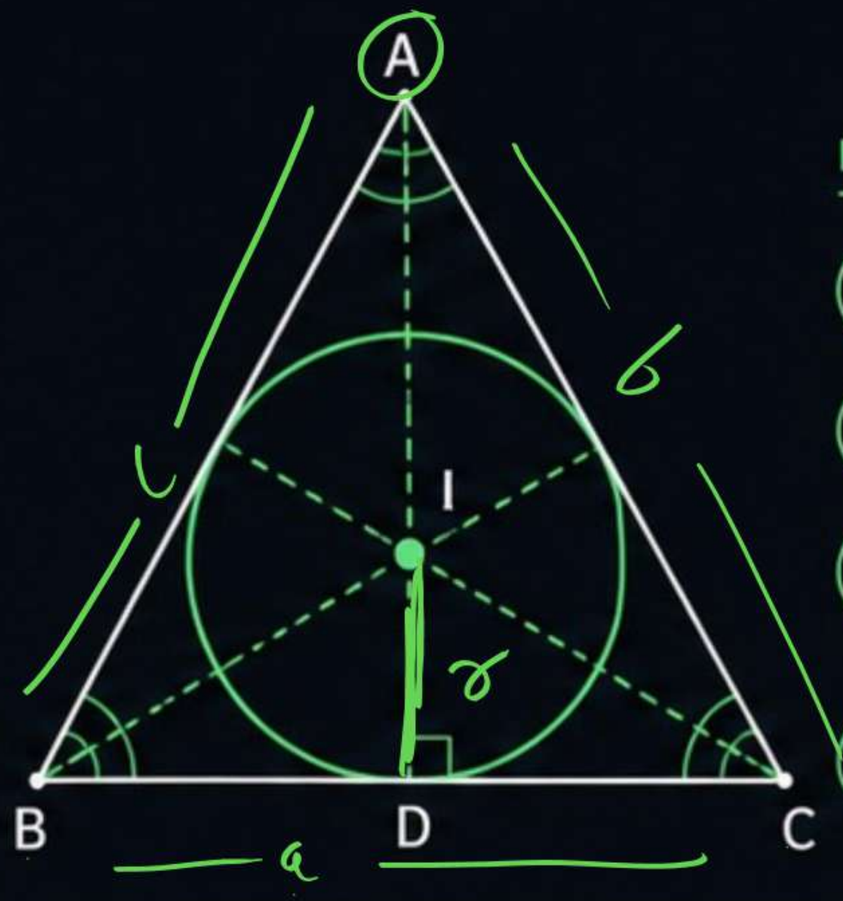


INCIRCLE

The circle that touches all three sides of a triangle from the inside.

CONSTRUCTION

1. Draw the angle bisectors of any two angles of the triangle.
2. Mark their intersection point I (incenter).
3. From I, drop a perpendicular to any side. Let the foot be D.
4. With center I and radius ID, draw the circle. It will touch all three sides.



I is called the incenter and $r = ID$ is the inradius.

IMPORTANCE / KEY PROPERTIES

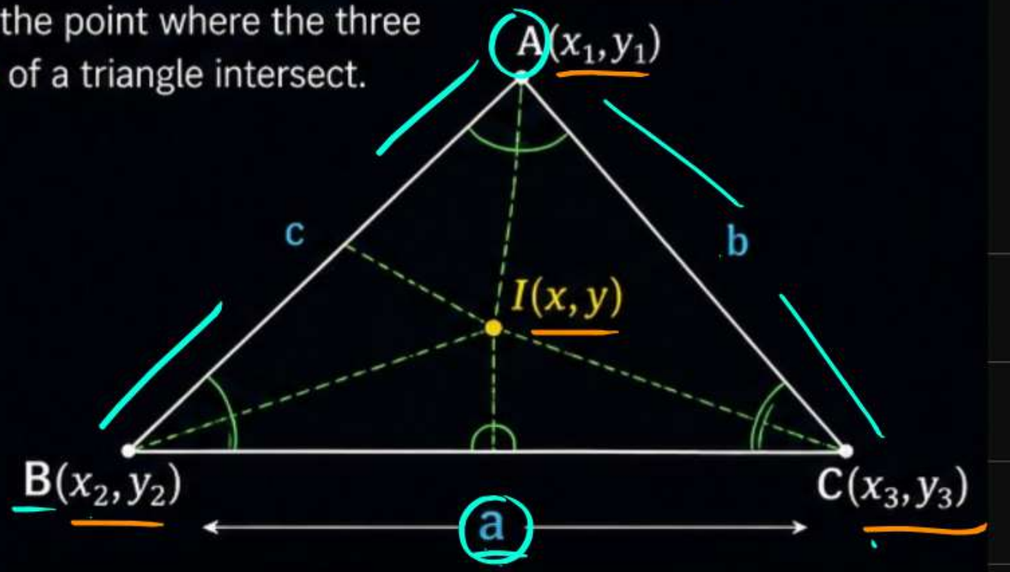
- I is equidistant from all three sides.
- The incircle touches each side at exactly one point.
- Area of triangle = r × s where r = inradius, s = semiperimeter.
- The incircle is unique for every triangle.

$s = \frac{a+b+c}{2}$

$A_\Delta = r \cdot s$

INCENTER OF A TRIANGLE

The incenter is the point where the three angle bisectors of a triangle intersect.



FORMULA

Let the sides opposite to vertices $A(x_1, y_1)$, $B(x_2, y_2)$, $C(x_3, y_3)$ be a = BC, b = CA, c = AB.

Coordinates of Incenter (I) :

$$I(x, y) = \left[\frac{ax_1 + bx_2 + cx_3}{a + b + c} , \frac{ay_1 + by_2 + cy_3}{a + b + c} \right]$$

where a = BC, b = CA, c = AB

Q. The three vertices of a triangle are (8,-4), (10,10) and (-4,8) then find the coordinates of the incentre of the triangle.

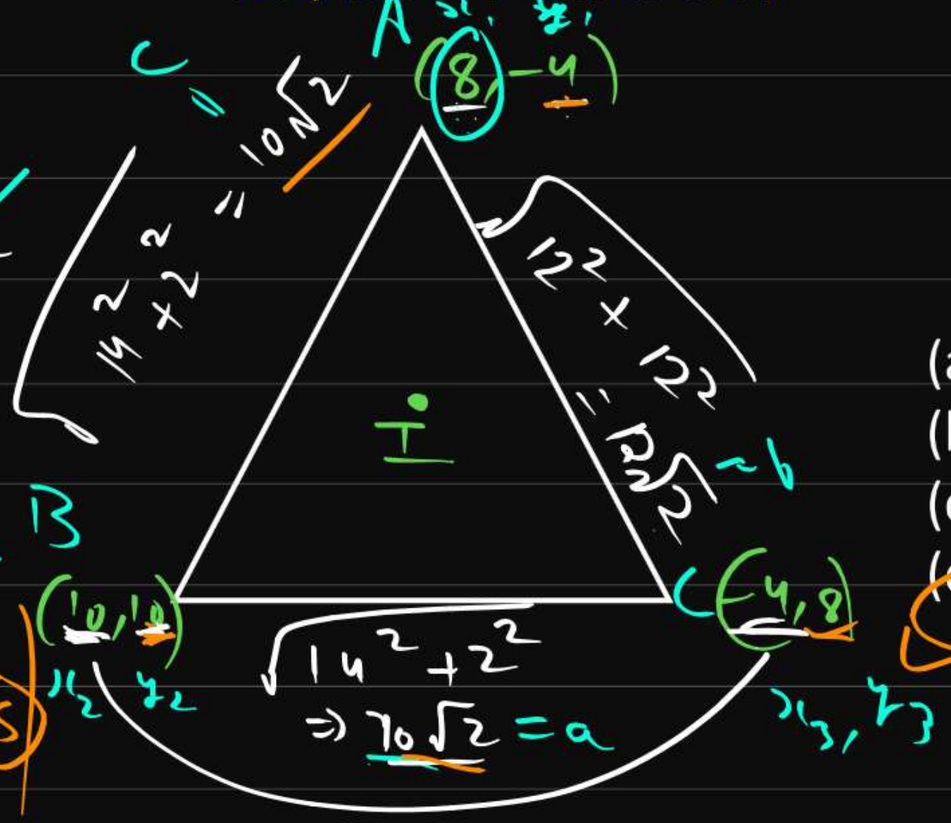
त्रिभुज के तीन शीर्षांक हैं (8, -4), (10,10) और (-4,8) फिर त्रिभुज के अन्तः केंद्र के निर्देशांक का पता लगाएं।

$$x = \frac{ax_1 + bx_2 + cx_3}{a+b+c}$$

$$= \frac{8 \times 10\sqrt{2} + 10 \times 12\sqrt{2} + -4 \times 10\sqrt{2}}{10\sqrt{2} + 12\sqrt{2} + 10\sqrt{2}}$$

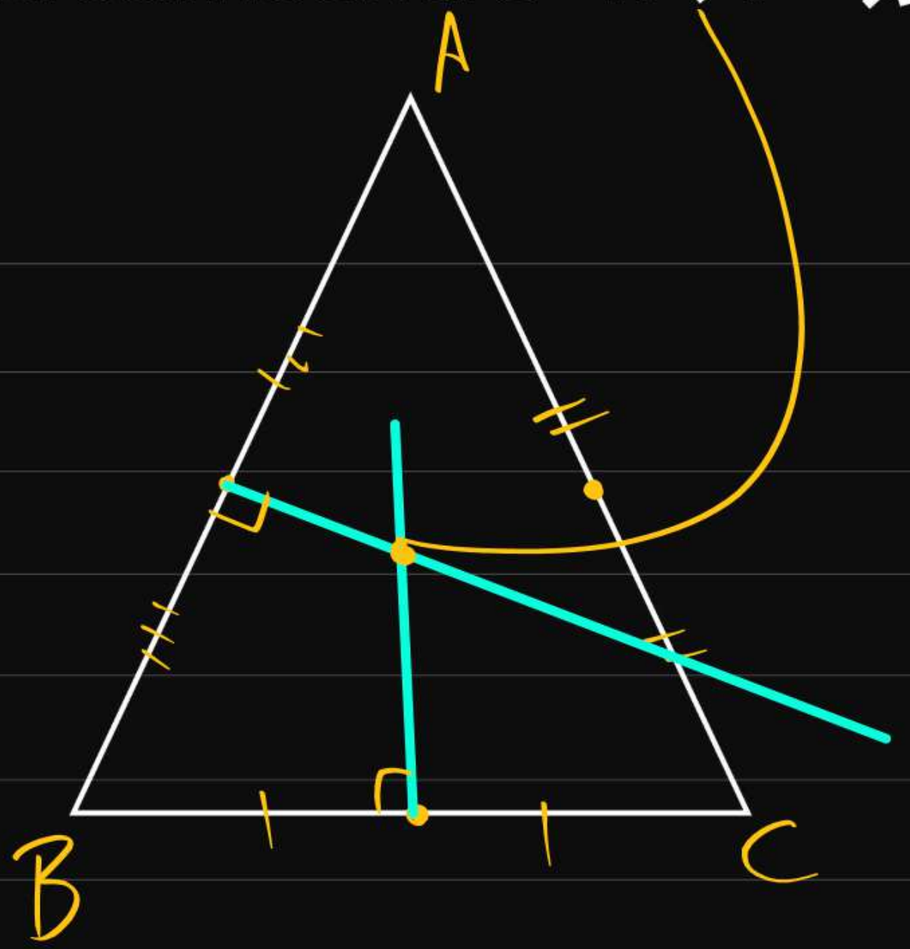
$$\Rightarrow \frac{160}{32} = 5$$

$$y = \frac{-40\sqrt{2} + 120\sqrt{2} + 80\sqrt{2}}{32\sqrt{2}} \Rightarrow \frac{160}{32} = 5$$



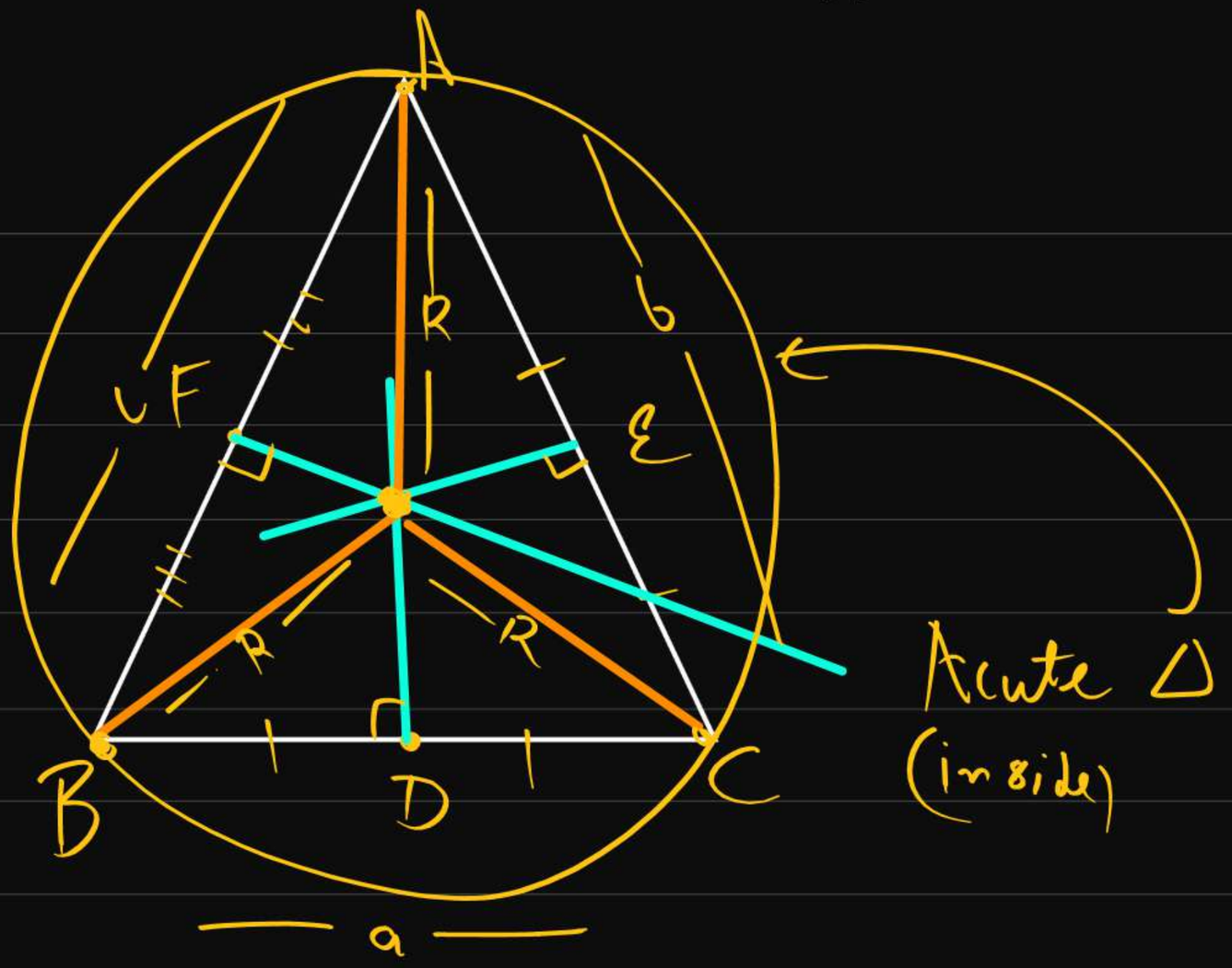
(a) (6,4)
(b) (4,5)
(c) (5,3)
(d) (5,5)

Circumcentre परिकेन्द्र



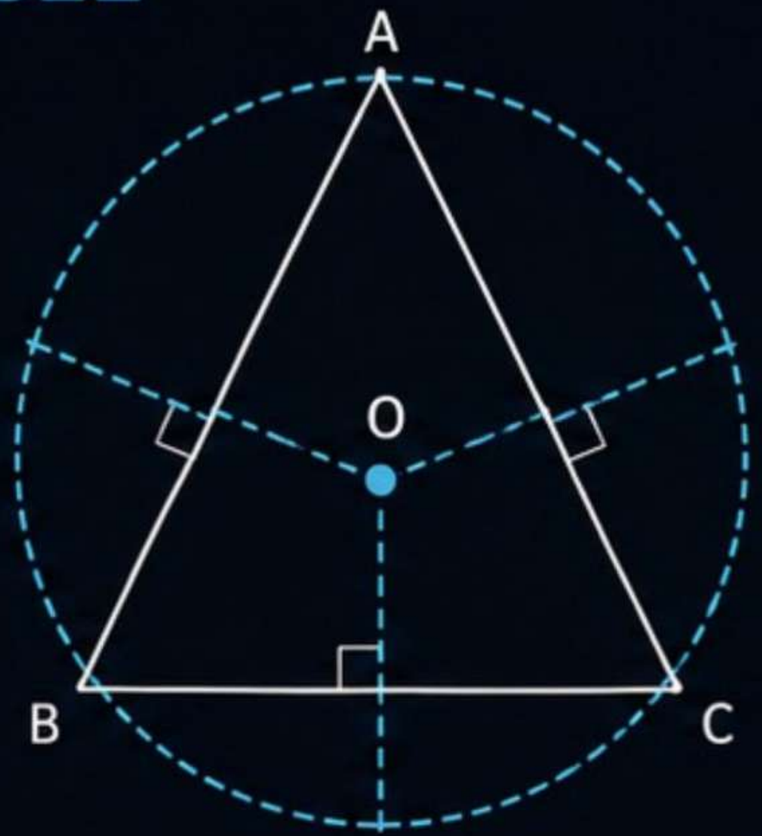
Circumcentre परिकेन्द्र

$$A = \frac{abc}{4R}$$



CIRCUMCIRCLE

The circle that passes through all three vertices of a triangle.



CONSTRUCTION

1. Draw the perpendicular bisectors of any two sides of the triangle.
2. Mark their intersection point O (circumcenter).
3. With center O and radius OA (or OB or OC), draw the circle.

IMPORTANCE / KEY PROPERTIES

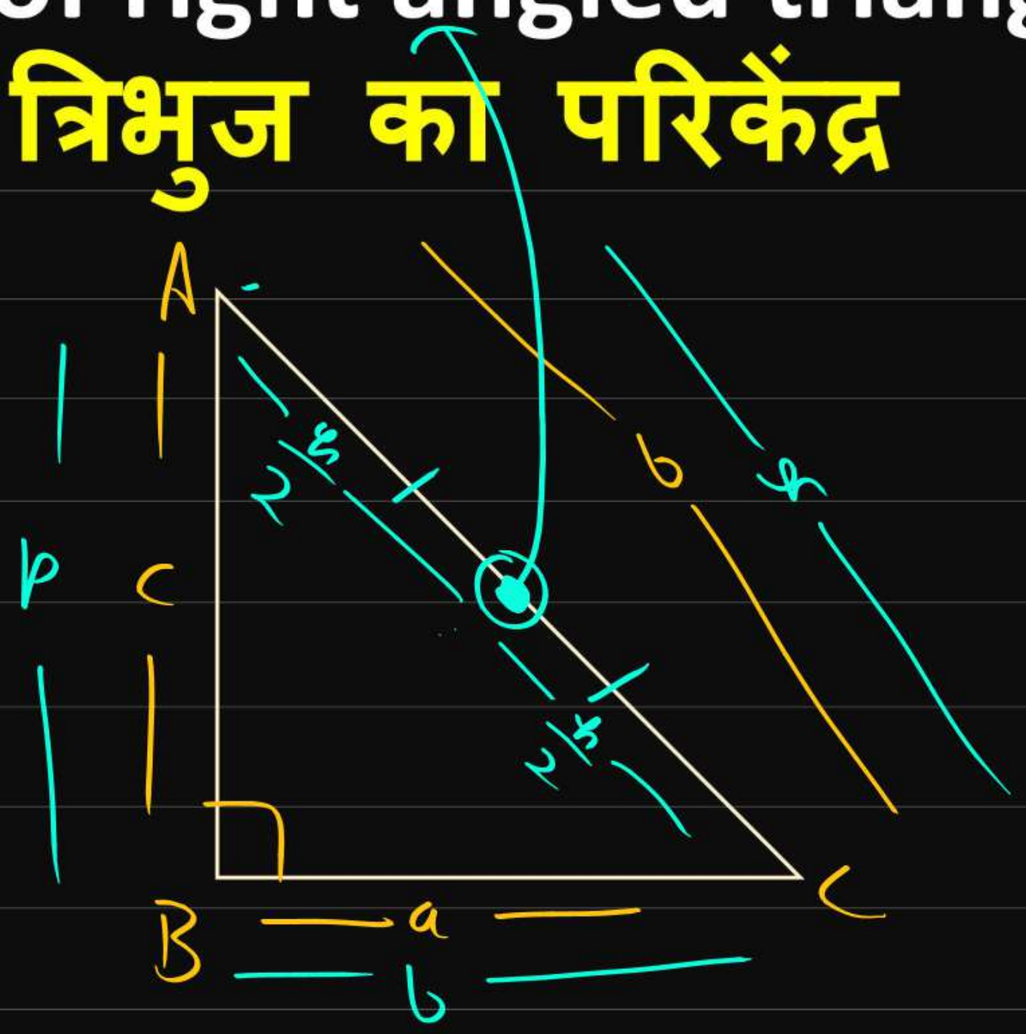
- O is equidistant from all three vertices.
- The circumcircle passes through A, B and C.
- Radius $R = OA = OB = OC$.
- For an acute triangle, O is inside the triangle.
- For a right triangle, O is the midpoint of the hypotenuse.
- For an obtuse triangle, O is outside the triangle.

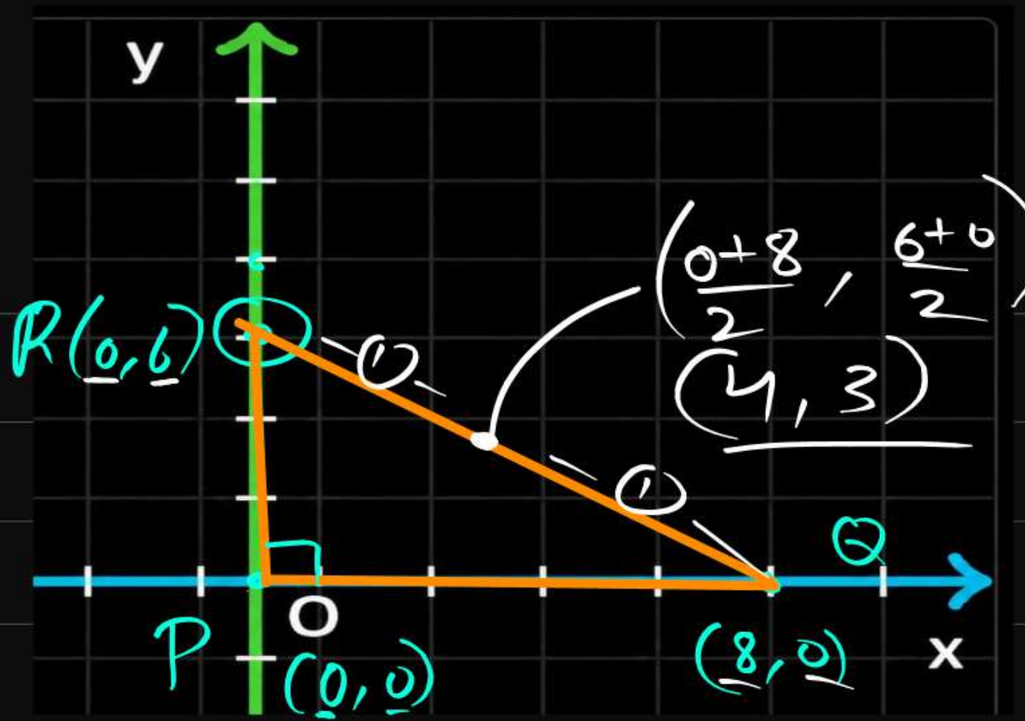
O is called the circumcenter and R is the circumradius.

Circumcentre of right angled triangle

समकोण त्रिभुज का परिकेंद्र

$$R = \frac{h}{2}$$





Q. A right triangle has vertices at P(0,0), Q(8,0), and R(0,6). What are the coordinates of its Circumcenter?

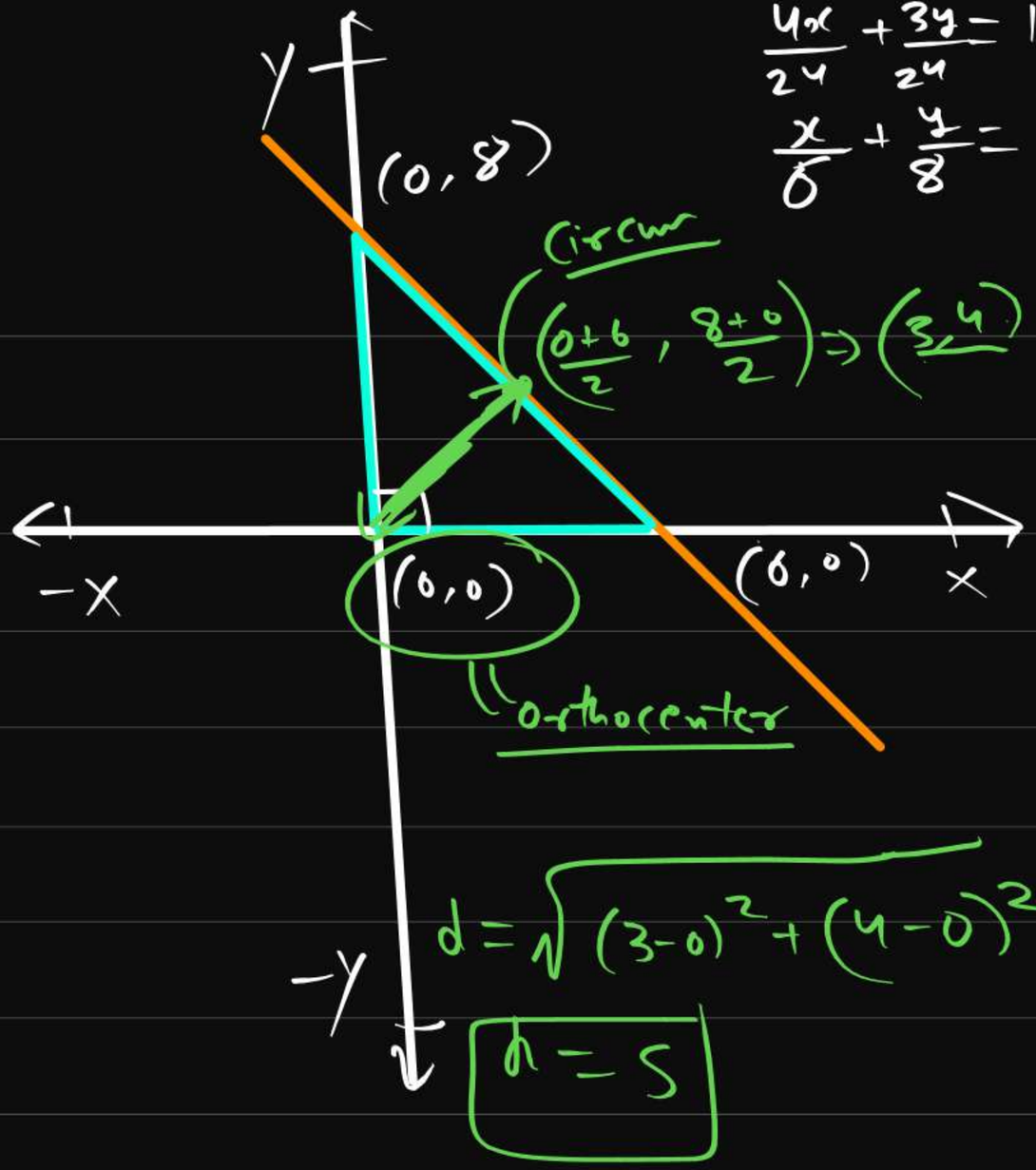
एक समकोण त्रिभुज के शीर्ष P(0,0), Q(8,0) और R(0,6) पर हैं। इसके परिकेंद्र के निर्देशांक क्या हैं?

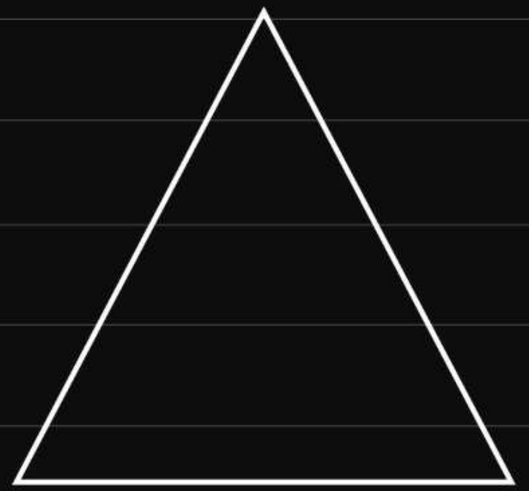
SSC CGL 2025

(a) (4, 3) (b) (3, 4)
(c) (4, 4) (d) (2, 2)

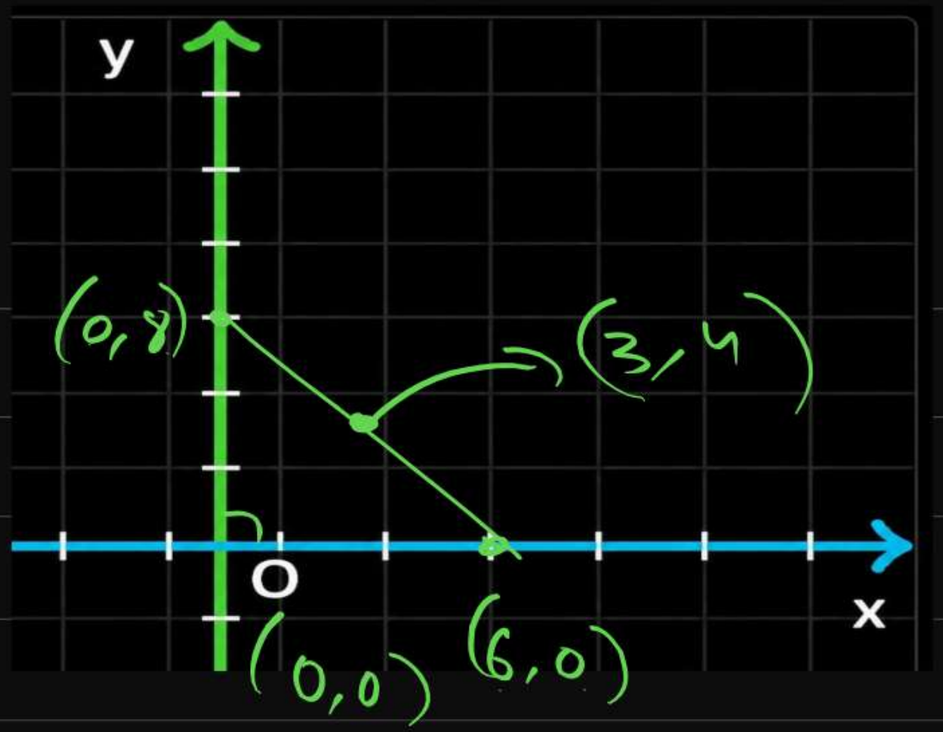
Q. A straight line $4x + 3y = 24$ forms a triangle with the coordinate axes. What is the distance between the orthocentre of the triangle and the centre of the circle that circumscribes the triangle?

एक सीधी रेखा $4x + 3y = 24$ समन्वित अक्षों के साथ एक त्रिकोण बनाती है। त्रिभुज के परिकेन्द्र और लंबकेन्द्र के बीच की दूरी क्या है?





A. 10 units
B. 5 units
C. 13 units
D. 12 units



Q. A triangle has vertices at A(0, 0), B(6, 0), and C(0, 8). What are the coordinates of its Circumcenter ?

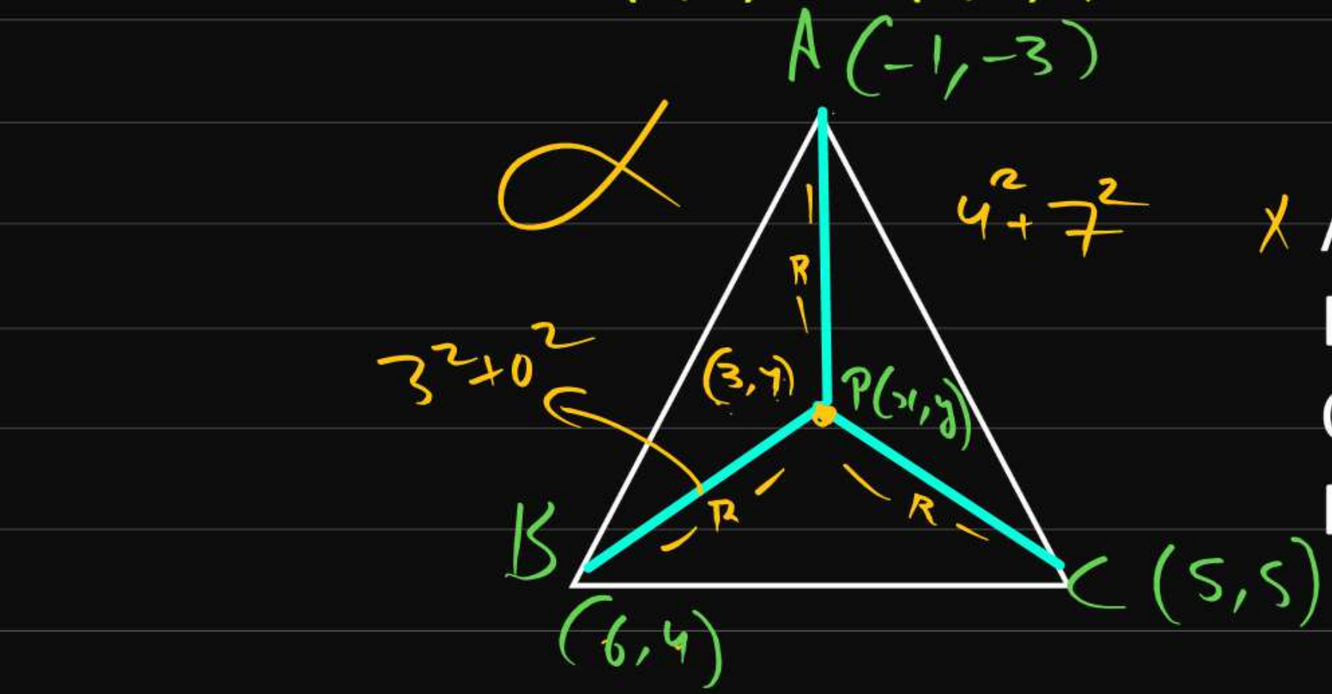
एक त्रिभुज के शीर्ष A(0, 0), B(6, 0) और C(0, 8) पर हैं। इसके परिकेन्द्र के निर्देशांक क्या हैं?

SSC CPO 2025

(a) (4, 3) (b) (3, 5) (c) (3, 4) (d) (4, 5)

Q. Find the coordinates of circumcentre of a triangle whose vertices are (-1,-3), (6,4) and (5,5).

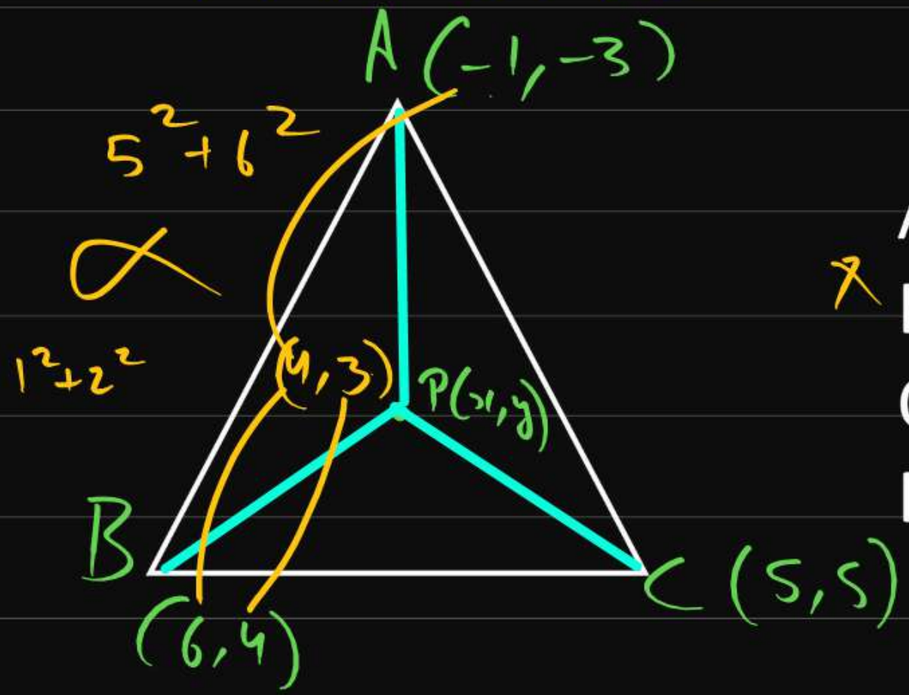
त्रिभुज की परिकेन्द्र के निर्देशांक ज्ञात कीजिए जिनके शीर्षांक (-1,-3), (6,4)और (5,5) हैं।



✗ A) (3,4)
B) (4,3)
C) (2,1)
D) (1,2)

Q. Find the coordinates of circumcentre of a triangle whose vertices are (-1,-3), (6,4) and (5,5).

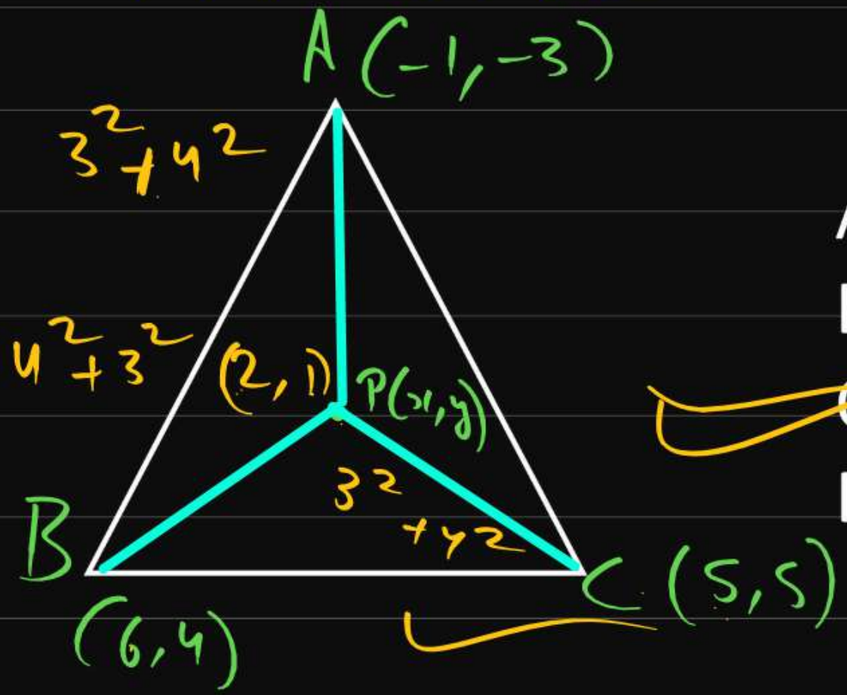
त्रिभुज की परिकेन्द्र के निर्देशांक ज्ञात कीजिए जिनके शीर्षांक (-1,-3), (6,4)और (5,5) हैं।



A) (3,4)
B) (4,3)
C) (2,1)
D) (1,2)

Q. Find the coordinates of circumcentre of a triangle whose vertices are (-1,-3), (6,4) and (5,5).

त्रिभुज की परिकेन्द्र के निर्देशांक ज्ञात कीजिए जिनके शीर्षांक (-1,-3), (6,4)और (5,5) हैं।



A) (3,4)
B) (4,3)
C) (2,1)
D) (1,2)

Q. Find the coordinates of circumcentre of a triangle whose vertices are (-1,-3), (6,4) and (5,5).

त्रिभुज की परिकेन्द्र के निर्देशांक ज्ञात कीजिए जिनके शीर्षांक (-1,-3), (6,4)और (5,5) हैं।

$R^2 = (x-6)^2+(y-4)^2 = (x+1)^2+(y+3)^2 = (x-5)^2+(y-5)^2$

x^2+y^2

$36 - 12x + 16 - 8y = 1 + 2x + 9 + 6y$

x^2+y^2

$25 - 10x + 25 - 10y$

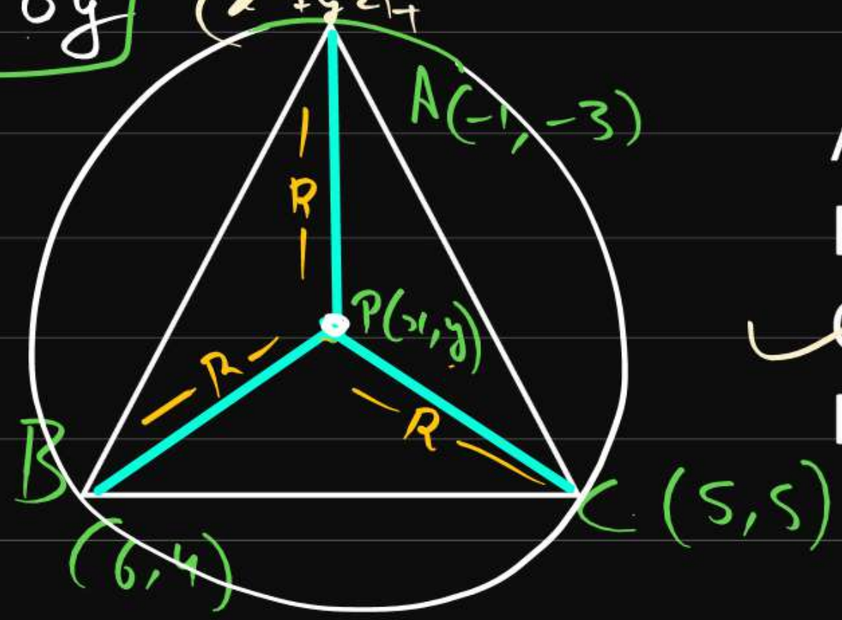
$14x + 14y = 42$

$x + y = 3$) $\times 4$

$12x + 12y = 40$) $(-)$

$3x + 4y = 10$

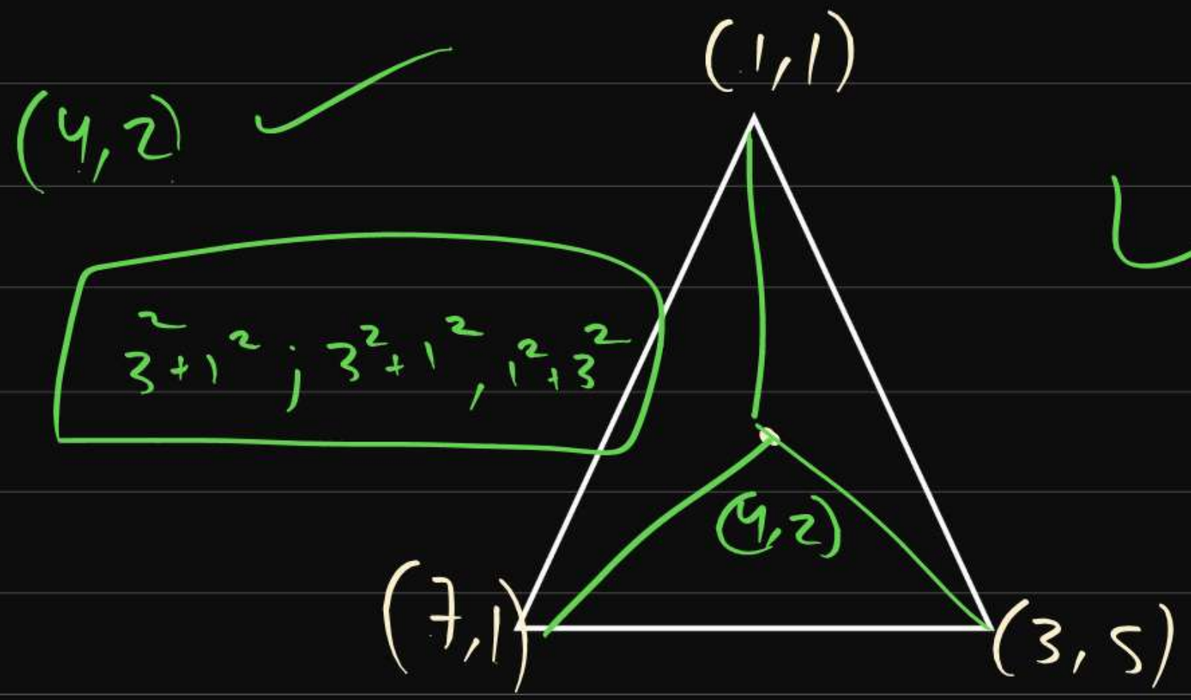
$x = 2 ; y = 1$



A) (3,4)
B) (4,3)
C) (2,1)
D) (1,2)

Q. Find the coordinates of the circumcentre of a triangle whose vertices are (1, 1), (7, 1) and (3, 5).

उस त्रिभुज के परिकेंद्र के निर्देशांक ज्ञात कीजिए जिसके शीर्ष (1, 1), (7, 1) और (3, 5) हैं।



(a) (4, 2)
(b) (4, 1)
(c) (3, 2)
(d) (3, 1)

INCIRCLE

The circle that touches all three sides of a triangle from the inside.

CONSTRUCTION

1. Draw the angle bisectors of any two angles of the triangle.
2. Mark their intersection point I (incenter).
3. From I, drop a perpendicular to any side. Let the foot be D.
4. With center I and radius ID, draw the circle. It will touch all three sides.

IMPORTANCE / KEY PROPERTIES

- I is equidistant from all three sides.
- The incircle touches each side at exactly one point.
- Area of triangle = r × s where r = inradius, s = semiperimeter.
- The incircle is unique for every triangle.

I is called the incenter and $r = ID$ is the inradius.

CIRCUMCIRCLE

The circle that passes through all three vertices of a triangle.

CONSTRUCTION

1. Draw the perpendicular bisectors of any two sides of the triangle.
2. Mark their intersection point O (circumcenter).
3. With center O and radius OA (or OB or OC), draw the circle.

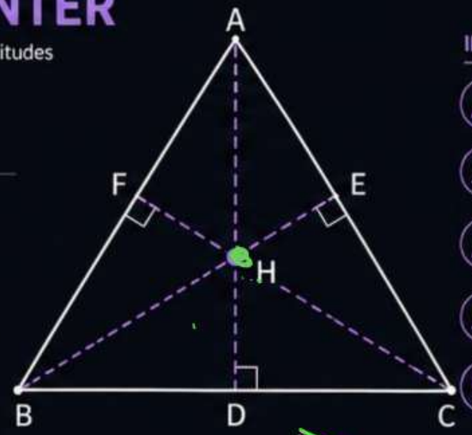
IMPORTANCE / KEY PROPERTIES

- O is equidistant from all three vertices.
- The circumcircle passes through A, B and C.
- Radius $R = OA = OB = OC$.
- For an acute triangle, O is inside the triangle.
- For a right triangle, O is the midpoint of the hypotenuse.
- For an obtuse triangle, O is outside the triangle.

O is called the circumcenter and R is the circumradius.

ORTHOCENTER

The point where all three altitudes of a triangle intersect.



CONSTRUCTION

1. From vertex A, draw an altitude to side BC.
2. From vertex B, draw an altitude to side AC.
3. The intersection of these two altitudes is H (orthocenter).
4. (The third altitude from C will also pass through H.)

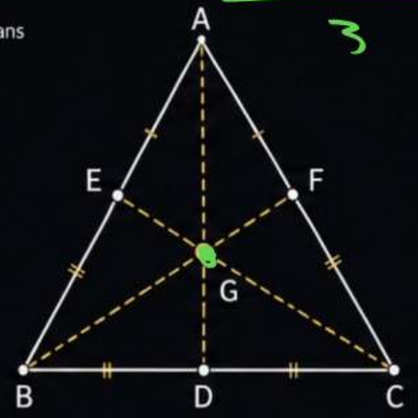
IMPORTANCE / KEY PROPERTIES

- H is the intersection point of the three altitudes.
- In an acute triangle, H is inside the triangle.
- In a right triangle, H is at the right-angled vertex.
- In an obtuse triangle, H is outside the triangle.
- The orthocenter need not lie inside the triangle.

H is called the orthocenter.

CENTROID

The point where all three medians of a triangle intersect.



CONSTRUCTION

1. Find the midpoint of side BC. Let it be D.
2. Draw the median AD.
3. Find the midpoint of side AC. Let it be E.
4. Draw the median BE.
5. Their intersection point is G (centroid). The third median from C also passes through G.

IMPORTANCE / KEY PROPERTIES

- G divides each median in the ratio 2:1.
- $AG = \frac{2}{3}AD, \quad GD = \frac{1}{3}AD$ (similarly for the other medians).
- The centroid is always inside the triangle.
- It is the center of gravity of the triangle (balance point).
- The centroid is unique for every triangle.

G is called the centroid.

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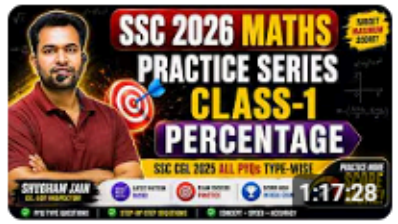

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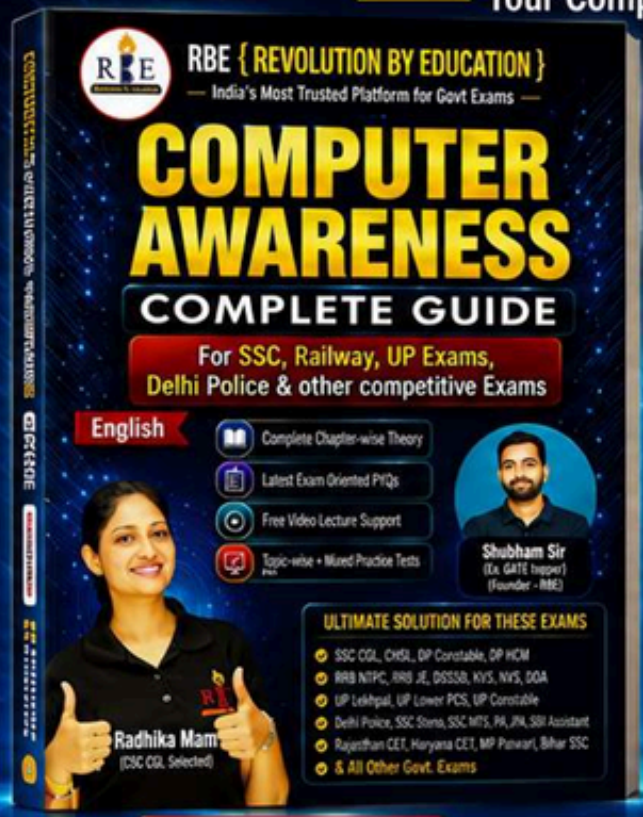

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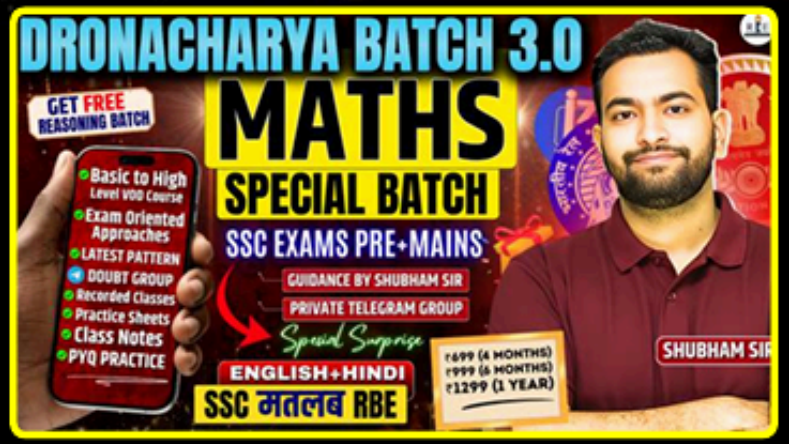

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